



TrustPLC CTSC-200 Series PLC

User Manual

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Shenzhen co-trust Technology Co.,Ltd.

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We have checked the contents of this manual for agreement with the hardware and software described. Since deviations cannot be precluded entirely, we cannot guarantee full agreement. However, the data in this manual are reviewed regularly and any necessary corrections included in subsequent editions. Suggestions for improvement are welcomed.

Safety Guidelines

Only qualified person are allowed to install, operate and maintenance on TrustPLC CTSC-200 Series PLC. Qualified persons are defined as persons who are authorized to commission, ground, tag circuits, equipment, and systems in accordance with established safety practices and standards.

COTRUST has no responsibility for any consequence caused by using this product.

This manual contains notices that you should observe to ensure your personal safety, as well as to protect the product and connected equipment. These notices are highlighted in the manual by a warning triangle and are marked as follows according to the level of danger:



Warning

Warning indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



Caution

Caution used with the safety alert symbol indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.



Notice

Notice indicates the supplementation and description for specific operation.

Precautions for Use

Safety circuit must be used to ensure the operation security of PLC system when the external power off or PLC failed. Notices in design for use including:

Emergency brake circuit, safety circuit, Positive and reverse operation interlock circuit and the Up/Down threshold interlock switch of position for preventing device damage must be implemented in the external circuit of PLC.

The external safeguard circuit and safety mechanism must be designed for the output signal which indicating major accident, to ensure the device operation security.

The PLC can detect system exceptions which resulting all output turned off, to ensure the device operation security, add appropriate external control circuits.

When the Relay or Resistor Unit in PLC was damaged, the ON/OFF for PLC Outputs cannot be controlled.

The Power system should fitted with lightning protection device, to prevent the PLC interfaces, like power/signal inputs and controller outputs from being damaged by lightning over-voltage.

Installation Notes

Do not install PLC at places as: dust, smut, conductive dust, corrosive gas, combustible gas, vibration and shock. Do not expose the PLC into high temperature, moisture condensation, raining, thunderstrike, fire or disoperation can also cause damage to the product.

Prevent the metal filing and cable outlet falling into the PLC ventilation hole when wiring and tightening the screws, or other which could cause fire, failure and misoperation.

After installing PLC, there must be no sundries in case of fire, fault and misoperation caused by bad heat dissipation.

Hot plugging is not allowed, shield cable must be used to increase immunity of inference.

Wiring Notes

Before Installation and wiring, all external powers must be cut-off, otherwise there maybe electric shock and device damage.

Cover the terminal plate before powering on, follow the instructions in this manual to connect power. PLC in/out signals wiring cannot be parallel with other High voltage or inference lines, they should be layout in separate slots.

Separate the PG terminal on CPU with high voltage.

Operation and Maintenance

Do not touch the terminals with power on, in case of electric shock and device damage.

Turn off the power and then clean and tighten screws, connect and disconnect communication cable.

Do not dismantle the controller in case of damaging the internal parts.

Please reading this manual before modifying program, start, test and stop the CPU.

Product Obsolescence

The combustion of electrolytic capacitor on PCB can cause explosion, the main material on PLC is plastic which would generate toxic gas when burnt.

Please follow the local environment regulations to process the abandoned products.

Preface

Thank you for choosing TrustPLC CTSC-200 PLC. Please reading this document carefully before using, so that you could use it more securely and utilizing the abundant functions.

This manual concludes hardware specifications, features and installation of TrustPLC CTSC-200 PLC and its expansion modules, besides, you can also refer to the instructions set and order number.

Applicable Objects

All information about TrustPLC CTSC-200 PLC installing and debugging are oriented for Engineers, Installation technician, maintainer and automation qualified electrician.

Terminology

Abbreviation	Description
PLC	Programmable Logic Controller
STL	Statement List
LAD	Ladder Logic
FBD	Function Block Diagram
AI	Analog input
AQ	Analog output
HC	High-speed counter
RTD	Resistance Temperature Detector

Online Help

For related product materials and technical support, please visit: <http://www.co-trust.com>

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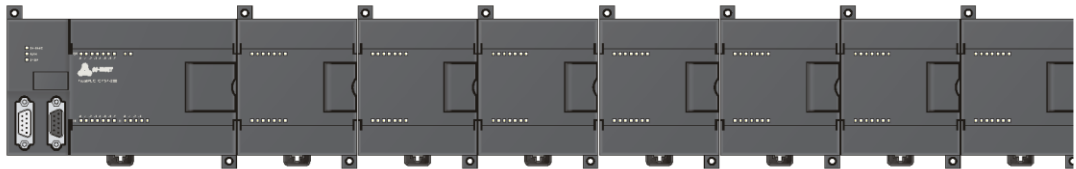
1 Product Overview

1.1 System Overview

TrustPLC CTSC-200 is a powerful PLC system, which can meet various needs for automatic control. Thanks to its compact design, good scalability, low cost, and powerful instructions, which allows TrustPLC CTSC-200 to meet the small and medium-sized control needs perfectly.

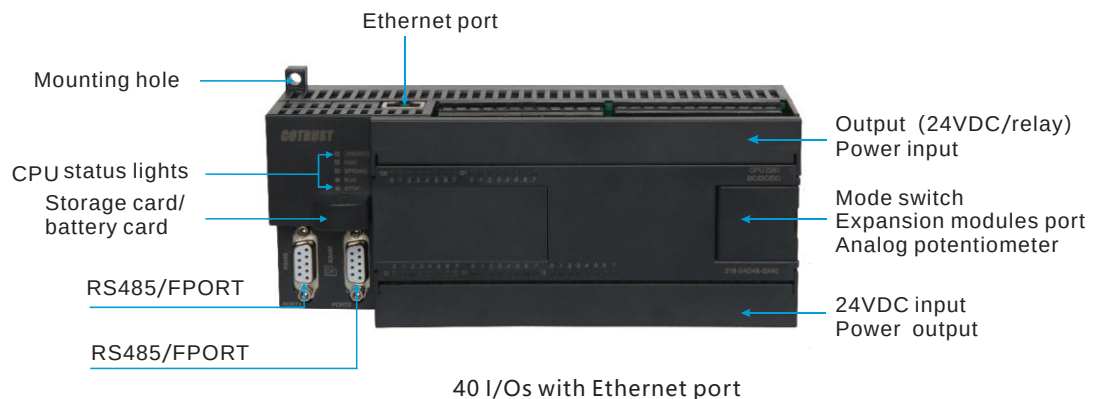
TrustPLC CTSC-200 series products offers you various CPU and modules, the abundant CPU types and voltage classes help you to solve industrial automation problems.

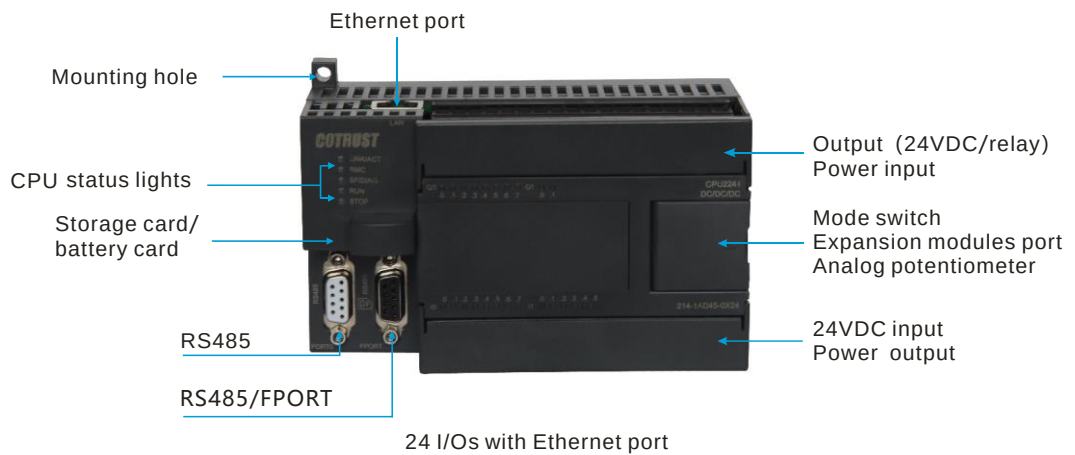
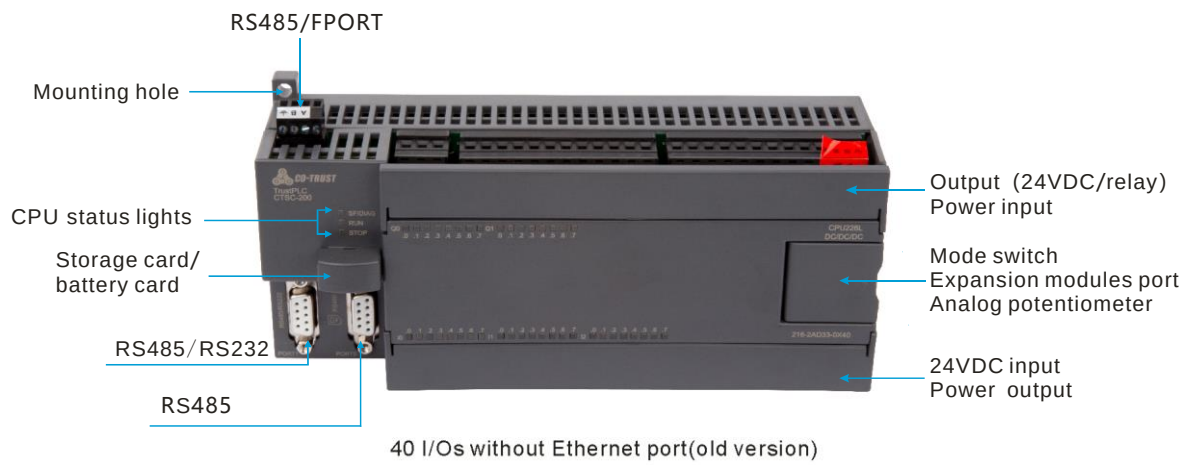
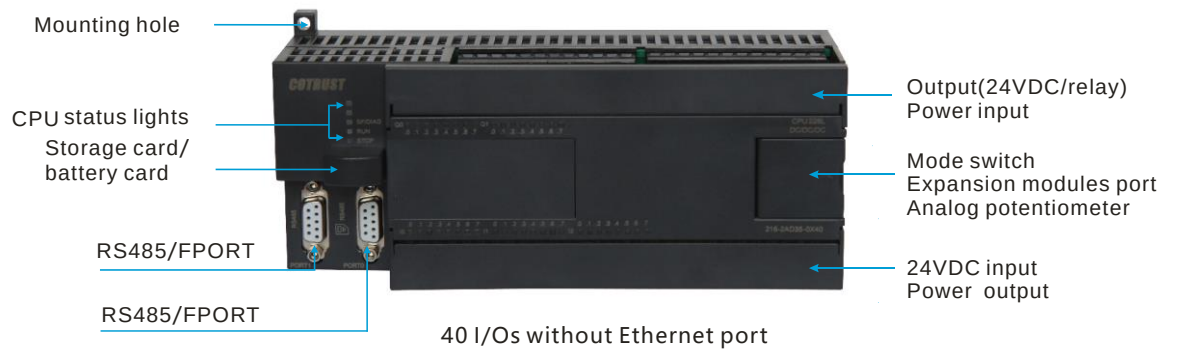
TrustPLC CTSC-200 system can be composed of a single CPU, or composed of one CPU and seven extension modules.



1.2 Introduction of the CPU

TrustPLC CTSC-200 CPU is a small robust programmable controller that encapsulates microprocessor, power supply and multiple I/Os, for which can extend up to 7 modules can be extended.





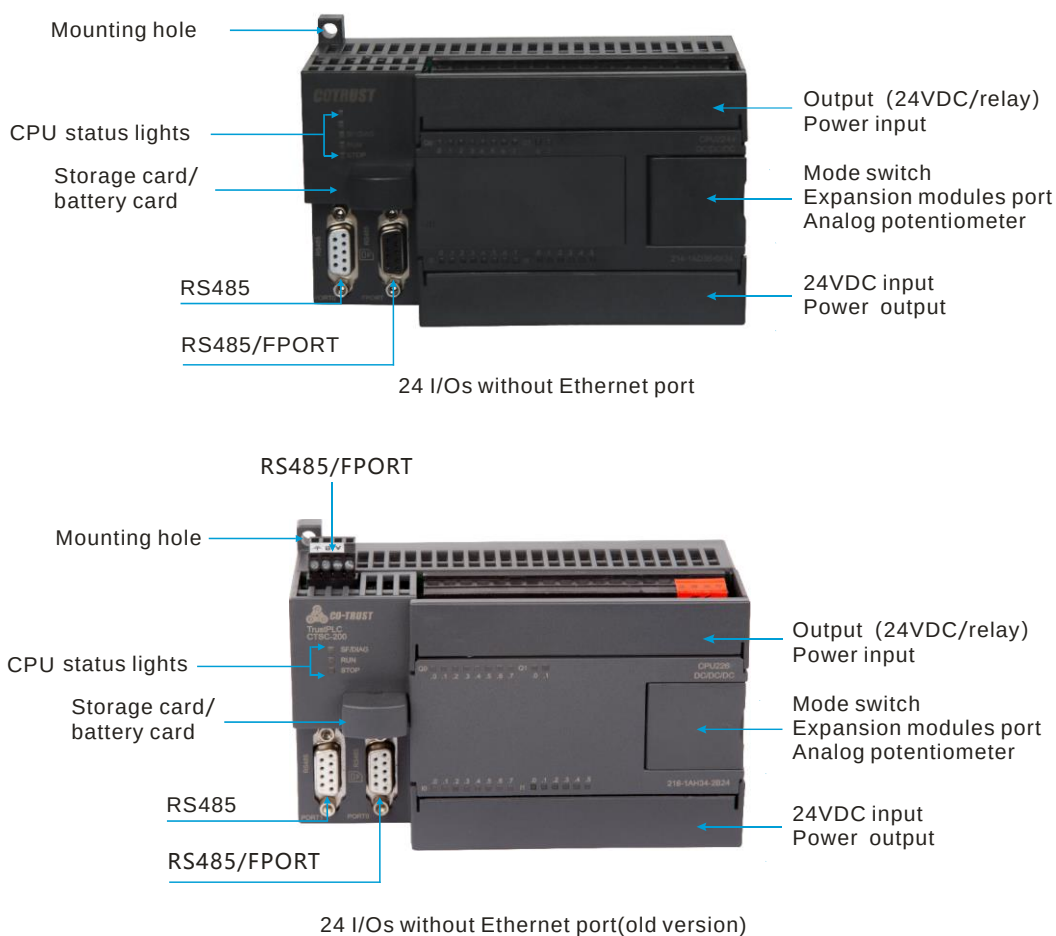


Figure 1-1 Structure of CTSC-200 CPU

Table 1-1 Introduction of CPU

Name	Order No.	Description
CPU 224+	Transistor: CTS7 214-1AD33-0X24	Integrates two COM ports: one PPI port, one Freeport (FPORT), 14DI/10DO; 16KB program/110KB data (permanent retention); six isolated 30KHz HSCs, two isolated 50KHz motion outputs or two 20KHz PTO/PWM outputs (not support synchronously use)
	Relay: CTS7 214-1BD33-0X24	
CPU 224+ V5 (Upgrade)	Transistor: CTS7 214-1AD35-0X24	Integrates two COM ports: one PPI port, one FPORT, integrates communication protocols, such as MODBUS; 14DI/10DO; 16KB program/110KB data; six isolated 30KHz HSCs, two 50KHz outputs or two isolated 20KHz PTO/PWM outputs(not support synchronously use) , support extension board.
	Relay: CTS7 214-1BD35-0X24	
CPU 224E	Transistor: CTS7 214-1AE33-0X24	Integrates two COM ports: one PPI port, one FPORT; 14DI/10DO; 16KB program/108KB data; 72KB program/110KB data; two isolated 30KHz HSCs, two isolated 50KHz outputs.
	Relay: CTS7 214-1BE33-0X24(shut down))	
CPU 224I	Transistor: CTS7 214-1AD41-0X24	Integrates three COM ports: one PPI port, one Freeport (FPORT), 1 Ethernet port; 14DI/10D; 16KB

	Relay: CTS7 214-1BD41-0X24	program/108KB data; six isolated 30KHz HSCs, two isolated 50KHz HSP outputs
CPU224XP	Transistor: CTS7 214-2AD41-0324	Integrates three COM ports: two PPI/FPORT, one Ethernet port; 14DI/10DO; 2AI/1AQ, 72KB program/110KB data; six isolated 30KHz HSCs, two 50KHz outputs (Pulse/Dir or PTO/PWM)
	Relay: CTS7 214-2BD41-0324	
CPU226I	Transistor: CTS7 216-2AD41-0X40	Integrates three COM ports: two PPI/FPORT, one Ethernet port; 24DI/16DO; 72KB program/110KB data; six isolated 30KHz HSCs, two isolated 50KHz HSP outputs.
	Relay: CTS7 216-2BD41-0X40	
CPU 226M	Transistor: CTS7 216-1AD33-0X24	Integrates three COM ports: one PPI port, one PPI/FPORT, one FPORT; 14DI/10DO; 72KB program/110KB data; six isolated 30KHz HSCs, two isolated 20KHz HSP outputs.
	Relay: CTS7 216-1BD33-0X24	
CPU 226M V5 (Upgrade)	Transistor: CTS7 216-1AD35-0X24	Integrates two COM ports: two PPI/FPORT; 24DI/16DO; 72KB program/110KB data; six isolated 30KHz HSCs, two isolated 20KHz HSP outputs, support extension board.
	Relay: CTS7 216-1BD35-0X24	
CPU 226L	Transistor: CTS7 216-2AD33-0X40	Integrates three COM ports: one PPI port, one PPI/FPORT, one FPORT; 24DI/16DO; 72KB program/110KB data; six isolated 30KHz HSCs, two isolated 20KHz HSP outputs
	Transistor: CTS7 216-2AF33-0X40(shut down)	
	Relay: CTS7 216-2BD33-0X40	
	Relay: CTS7 216-2BF33-0X40(shut down)	
CPU 226L V5 (Upgrade)	Transistor: CTS7 216-2AD35-0X40	Integrates two COM ports: two PPI/FPORT; 24DI/16DO; 72KB program/110KB data; six isolated 30KHz HSCs, two isolated 20KHz HSP outputs, support extension board.
	Relay: CTS7 216-2BD35-0X40	
CPU 226H	Transistor: CTS7 216-1AH33-0X24(shut down)	Integrates three COM ports: one PPI port, one PPI/FPORT, one FPORT; 14DI/10DO; 72KB program/110KB data; six isolated 200KHz HSCs, 4-axis 200KHz HSP outputs (Pulse/Dir). Support relative/absolute position, 2-axis liner/circular interpolation, Homing instructions etc., Which can be used for certain complex control functions.
	Transistor: CTS7 216-1AH33-0A24(shut down)	
	Transistor: CTS7 216-1AH33-0B24(shut down)	
	Transistor: CTS7 216-1AH33-1A24(shut down)	
	Transistor: CTS7 216-1AH33-1B24(shut down)	
	Transistor: CTS7 216-1AH34-0X24(shut down)	

	Transistor: CTS7 216-1AH34-0B24	
	Transistor: CTS7 216-1AH34-1B24	
	Transistor: CTS7 216-1AH34-2B24	
CPU 226M-CAN	Transistor: CTS7 216-1AC33-0X24	
CPU 226H V5 (Upgrade)	Transistor: CTS7 216-1AH35-2B24	Integrates two COM ports: two PPI/FPORT; 14DI/10DO; 72KB program/110KB data; six isolated 200KHz HSCs, four isolated 200KHz HSP outputs, support extension board but not PLS instruction.
	Transistor: CTS7 216-1AH35-0B24	
Dedicated controller	Transistor: CTSC 266-3AT33-2424	TrustPLC CTSC-200 PSC266 Dedicated controller, 72KB program/110KB data, 24VDC, 20TC, 4AQ, 14DI/10DO, integrates three COM ports.

TrustPLC CTSC-200 CPU Features:

Reliability and Stability

- High ESD/EFT, tested by strict industrial environment, CE certified
- Three proofings processed, suited for various harsh industrial environment
- User program and data can be kept permanently
- Multiple password protection, one-way downloading for kernel program, permanent confidential
- Anti-thunder for COM ports, with high reliability
- Build-in real-time clock

Robust communication capability

- Integrate 2-3 COM ports
- Support MPI, PPI, DP, Freeport, ModBus and Ethernet (only for CPU224I) communication protocols
- Exchanging 200Bytes per read-write operation, enhanced networking communication

High intelligence

- Integrated parameter self-tuning fuzzy logic algorithm and temperature control PID library, with high accuracy and dynamic performance
- Build-in plenty of integrated functions with simplified programming and flexible control
- High-speed closed loop can be delivered to support some high-speed system application

- Abundant of motion control functions, suitable for some synchronization and positioning applications

High speed and high-capacity

- Logical operation rate: 0.15μs
- Floating-point calculation rate: 8μs
- High program capacity: 12K—72Kbytes
- Large data space: 8K—10K, max 110Kbytes
- High-speed Input counter: 6, max. 200KHz
- High speed pulse output: Quad, max. 200KHz

Ultra system expansion

- Analog I/O: up to 56AI/28AQ
- Digital I/O: up to 248 I/Os
- Up to 7 Extended I/O modules

Fast Programming

- Support IEC61131 programming language, LAD, STL and FBD
- Variety programming language.
- Intelligent help.

Abundant instructions and wizards

- 144 Basic Instructions; 102 Applied instructions
- Floating-point operation instruction
- PID instruction wizard
- Motion Control wizard
- High-speed input wizard
- Wizard for data exchanging between CPUs
- HMI wizard

1.3 Expansion Module

TrustPLC CTSC-200 CPU offers local I/Os, but various expansion modules can also be utilized to provide more additional I/Os and communication functions. The expansion modules include digital I/O modules, analog I/O modules, temperature collection modules and DP communication modules etc. These modules each consists of different I/O number, used for configuring various scale of I/Os with high cost performance. All modules are installed with standard DIN35 rail.

Table 1-2 TrustPLC CTSC-200 expansion module

Expansion module	Types	Specification
Digital	Input module	8 DI, 16 DI, 32 DI
	Output module	8 DO, 16 DC outputs, 32 DC outputs, 8 RLY outputs, 16 RLY outputs
	In/Out module	4 DI/4DO, 8DI/8DO, 16DI/16DO 4 DI/4 RLY outputs, 8 DI/8 RLY outputs, 16 DI/16 RLY outputs
Analog	Input module	4AI×12BIT, 2AI×16BIT, 4AI×16BIT, 8AI×16BIT, 8AI × 12BIT
	Output module	2AQ×12BIT, 4AQ×12BIT
	In/Out module	4AI/1AQ×12BIT
	TC input module	4AI×TC, 8AI×TC
	RTD input module	2AI×RTD, 4AI×RTD, 2Pt100/NTC+2AI×16BIT, 8AI×Pt100/NTC
Others	PID module	4AI×TC PID, 8AI×TC PID, 8AI×PID
	Communication module	EM277A DP slave module, EM277B DP slave module
	Weight module	EM231-7WA, EM231-7WB
	Motion control module	EM253-1BH

Specification of TrustPLC CTSC-200 series modules:

Bus reliability

All modules implement bus photoelectric isolating technology, which highly improved the stability of communication bus.

Expandability

CTSC-200 series CPU can be expanded up to 7 modules.

Max capacity of I/O

Digital I/O: 128DI&128DO; Analog I/O: 56AI&28AQ. However with the introduction of some more density modules, the max capacity of I/O will be larger.

Filter technique

All Analog modules have integrated CPU, with advanced filter technique, they can provide excellent sampling stability.

Digital modules

ALL I/Os of the Digital modules have implemented photoelectric isolation and disturbance rejection.

Temperature acquisition modules

With the three-isolation technology of bus, power supply and channel isolation as well as 24-bit sampling AD, the anti-interference ability is very strong and the sampling precision is very high. And more intelligent fault diagnosis function, high security.

High speed and high precision modules

16 bit of precision, the sampling time for each channel is less than 200us, satisfy many occasions of high speed sampling and closed loop control.

Intelligent modules

The main temperature control PID module, weighing module, motion control module, module which integrate powerful CPU, control in the module execution, greatly shorten the response time of execution.

Communication module

The DP slave station module greatly improves the interconnection and communication capability of the system

1.4 Max System Configuration

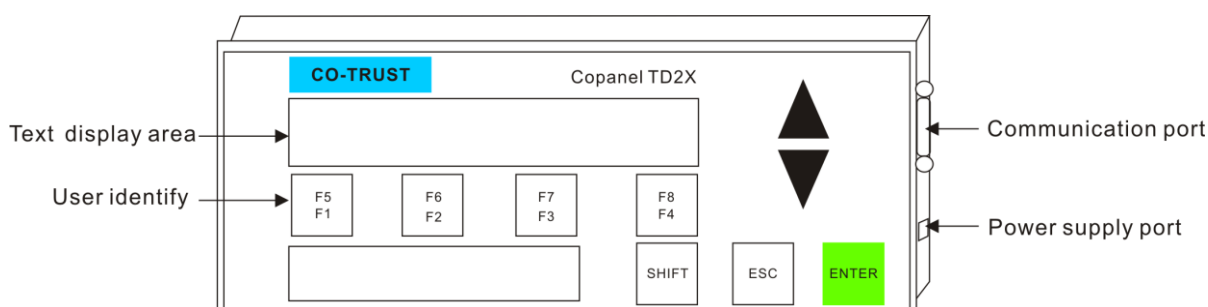
Up to 7 expansion modules can be connected to each PLC.

Digital image register: 128DI/128DQ for each CPU. With these logical I/Os are allocated as 8-points modules, some couldn't be addressed. For instance, the CPU224+ has local 14DI/10DO, but it needs 16 I/Os.

Analog image register: max.56AI/28AQ (not used for 8AI module).

1.5 Display Panels

Copanel TD series include TD2X, TD4X and TD4S, in which the TD2X supports 20 ASCII characters displayed in dual-line, Copanel TD4X and TD4S support four-line text display, all these displays can communicate with CTSC-200 PLC. It can also display text messages and other application data via TD 200 wizard.



TD2X

Copanel TD2X is a text display of TrustPLC CTSC-200 series, which supports MagicWorks PLC and Siemens Micro/Win V4.0 program software.

Features:

- Display CPU information

- Display and modify values stored in CPU V memory
- Forced/Unforced I/Os
- Menus and Tips in 6 languages German, English, Chinese, French, Italian, Spanish)
- Multiple character sets
- Powered by independent supply or directly powered by CPU with the TD2X dedicated cable
- Set date and time for CPU with real-time clock

TD4X

Copanel TD4X is a text display of TrustPLC CTSC-200 series, which supports MagicWorks PLC and Micro/Win V4.0 SP5 or higher.

Features:

- Menus and Tips in 6 languages (German, English, Chinese, French, Italian, Spanish)
- 9 types of character sets
- Display CPU information
- Custom alarms
- Custom keys
- Custom Screen saver time
- Embedded variable into SCREEN and ALARM
- Change PLC modes
- Forced/Unforced I/Os
- Display and modify values stored in CPU V memory
- Set date and time for CPU with real-time clock
- Copy program into memory card
- Password protection
- Powered by independent supply or directly powered by CPU with the TD4X dedicated cable
- Access screen and alarms
- Set TD Device

TD4S

Copanel TD4S is a TrustPLC CTSC-200 series Text Display, which is compatible with both MicroWin and MagicWorks PLC, can be used for operators to interact with program.

Features:

- Menus and Tips in 6 languages (German, English, Chinese, French, Italian, Spanish)

- 9 types of character sets
- Display CPU information
- Custom alarms
- Custom keys
- Custom screen and menu
- Embedded variables into SCREEN and ALARM
- Password protection
- Access screen and alarms
- Edit Time and Date
- Change language set
- Set TD Device
- View TD information stored in CTSC-200CPU
- Force I/Os in CTSC-200 CPU
- Change the operation mode of CTSC-200 CPU
- Copy program into memory card
- Edit CPU memory

1.6 Introduction of Communication

TrustPLC CTSC-200 PLC is compatible with Siemens S7-200 PLC and supports network communication interfaces like Siemens PPI cable and CP communication card. TrustPLC CTSC-200 PLCs have RS485 and RS232 ports, but in which the CPU224I has a Ethernet port in addition, for detailed port definition, please refer to Chapter [5 Network](#).

TrustPLC CTSC-200 PLCs support PPI, Freeport, MPI, Profibus-DP (used with EM277 module) and Ethernet protocols, please refer to Chapter [5 Network](#).

2 Getting Started

This chapter introduces how to connect CTSC-200 PLCs and use MagicWorks PLC software to program and run the PLCs.

2.1 Connecting to CTSC-200 PLC

- Before powered up, the CTSC-200 PLC needs to be connected with Programming device by using RS485 cable.
- CTSC-200 PLC (CPU224I) supports establishing Ethernet communication via standard network cable.

Power supply connection

CTSC-200 CPU supports DC and AC supply as shown below:

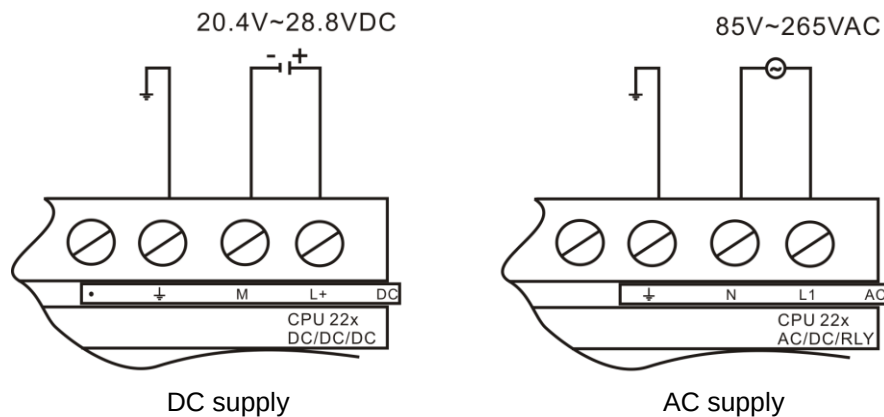


Figure 2-1 CTSC-200 power supply



Warning

Don't install and wire the CTSC-200 PLC with power on, faulty operation can cause serious damage for machines and personal death. Ensure disconnecting power while install or disassemble any electric device

Connecting with RS485 cable

Figure 2-2 illustrates how to use RS485 cable to connect CTSC-200 CPU and program device.



Notice

Refer to Appendix [L Usage of CTSC-200 Storage Card and Battery Card](#). Download address for drive: www.co-trust.com.

1 Connect the USB port on program cable with the COM port on PC (create a virtual serial

port).

2 Connect the USB port on program cable with the COM 0/COM 1 on CTSC-200 CPU.

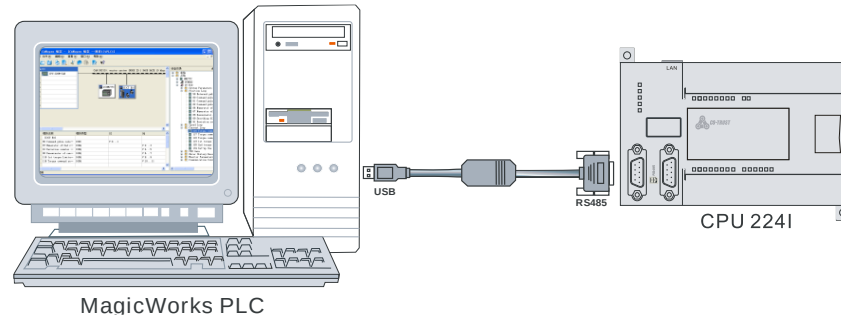


Figure 2-2 Using program cable to connect CTSC-200 PLC with PC

PLC connection with net cable

Figure 2-3 illustrates how to use standard net cable to connect CTSC-200 PLC with PC.

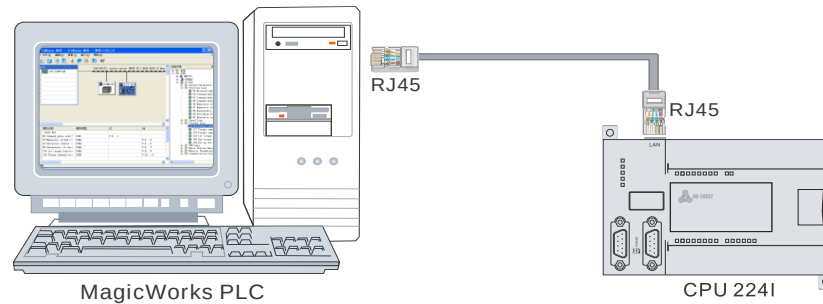


Figure 2-3 Using standard net cable to connect CTSC-200 PLC with PC

MagicWorks PLC

Double-click the icon  to start MagicWorks PLC software, select "File ->New" to create a new project. As shown in figure 2-4, this is a project tree, in which you can click icons to open MagicWorks PLC components.

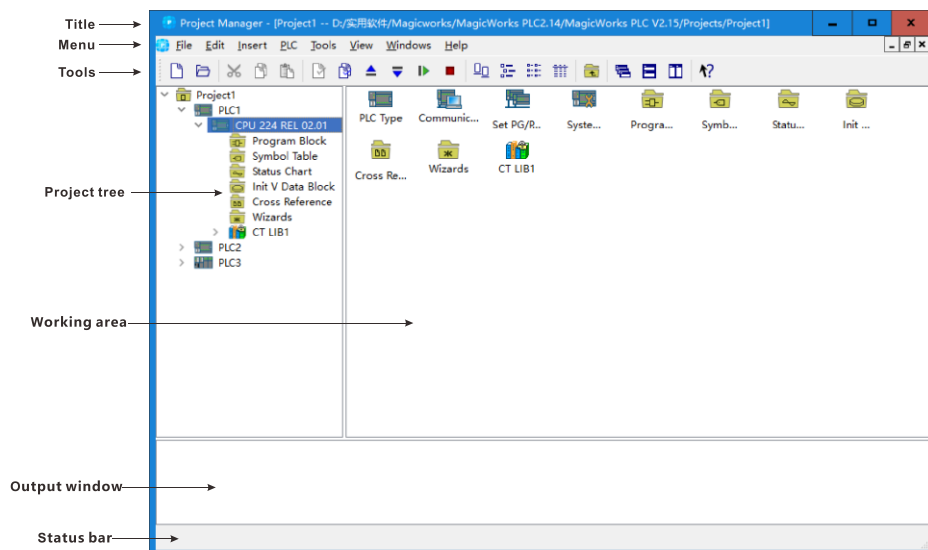


Figure 2-4 MagicWorks PLC Operation interface

2.2 Communication Settings

CTSC-200 PLCs all support serial communication, in which the CPU224I supports both serial and Ethernet communication. In this section, we use CPU224I as example to introduce the communication configurations.

2.2.1 Serial Port Communication

The steps for PPI communication with MagicWorks PLC:

1. Click the “Set PG/PC Interface” in the “Communications” screen. Select the interface, the default is “PC/PPI Cable (PPI)”.

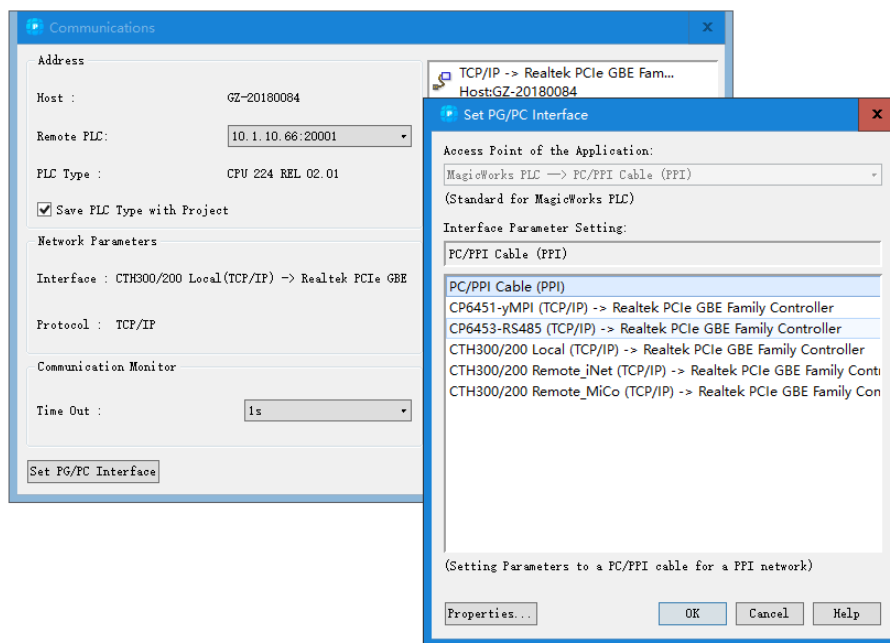
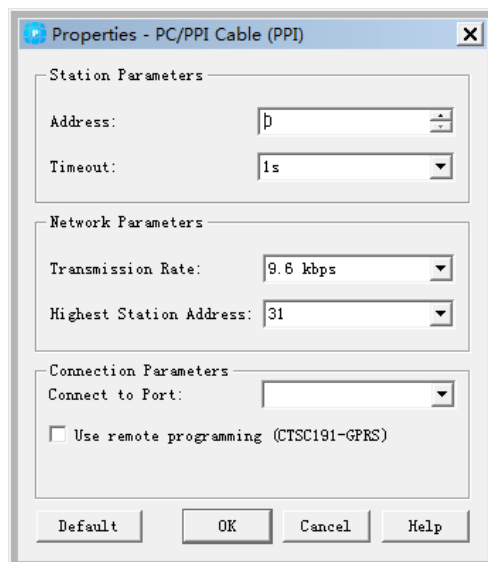


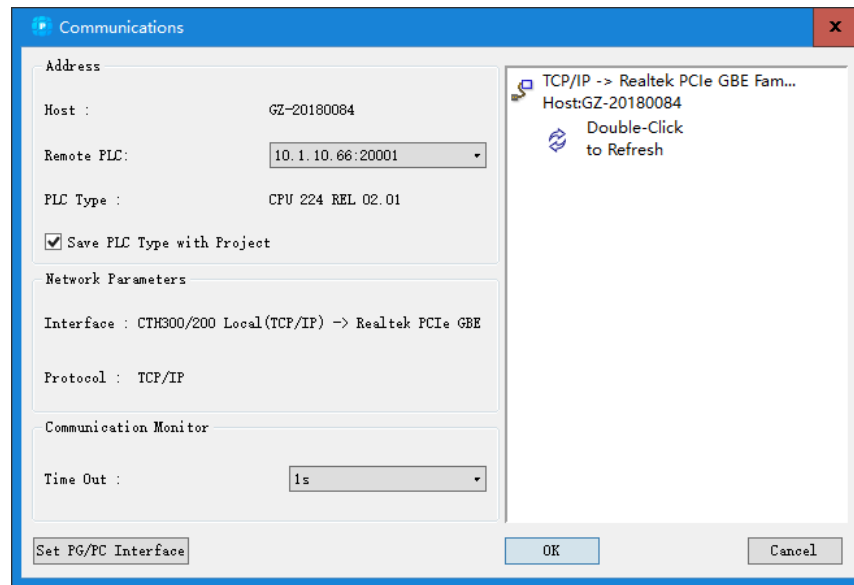
Figure 2-5 MagicWorks PLC serial port communication interface

2. Select the “PC/PPI Cable (PPI)” in “Set PG/PC Interface”, then click the “Properties” button in this dialog to set the communication parameters



- 1) PC/PPI Address: 0
- 2) Interface: COM1
- 3) Transmission rate: 9.6Kbps

3. Click OK to return to the “Communications” page. Then double-click Refresh to find the PLC.



If communication failed, Please do examine as the following steps:

1) Check wiring

Please use the cable provided by Co-trust (Order No. CTS7-191-USB) and ensure the connector remained intact. Switch PLC to STOP when the communication protocol had changed previously.

2) Check the drive

The cable provided by Co-trust must have matched drive, please obtain the drive from the following site: <http://www.co-trust.com>

3) Check the communication settings

- Select the PPI protocol (PC/PPI Cable).
- Ensure the selected COM port not hold by other programs.
- Select the suitable baud rate.

2.2.2 Ethernet Communication

The steps for TCP/IP communication with MagicWorks PLC:

- 1) Click the “Set PG/PC Interface” in the “Communications” screen
- 2) Choose “CPU300/200 Local (TCP/IP) -> Realtek PCIe GBE Family Controller”

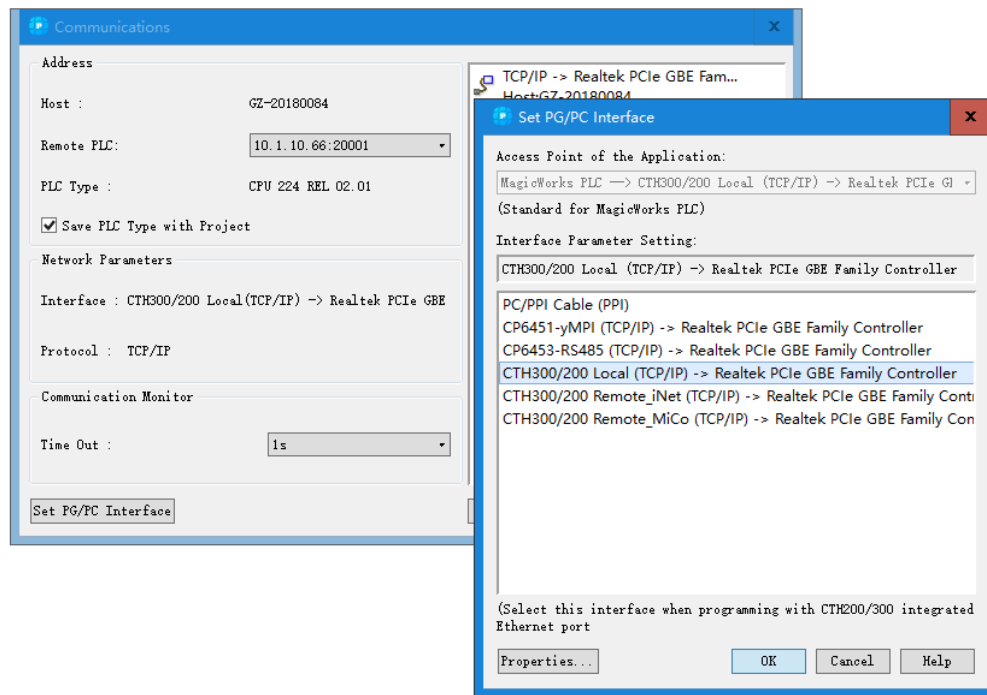
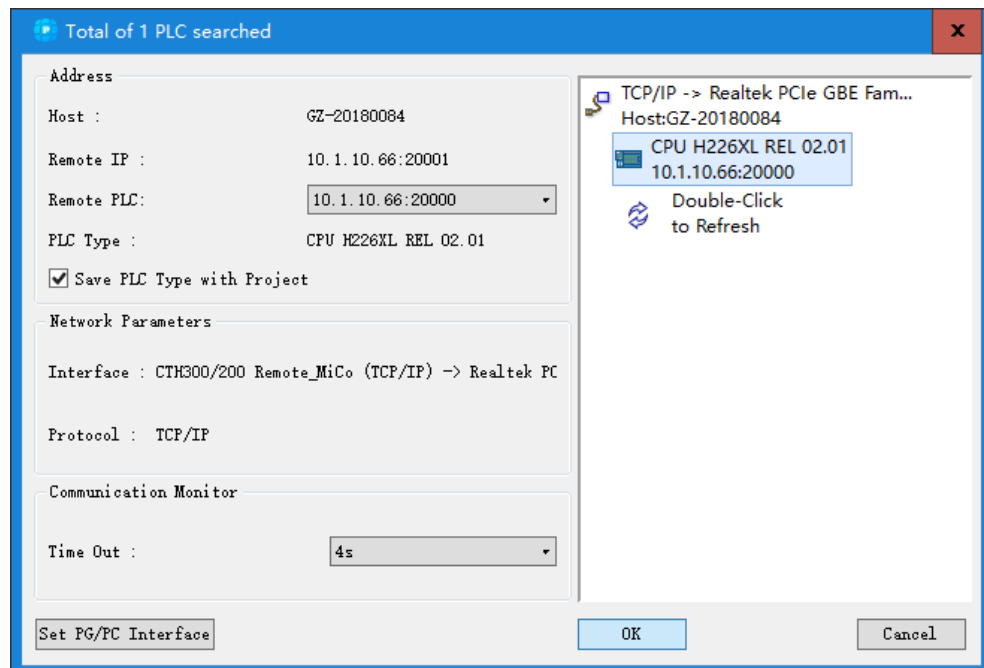


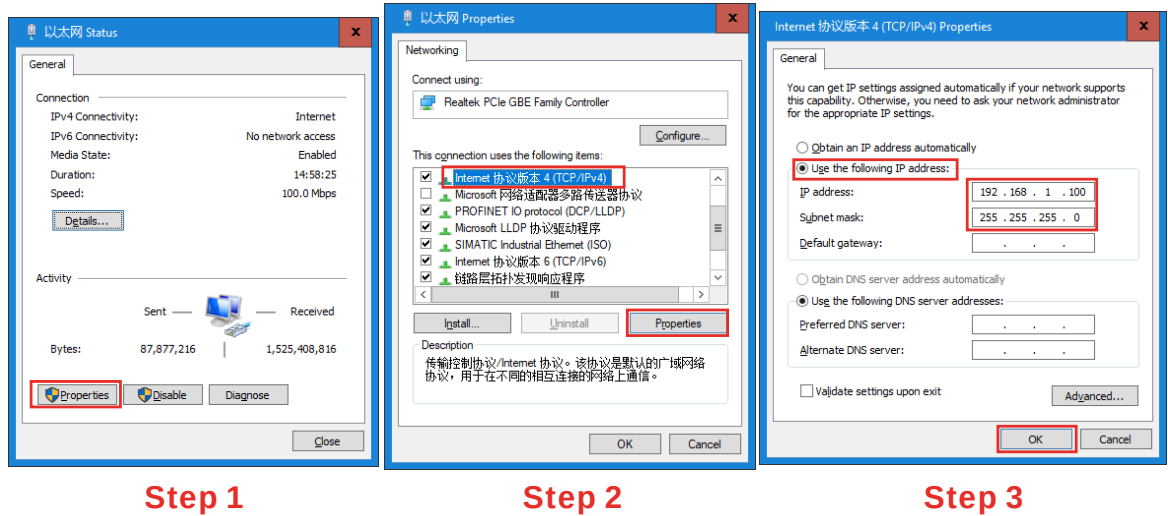
Figure 2-6 MagicWorks PLC Ethernet communication interface

3) Establish communication with CTSC-200 PLC: Double-Click the Refresh icon in the following dialog, MagicWorks PLC will search for and show the connected CTSC-200 PLC.



Users can choose the CTSC-200 station and click OK after searching. If the MagicWorks PLC hasn't found the CTSC-200 CPU, please check wiring first and then validate the Communication settings, after which repeat the above steps. When the PLC communication has established, The IPs for PC and PLC should be in same segment but can not be identical, as shown in the following figure, IP for PLC searched by PC is 192.168.1.202, then the PC IP should be 192.168.1.XXX (in which XXX ranges from 1

to-255). Please set PC IP as following procedures:



Caution

CPU224I supports Ethernet communication with an Ethernet port and can establish a remote connection with PC. Only the remote IP for PLC be in the same domain with PC IP, then the PLC can be searched. For details about Ethernet connection, please refer to Chapter [5 Network](#).

2.3 Create Program

To help you know how to use CTSC-200 PLC, this section give an example to illustrate the creation, download and running of PLC program. The following program example uses seven instructions in a program segment to move the numeric byte transferred to VB0 one byte to the left at 1s intervals. This example uses the ladder diagram editor to enter the program, and describes the program compilation, download and operation process.

The following is the ladder diagram and statement table program, statement table lists the program notes, explained in detail the logical relationship of the program.

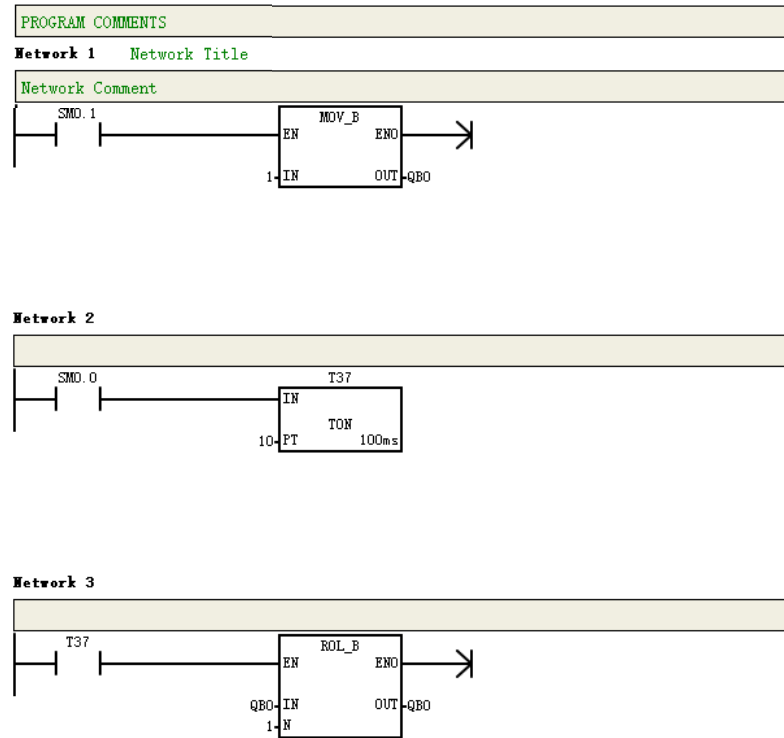


Figure 2-7 Example of CTSC-200 program

Segment comment:

Network 1:
 LD SM0.1 // Enable by first scan of SM0.1
 MOV_B 1, QBO // Transfer 1 into QBO
 Network 2:
 LD SM0.0 // Enable signal
 AN T37 // Circulate timer signal
 TON T37, 10 // Set timer T37, Time 100ms x 10 = 1s
 Network 3:
 LD T37 // Set enable pulse via T37
 RLB QBO, 1 // QBO shift 1 bit left, with the PLC LEDs light on at 1s interval.

2.3.1 Edit Program

Click the Program Block to open the program editor, as shown in figure 2-8. User can drag&drop the ladder instructions into program editor, or use the shortcut for available instructions directly.

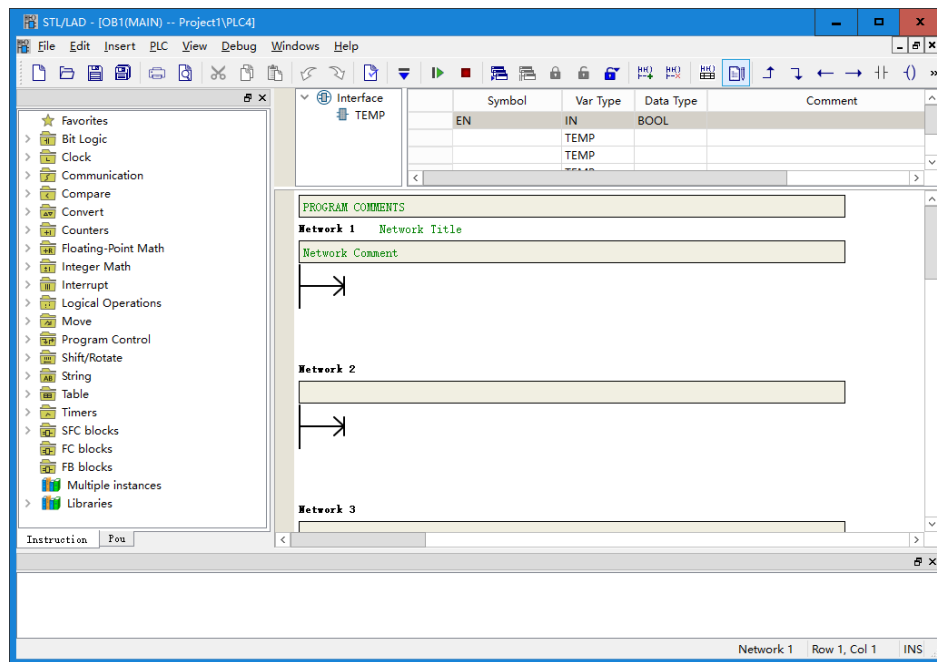


Figure 2-8 Program editor

2.3.2 Compile Program

The program needs to be compiled when editing completely:

- 1) Select menu command "PLC -> Compile" or click Compile button 
- 2) The output window at bottom will show status for compiling, if there occurs error, users can click the prompt to view details

```
Compiling PLC4 - Program Block...
OB1(MAIN)
Block Size = 93 (bytes), 0 errors, 0 warnings.
```

Figure 2-9 Program compile result

Save project:

- 1) Select menu command "File -> Save as".
- 2) Enter the required project name in "Name" dialog.
- 3) Select the required project Storage.

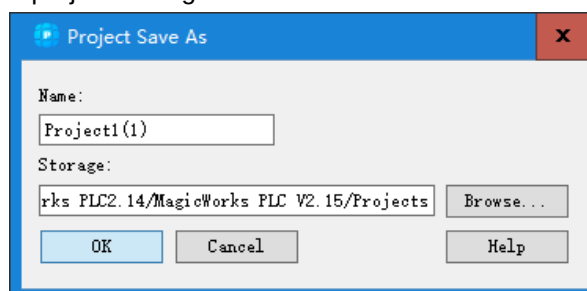


Figure 2-10 Project Storage

2.3.3 Download Program

Click the download icon  in toolbar or select the menu command “PLC -> Download” to download the program into CTSC-200, as shown in Figure 2-11 and 2-12. CTSC-200 PLCs support read/write online, which means user can write instructions or parameters into PLCs during Running, without switching to STOP.

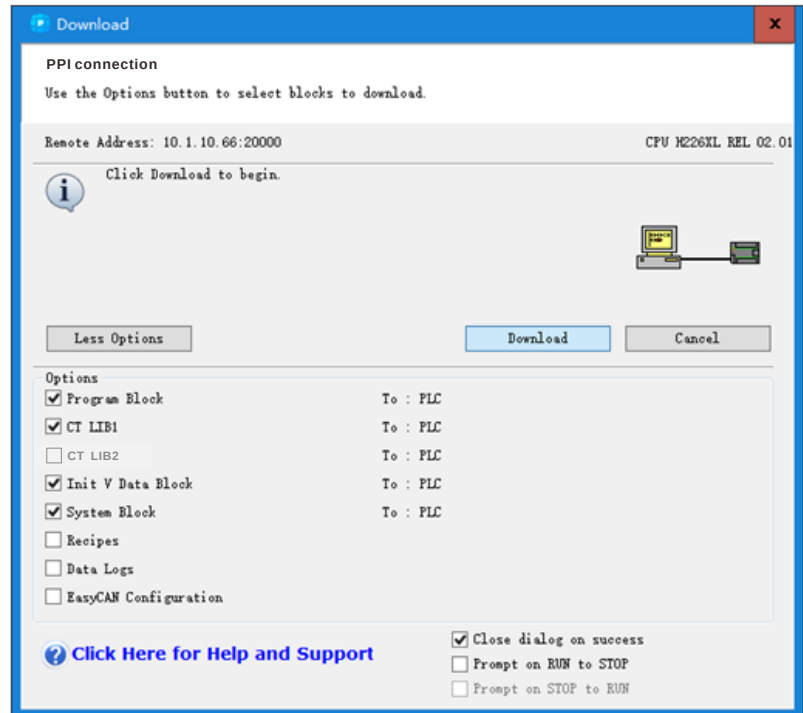


Figure 2-11 PPI connection to download program

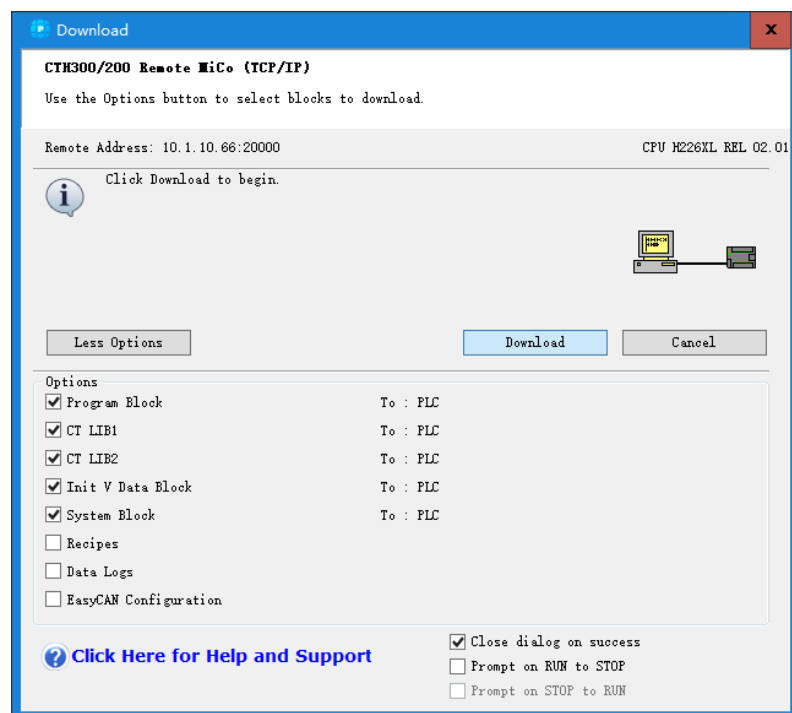


Figure 2-12 Ethernet connection to download program

2.3.4 Run PLC

MagicWorks PLC can switch the CTSC-200 PLC into RUN mode and execute program when the Mode Switch of PLC set to RUN:

- 1) Click the RUN icon  in toolbar or select the menu command "PLC -> RUN".
- 2) Click Yes to enter RUN mode and the CPU would execute program, Q0.0-Q0.7 will light on circularly at 1s interval, which means left shaft the value circularly in Q memory.

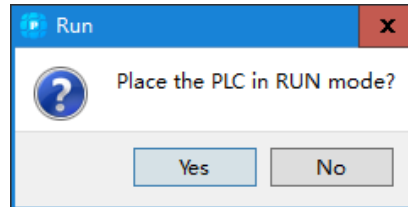


Figure 2-13 Run program

Users can monitor the program via project tree -> state table. MagicWorks PLC would show the appropriate results. Click the icon  or select menu command "PLC -> STOP" would stop the PLC.

3 Installation

TrustPLC CTSC-200 PLCs have been designed convenient for installation, they can be fixed into backplane of the cabinet via mounting hole, or use the DIN clamp to mount on a standard DIN rail. It's compact structure allow users utilizing space efficiently.

This chapter will show user the way to install and wiring the CTSC-200 PLC.

3.1 Important Notices

TrustPLC CTSC-200 PLCs can be installed on the backplane of cabinet or on the standard DIN rail, vertically or horizontally. Users must observe the following notices:

Isolate the PLC with Heat, HV and Electronic noise

According to the general conventions, PLC with low voltage must be isolated with HV and electronic noise sources.

When mounted on the backplane of cabinet, the PLC should be arranged into lower temperature area of the cabinet to extend its lifetime.

The AC power line, high energy and high switching frequency DC line, low voltage signal line and communication cable should be avoided putting into one slot.

Make room for heat dissipation and wiring

CTH200TrustPLC CTSC-200 PLCs are designed with natural ventilation and heat dissipation, with at least 30mm space above and below the module. Distance between the front plane and back plane must be at least 80mm.



Notice

Comparing with horizontal installation, the max. Ambient temperature allowed with vertical installation should lack for 10°C, and CPU should be installed below all expansion modules.

Enough spaces should be left for cable wiring and connecting when installing TrustPLC CTSC-200 PLC.

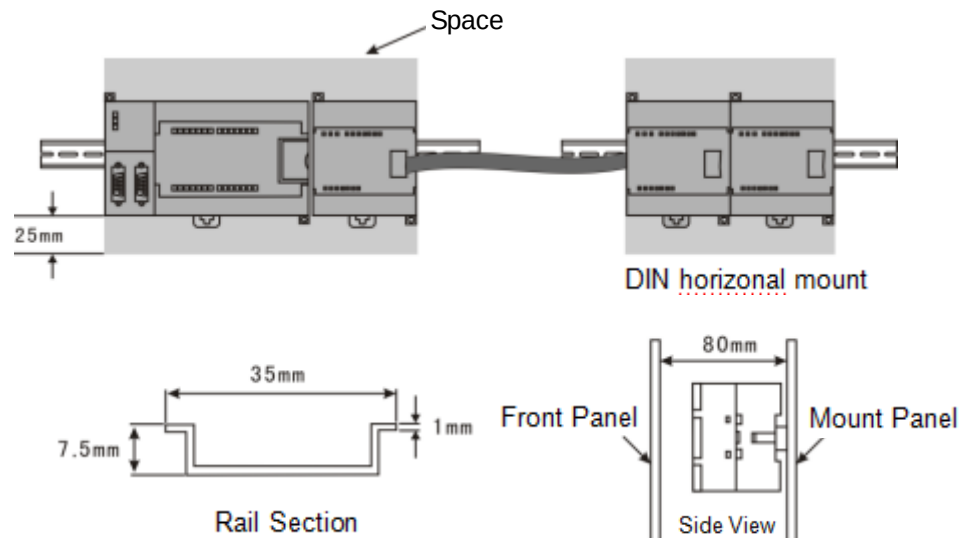


Figure 3-1 Installation

Power budget

CTSC -200 series CPU has an internal power supply offer 24VDC for itself, the expansion modules and other device.

It also provide 5VDC logic power supply, which can be used for power supply of any expansion in the system. Pay close attention to the system configuration to ensure that the CPU can provide the 5V power for the selected expansion module. If the configuration requires more power than the CPU provides, some expansion modules must be removed or select CPU that can provide more power.

CTSC-200 series CPU also provide 24VDC sensor power supply (except PSC266), which can provide 24VDC for input, relay coil on expansion module or other power supply. If the required power exceeds budget, an external 24VDC power supply must be added. For specific CTSC-200 CPU 24VDC sensor power budget, refer to Chapter [6.3 24VDC Power](#).

If you need an external 24VDC power supply, make sure that the power supply is not connected in parallel to the sensor power supply of the CTSC -200 series CPU. To improve electrical noise protection, please connect to the common terminal (M) of different power sources.



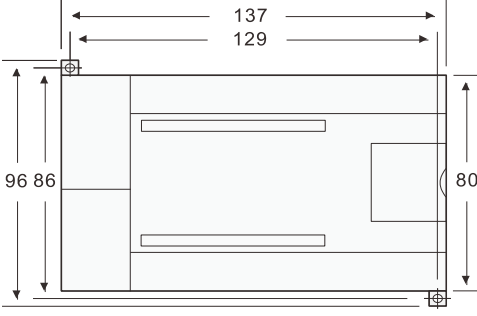
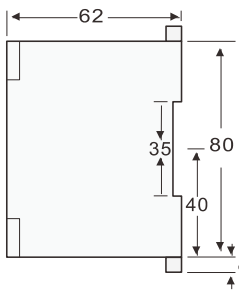
Warning

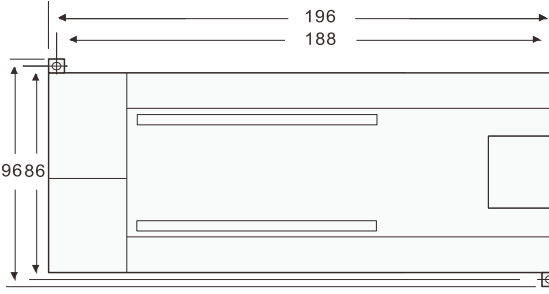
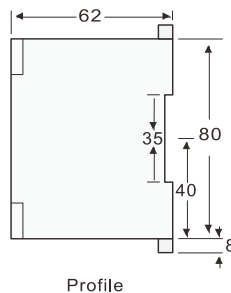
An external 24VDC power supply connected in parallel to CTSC-200DC sensor power supply can cause a conflict because each power supply seeks to establish its own potential output. This conflict will shorten the service life of one or two power supplies or cause an immediate failure, then uncertain operation of the PLC system. The unpredictable operation may cause personal injury and equipment damage.

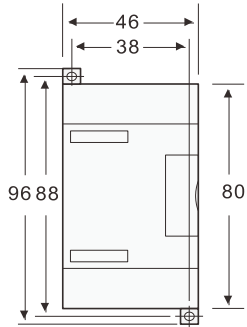
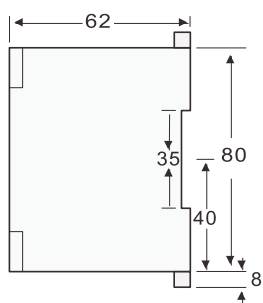
CTSC-200 CPU's 24VDC sensor and any external power supply should be supplied at different points. Allow separate connections to the utility power.

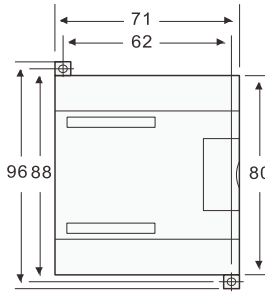
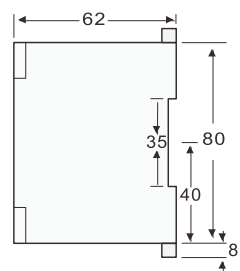
3.2 Installation Dimension Description

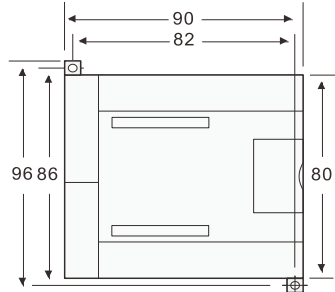
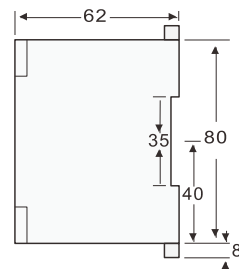
TrustPLC CTSC-200 PLCs and all expansion modules have mounting holes, with which can be installed at back plane conveniently. The following figure shows the Installation Dimension for all PLC and expansion modules.

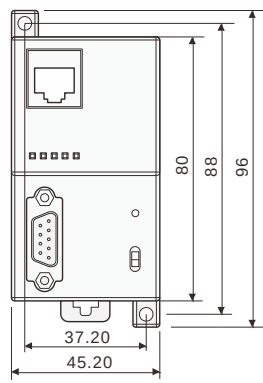
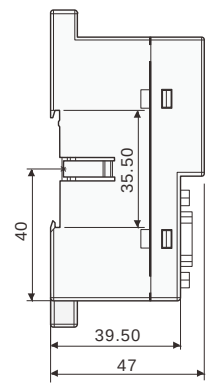
Type	Unit: mm	
CPU 224+ CPU 224+ V5(upgrades) CPU 224E CPU224I CPU224XP CPU 226H CPU 226HV5(upgrades) CPU 226M CPU 226M V5(upgrades) CTS7 221-1BL32 CTS7 222-1BL32 CTS7 223-1BL32 CTS7 223-1PL32		
	Front	Profile

Type	Unit: mm	
CPU226I CPU 226L CPU 226L V5(upgrades) PSC 266		
	Front	Profile

Type	Unit: mm	
CTS7 221-1BF32 CTS7 222-1BF32 CTS7 222-1HF32 CTS7 223-1BF32 CTS7 223-1HF32 CTS7 232-0HB32		
	Front	Profile

Type		Unit: mm	
CTS7 221-1BH32 CTS7 221-2BL32 CTS7 222-1BH32 CTS7 222-1HH32 CTS7 223-1BH32 CTS7 223-1PH32 CTS7 231-0HC32 CTS7 231-7HB32 CTS7 231-7HC32 CTS7 231-0HF32 CTS7 231-1HF32 CTS7 231-7PA32 CTS7 231-7PB32	CTS7 231-7PC32		
	CTS7 231-7PD32		
	CTS7 231-7PF32		
	CTS7 231-7TF32		
	CTS7 231-7TD32		
	CTS7 231-7HF32		
	CTS7 231-7ND32		
	CTS7 231-7NF 32		
	CTS7 231-7WA32		
	CTS7 231-7WB32		
	CTS7 232-0HF32		
	CTS7 235-0KD32		
	CTS7 253-1BH32		
	CTS7 277-0AA32		
	CTS7 277-0AB32		
	CTS7 277-0AC32		

Type	Unit: mm	
CTS7 277-0AC32		
	Front	Profile

Type	Unit: mm	
CTSE 645-1AA10 CTSE 645-1AB10 CTSE 645-1AJ10		
	Front	Profile

3.3 Installation Method

TrustPLC CTSC-200 can be installed on standard DIN35 rail or panel.

Prerequisites

Make sure the equipment has been powered off before assembling and disassembling, meanwhile, all related devices must also be powered off.

**Warning**

Don't assemble and disassemble the CTSC-200 PLC and related devices with power, otherwise can cause electric shock or malfunction, even serious damage, injury or death.

With the CTSC-200 PLC powered off, Correct or equivalent modules need to be used for replacing and installing, meanwhile the direction and location must be correct when replacing modules, or it can result in damage, injury or death.

**Notice**

Incorrect modules would cause CTH200 PLC program failure.

Install/Disassemble CPU and expansion modules

Following the procedures below:

✧ Mounting on panel

- 1) Locate the open holes according to the dimensions on Figure 3-2.
- 2) Fix the modules on backplane by using appropriate screws.
- 3) With expansion modules, connect the flat cable for expansion modules to the extended port below the front cover.

✧ Mounting on DIN rail

- 1) Fix the rail on backplane with 80mm distance.
- 2) Open the DIN clamp below the module, lock the module back on DIN rail.
- 3) When using expansion modules, connect the flat cable for expansion modules to the extended port below the front cover.
- 4) Spin the module to approach the DIN rail, then close the DIN clamp.
- 5) Check if the DIN clamp fit the rail closely.
- 6) Don't push the front of module, you can push the open holes instead to prevent damage.

**Notice**

When mounting the CTSC-200 PLC vertically in high vibration environment, the DIN rail blocks should be used, thus backplane mounting is highly recommended to achieve high grade of vibration protection.

✧ Disassemble CPU or expansion modules

Remove CPU and related expansion modules as following:

- 1) Remove the power of CTSC-200 PLC
- 2) Dismantle all cables and wires on module.

3) Open the front cover to pull up the extended flat cable from neighbor expansion module (if there has).

4) Remove the mounting screw or open the DIN clamp.

5) Remove the CPU and modules.

✧ Disassemble terminal strip

1) Open the cover on terminal strip.

2) Plug the screwdriver into the slot of terminal block as shown in the following picture.

3) Pull down and pray the strip out.

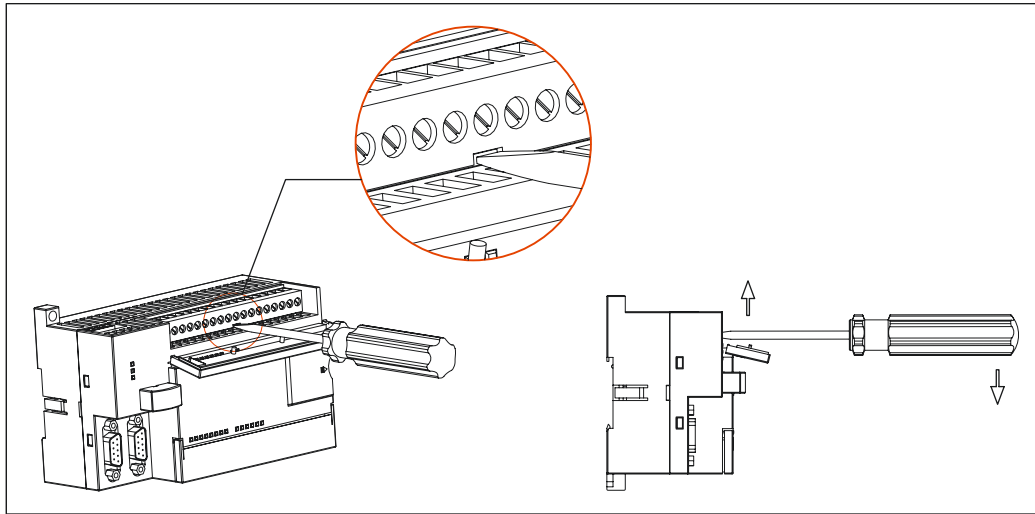


Figure3-3 Dismantle the terminal strip

✧ Mounting terminal strip

1) Open the front cover on the terminal position.

2) Make sure the module pins are alignment with the holes on terminal strip.

3) Push the terminal strip down into module and lock it up.

3.4 Grounding and Wiring

It's important to implement wire and ground correctly for all electrical equipments, which can ensure the best system operation features and provide better noise protection.

CTSC-200 PLCs and related devices must all be powered off before grounding and wiring. The available Electrical coding rules must be obeyed and related safety standards should be followed while installing and operating all devices. Keep contact with regional standard request.

Please always following appropriate safety precautions and ensure that power of CTH200 PLC is cut-off before attempting to install or remove the CTH200 PLC or related equipments.

**Warning**

Grounding or wiring with power could result in death or serious injury to personnel, and/or damage to equipment.

Safety must be considered when designing ground and wire for CTSC-200 system in case triggering the device malfunction. Therefore, all safety rules should be implemented to protect operator and device.

**Warning**

The control equipment may cause the misoperation of the equipment it controls, which may lead to death or serious personal injury and equipment damage. Therefore, the TrustPLC CTSC-200 system must have the emergency stop function, electromechanical interlock or other redundant safety facilities independent of the TrustPLC CTSC-200 system.

3.5 Suppression Circuit

Inductive loads should be implemented with suppression circuits to limit voltage rise when the control output turns off. Suppression circuits protect your outputs from premature failure due to high inductive switching currents. In addition, suppression circuits limit the electrical noise generated during switching inductive loads.

**Notice**

The effectiveness of a given suppression circuit depends on the application, and you must verify it for your particular use. Ensure that all components used in your suppression circuit are rated for use in the application.

DC Outputs and Relay Outputs which Control DC Loads

The DC outputs have internal protection that is adequate for most applications. Since the relays can be used for either a DC or an AC load, internal protection is not provided.

Figure 3-4 shows a sample suppression circuit for a DC load. In most applications, the addition of a diode (A) across the inductive load is suitable, but if your application requires faster turn-off times, then the addition of a Zener diode (B) is recommended. Make sure the Zener diode can provide enough current for output circuit.

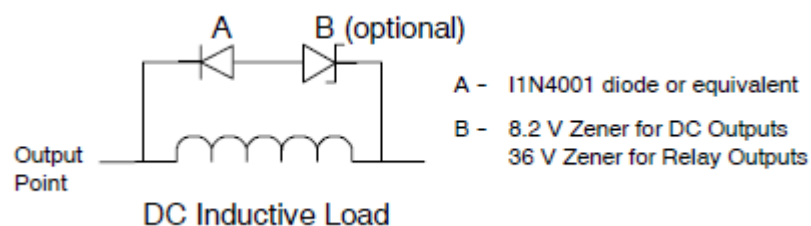


Figure 3-4 Suppression Circuit for DC Load

AC Outputs and Relay Outputs for AC load

The AC outputs have internal protection that is adequate for most applications. Since the relays can be used for either DC or AC load, internal protection is not provided.

Figure 3-5 shows an instance for the suppression circuit of AC load. In most applications, the additional MOV resistor can be used for limit the peak voltage and protect the internal circuit in CTSC-200 PLC. Make sure the operation voltage for MOV is at least 20% greater than regular line voltage.

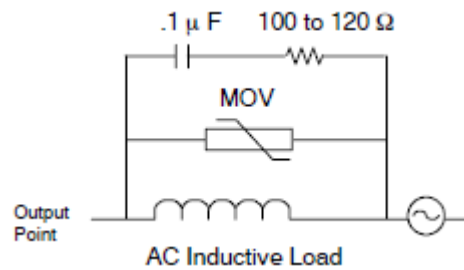


Figure 3-5 Suppression Circuit for an AC Load

4 Technical Specifications

4.1 General Specifications

Table 4-1-1 Electrical and environmental specifications

Items	Description
Rated input voltage	AC: 220V, DC: 24V
Voltage range	AC: 85 V~265 V, DC: 20.4 to 28.8V
Operation temperature	0 ~ 55℃
Storage temperature	-25 ~ 70 ℃
Ventilation	Inlet Air 50.8 mm below unit
Humidity	5 ~ 95% (non-condensing)
Electrical Interference	Pulse width 50ns, repetition frequency 5kHz, 2,000V voltage peak
Vibration	Frequency 10~57Hz, amplitude 0.1mm, acceleration 1.0g, 10 times each dimension
impact resistance	15g for 11ms, 3 times at 3 dimensions
High voltage insulation	Terminal to ground 2200VDC, I/O port to other terminal 1500VAC, dwell time 1 minute
Ground connection	Type 3 (single point grounding with multi-branch)
Operating environment	Dust proof, Noncorrosive environment
Toppling	100 mm, 4 drops, unpacked
Free Fall	1 m, 5 times, packed for shipment



Notice

- TrustPLC CTSC-200 must be installed on a grounded metal rack with its ground connector connected directly to the mounting metal cable and wired along the metal support. The unit must be installed in a grounded metal housing. The AC input power cord must be installed with EPCOS b84115-e-a3 filter or equivalent equipment and the lead length between the filter and TrustPLC CTSC-200 must not exceed 25CM. The 24VDC power supply, the cord of sensor power must be shielded
- 3 grounding modes: Connect the GND terminal on each device separately to the grounding device.

4.2 CPU

4.2.1 Specifications of CPU

This section introduces the whole general specifications for CTSC-200 CPUs, including Inputs, Outputs and Power Supply.

Table 4-2-1 CPU Power supply

Order No.	Name	Power supply	Digital input (VDC)	Digital output	Port	Analog input	Analog output
CTS7214-1AD33-0X24	CPU 224+	24VDC	14*24	10*24VDC	2	-	-
CTS7214-1BD33-0X24		120~240VAC	14*24	10*relay	2	-	-
CTS7214-1AD35-0X24	CPU 224+ V5(upgrade)	24VDC	14*24	10*24VDC	2	-	-
CTS7214-1BD35-0X24		120~240VAC	14*24	10* relay	2	-	-
CTS7214-1AE33-0X24	CPU 224E	24VDC	14*24	10*24VDC	2	-	-
CTS7214-1BE33-0X24		120~240VAC	14*24	10* relay	2	-	-
CTS7214-1AD41-0X24	CPU 224I	24VDC	14*24	10*24VDC	3	-	-
CTS7214-1BD41-0X24		120~240VAC	14*24	10* relay	3	-	-
CTS7214-2AD41-0324	CPU224XP	24VDC	14*24	10*24VDC	3	2	1
CTS7214-2BD41-0324		120~240VAC	14*24	10* relay	3	2	1
CTS7216-2AD41-0X40	CPU 226I	24VDC	24*24	16*24VDC	3	-	-
CTS7216-2BD41-0X40		120~240VAC	24*24	16* relay	3	-	-
CTS7216-1AD33-0X24	CPU 226M	24VDC	14*24	10*24VDC	3	-	-
CTS7216-1BD33-0X24		120~240VAC	14*24	10* relay	3	-	-
CTS7216-1AD35-0X24	CPU 226M V5(upgrade)	24VDC	14*24	10*24VDC	2	-	-
CTS7216-1BD35-0X24		120~240VAC	14*24	10* relay	2	-	-
CTS7216-2AD33-0X40	CPU 226L	24VDC	24*24	16*24VDC	3	-	-
CTS7216-2AF33-0X40		24VDC	24*24	16*24VDC	3	-	-
CTS7216-2BD33-0X40		120~240VAC	24*24	16* relay	3	-	-
CTS7216-2BF33-0X40		120~240VAC	24*24	16* relay	3	-	-
CTS7216-2AD35-0X40	CPU 226L V5(upgrade)	24VDC	24*24	16*24VDC	2	-	-
CTS7216-2BD35-0X40		120~240VAC	24*24	16* relay	2	-	-
CTS7216-1AH33-0X24	CPU 226H	24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH33-0A24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH33-0B24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH33-1A24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH33-1B24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH34-0X24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH34-0B24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH34-1B24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH34-2B24		24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AH35-0B24	CPU 226H	24VDC	14*24	10*24VDC	2	-	-

CTS7216-1AH35-2B24	V5(upgrade)	24VDC	14*24	10*24VDC	2	-	-
CTS7216-1AH34-2B24	CPU 226H-CAN	24VDC	14*24	10*24VDC	3	-	-
CTS7216-1AC33-0X24	CPU 226M-CAN	24VDC	14*24	10*24VDC	3	-	-
CTSC266-3AT33-2424	PSC 266	24VDC	14*24	10*24VDC	3	-	4

Table 4-2-2 CPU Summary

Types	Dimension (mm)	Power (W)	Available VDC (mA)	
			+5VDC	+24VDC
CPU224E DC/DC/DC, 14DI/10DO	137×80×62	7	660	280
CPU224E AC/DC/RLY, 14DI/10DO	137×80×62	7	660	280
CPU224+ DC/DC/DC, 14DI/10DO	137×80×62	7	660	280
CPU224+ AC/DC/RLY, 14DI/10DO	137×80×62	7	660	280
CPU224+ V5(upgrade) DC/DC/DC, 14DI/10DO	137×80×62	7	660	280
CPU224+ V5(upgrade) AC/DC/RLY, 14DI/10DO	137×80×62	7	660	280
CPU224I DC/DC/DC, 14DI/10DO	137×80×62	7	660	280
CPU224I AC/DC/RLY, 14DI/10DO	137×80×62	7	660	280
CPU 224XP DC/DC/DC, 14DI/10DO	137×80×62	7	660	280
CPU 224XP AC/DC/RLY, 14DI/10DO	137×80×62	7	660	280
CPU226I DC/DC/DC, 24DI/16DO	196×80×62	11	1000	400
CPU226I AC/DC/RLY, 24DI/16DO	196×80×62	11	1000	400
CPU226M DC/DC/DC, 14DI/10DO	137×80×62	9	660	280
CPU226M AC/DC/RLY, 14DI/10DO	137×80×62	9	660	280
CPU226M V5(upgrade) DC/DC/DC, 14DI/10DO	137×80×62	7	660	280
CPU226M V5(upgrade) AC/DC/RLY, 14DI/10DO	137×80×62	7	660	280
CPU226L DC/DC/DC, 24DI/16DO	196×80×62	11	1000	400
CPU226L AC/DC/RLY, 24DI/16DO	196×80×62	11	1000	400
CPU226L V5(upgrade) DC/DC/DC, 24DI/16DO	196×80×62	7	1000	400
CPU226L V5(upgrade) AC/DC/RLY, 24DI/16DO	196×80×62	7	1000	400
CPU226H DC/DC/DC, 14DI/10DO	137×80×62	11	660	280
CPU226H V5(upgrade) DC/DC/DC, 14DI/10DO	137×80×62	7	660	280
PSC266 DC/DC/DC/TC/AQ 14DI/10DO, 20TC, 4AQ	196×80×62	11	1000	-

Table 4-2-3 Specifications of CTSC-200 CPU power supply

Power supply	DC		AC	
Input voltage	20.4~28.8 VDC		85~264 VAC	
Input current	CPU only	Maximum load	CPU only	Maximum load
CPU224+/ 224+V5(upgrade)/224 E/224I/224XP/ 226M/226H/226H V5(upgrade) /226M V5(upgrade)	110 mA	700 mA	60/30 mA in 120/240 VAC	200/100 mA in 120/240 VAC
CPU226L/226I/ 226L V5(upgrade)	150 mA	1050 mA	80/40 mA in 120/240VAC	320/160 mA in 120/240 VAC
PSC266	150 mA	1050 mA	80/40 mA in 120/240VAC	320/160 mA in 120/240 VAC
Surge current	10 A in 28.8 VDC		20 A in 264 VAC	
Isolation (field and logic)	Not isolated		1500 VAC	
Protection time(power consumption)	Regular more than 10ms in 24VDC		20/80ms in 120/240 VAC V5(upgrade): more than 10ms/20ms in 120/240VAC	
Fuse(not replaceable)	3A, 250 V fusing slowly		2A, 250 V fusing slowly	
24 VDC sensor power supply				
Sensor voltage	L+ -5V		20.4~28.8 VDC	
Current limit	Peak current 1.5 A, no destructive heat limitation(rated load)			
Current ripple noise	Derive from input current		The peak value is less than 1V	
Isolation (field and logic)	Not isolated			

**Notice**

When the mechanical contact is connected to the output power of TrustPLC CTSC-200 CPU or any digital expansion module, it will send a "1" signal lasting about 50ms to the digital output. You must plan for this, especially when using devices that respond to short duration pulses.

Table 4-2-4 General specifications

Items	Specification
Programming tools	MagicWorks PLC/Step7 MicroWIN
Programming language	STL/ LD
User program (EEPROM)	CPU 224E/224+/224+V5(upgrade)/224I: 12KB, extend to 16KB CPU 224XP/226M/226L/226I/226H: 24KB, extend to 72KB 226 V5(upgrade): 24KB, extend to 72KB
User data(EEPROM)	CPU 224E/224+/224I: 8KB, extend to 108KB CPU224+V5(upgrade): 8KB, extend to 108KB, everlasting

	CPU 224XP/226M/226L/226I/226H: 10KB, extend to 110KB 226 V5(upgrade): 10KB, extend to 110KB, everlasting
Backup (super capacitor)	About 112 hours (typical value)
Power-off protection time	About 200 days (typical value)
Programming tools	Supercapacitor 112 hours(typical value), power-off protection time 200 days (typical value)
Programming language	CPU224+V5 everlasting Other CPU: Supercapacitor 112 hours (typical value), power-off protection time 200 days (typical value)
Features of I/O	
Digital input/output	CPU 226L/226I/226L V5(upgrade): 24 input/16 output Other CPU: 14 input/10 output
Analog expansion board accessory	Only CPU 224XP: 2AI/1AQ, AIW addressing
Optional expansion board	CPU 224I/226I: support CAN expansion board and analog expansion board
Digital I/O image	256 (128 in/128 out)
Analog I/O image	64 (32 in /32 out)
Maximum expansion module	7
HSC	CPU 224E: 2 Other CPU: 6
Single phase	CPU 224E: 2x30kHz CPU 226H/226H V5(upgrade): 6x200kHz Other CPU: 6x30kHz
Diphase	CPU 224E: 2x20kHz CPU 226H V5(upgrade)/CPU 226H: 4x200kHz Other CPU: 4x20kHz
Pulse acquisition input	CPU 226L/226I/226L V5(upgrade): 24 Other CPU: 14
HSPSO	CPU 224E: 2 axes 50KHz CTSC motion instructions(for DC output) CPU 226H: 4 axes 200KHz CTSC motion instructions CPU 224+/226M/226L: PTO/PWM 2 axes 20KHz (for DC output) Other CPU: PTO/PWM 2 axes 20KHz; PTP/Speed_ctrl CTSC motion instructions: 2 axes 50KHz(for DC output)
Timer	256 timers in all: 1ms – 4; 10ms – 16; 100ms - 236
Counter	256(support by supercapacitor)
Internal memory	256(support by supercapacitor)
Power-off save	112(save to EEPROM)
Timing interruption	2 with 1ms resolution
Edge interrupts	4 up and 4 down

Analog potentiometer	2 with 8 bits resolution
Boolean execution speed	$\leq 0.15\mu s$
Float execution speed	$\leq 8\mu s$
Calendar clock	Build-in
Built-in library	MODBUS RTU master-slave communication library MODBUS TCP communication library PID_T communication library (support 64 loops) PTP, SPEED, SET_POS, READ_POS instructions in Motion control library
LED indicator	SF (RED): ON - system error, OFF - normal
	RUN (Green): ON - Run, OFF - Stop
	STOP (Orange): ON - Stop, OFF - Run
	RMC (Green): ON - Enable RMC, OFF - Disable RMC
	LINK/ACT (Green): ON - connected, flash - transfer, OFF - disconnected
Communication features	
CPU224E/224+ V5(upgrade)	2 COM ports: PPI(PORT0): RS485; FPORT: RS485
CPU224+	2 COM ports: PPI(PORT0): RS485; FPORT: RS232/RS485
CPU224I	3 COM ports: PPI(PORT0): RS485; FPORT: RS485; LAN: RJ45
CPU224XP/226I	3 COM ports: 2 PPI/FPORT (PORT0 and PORT1) : RS485; 1 LAN: RJ45
226 V5(upgrade)	2 COM ports: PPI/FPORT(PORT0 and PORT1): RS485
CPU226M CPU226L CPU226H	3 COM ports: 1 PPI(PORT0): RS485; 1 PPI/FPORT(PORT1): RS232/RS485; 1 FPORT: RS485 Or 1 PPI (PORT0) : RS232/RS485; 2 FPORT0 and FPORT1: RS485 Or 1 PPI(PORT0): RS485; 1 PPI/FPORT(PORT1): RS485; 1 FPORT: RS485
Serial Port Communication (RS485)	
Protection	2000V EFT shock resistance
PPI/MPI baud rate	9.6, 19.2, 187.5 Kbps
FPORT baud rate	1.2 Kbps ~ 115.2 Kbps
Max Cable length	With Isolated repeater: up to 187.5 Kbps for 1000m Up to 38.4 Kbps for 1200m; Without Isolated repeater: 50m
Max stations	32 for each segment, 126 for each network
Max master stations	32
PPN (PPI master mode)	Yes (NETR/NETW), 8, in which 2 reserved
MPI connections	4, 2 reserved (1 for PG, 1 for OP)
Ethernet communication (CPU224I/224XP/226I)	

Port		1 standard Ethernet port
Communication standard		IEEE802.3
Transmission rate		10Mbps and 100Mbps adaptive
Adaptive cross-over connection		-
Port for industrial Ethernet (10/100Mbps)		RJ45 plug
Protocol type		UDP/TCP protocol, Support iNet Ethernet programming and communications between PLCs
Configuration		Support searching PLC and downloading system configuration by Ethernet port on PC
Max connections		Each PLC supports up to 8 UDP and 4 TCP
IT max connections		8 UDP_PPI, 4 master stations and 4 slave stations 4 ModBus_TCP, 2 master stations and 2 slave stations
User data		Up to 200 bytes for TCP/IP transmission. Up to 240 bytes for ModBus_TCP transmission.
Server port		Local: 1~65535 Remote: 8888
Starting time/ restarting time		About 3s
Indicator light	Green RMC	On: remote control on, off: remote control off
	Green Ethernet	On: Connected; flashing: transfer; off: Not connected
Max cable length		100m
Isolation		Communication port isolated

Table 4-2-5 Specifications of CPU PSC266

Items	PSC266 dedicated controller
Memory	
Program memory	24KB, extend to 72KB
Data memory	10KB, extend to 110KB
Power off hold time (supercapacitor)	About 112h(typical value)
I/O	
Digital input/output	14 input/10 output
Thermocouple input	20 with K type (-150~1370℃)
Analog output	4 with 0~10VDC
Digital I/O image	256 (128 in/128 out)
Analog I/O image	64 (32 in /32 out)
Maximum expansion module	7
Pulse acquisition input	14
HSC	6 timer in all
Single phase	6, each work in 30 KHz

Diphase	4, each work in 20 KHz
Pulse output	2, each work in 20 KHz(for DC output)
Features	
Timers	256 timers in all; 1ms – 4; 10ms – 16; 100ms - 236
Counters	256(supported by supercapacitor)
Internal memory	256(supported by supercapacitor)
Power-off save	112(save to EEPROM)
Timing interruption	2 with 1ms resolution
Edge interrupts	4 up and 4 down
Analog integrate	2 with 8 resolution
Execution speed of Boolean instruction	0.15μs/instructions
Calendar clock	Build-in
Communication features	
Communication ports	3 COM ports (2 PPI ports, 1 Freeport) PORT1: RS485 PORT0: RS485 FPORT(Freeport): RS485
Ports protection	2000V EFT shock resistant
PPI, MPI baud rate	9.6, 19.2, 187.5 KB
Freeport baud rate	1.2 KB~115.2 KB
Max cable length	With isolated repeater: up to 187.5 KB for 1000m; up to 38.4 KB for 1200m; Without isolated repeater: 50m
Maximum stations	32 for each segment, 126 for each Internet
Maximum equipments	32
Equal net (PPI master equipment mode)	Yes (NETR/NETW)
MPI connections	4 in all, 2 reserved(1 for PG, 1 for OP)

**Notice**

Power distribution must be calculated in order to determine the power or current provided by TrustPLC CTSC-200 CPU for your configuration. If the power has exceeded what provided by CPU, the max number of modules cannot be connected. Please refer to Chapter [6 Power Budget](#) for Power calculation.

Table 4-2-6 Digital input&output specifications of CPU

CPU224+/224E/224I/226M/226H/226L/226I/V5 upgrade	
Items	Digital input
Type	Source/drain (IEC type 1 source)
Rated voltage	Typical 24 VDC at 1mA
Max allowable continuous voltage	30 VDC

Surge voltage	35 VDC for 0.5 s	
Logic 1 (min)	15 VDC at 2.5mA	
Logic 0 (max)	5VDC at 1mA	
Input delay	Selective (0.2~12.8ms)	
Max allowable leakage current (Bero)	1mA	
Input filter	Configurable, support 0.2ms, 0.4ms, 0.8ms, 1.6ms, 3.2ms, 6.4ms, 12.8ms. Default as 6.4ms	
Isolation (field to logic)	Yes	
Optical (current)	500 VAC, 1 minute	
Isolated groups	Refer to the wiring diagram	
Simultaneous inputs	All for 55°C	
Max cable length	Shielded: 500m for standard inputs, 50m for HSC inputs (which need shield twisted pair and ground the shield)	
	Unshielded: 300m standard inputs	
Items	Digital output	Relay output
Type	Solid	Dry contact
Rated voltage	24 VDC	24 VDC or 110/220 VAC
Voltage range	20.4~28.8 VDC	5~30 VDC or 5~250 VAC
Max surge current	8A, 100ms	7A, closed contact
Logic 1 (min)	0.1 VDC, 10 kΩ load	---
Logic 0 (max)	20 VDC, max current	---
Rated current for each point (max)	0.75 A	2.0 A
Rated current for each common (max)	6A	10 A
Leakage current (max)	10 μA	---
Lighting load (max)	5W	30 W DC, 200 W AC
Sensitive clamp voltage	L+ -48 VDC, 1W loss	---
On-resistance	0.3Ω(<0.6), typical value	0.2Ω(max at new use)
Isolation	500 VAC, 1min --- --- Refer to the wiring diagram	--- 1500 VAC, 1min 100MΩ Refer to the wiring diagram
Optical (field0 logic)		
Logic-terminal		
Resistance (logic-terminal)		
Isolated group		
Delay (max)	2μs(Q0.0&Q0.1), 15μs for others	--- --- 10ms
Off-ON	10μs(Q0.0&Q0.1), 130 μs for others	
ON-Off	---	
Switch	---	
Pulse frequency (max)	Max 200 kHz	1 Hz
mechanical lifetime	---	10,000,000 (no-load)

Contact lifetime	---	100,000 (rated load)
Simultaneous outputs	All at 55°C	
Two Parallel outputs	Yes	No
Max cable length	Shielded	500m
	Unshielded	150m

**Notice**

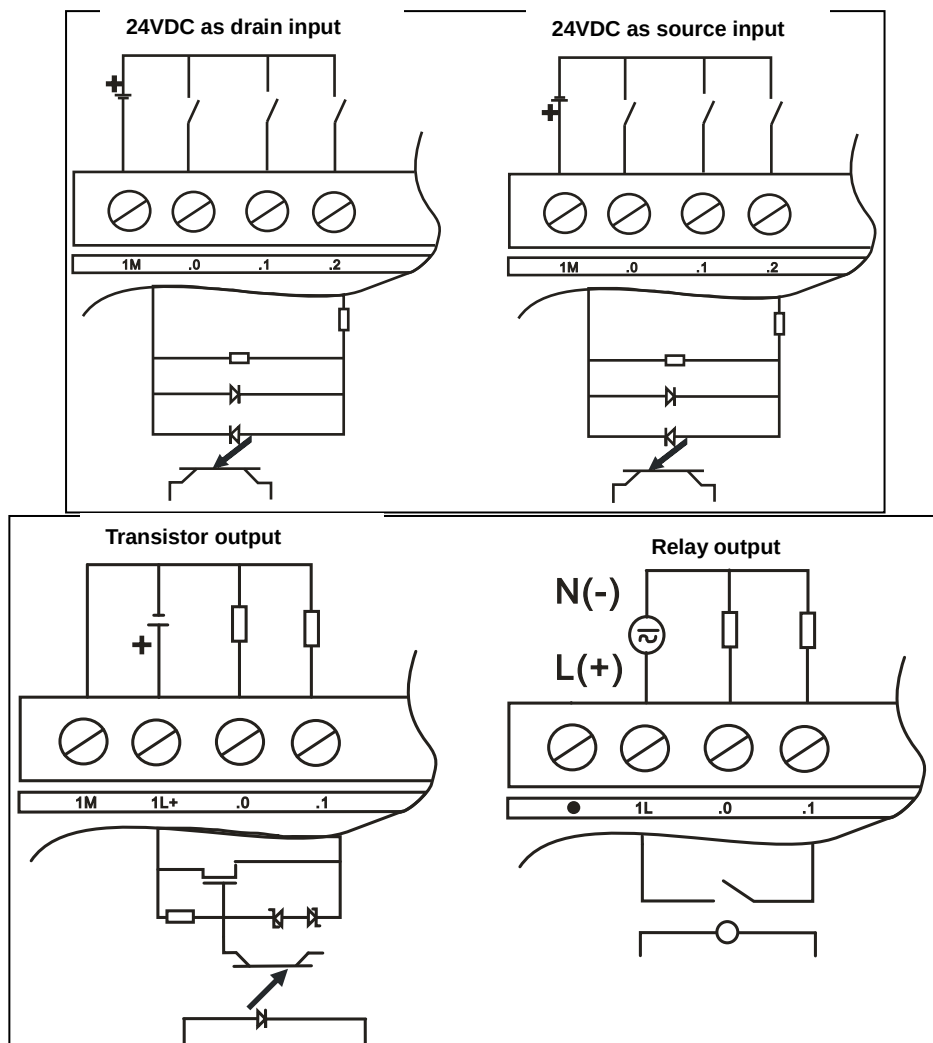
Shielded twisted pair is recommended for HSC inputs.

Table 4-2-7 Analog input&output specifications of CPU224XP

Analog input specifications			
Matched analog expansion board		CTH2 AMS-03S1-EB(standard)	CTH2 AMS-06S1-EB (optional)
Addressing mode		AIW Local 2AI/1AQ use is fixed address AIW0, AIW2/AQW0, AQW2	SM (refer to Chapter 5.7 CANopen Communication Sample)
Build-in analog inputs		2	4
Input type		Single-ended input	
Input voltage range		+/-10V DC	
Data word format		-32000~+32000	
Input impedance		>100K	
Max input voltage		30V	
Resolution		11bit plus 1 sign bit	
Isolation		-	
Accuracy	Worst case (0-55 °)	±2.5% full scale	
	Typical case (25 °)	±1.0% full scale	
	Repeatability	±0.05% full scale	
Analog-digital conversion time		125ms	
Transduction principle		Σ-Δ	
Input response time		250ms	
Noise suppression		Typical: -20dB◎ 50HZ	
Analog output specifications			
Build-in analog outputs		1	2
Output resolution		12 bits	
Isolation		-	
LSB value	Voltage	2.50mV	
	Current	5.00μA	-
Output range	Voltage	0~10V	
	Current	0~20mA	-

Data word format	Voltage	0~32767	
	Current	0~32000	-
Accuracy	Voltage	Worst case (0 ~ 55 °C): $\pm 2\%$ full scale Typical (25 °C): $\pm 1\%$ full scale	
	Current	Worst case (0 ~ 55 °C): $\pm 3\%$ full scale Typical (25 °C): $\pm 1\%$ full scale	Worst case (0 ~ 55 °C): $\pm 3\%$ full scale Typical (25 °C): $\pm 1\%$ full scale
Steady time	Voltage	100us	
	Current	2ms	-
Max drive	Voltage	Min 5000 Ω	
	Current	Max 500 Ω	-

4.2.2 CPU Wiring Diagrams



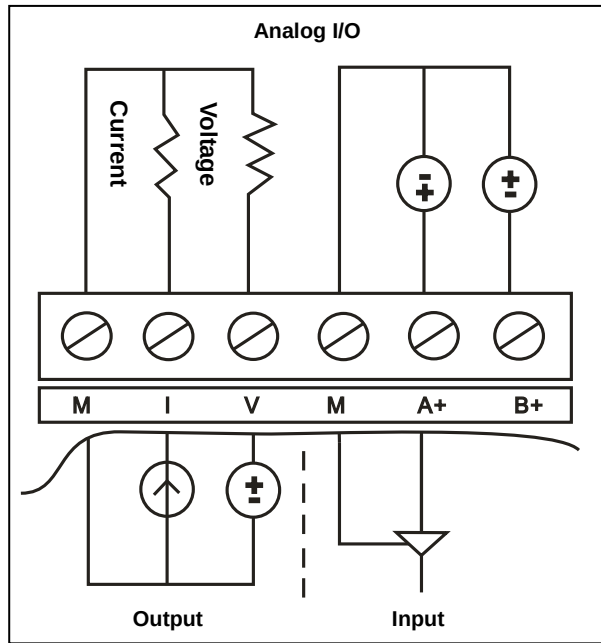
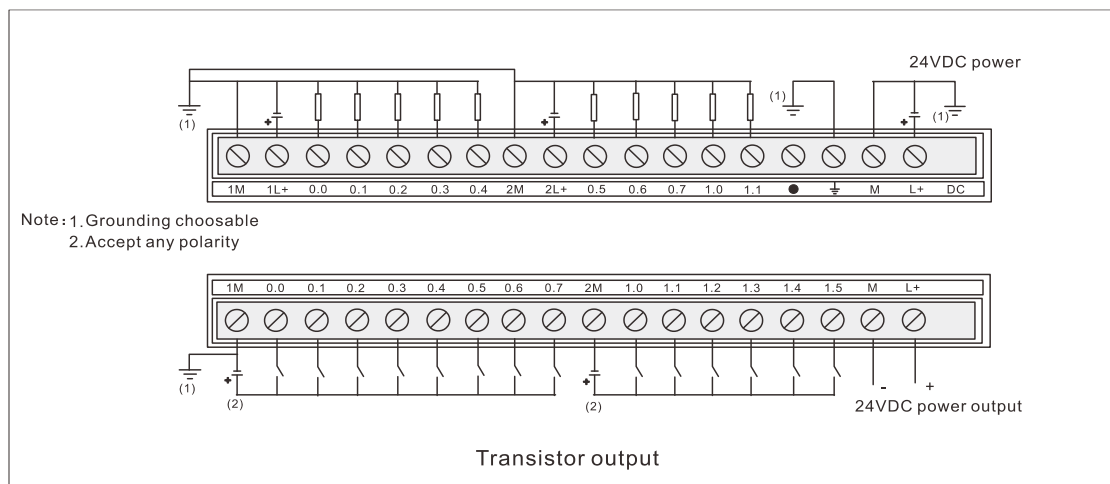
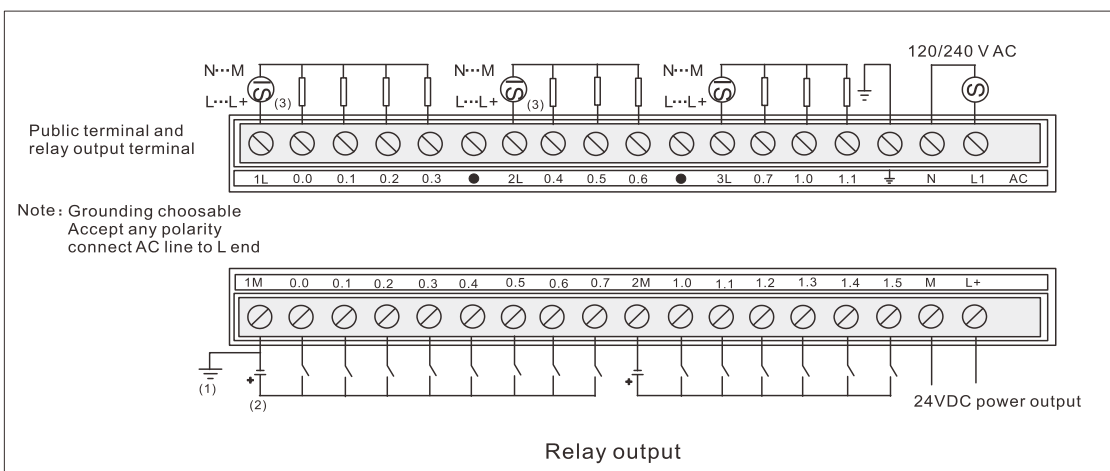


Figure 4-1 CPU Inputs and Outputs



Transistor output



Relay output

Figure 4-2 CPU 224E/224+/224+ V5 (upgrade)/224I/224XP/226M/226M V5 (upgrade)/226M-CAN wiring diagrams

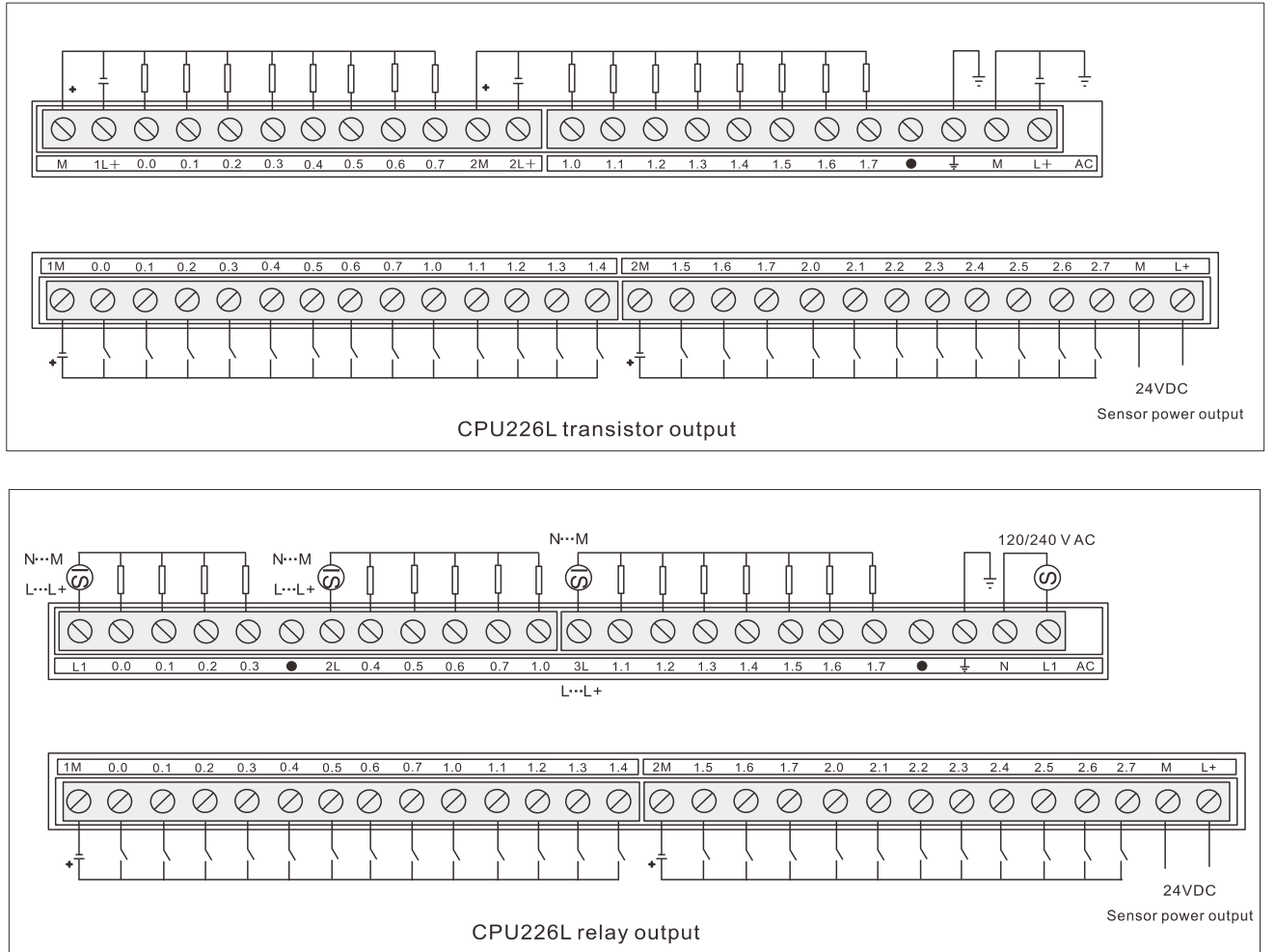


Figure 4-3 CPU 226L wiring diagrams

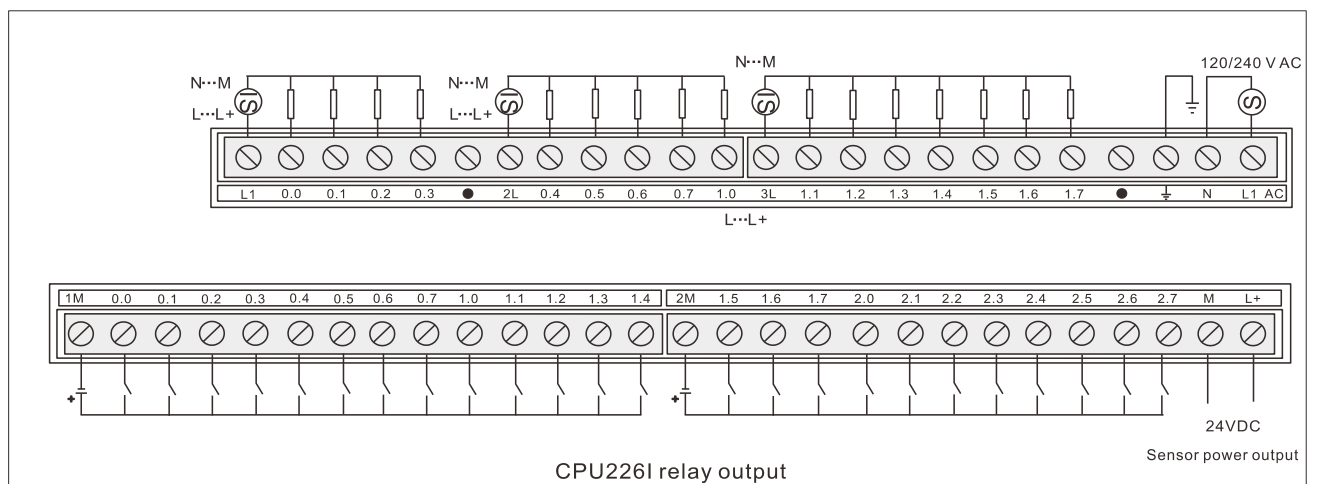
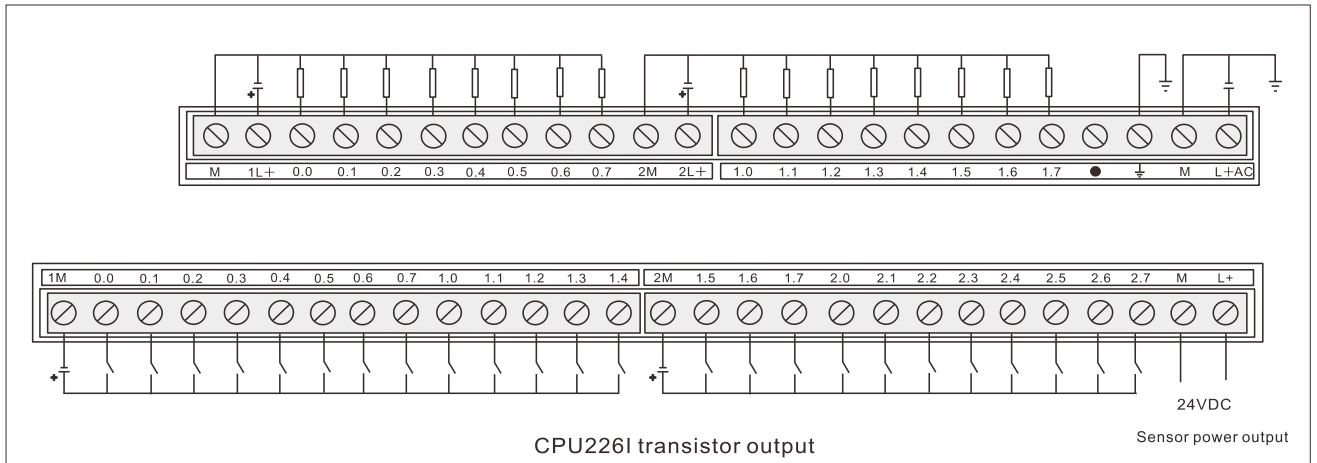


Figure 4-4 CPU 226L V5 (upgrade) wiring diagrams

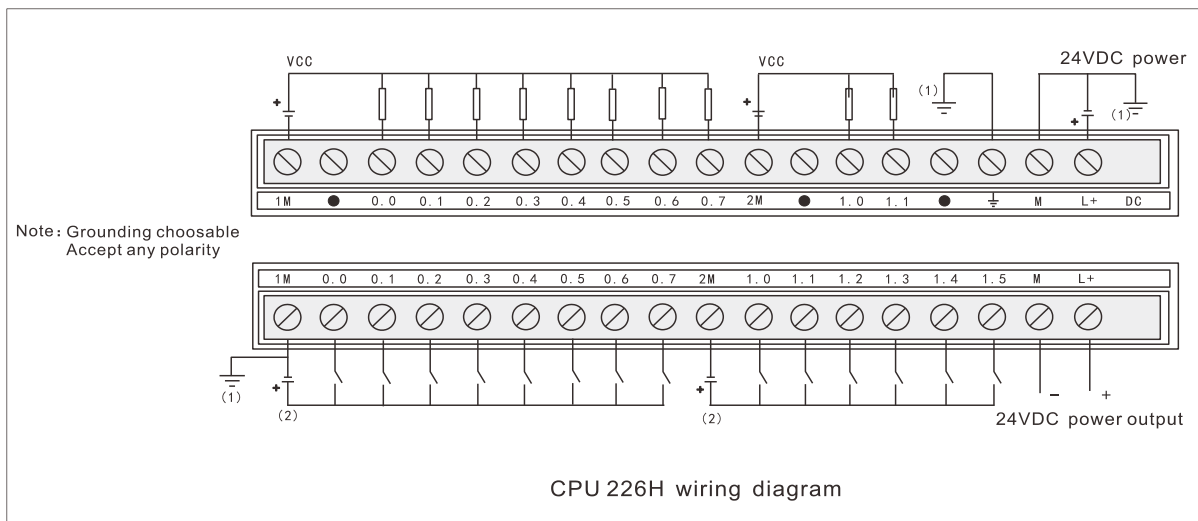


Figure 4-5 CPU 226H/226H V5 (upgrade) wiring diagrams

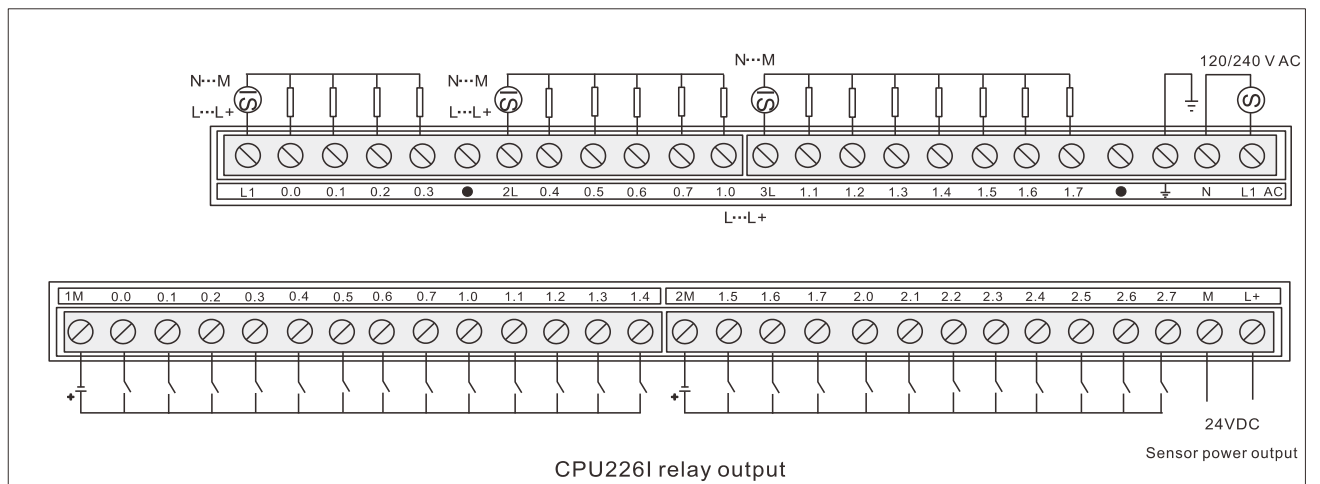
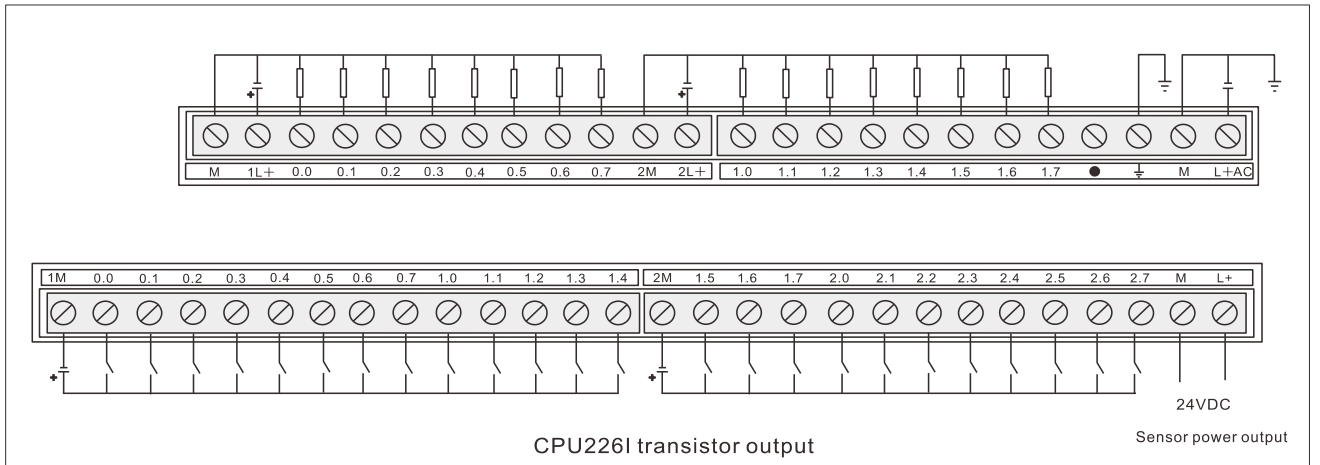


Figure 4-6 CPU 226I wiring diagrams

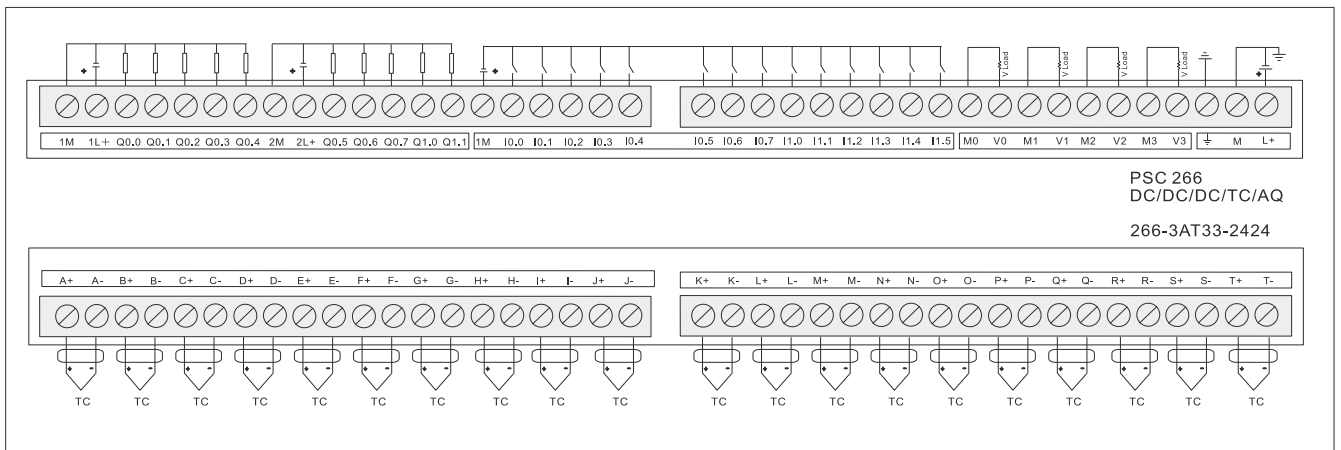


Figure 4-7 PSC266 wiring diagrams

4.3 Digital Module Specifications

Table 4-3-1 Digital module Order No.

Description	Order No.
TrustPLC CTSC-200 EM221 digital input module, 8 inputs	CTS7 221-1BF32
TrustPLC CTSC-200 EM221 digital input module, 16 inputs	CTS7 221-1BH32
TrustPLC CTSC-200 EM221 digital input module, 32 inputs	CTS7 221-1BL32
TrustPLC CTSC-200 EM221 digital input module, 32 inputs, compact	CTS7 221-2BL32
TrustPLC CTSC-200 EM222 transistor-type output module, 8 outputs	CTS7 222-1BF32
TrustPLC CTSC-200 EM222 transistor-type output module, 16 outputs	CTS7 222-1BH32
TrustPLC CTSC-200 EM222 transistor-type output module, 32 outputs	CTS7 222-1BL32
TrustPLC CTSC-200 EM222 relay-type output module, 8 outputs	CTS7 222-1HF32
TrustPLC CTSC-200 EM222 relay-type output module, 16 outputs	CTS7 222-1HH32
TrustPLC CTSC-200 EM223 digital I/O module, 4 inputs/4 outputs, 24V DC	CTS7 223-1BF32
TrustPLC CTSC-200 EM223 digital I/O module, 8 inputs/8 outputs, 24V DC	CTS7 223-1BH32
TrustPLC CTSC-200 EM223 digital I/O module, 16 inputs/16 outputs, 24V DC	CTS7 223-1BL32
TrustPLC CTSC-200 EM223 digital I/O module, 4DI×24V DC, 4DO×Relay	CTS7 223-1HF32
TrustPLC CTSC-200 EM223 digital I/O module, 8DI×24V DC, 8DO×Relay	CTS7 223-1PH32
TrustPLC CTSC-200 EM223 digital I/O module, 16DI×24V DC, 16DO×Relay	CTS7 223-1PL32

4.3.1 Digital Input Module Specifications

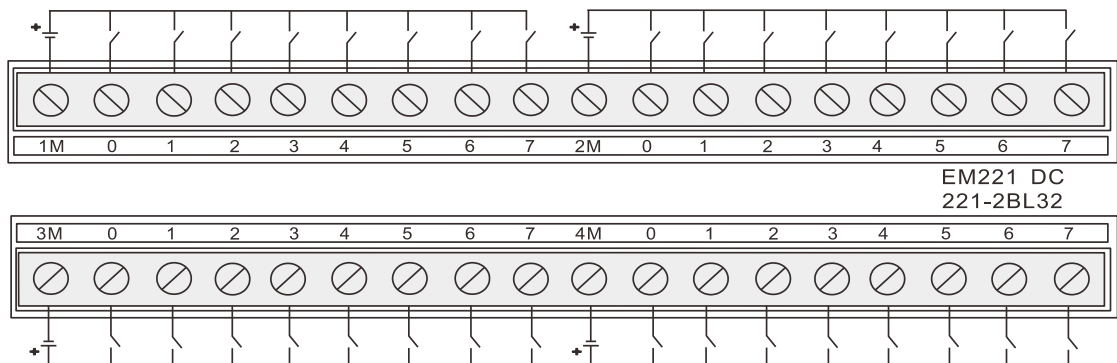
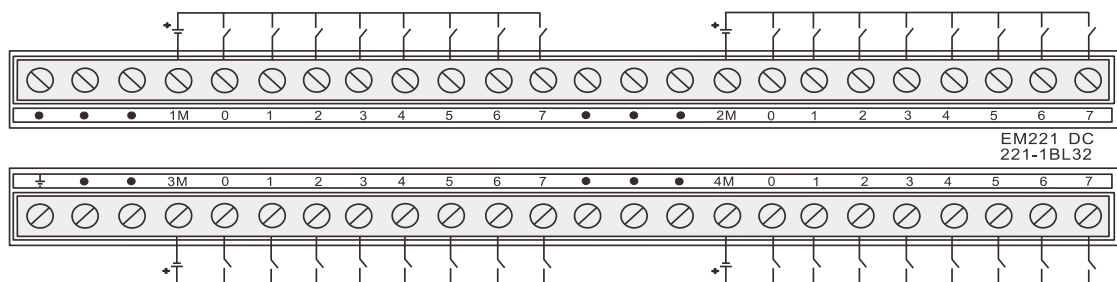
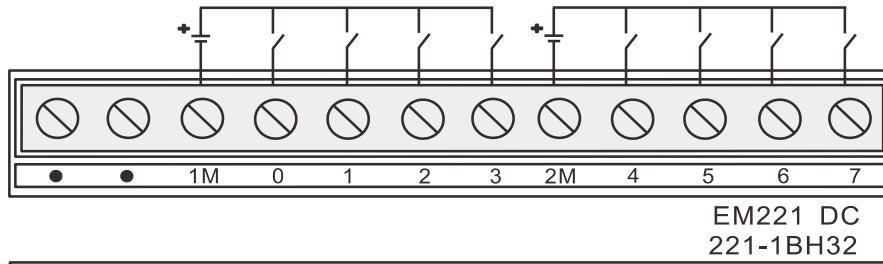
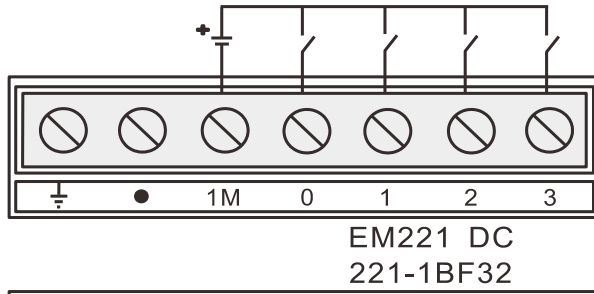
Table 4-3-2 Digital input module Order No.

Description	Order No.
TrustPLC CTSC-200 EM221 digital input module, 8 inputs	CTS7 221-1BF32
TrustPLC CTSC-200 EM221 digital input module, 16 inputs	CTS7 221-1BH32
TrustPLC CTSC-200 EM221 digital input module, 32 inputs	CTS7 221-1BL32
TrustPLC CTSC-200 digital input module, 32 inputs, compact	CTS7 221-2BL32

Table 4-3-3 Digital input module specifications

Inputs	8 inputs, 24V DC	16 inputs, 24V DC	32 inputs, 24V DC	32 inputs, 24V DC
Dimensions				
(W×H×D)	46×80×62mm	71×80×62mm	138×80×62mm	71×80×62mm
Power dissipation				
Power	2W	3W		
+5V DC current	40 mA	65 mA	111 mA	110 mA
Input features				
Inputs	8	16	32	
Type	Source/Drain			
Input voltage				
Rating	24V DC			
Max allowable voltage	30V DC			
Surge voltage	35V DC for 0.5s			
Signal "1"	15~30V			
Signal "0"	0~5V			
Isolation				
Photoelectric isolation	500V AC for 1 minute			
Points in each group	4	Points in each group	4	Points in each group
Max input delay	4.5ms			
Leakage current	Max 1mA AC			
Cable length (max)				
Unshielded	300 m			
Shielded	500 m			
Inputs with simultaneous ON				
40℃	8	16	32	
50℃	8	16	32	

Terminal connections



4.3.2 Digital Output Module Specifications

4.3.2.1 Transistor Output

Table 4-3-4 Digital output modules Order No.

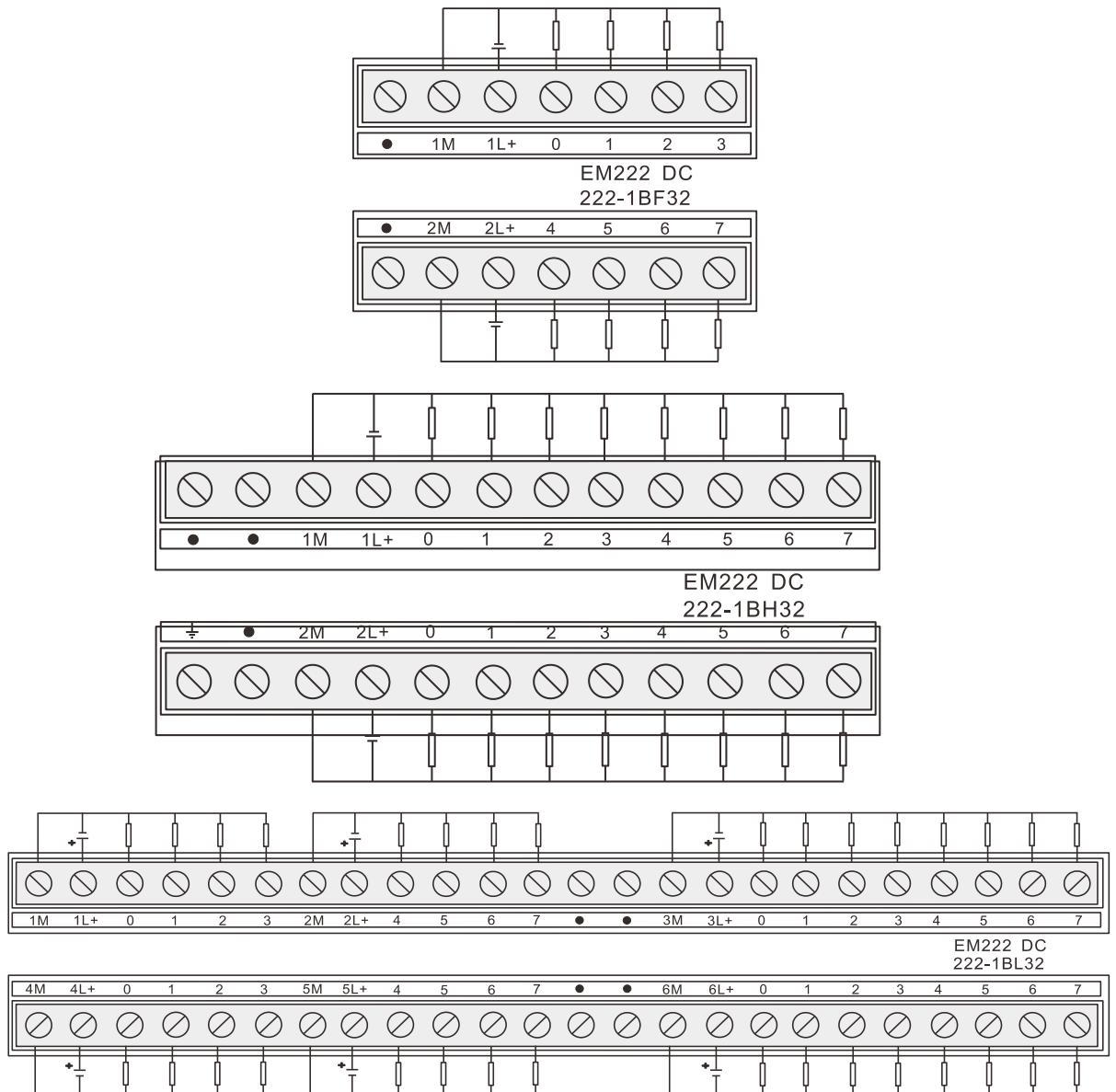
Description	Order No.
TrustPLC CTSC-200 EM222 Transistor output module, 8 outputs	CTS7 222-1BF32
TrustPLC CTSC-200 EM222 Transistor output module, 16 outputs	CTS7 222-1BH32
TrustPLC CTSC-200 EM222 Transistor output module, 32 outputs	CTS7 222-1BL32

Table 4-3-5 Digital output modules specifications

Features	8, 24V DC	16, 24V DC	32, 24V DC
Dimensions			
(W×H×D)	46×80×62mm	71×80×62mm	138×80×62mm
Power dissipation			
Power	2W	3W	
+5V DC current	74.4mA	130 mA	245 mA
Outputs			
Counts	8	16	32
Type	Solid – MOSFET		
Output voltage			
Rated voltage	24V DC		
Allowable voltage range	20.4~28.8V DC		
Signal "1"	Min 20V DC		
Signal "0"	Max 0.1V DC (10KΩ load)		
Output current			
Signal "1"	0.5A		
Groups	2		6
Points in each group	4	8	4/8
Outputs with simultaneous ON	8	16	32
Max current for each group	2A	4A	2/4A
Lamp load	5W		
On resistance	0.3Ω		
Max Leakage current on each point	10μA		

Surge current (max)	8A, 100ms		
Isolation			
Optical coupler isolation	Opto-isolator, 500V AC for 1 minute		
Isolated points in each group	4	8	4/8
Output delay			
OFF—ON	Max 50μs		
ON—OFF	Max 200μs		
Cable length (max)			
Unshielded	150 m		
Shielded	500 m		

Terminal connections



4.3.2.2 Relay Output

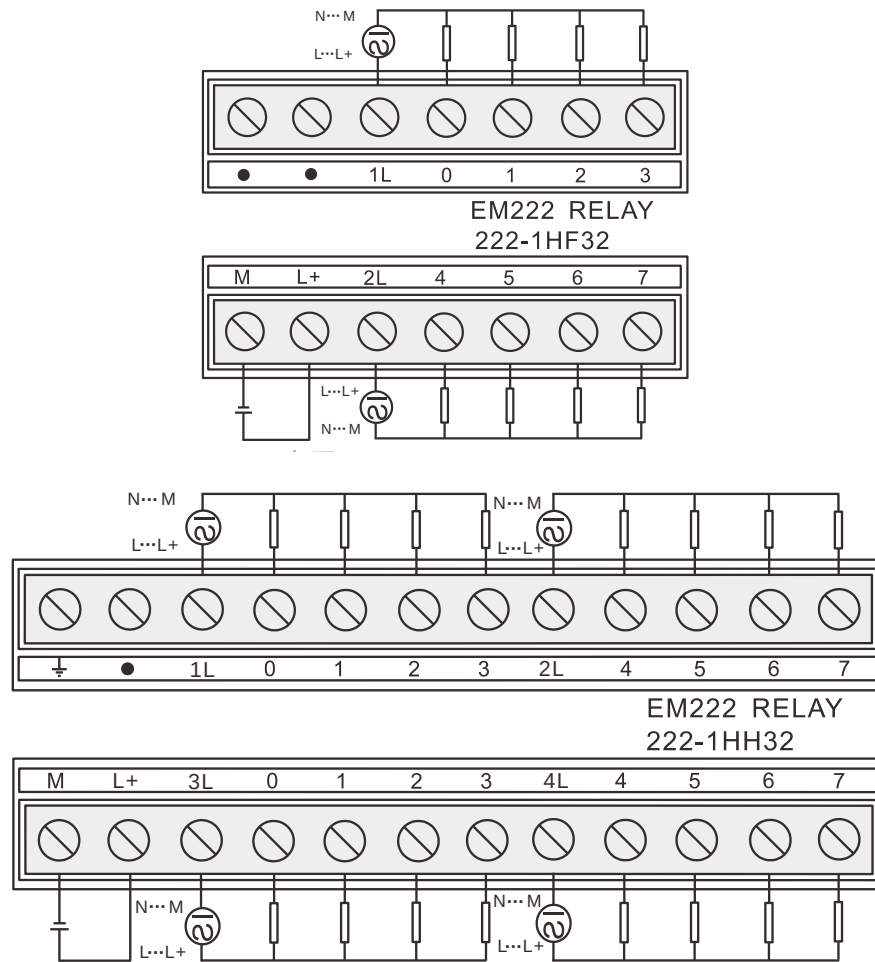
Table 4-3-6 Digital outputs modules Order No.

Description	Order No.
TrustPLC CTSC-200 EM222 relay output module, 8 outputs	CTS7 222-1HF32
TrustPLC CTSC-200 EM222 relay output module, 16 outputs	CTS7 222-1HH32

Table 4-3-7 Digital outputs modules specifications

Features	8, relay	16, relay
Dimensions		
(W×H×D)	46×80×62mm	71×80×62mm
Power dissipation		
Power	2W	3W
Current (+5V DC)	56 mA	80 mA
Output features		
Counts	8	16
Type	Dry-contact	
Output voltage	DC: 5~30V, AC: 5~250V	
Output current		
Signal "1"	2A	
Groups	2	4
Points in each group	4	
Points with simultaneous ON	8	16
Max current for each group	8A	
lamp load	5W	
ON resistance	0.2Ω	
Max surge current	7A at closed contact	
Short protection	Supplied externally	
Isolation		
Points in each group	4	
Coil - logic power	none	
Coil - contact	1500V AC for 1 minute	
Resistor (Coil - contact)	Min 100MΩ	
Relay features		
ON-OFF delay	15ms	
ON/OFF frequency	1HZ	
Mechanical lifetime (no load)	30,000,000	
Contact lifetime (rated load)	300,000	
Cable length (max)		
Unshielded	150 m	
Shielded	500 m	

Terminal connections



4.3.3 Digital I/O Module Specifications

4.3.3.1 Transistor Output

Table 4-3-8 Transistor output modules Order No.

Description	Order No.
TrustPLC CTSC-200 EM223 digital I/O module, 4 inputs/4 outputs, 24V DC	CTS7 223-1BF32
TrustPLC CTSC-200 EM223 digital I/O module, 8 inputs/8 outputs, 24V DC	CTS7 223-1BH32
TrustPLC CTSC-200 EM223 digital I/O module, 16 inputs/16 outputs, 24V DC	CTS7 223-1BL32

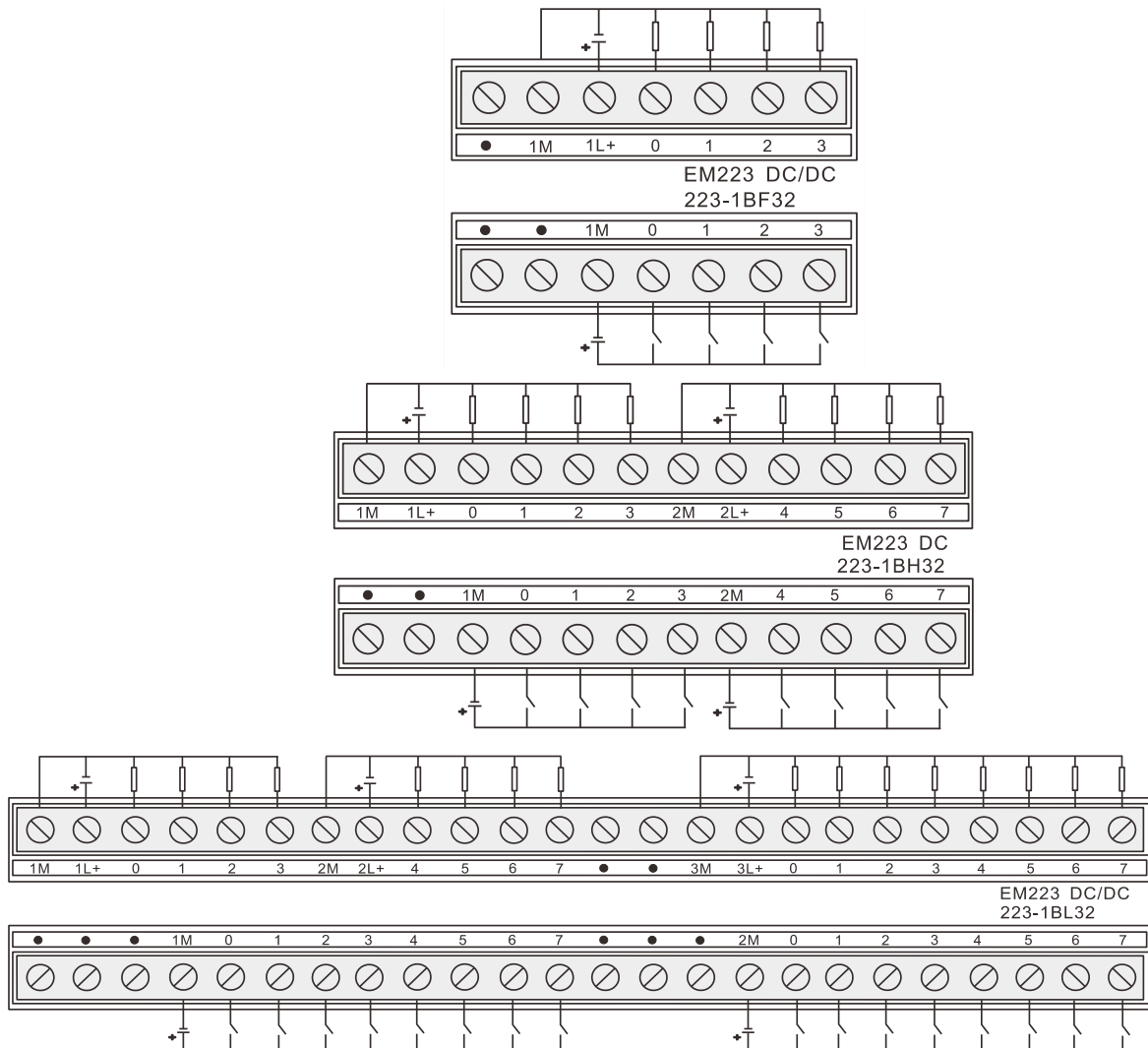
Table 4-3-9 Transistor output modules specifications

Features	4, 24V DC	8, 24V DC	16, 24V DC
Dimensions			
(W×H×D)	46×80×62mm	71×80×62mm	138×80×62mm
Power dissipation			

Power	2W	3W	
+5VDC current	54mA	63mA	173mA
Inputs			
Counts	4	8	16
Type	Source/Drain		
Input voltage			
Rated voltage	24V DC		
Max allowable voltage	30V DC		
Surge voltage	35V DC, 0.5s		
Signal “1”	15~30V		
Signal “0”	0~5V		
Isolation			
Photoelectric isolation	500V AC for 1 minute		
Counts in each group	4	8	
Max input delay	4.5ms		
Leakage current	Max 1mA AC		
Max cable length			
Unshielded	300 m		
Shielded	500 m		
Inputs with simultaneous ON			
40℃	4	8	16
50℃	4	8	16
Output			
Counts	4	8	16
Type	Solid—MOSFET		
Output voltage			
Rated load voltage	24V DC		
Allowable voltage range	20.4~28.8V DC		
Signal "1"	Min 20V DC		
Signal "0"	Max 0.1V DC (10K Ω load)		
Output current			
Signal "1"	0.5A		
Output groups	1	2	3
Points in each group	4		4/4/8
Outputs with simultaneous ON	4	8	16
Max current for each group	2A	4A	2/2/4A

Lamp load	5W	
Contact resistance	0.3Ω	
Peak current for each point	10μA	
Max surge current	8A, 100ms	
Isolation	Opto-coupler isolation, 500V AC for 1 minute	
Isolated points in each group	4	4/4/8
Output delay		
OFF—ON	Max 50μs	
ON—OFF	Max 200μs	
Max cable length		
Unshielded	150 m	
Shielded	500 m	

Terminal connection



4.3.3.2 Relay Output

Table 4-3-10 Relay output modules Order No.

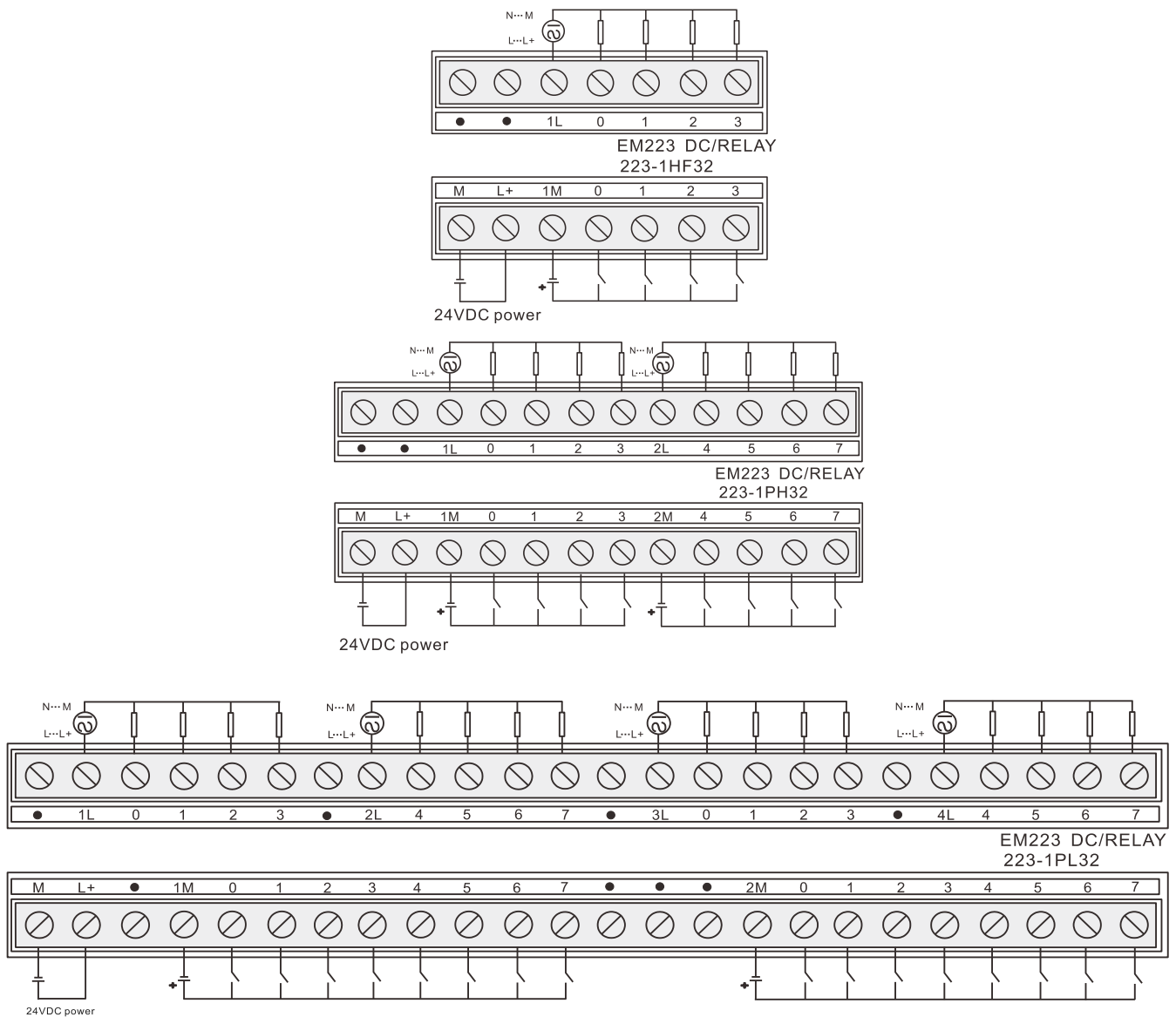
Description	Order No.
TrustPLC CTSC-200 EM223 digital I/O module, 4DI×24V DC, 4DO×Relay	CTS7 223-1HF32
TrustPLC CTSC-200 EM223 digital I/O module, 8DI×24V DC, 8DO×Relay	CTS7 223-1PH32
TrustPLC CTSC-200 EM223 digital I/O module, 16DI×24V DC, 16DO×Relay	CTS7 223-1PL32

Table 4-3-11 Relay output modules specifications

Features	4, relay	8, relay	16, relay
Dimension			
(W×H×D)	46×80×62mm	71×80×62mm	138×80×62mm
Power dissipation			
Power	2W	3W	6W
+5VDC Current	49mA	79mA	138mA
L+ Current	9mA per output at ON		
L+ voltage range	20.4~28.8V DC		
Input			
Counts	4	8	16
Type	Drain/Source		
Input voltage			
Rating	24V DC		
Max allowable voltage	30V DC		
Surge voltage	35V DC, 0.5s		
Signal "1"	15~30V		
Signal "0"	0~5V		
Isolation			
Photoelectric isolation	500V AC for 1 minute		
Points in each group	4	8	
Max input delay	4.5ms		
Leakage current	Max 1mA AC		
Cable length (Max)			
Unshielded	300 m		
Shielded	500 m		
Points for Simultaneous ON			
40℃	4	8	16
50℃	4	8	16

Output			
Counts	4	8	16
Type	Relay-dry contact		
Output voltage range	DC: 5~30V, AC: 5~250V		
Output Current			
Signal "1"	2A		
Output groups	1	2	4
Points in each group	4		
Max current for each group	4	8	16
Lamp load	8A		
Contact resistance	DC:30W, AC:200W		
Max surge current	0.2Ω		
Short protection	7A, contact off		
Signal "1"	Supplied externally		
Isolation			
Points in each group	4		
Coil - contact	None		
Coil - logic power	1500V AC for i minute		
Resistor (Coil - contact)	Min 100MΩ		
Relay features			
ON-OFF delay	15 ms		
ON/OFF frequency	1HZ		
Mechanical lifetime (with no load)	30,000,000		
Contact lifetime (rated load)	300,000		
Cable length (Max)			
Unshielded	150m		
Shielded	500m		

Terminal connection



4.4 Analog Expansion Module Specifications

Table 4-4-1 Analog expansion modules Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 high accuracy analog input module, 8 inputs, 16BIT, current input, Photoelectric isolation	CTS7 231-1HF32
TrustPLC CTSC-200 EM231 high speed and accuracy analog input module, quad voltage inputs, 16BIT, integrated dual 10V DC output	CTS7 231-7HC32
TrustPLC CTSC-200 EM231 high speed and accuracy analog input module, dual voltage inputs, 16BIT, integrated dual 10V DC output	CTS7 231-7HB32

TrustPLC CTSC-200 EM231 analog input module, current-voltage type, 4AI, 12BIT, Photoelectric isolation	CTS7 231-0HC32
TrustPLC CTSC-200 EM231 high accuracy analog input module, 8 AI 16BIT, voltage input, Photoelectric isolation	CTS7 231-0HF32
TrustPLC CTSC-200 EM231 high accuracy analog input module, 8AI x 12bit, current-voltage input	CTS7 231-5HF32
TrustPLC CTSC-200 EM232 analog output module., dual current-voltage outputs, 12BIT, Photoelectric isolation	CTS7 232-0HB32
TrustPLC CTSC-200 EM232 analog output module, quad current-voltage outputs, 12BIT, Photoelectric isolation	CTS7 232-0HF32
TrustPLC CTSC-200 EM235 analog I/O module, 4 AI/1AQ, 12BIT, current-voltage type, Photoelectric isolation	CTS7 235-0KD32

4.4.1 Analog Input Module

4.4.1.1 CTS7 231-1HF32 Specifications

COTRUST provides dedicated 8 channels analog module of current input type CTS7 231-1HF32 for current output type transmitter, which is easier to use and cost effective to achieve higher speed and better performance.

Table 4-4-2 Order No.

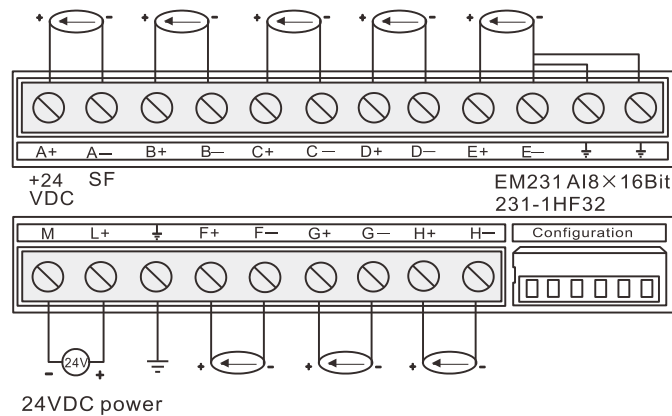
Description	Order No.
TrustPLC CTSC-200 EM231 high accuracy analog input module, 8 inputs, 16BIT, current input, Photoelectric isolation	CTS7 231-1HF32

Table 4-4-3 Analog input module specifications

Used for	CTSC-200 CPU224/226/PSC266	S7-224/226
Features		
(W×H×D)	120.5×80×62mm	
Weight	210 g	
Plug-in I/O terminal	No	
Power dissipation		
Bus (5V DC)	25mA	
L+	30MA	
L+ current range	20.4~28.8V DC	
Power	1W	
Analog input		
Counts	8	
Types	Current	
Scale range	0~20mA; 4~20mA	
Data word format	0~32000	
Resolution	0.000625 mA (0~20mA) 0.0005 mA (4~20mA)	
Measure accuracy	0.1%	

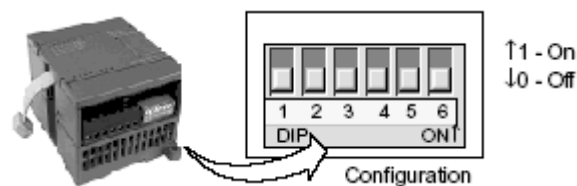
Input impedance	250 Ω
Update time	100ms (8 channels)
Common voltage	120V AC
Rejection	90dB, 60Hz @ DC
Input filtering attenuation	-3db @ 325 HZ
Diagnose	
Diagnose program	+24VDC power light, On: normal supply, Off: No power, all channels turn to 32766 wrong switch setting, all channels turn to 32767
Isolation	
On-site —logic	500V AC
On-site—24VDC	500V AC
24VDC—logic	500V AC

Terminal connection



Select range and configure software

【Select DIP switch】



【Select range】

The following table shows how to set the range of EM231 8AI with DIP switch. Use 1, 2 and 3 to select the analogy input range which can be set to the same.

Attention: the unused DIP switch SW4 ~ SW6 must be set to OFF.

Unipolar			Full range input	Resolution
SW1	SW2	SW3		
OFF	OFF	OFF	0-20mA	0.000625mA
	OFF	ON	4-20mA	0.0005mA

【Configure software】

For EM231 8AI×16 bits, reading in VW instead of AIW. The relative location of the module is different, so is the corresponding address. Calculation formula:

$$x(\text{VW}x) = \text{slot No.} \times 64 + \text{input channel} \times 2$$

The slot number of the address quick reference table corresponds to the installation location of the module. The first expansion module close to the CPU has the slot number of 0, the second expansion module has the slot number of 1, and so on. There are 8 input channels, from A to G, and the corresponding number is 0 to 7.

Quick reference of address based on formula above:

VWxx	channel 0	channel 1	channel 2	channel 3	channel 4	channel 5	channel 6	channel 7
slot 0	VW0	VW2	VW4	VW6	VW8	VW10	VW12	VW14
slot 1	VW64	VW66	VW68	VW70	VW72	VW74	VW76	VW78
slot 2	VW128	VW130	VW132	VW134	VW136	VW138	VW140	VW142
slot 3	VW192	VW194	VW196	VW198	VW200	VW202	VW204	VW206
slot 4	VW256	VW258	VW260	VW262	VW264	VW266	VW268	VW270
slot 5	VW320	VW322	VW324	VW326	VW328	VW330	VW332	VW334
slot 6	VW384	VW386	VW388	VW390	VW392	VW394	VW396	VW398



Notice

Since the address assigned by COTRUST's TD2X text display panel and SIEMENS 'TD200 text display panel is fixed at VW0, if you need to use both TD2X or TD200 and EM231 8AI modules in your system, the EM231 8AI module cannot be installed in the first position (Slot 0), otherwise it will not work properly.

4.4.1.2 CTS7 231-7HC32/ 7HB32 Specifications

Table 4-4-4 Order No.

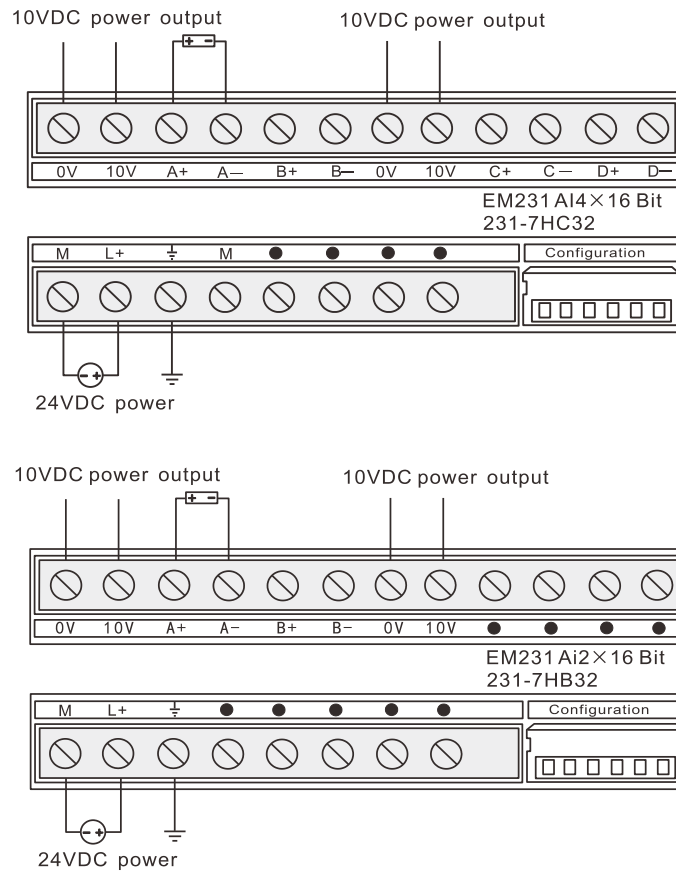
Description	Order No.
TrustPLC CTSC-200 EM231 high speed and accuracy analog input module, 4 voltage inputs, 16BIT, integrated 2 10VDC outputs	CTS7 231-7HC32
TrustPLC CTSC-200 EM231 high speed and accuracy analog input module, 2 voltage inputs, 16BIT, integrated 2 10VDC outputs	CTS7 231-7HB32

Table 4-4-5 Analog input modules specifications

Types	CTS7 231-7HC32	CTS7 231-7HB32
Features		
(W×H×D)	71×80×62mm	
LED indicator lights	24V, On: Power supply normal, Off: Abnormal power supply	
Current dissipation		
+5VDC current	40mA	
L+	35 mA	

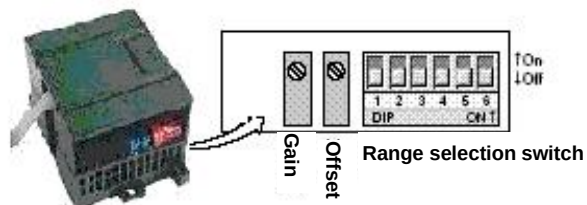
L+ coil voltage	20.4~28.8V DC	
Power	3.5W	
Power supply		
Output voltage	10V DC	
Rated output current	10mA	
Short circuit protection	Yes	
Analog input features		
Counts	4	2
Isolation(field and logic)	None	
Type of input	Different input	
Range	±10V, ±5V	
Max input voltage	30V	
Max input current	25mA	
Reverse polarity protection	Yes	
Allowable different voltage		
UCM voltage	3V	
Max different voltage between input ground and power ground	75V DC	
Input current	0~20mA	
Measurement	Continuous close	
Shifting time for each channel	200μs	
Max shifting time for 4 channels	1ms	
Precision	15 bits + sign bit	
CMR	More than 85dB, DC - 60Hz	
Voltage measurement error(limit value)	Less than 0.1%	
Linear error(all range)	± 0.05 %	± 0.025 %
Resolution of AD converter	16 bits	

Terminal connection



Calibration and range

【Calibration and the position of range switch】



【Input calibration】

Because the calibration adjustment affects the OPAMP behind the analog multiplexer, it affects all user input channels. In addition, because the component parameters of the input channels in front of the multiplexer may differ, there is a slight difference of the same input signal on different channels even after calibration.

The input has been filtered in the module, so the measurements are stable. Start the analog input filter for all input of the module for better performance parameters. When calculating the average value, select the sampling times above 64.

Calibration steps:

- 1) Cut off external power of modules, configure DIP switch to select needed range.
- 2) Connect CPU with module power, wait for 15min above.

- 3) Use a transmitter, a voltage input source or a current input source to add the zero signal to module input.
- 4) Read measurement of the input in CPU.
- 5) Adjust OFFSET potentiometer till the reading is zero, finish zero set calibration.
- 6) Input a full scale signal to read the measurement in CPU.
- 7) Adjust GAIN potentiometer till the reading is 32000.
- 8) Repeat 3~7 when necessary.

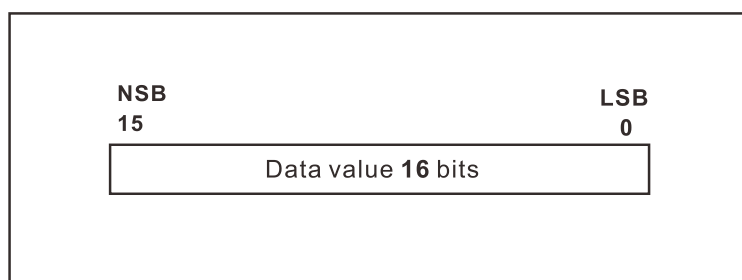
【Select range】

The following table shows how to set the range of EM231 4AI with DIP switch. Use 1, 2 and 3 to select the analogy input range which can be set to the same.

Attention: the unused DIP switch SW4 ~ SW6 must be set to OFF.

SW1	SW2	SW3	Full scale	Resolution
OFF	OFF	ON	$\pm 10\text{ V}$	0.3mV
	ON	OFF	$\pm 5\text{ V}$	0.15mV

Input data format



4.4.1.3 CTS7 231-0HC32 Specifications

Table 4-4-6 Order No.

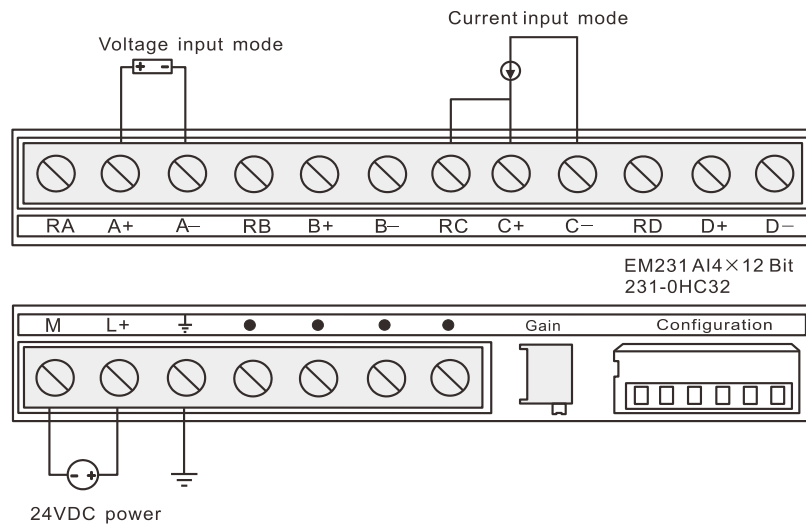
Description	Order No.
TrustPLC CTSC-200 EM231 analog input module, 4 current and voltage inputs, 12BIT, opto-isolator	CTS7 231-0HC32

Table 4-4-7 Analog input module specifications

Features	
(W×H×D)	71×80×62mm
LED indicator lights	24V, on: power supply normal, off: abnormal power supply
Current dissipation	
+5V DC current	34mA
L+	17 mA
L+ coil voltage	20.4~28.8V DC
power	2W

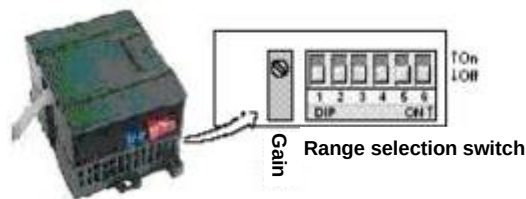
Analog input features	
Counts	4
Isolation(field and logic)	opto-isolator: 500VAC, 1min
Type of input	Different input
Range	
Voltage input(unipolarity)	0~10V, 0~5V
Voltage input(bipolar)	±5V, ±2.5V
Current input	0~20 mA
Data Word format	
Unipolarity, full range	0~32000
Bipolar, full range	-32000~32000
Input resolution	
Voltage input(unipolarity)	2.5 mV (0~10V) 1.25 mV (0~5V)
Voltage input(bipolar)	2.5 mV (±5V) 1.25 mV (±2.5V)
Current input	5μA (0~20mA)
A/D conversion time	Less than 300μs
Analog input response time	1.5ms
CMR	40dB, DC~60Hz
CMV	Signal voltage+CMV (must be less than or equal to 12V)
Input impedance	More than 10MΩ
Max input voltage	30V
Max input current	30mA

Terminal connection



Calibration and range

【Calibration and the position of range switch】



【Input calibration】

Because the calibration adjustment affects the OPAMP behind the analog multiplexer, it affects all user input channels. In addition, because the component parameters of the input channels in front of the multiplexer may differ, there is a slight difference of the same input signal on different channels even after calibration.

The input has been filtered in the module, so the measurements is stable. Start the analog input filter for all input of the module for better performance parameters. When calculating the average value, select the sampling times above 64.

Calibration steps:

- 1) Cut off external power of modules, configure DIP switch to select needed range.
- 2) Connect CPU with module power, wait for 15min above.
- 3) Use a transmitter, a voltage input source or a current input source to add the zero signal to module input.
- 4) Read measurement of the input in CPU.
- 5) Adjust GAIN potentiometer till the reading is 32000.

【Range selection】

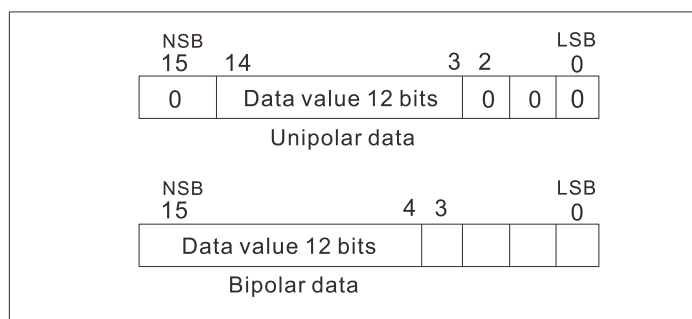
The following table shows how to set the range of EM231 4AI with DIP switch. Use 1, 2

and 3 to select the analogy input range which can be set to the same.

Attention: the unused DIP switch SW4 ~ SW6 must be set to OFF.

Unipolarity			Full scale input	Resolution
SW1	SW2	SW3		
ON	OFF	ON	0 - 10V	2.5mV
	ON	OFF	0 -5V	1.25mV
			0 - 20mA	5 uA
Bipolar			Full scale input	Resolution
SW1	SW2	SW3		
OFF	OFF	ON	± 5V	2.5mV
	ON	OFF	± 2.5 V	1.25mV

Input data format



Notice

A 12-bit reading of A/D converter (ADC) whose data format is left-aligned. The most significant bit is the sign bit (0 represents positive number). For unipolar format, 3 consecutive zeroes make the ADC count value change by 1 unit, then the change of data word is changed by 8 units. For bipolar format, four consecutive zeroes make the ADC count value to change by 1 unit, and the change of data word is changed by 16 units.

4.4.1.4 CTS7 231-0HF32 Specifications

Table 4-4-8 Order No.

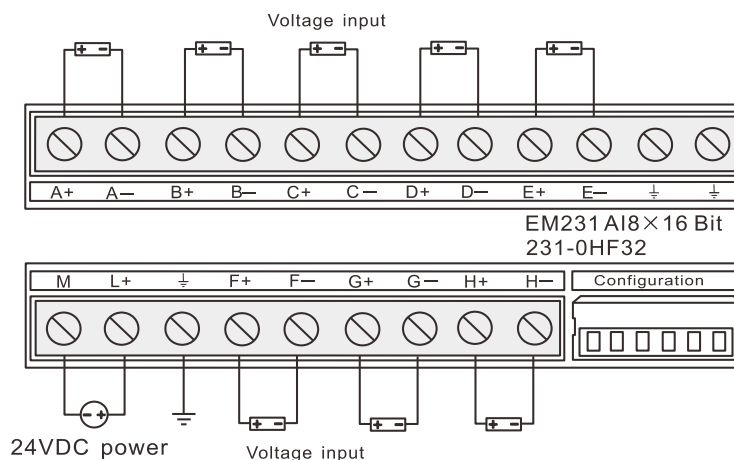
Description	Order No.
TrustPLC CTSC-200 EM231 high accuracy analog input module, 8 inputs, 16BIT, opto-isolator	CTS7 231-0HF32

Table 4-4-9 Analog inputs module specifications

Features	
(W×H×D)	71×80×62mm
LED indicator lights	24V, on: power supply normal, off: abnormal power supply
Current dissipation	
+5V DC current	44 mA
L+	31mA
L+ coil voltage	20.4~28.8V DC
Power	2.5 W

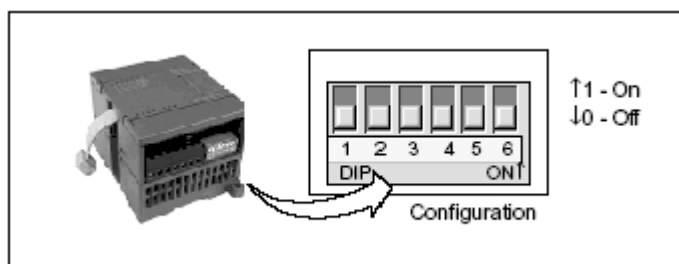
Analog input features	
Counts	8
Isolation(field and logic)	Opto-isolator: 500VAC, 1min
Type of input	Different input
Range	
Voltage input(unipolarity)	0~10V, 0~5V
Voltage input(bipolar)	±5V,±2.5V
Data Word format	
Unipolarity, full range	0~32000
Bipolar, full range	-32000~32000
Input resolution	
Voltage input(unipolarity)	300μV (0~10V) 150μV (0~5V)
Voltage input(bipolar)	150μV (range ±5V) 75μV (range ±2.5V)
A/D conversion time	About 10ms
Conversion time for 8channels	100ms
CMR	40dB, DC - 60Hz
CMV	Signal voltage+CMV (must be less than or equal to 12V)
Input impedance	More than 10MΩ
Max input voltage	30V
Max input current	30mA
A/D conversion resolution	16 bits

Terminal connection



Range selection and software configuration

【Range selection switch position】



【Range selection】

The following table shows how to set the range of EM231 8AI with DIP switch. Use 1, 2 and 3 to select the analogy input range which can be set to the same.

Attention: The unused DIP switch SW4 ~ SW6 must be set to OFF.

Unipolarity			Full range input	Resolution
SW1	SW2	SW3		
ON	OFF	ON	0-10V	300μV
	ON	OFF	0 - 5V	150μV
Bipolar			Full range input	Resolution
SW1	SW2	SW3		
OFF	OFF	ON	± 5V	150μV
	ON	OFF	± 2.5 V	75μV

【Software configuration】

For EM231 8AI×16 bits, reading in VW instead of AIW. The relative location of the module is different, so is the corresponding address. Calculation formula:

$$x(\text{VWx}) = \text{slot no.} \times 64 + \text{input channel} \times 2$$

The slot number of the address quick reference table corresponds to the installation location of the module. The first expansion module close to the CPU has the slot number of 0, the second expansion module has the slot number of 1, and so on. There are 8 input channels, from A to G, and the corresponding number is 0 to 7.

Quick reference of address based on formula above:

VWxx	channel0	channel 1	channel 2	channel 3	channel 4	channel 5	channel 6	channel 7
Slot 0	VW0	VW2	VW4	VW6	VW8	VW10	VW12	VW14
Slot 1	VW64	VW66	VW68	VW70	VW72	VW74	VW76	VW78
Slot 2	VW128	VW130	VW132	VW134	VW136	VW138	VW140	VW142
Slot 3	VW192	VW194	VW196	VW198	VW200	VW202	VW204	VW206
Slot 4	VW256	VW258	VW260	VW262	VW264	VW266	VW268	VW270
Slot 5	VW320	VW322	VW324	VW326	VW328	VW330	VW332	VW334
Slot 6	VW384	VW386	VW388	VW390	VW392	VW394	VW396	VW398

**Notice**

Since the address assigned by COTRUST's TD2X text display panel and SIEMENS 'TD200 text display panel is fixed at VW0, if you need to use both TD2X or TD200 and EM231 8AI modules in your system, the EM231 8AI module cannot be installed in the first position (Slot 0), otherwise it will not work properly.

4.4.1.5 CTS7 231-5HF32 Specifications

Table 4-4-10 Order No.

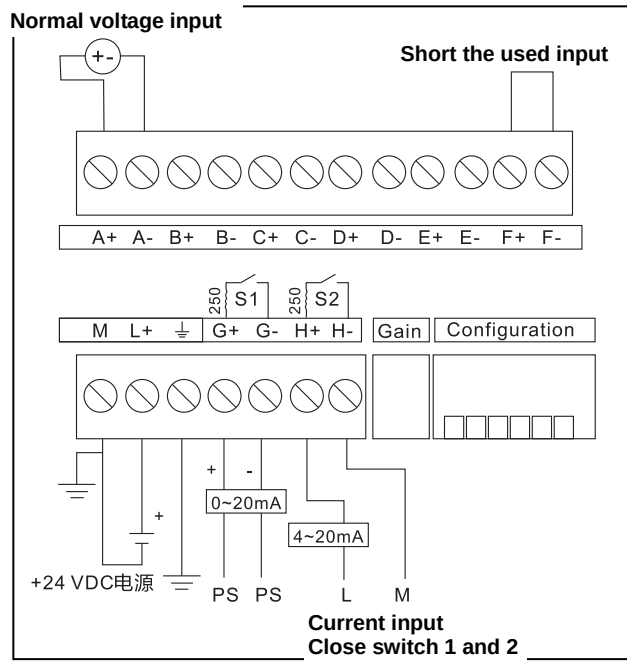
Description	Order No.
TrustPLC CTSC-200 EM231 high accuracy analog input module, 8AI x 12bit, for both current and voltage input	CTS7 231-5HF32

Table 4-4-11 Analog inputs module specifications

Features	
(W×H×D)	71×80×62mm
LED indicator lights	24V, on: power supply normal, off: abnormal power supply SF, ON: Abnormal, OFF: Normal
Current dissipation	
Rated input voltage	24V DC
Input voltage range	20.4V~28.8V DC
Input current	50mA
Reverse polarity protection	Yes
Bus supply voltage	+5V DC
Bus supply current	50mA
Inputs	
Type	Differential voltage, 2 channels for current selection
Counts	8
Range:	
Voltage	0~5V, 0~10V, $\pm 2.5V$ (channel 0~7)
Current	0~20mA (channel 6~7)
Max input voltage	30V DC
Max input current	32mA
Input impedance:	
voltage	$\geq 2M \Omega$
current	250 Ω
Data format	Voltage: 0~+32000(Unipolarity), -32000~+32000(Bipolar) Current: 0~+32000
Input step response	1.5ms to 95%
CMR	40dB, DC to 60Hz
CMV	-12V $\leq$ signal voltage+CMV $\leq$ +12V
Resolution	Unipolarity: 12bit Bipolar: 11bit+ sign bits

Measure principle	Successive approximation
Measure error	0.5%(max)
Isolation	
Field to logic	500V AC

Terminal connection



DIP switch

The following table shows how to configure the module using DIP switches. Switches 3, 4 and 5 select the analog input range, switches 1 and 2 select the current input mode. ON of the switch 1 select the channel 6 current input mode; OFF select the voltage mode. ON of switch 2 select the channel 7 current input mode; OFF select the voltage mode.

Unipolar			Full-scale Input	Resolution
SW3	SW4	SW5		
ON	OFF	ON	0~10V	2.5mV
	ON	OFF	0~5V	1.25mV
			0~20mA	5μA
Bipolar			Full-scale Input	Resolution
SW3	SW4	SW5		
OFF	OFF	ON	±5V	2.5mV
	ON	OFF	±2.5V	1.25mV

4.4.2 Analog Output Modules Specifications

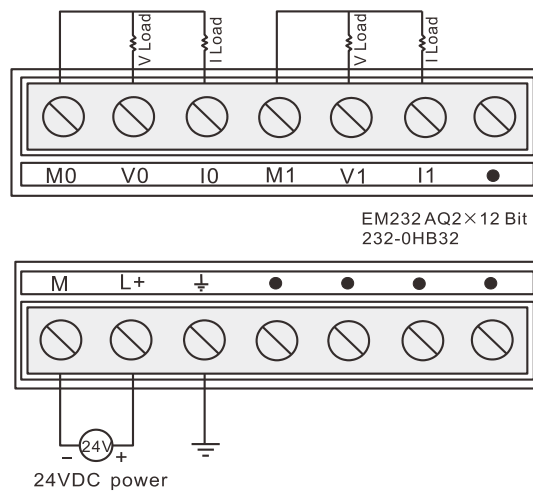
Table 4-4-12 Order No.

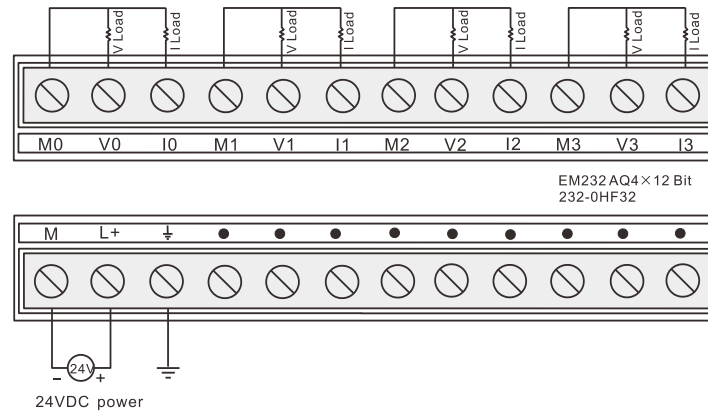
Description	Order No.
TrustPLC CTSC-200 EM232 analog output modules, 2 outputs, 12BIT, optocoupler	CTS7 232-0HB32
TrustPLC CTSC-200 EM232 analog output modules, 4 outputs, 12BIT, optocoupler	CTS7 232-0HF32

Table 4-4-13 Analog output modules specifications

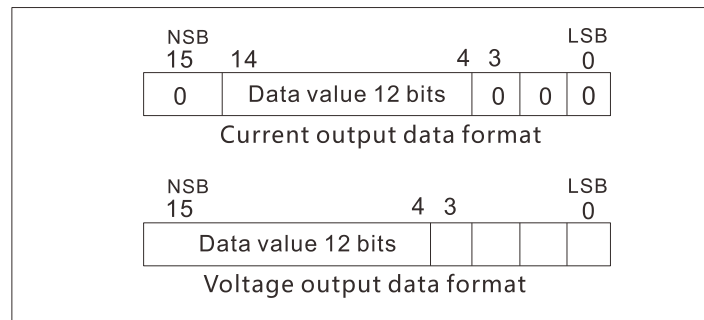
General	EM232: 2 AQ × 12 bits	EM232: 4 AQ × 12 bits
Features		
(W×H×D)	46×80×62mm	71×80×62mm
LED indicator lights	24V, On: Power supply normal, Off: Abnormal power supply	
Current dissipation		
+5V DC current	24mA	31mA
L+	61mA	112mA
L+ coil voltage	20.4~28.8V DC	
Power	2W	
Analog output features		
Counts	2	4
Isolation(A to D)	Optocoupler: 500VAC, 1min	
Output range		
Voltage	±10V	
Current	0~20mA	
Output resolution		
Voltage	12BIT	
Current	11BIT	
Data Word format		
Voltage	-32000~+32000	
Current	0~32000	
Measuring error	Typical: ±0.5% of full-scale; worst case: ±2% of full-scale	
Stable time		
Voltage	100μs	
Current	2ms	
Maximum drive @24V supply		
Voltage	Minimum 5000Ω	
Current	Maximum 500Ω	

Terminal connection





Output data format



Notice

The 12-bit reading of D/A converter, its output data format is left-aligned and its highest significant bit is the symbol bit (0 means positive). Four consecutive zeroes are truncated before the data is loaded into the DAC register, and these bits do not affect the output signal value.

4.4.3 Analog IO Modules Specifications

Table 4-4-14 Order No.

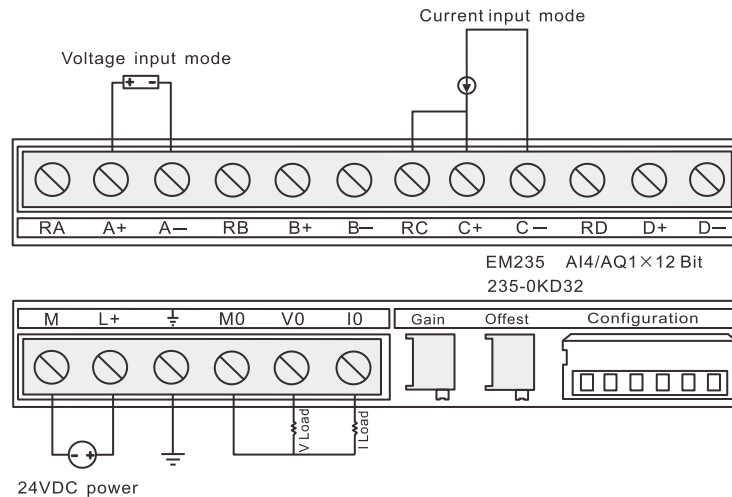
Description	Order No.
TrustPLC CTSC-200 EM235 Analog input&output modules, 4 inputs/1outputs, 12BIT, both voltage¤t, optocoupler	CTS7 235-0KD32

Table 4-4-15 Analog input&output modules specifications

Features	
(W×H×D)	71×80×62mm
LED indicator lights	24V, on: power supply normal, off: abnormal power supply
Current dissipation	
+5V DC current	34 mA
L+	48 mA
L+ coil voltage	20.4~28.8V DC
Power	2W
Analog input features	
Counts	4
Isolation(field and logic)	optocoupler: 500VAC, 1min

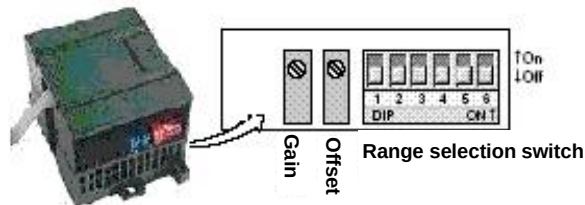
Type	Differential input
Range	
Voltage input(unipolarity)	0-10V, 0-5V, 0-1V, 0-500mV, 0-100mV, 0-50mV
Voltage input(bipolar)	±10V,±5V,±2.5V,±1V,±500mV, ±250mV, ±100mV,±50mV,±25mV
Current input	0~20mA
Data Word format	
Unipolarity, full range	0~32000
Bipolar, full range	-32000~32000
Input resolution	
Voltage input(unipolarity)	2.5mV(0~10V) 1.25mV(0~5V)
Voltage input(bipolar)	2.5mV(±5V range) 1.25mV(±2.5V range)
Current input	5μA (0~20mA range)
AD conversion time	Less than 300μs
A/D input response time	1.5ms
CMR	40dB, DC~60Hz
CMV	Signal voltage+CMV (must be less than or equal to 12V)
Input impedance	More than 10MΩ
Max input voltage	30V
Max input current	30mA
Resolution of AD converter	12 bits
Analog output features	
Counts	1
Voltage	±10V
Current	0~20mA
Output resolution	
Voltage	12BIT
Current	11BIT
Data Word format	
Voltage	-32000~+32000
Current	0~32000
Precision	
Voltage	Typical: ±0.5% of full-scale; worst case: ±2% of full-scale
Current	
Stable time	
Voltage	100μs
Current	2ms
Maximum drive @24V supply	
Voltage	Minimum 5000Ω
Current	Maximum 500Ω

Terminal connection



Calibration, range and gain selection

【Calibration and the position of range switch】



【Input calibration】

Because the calibration adjustment affects the OPAMP behind the analog multiplexer, it affects all user input channels. In addition, because the component parameters of the input channels in front of the multiplexer may differ, there is a slight difference of the same input signal on different channels even after calibration.

The input has been filtered in the module, so the measurements is stable. Start the analog input filter for all input of the module for better performance parameters. When calculating the average value, select the sampling times above 64.

Calibration steps:

- 1) Cut off external power of modules, configure DIP switch to select needed range.
- 2) Connect CPU with module power, wait for 15min above.
- 3) Use a transmitter, a voltage input source or a current input source to add the zero signal to module input.
- 4) Read measurement of the input in CPU.
- 5) Adjust OFFSET potentiometer till the reading is zero, finish zero set calibration.
- 6) Input a full scale signal to read the measurement in CPU.
- 7) Adjust GAIN potentiometer till the reading is 32000.
- 8) Repeat 3~7 when necessary.

【Range and gain selection】

Table 4-4-16 shows how to configure the EM235 module using DIP switches. Use SW1 and SW6 to select the analog input range and resolution. Set all inputs the same analog input range and format. Table 4-4-17 shows how to select unipolar/bipolar (SW6), GAIN (SW4 and SW5) and attenuation (SW1, SW2 and SW3). ON is closed, OFF is open.

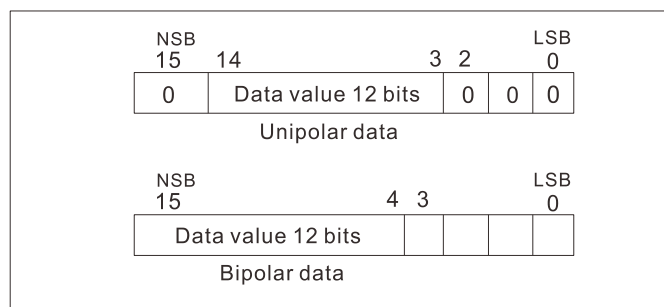
Table 4-4-16 EM235 analog input range and resolution selection

Unipolar						Full-scale input	Resolution
SW1	SW2	SW3	SW4	SW5	SW6		
ON	OFF	OFF	ON	OFF	ON	0- 50 mV	12.5 μ V
OFF	ON	OFF	ON	OFF	ON	0 - 100 mV	25 μ V
ON	OFF	OFF	OFF	ON	ON	0 - 500 mV	125 μ V
OFF	ON	OFF	OFF	ON	ON	0 -1 V	250 μ V
ON	OFF	OFF	OFF	OFF	ON	0 - 5 V	1.25 μ V
ON	OFF	OFF	OFF	OFF	ON	0 -20 mA	5 μ A
OFF	ON	OFF	OFF	OFF	ON	0 -10 V	2.5 mV
Bipolar						Full-scale input	Resolution
SW1	SW2	SW3	SW4	SW5	SW6		
ON	OFF	OFF	ON	OFF	OFF	$\pm$ 25 mV	12.5 μ V
OFF	ON	OFF	ON	OFF	OFF	$\pm$ 50 mV	25 μ V
OFF	OFF	ON	ON	OFF	OFF	$\pm$ 100 mV	50 μ V
ON	OFF	OFF	OFF	ON	OFF	$\pm$ 250 mV	125 μ V
OFF	ON	OFF	OFF	ON	OFF	$\pm$ 500 mV	250 μ V
OFF	OFF	ON	OFF	ON	OFF	$\pm$ 1 V	500 μ V
ON	OFF	OFF	OFF	OFF	OFF	$\pm$ 2.5 V	1.25 mV
OFF	ON	OFF	OFF	OFF	OFF	$\pm$ 5 V	2.5 mV
OFF	OFF	ON	OFF	OFF	OFF	$\pm$ 10 V	5 mV

Table 4-4-17 EM235 unipolar/bipolar, GAIN and attenuation selection

EM 235 switches						Unipolar/bipolar	GAIN	Attenuation
SW1	SW2	SW3	SW4	SW5	SW6			
---	---	---	---	---	ON	Unipolar	---	---
---	---	---	---	---	OFF	Bipolar	---	---
---	---	---	OFF	OFF	---	---	x1	---
---	---	---	OFF	ON	---	---	x10	---
---	---	---	ON	OFF	---	---	x100	---
---	---	---	ON	ON	---	---	---	---
ON	OFF	OFF	---	---	---	---	---	0.8
OFF	ON	OFF	---	---	---	---	---	0.4
OFF	OFF	ON	---	---	---	---	---	0.2

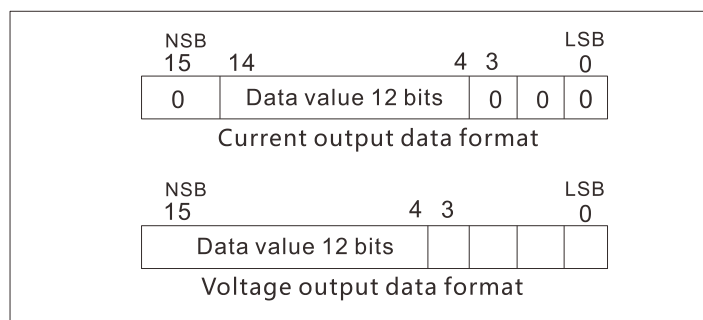
Input data format



Notice

The 12-bit reading of D/A converter, its output data format is left-aligned and its highest significant bit is the symbol bit (0 means positive). For unipolar format, 3 consecutive zeroes make the ADC count value change by 1 unit, so data word changes by 8 units. For bipolar format, 4 consecutive zeroes cause the ADC count value to change by 1 unit, so the data word changes by 16 units.

Output data format



Notice

The 12-bit reading of D/A converter, its output data format is left-aligned and its highest significant bit is the symbol bit (0 means positive). 4 consecutive zeroes are truncated before the data is loaded into the DAC register, and these bits do not affect the output signal value.

4.5 Temperature Measuring Module Specifications

Table 4-5-1 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 thermal resistance input modules, 2 inputs	CTS7 231-7PB32
TrustPLC CTSC-200 EM231 thermal resistance input modules, 4 inputs	CTS7 231-7PC32
TrustPLC CTSC-200 EM231, 4AI×TC, J/K/S/T/R/E/N, +/-80mV	CTS7 231-7PD32
TrustPLC CTSC-200 EM231, 8AI×TC, J/K/S/T/R/E/N, +/-80mV	CTS7 231-7PF32
TrustPLC CTSC-200 EM231 hybrid input modules, 2 NTC or PT100, 2inputs, isolated 16 bits accuracy	CTS7 231-7ND32
TrustPLC CTSC-200 EM231 analog input modules 8AI×Pt100/NTC	CTS7 231-7NF32

4.5.1 Thermal Resistance Measuring Module Specifications

Table 4-5-2 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 thermal resistance input modules, 4 inputs	CTS7 231-7PB32
TrustPLC CTSC-200 EM231 thermal resistance input modules, 4 inputs	CTS7 231-7PC32

Table 4-5-3 RTD modules specifications

Types	EM231 2AI×RTD		EM231: 4AI×RTD
Features			
(W×H×D)	71×80×62mm		71×80×62mm
LED indicator lights	24V, ON: power supply normal, OFF: abnormal power supply SF, ON: Abnormal, OFF: Normal, BLINK: wrong input signal		
Current dissipation			
+5V DC current	35 mA		
L+	34 mA	37 mA	
L+ coil voltage	20.4~28.8V DC		
Power	1.7W		
Input specifications			
Type	Referenced ground thermal resistance		
Counts	2	4	
Range	Type of thermal resistance (pick one): Pt-100Ω, 200Ω, 500Ω, 1000Ω(α=3850ppm, 3920ppm, 3850.55ppm, 3916ppm, 3902ppm) Pt-10000Ω(α=3850ppm) Cu-9.035Ω(α=4720ppm) Ni-100Ω, 120Ω, 1000Ω(α=6720ppm, 6178ppm) R-150Ω, 300Ω, 600Ω		
Temperature measuring range	Pt-100Ω, 200Ω, 500Ω, 1000Ω: -200℃~850℃ Pt-10000Ω: -200℃~600℃ NI-0.00672: -80℃~260℃ NI-0.006178: -60℃~300℃ Cu-0.004270: -200℃~260℃ <Note> Measuring out of range report an error, refer to table 4-5-2.		
Isolation			
Field to logic	500V AC		
Field to 24V DC	500V AC		
24V to logic	500V AC		
Common mode input range(input channel to input channel)	0		
CMR	>120dB@120V AC		
Input resolution			

Temperature	0.1℃/0.1℉	
Voltage	15 bits+sign bit	
Measuring principle	Sigma-Delta	
Turnover time(all channel)	425ms	825ms
Cable length to sensor	Maximum 100m	
Loop resistance	20Ω, Cu 2.7Ω	
Noise suppression	85dB@ 50Hz/60Hz/400Hz	
Data Word format		
Input impedance	>10MΩ	
Maximum input voltage	30V DC(detection), 5V DC	
Resolution	15 bits+sign bit	
Input filter attenuation	-3dB@ 21kHz	
Basic error	0.1% FS(resistance)	
Repeatability	0.05% FS	

Terminal connection

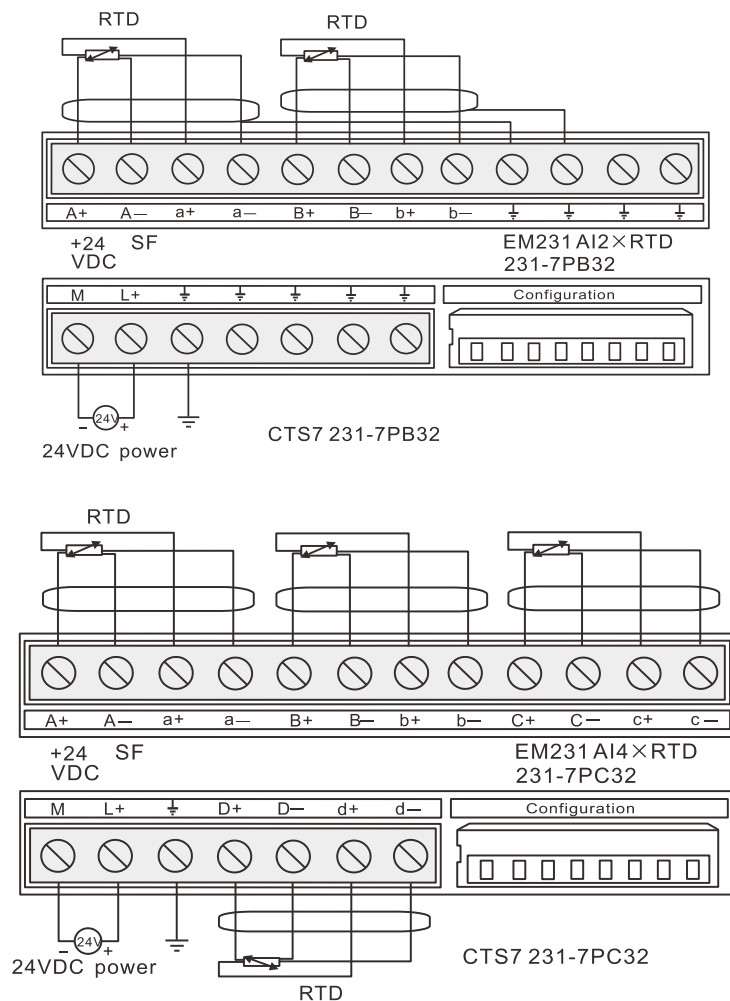


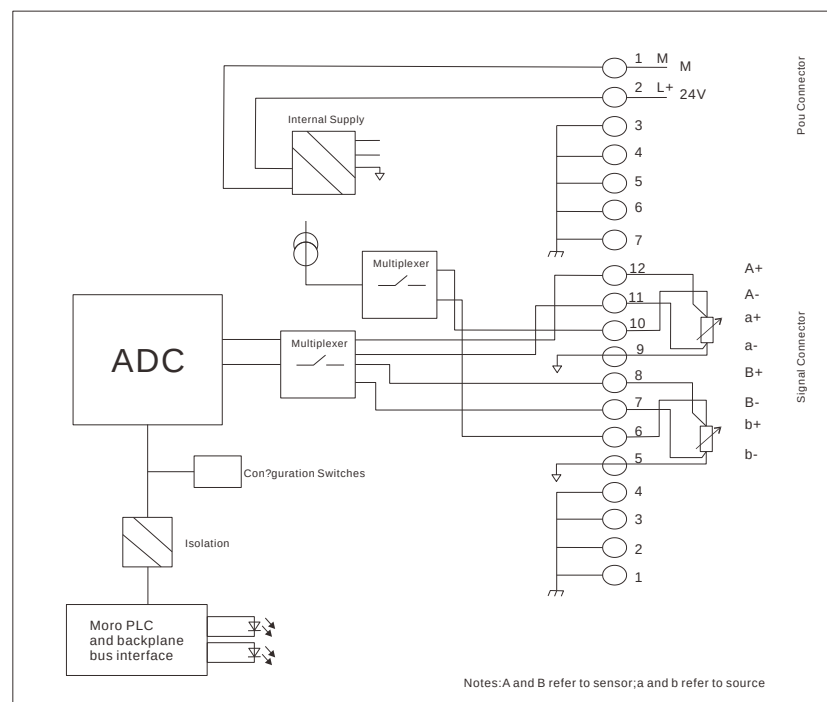
Table 4-5-4 Thermal resistance modules diagnostic

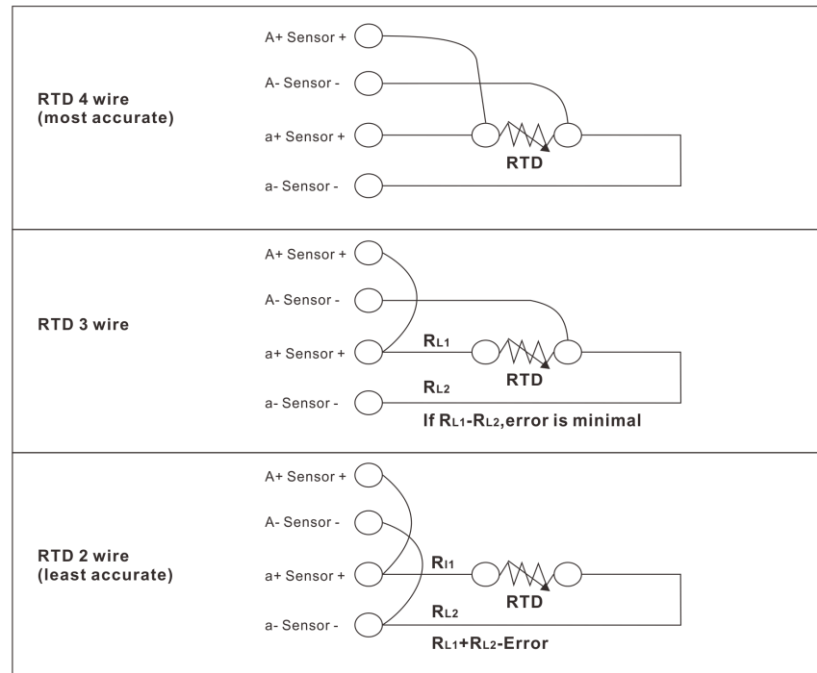
Error type	Channel data	SF LED	24V LED	Range status bit	24V power failure
No module power	32766	OFF	OFF	0	1
Offline	32767(positive calibration) -32768(negative calibration)	BLINK	ON	1	0
Temperature out of range	32767(positive calibration) -32768(negative calibration)	BLINK	ON	1	0

<Note> Range status bit is the bit in the byte of the error register of the module. You can check SMB8~21 according to the order of the module and query the error code of the corresponding module.

SMB8	Module 1 Identify Register
SMB9	Module 1 Error Register
SMB10	Module 2 Identify Register
SMB11	Module 2 Error Register
SMB12	Module 3 Identify Register
SMB13	Module 3 Error Register
SMB14	Module 4 Identify Register
SMB15	Module 4 Error Register
SMB16	Module 5 Identify Register
SMB17	Module 5 Error Register
SMB18	Module 6 Identify Register
SMB19	Module 6 Error Register
SMB20	Module 7 Identify Register
SMB21	Module 7 Error Register

Connection between RTD and sensor





Notes: A refer to sensor, a refer to source

Users can directly connect the thermal resistance sensor to the CTSC-200 EM231 module or use the extended wiring mode. Use shielded wire for the best noise resistance. If you use shielded wire, connect the shielded wire to the 1-4 pin connection point of the signal connector. The ground is in common with 3-7 pins of the power connector. For the unused thermal resistance input channels, you should connect a resistor to the unused input channel to prevent errors due to floating-ground input signals from affecting the display of the active channel.

Connect the power supply to pins 1 and 2 of the power connector. Connect pin 3 of the power connector to a nearby housing ground. You may choose one of three wiring methods to connect the thermal resistance module to the sensor. 4-wire for highest accuracy, 2-wire for lowest accuracy. It is recommended to use 2-wire connection only when you care less about the wire error in the application.

【Select thermal resistance type】

EM 231 modules make CTSC-200 PLC connect with massive thermal resistance sensor conveniently. You can select the type of hot resistance, wiring mode, measurement unit and direction of open circuit fault by DIP switch. All thermal resistors connected to the same module must be same type.

As following figure shows, the DIP switch is at the lower right of modules. In order for the DIP switch setting to work, you need to Power off then on for the PLC or the 24VDC.

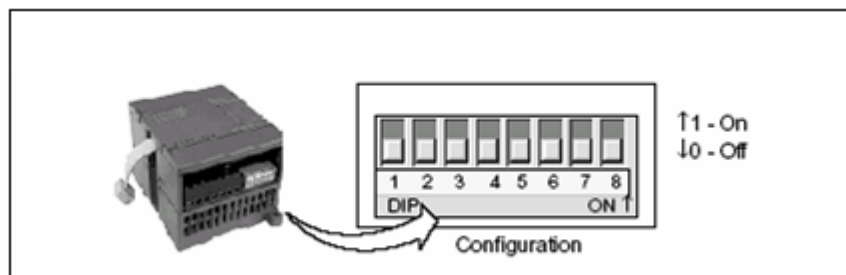


Table 4-5-5 Configuration of DIP switch

Thermal resistance	SW1	SW2	SW3	SW4	SW5
100 Pt 0.003850(Default)	0	0	0	0	0
200Ω Pt 0.003850	0	0	0	0	1
500Ω Pt 0.003850	0	0	0	1	0
1000Ω Pt 0.003850	0	0	0	1	1
100Ω Pt 0.003920	0	0	1	0	0
200Ω Pt 0.003920	0	0	1	0	1
500Ω Pt 0.003920	0	0	1	1	0
1000Ω Pt 0.003920	0	0	1	1	1
100Ω Pt 0.00385055	0	1	0	0	0
200Ω Pt 0.00385055	0	1	0	0	1
500Ω Pt 0.00385055	0	1	0	1	0
1000Ω Pt 0.00385055	0	1	0	1	1
100Ω Pt 0.003916	0	1	1	0	0
200Ω Pt 0.003916	0	1	1	0	1
500Ω Pt 0.003916	0	1	1	1	0
1000Ω Pt 0.003916	0	1	1	1	1
100Ω Pt 0.00302	1	0	0	0	0
200Ω Pt 0.003902	1	0	0	0	1
500Ω Pt 0.003902	1	0	0	1	0
1000Ω Pt 0.003902	1	0	0	1	1
Retained	1	0	1	0	0
100Ω Ni 0.00672	1	0	1	0	1
120Ω Ni 0.00672	1	0	1	1	0
1000Ω Ni 0.00672	1	0	1	1	1
100Ω Ni 0.006178	1	1	0	0	0
120Ω Ni 0.006178	1	1	0	0	1
1000Ω Ni 0.006178	1	1	0	1	0
10000Ω Pt 0.003850	1	1	0	1	1
10Ω Cu 0.004270	1	1	1	0	0
150Ω FS Resistance	1	1	1	0	1
300Ω FS Resistance	1	1	1	1	0
600Ω FS Resistance	1	1	1	1	

Table 4-5-6 Configuration of SW6-SW8

SW6	Calibration direction	SW7	Unit of measuring	SW8	Wiring method
0	Positive (+3276.7°)	0	°C	0	3
1	Negative (-3276.8°)	1	°F	1	2 or 4

4.5.2 Thermal Resistance Measuring Modules Specifications

Table 4-5-7 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231, 4AI×TC, J/K/S/T/R/E/N, +/-80mV	CTS7 231-7PD32
TrustPLC CTSC-200 EM231, 8AI×TC, J/K/S/T/R/E/N, +/-80mV	CTS7 231-7PF32

Table 4-5-8 TC specifications

Types	EM231 4AI×TC	EM231 8AI×TC
Features		
(W×H×D)	71×80×62mm	71×80×62mm
LED indicator lights	24V, ON: power supply normal, OFF: abnormal power supply SF, ON: Abnormal, OFF: Normal, BLINK: wrong input signal	
Current dissipation		
+5V DC current	35 mA	47 mA
L+	30 mA	30 mA
L+ coil voltage	20.4~28.8V DC	
Power	1.7W	1.7W
Input specifications		
Type	Suspension type thermal resistance	
Counts	4	8
Range	Type of thermal resistance (pick one): S, T, R, E, N, K, J Voltage: +/-80mV <Note> Temperature measuring out of range report an error, refer to Chapter 4.5.2 Thermal Resistance Measuring Modules Specifications . Measuring out of range report an error, refer to table 4-5-2.	
Isolation		
Field to logic	500V AC	
Field to 24V DC	500V AC	
24V to logic	500V AC	
Common mode input range(input channel to input channel)	120V AC	
CMR	>120dB@120V AC	
Input resolution		
Temperature	0.1℃/0.1℉	
Voltage	15 bits+ sign-bit	
Measuring principle	Sigma-Delta	
Turnover time(all channel)	425ms	825ms
Cable length to sensor	Maximum 100m	
Loop resistance	Maximum 100Ω	
Noise suppression	85dB@ 50Hz/60Hz/400Hz	
Data word format	Voltage: -27648 to +27648	
Input impedance	>1MΩ	

Maximum input voltage	30V DC
Resolution	15 bits+ sign-bit
Input filter attenuation	-3dB@ 21kHz
Basic error	0.1% FS(voltage)
Repeatability	0.05% FS
CJ error	±1.5℃

Terminal connection

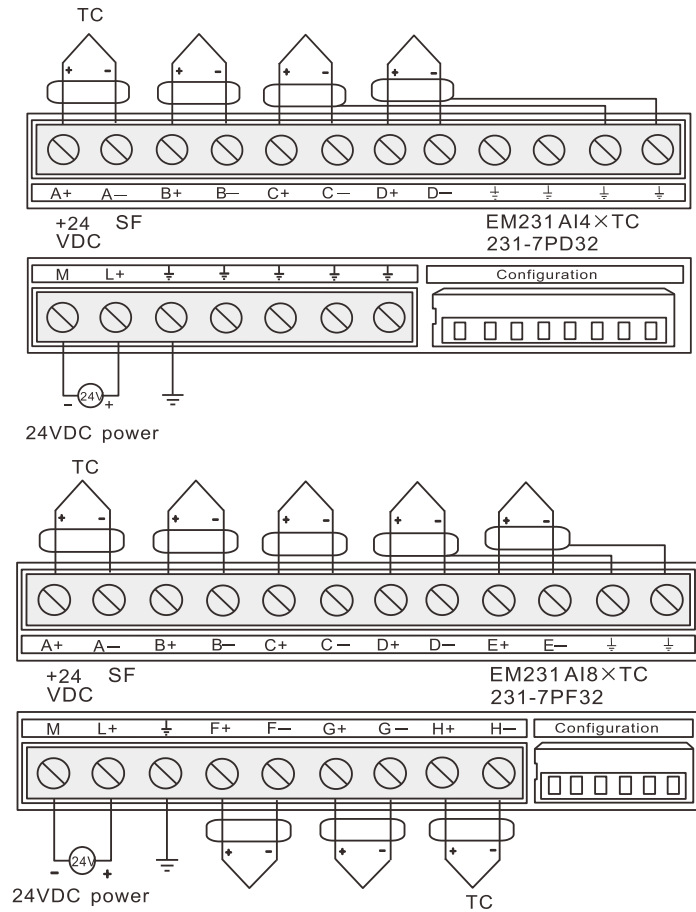


Table 4-5-9 Thermal resistance modules diagnostic

Error type	Channel data	SF LED	24V LED	Range status bit	24V power failure
No module power	32766	OFF	OFF	0	1
Offline	32767(positive calibration) -32768(negative calibration)	BLINK	ON	1	0
Temperature out of range	32767(positive calibration) -32768(negative calibration)	BLINK	ON	1	0

<Note> Range status bit is the bit in the byte of the error register of the module. You can check SMB8~21 according to the order of the module and query the error code of the corresponding module.

SMB8	Module 1 Identify Register
SMB9	Module 1 Error Register
SMB10	Module 2 Identify Register
SMB11	Module 2 Error Register
SMB12	Module 3 Identify Register
SMB13	Module 3 Error Register
SMB14	Module 4 Identify Register
SMB15	Module 4 Error Register
SMB16	Module 5 Identify Register
SMB17	Module 5 Error Register
SMB18	Module 6 Identify Register
SMB19	Module 6 Error Register
SMB20	Module 7 Identify Register
SMB21	Module 7 Error Register

Measuring range of thermal resistance

The temperature range(°C) and accuracy of multiple thermocouple

Data word (1 digit bit=0.1°C)		TypeJ	TypeK	TypeT	TypeE	TypeR,S	TypeN	±80mV	
Decimalism	Hexadecimal								
32767	7FFF	>1200.0°C	>1372.0°C	>400.0°C	>1000.0°C	>1768.0°C	>1300.0°C	>94.071mV	OF
↑	↑							↑	↑
32511	7EFF							97.071mV	
:	:								
27649	6C01							80.0029mV	OR
27648	6C00							80mV	
:	:								
17680	4510					↑			
:	:								
13720	3598		↑						
:	:		1372.0°C over range						
13000	32C8	↑	1300.0°C				↑		
:	:						1300.0°C		NR
12000	2EE0	1200.0°C							
:	:								
10000	2710			↑					
:	:								
4000	0FA0			-400.0°C					
:	:								
1	0001	0.1°C	0.1°C	0.1°C	0.1°C	0.1°C	0.1°C	0.0029mV	
0	0000	0.0°C	0.0°C	0.0°C	0.0°C	0.0°C	0.0°C	0.0mV	
-1	FFFF	-0.1°C	-0.1°C	-0.1°C	-0.1°C	-0.1°C	-0.1°C	-0.0029mV	
:	:								
-500	FE0C					Below the range -50.0°C			
-1500	FA24	-150.0°C							
:	:								
-2000	F830	Below the range	-200.0°C						
:	:								
-2100	F7CC	-210.0°C	Below the range						
:	:								
-2550	F60A			-255.0°C	-255.0°C				
:	:								
-2700	F574	↓	Below the range -270.0°C	Below the range -270.0°C	Below the range -270.0°C		-270.0°C		
:	:								
-27648	9400		↓	↓	↓		↓	-80mV	
-27649	93FF							-80.0029mV	
:	:								
-32512	8100							-94.071mV	UR
#	#							↓	↓
-32768	8000	<-210.0°C	<-270.0°C	<-270.0°C	<-270.0°C	<-50.0°C	<-270.0°C	<-94.07mV	UF
Full-range precision		S0.1%	S0.3%	S0.6%	S0.1%	S0.6%	S0.1%	S0.1%	
Precision		S1.5°C	S1.7°C	S1.4°C	S1.3°C	S3.7°C	S1.6°C	S0.10°C	
Cold end error		S1.5°C	S1.5°C	S1.5°C	S1.5°C	S1.5°C	S1.5°C	NA	
*OF=Upper overflow OR=Over range NR=rated range VR=Below the range UF=Lower overflow									
↑Indicates that all analog quantities greater than this value but less than the break threshold are reported as upper overflow values, 32767 (10x7FFF)									
↓Indicates that all analog quantities less than this value but greater than the break threshold are reported as lower overflow values, -32768 (0x8000)									

Switch and Software Configuration

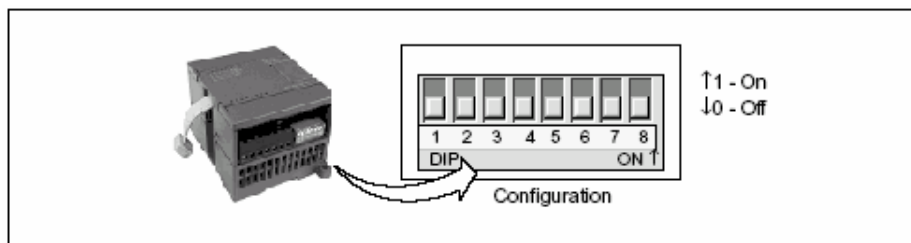
• DIP switch configuration

The EM231 module provides a convenient and isolated interface for the seven thermocouple types: J, K, E, N, S, T, and R. It also allows to connect tiny analog signals (±80mV). All thermocouples connected to the modules must be the same type, and it is best to use a shielded thermocouple sensor.

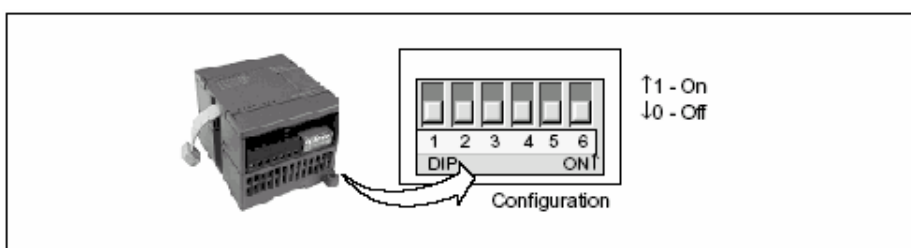
Thermocouple modules use DIP switch to select the thermocouple type, disconnect check, measurement unit, cold end compensation and open fault direction.

For the EM231 4TC module, SW1 ~ SW3 is used to select the thermocouple type, SW4 is

not used (set to the OFF position), SW5 is used to select the direction of wire breaking detection, SW6 is used to select whether to conduct wire breaking detection, SW7 is used to select the measuring unit, and SW8 is used to select whether to conduct cold end compensation.



For the EM231 8TC module, SW1 ~ SW3 is used to select the thermocouple type, SW4 is used to select the direction of wire breaking detection, SW5 is used to select the measuring unit, SW6 is used to select whether to conduct cold end compensation. wire breaking detection is fixed to be yes, no need to set.



In order for the DIP switch setting to work, you need to Power off then on for the PLC or the 24VDC.

Table 4-5-10 Type of thermal resistance configuration

Type	SW1	SW2	SW3
J (default)	0	0	0
K	0	0	1
T	0	1	0
E	0	1	1
R	1	0	0
S	1	0	1
N	1	1	0
+/- 80mV	1	1	1

Table 4-5-11 Other specifications

Items	EM231 4TC		EM231 8TC	
	Switch	setting	Switch	setting
breaking detection direction	SW5	0: positive(+3276.7°) 1: negative (-3276.8°)	SW4	0: positive(+3276.7°) 1: negative (-3276.8°)
Whether detect breaking	SW6	0: yes, 1: no	Fixed for breaking detection	
Select measuring unit	SW7	0: °C, 1: °F	SW5	0: °C, 1: °F
Whether CJC	SW8	0: yes, 1: no	SW6	0: yes, 1: no

- Software Configuration

For EM231 8AI×16 bits, reading in VW instead of AIW. The relative location of the module is different, so is the corresponding address. Calculation formula:

$$x(VWx) = \text{slot no.} \times 64 + \text{input channel} \times 2$$

The slot number of the address quick reference table corresponds to the installation location of the module. The first expansion module close to the CPU has the slot number of 0, the second expansion module has the slot number of 1, and so on. There are 8 input channels, from A to G, and the corresponding number is 0 to 7.

Table 4-5-12 Quick reference of address based on formula above:

VWxx	channel 0	channel 1	channel 2	channel 3	channel 4	channel 5	channel 6	channel 7
slot 0	VW0	VW2	VW4	VW6	VW8	VW10	VW12	VW14
slot 1	VW64	VW66	VW68	VW70	VW72	VW74	VW76	VW78
slot 2	VW128	VW130	VW132	VW134	VW136	VW138	VW140	VW142
slot 3	VW192	VW194	VW196	VW198	VW200	VW202	VW204	VW206
slot 4	VW256	VW258	VW260	VW262	VW264	VW266	VW268	VW270
slot 5	VW320	VW322	VW324	VW326	VW328	VW330	VW332	VW334
slot 6	VW384	VW386	VW388	VW390	VW392	VW394	VW396	VW398



Notice

Since the address assigned by COTRUST's TD2X text display panel and SIEMENS 'TD200 text display panel is fixed at VW0, if you need to use both TD2X or TD200 and EM231 8TC modules in your system, the EM231 8TC module cannot be installed in the first position (Slot 0), otherwise it will not work properly.

4.5.3 Hybrid Temperature Input Modules Specifications

4.5.3.1 CTS7 231-7ND32 specifications

EM231 NTC is analog expansion module of CTSC-200 PLC system, provide 4-channel analog acquisition, two for connecting thermistor NTC temperature sensor or thermal resistance PT100 temperature sensor, the other two for collecting voltage/current signal input, input accuracy of all channels (including the sign bit) are 16 bits. Mainly used for sterilization equipment or central air conditioning equipment, both temperature measurement and pressure signal measurement need occasions.

Table 4-5-13 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 hybrid input modules, 2 NTC or PT100, 2 inputs, isolated 16 bits accuracy	CTS7 231-7ND32

Table 4-5-14 EM231 modules specifications

Features	
(W×H×D)	71×80×62mm
LED indicator lights	24V , ON: power supply normal, OFF: abnormal power supply SF, ON: Abnormal, OFF: Normal, BLINK: wrong input signal

Current dissipation	
+5V DC current	87 mA
L+	60 mA
L+ coil voltage	20.4~28.8V DC
Power	1.7W
Inputs	
Thermal Resistance /thermistor Range	Thermal Resistance type(any type): Pt-100 (3850ppm,3920ppm,3850.55ppm,3916ppm 3902ppm) NTC(R25=10kΩ B=3950, R25=10kΩ B=3435)
Temperature measuring range	Pt-100: -50℃ ~ 850℃ NTC (R25=10K, B=3950) : -40℃~120℃ NTC (R25=10K, B=3435) : -40℃~150℃ <Note> Measuring out of range report an error, refer to table 4-5-2.
Voltage	±5V, ±10V, 0V~5V, 0V~10V
Current	0~20mA
Counts	2PT100/2NTC and 2AI
Isolation	
Field to logic	500VAC
Field to 24V DC	500VAC
24V to logic	500VAC
CMR	>120dB@120VAC
Sampling performance	
Temperature resolution	0.1℃/0.1°F
Voltage resolution	15 bits+ sign-bit
Measuring principle	Sigma-Delta
Turnover time(all channel)	425ms
Cable length to sensor	Maximum 100m
Loop resistance	Maximum 20Ω
Noise suppression	85dB@50Hz/60Hz/400Hz
Data word format	Temperature (NTC: R25=10kΩ, B=3950K) : -400~1200(only for channel 1&2) Temperature (NTC: R25=10kΩ, B=3435K) : -400~1500(only for channel 1&2) Temperature (PT100) : -500~2000(only for channel 1&2) voltage/current: unipolar 0~32000, bipolar -32000~+32000(only for channel 3&4)
Input impedance	Voltage input >10MΩ; current input =250Ω; NTC input >10MΩ
Maximum input voltage	30VDC(detection), 5VDC
Resolution	0.1℃/0.1° F

Input filter attenuation	-3dB@21kHz
Basic error	0.1%FS(resistance)
Repeatability	0.05%FS

Work condition

- Temperature: 0-55℃ for horizontal installation, 0-45℃ for vertical installation
- Humidity: 95% not condensing humidity

Usage

EM231 NTC modules use to extend the analog measuring ability of CTSC-200 PLC CPU by connecting the bus port to CPU modules.

【Configuration】

Input type	SW1	SW2	SW3	SW4	SW5
	Channel 1.2 valid			Channel 3.4 valid	
100ΩPT0.003850(default)	0	0	0	—	—
100ΩPT0.003920	0	0	1	—	—
100ΩPT0.00385055	0	1	0	—	—
100ΩPT0.003916	0	1	1	—	—
100ΩPT0.003902	1	0	0	—	—
NTC: R25=10kΩ, B=3950K	1	0	1	—	—
NTC :R25=10kΩ, B=3435K	1	1	0	—	—
Disable	1	1	1	—	—
0~5V	—	—	—	0	0
0~20mA	—	—	—	0	0
0~10V	—	—	—	0	1
-10V~10V	—	—	—	1	0
-5V~5V	—	—	—	1	1

SW6	Calibration direction(all channels valid)	SW7	Measuring unit	SW8	Wiring method
0	Positive(+3276.7°)	0	℃	0	3
1	Negative(−3276.8°)	1	°F	1	2 /4

【Starting Steps】

- 1) Connect DC24V supply to L+ and M as wiring diagram
- 2) Connect to CPU modules through I/O bus.
- 3) Connect CPU supply and communication port.
- 4) Connect the input signal as wiring diagram, leave the free channels port empty, single-point grounding for EARTH as specifications.
- 5) Configure sensor type, calibration direction, etc. As the configuration requirement.
- 6) Start power of CPU and modules.

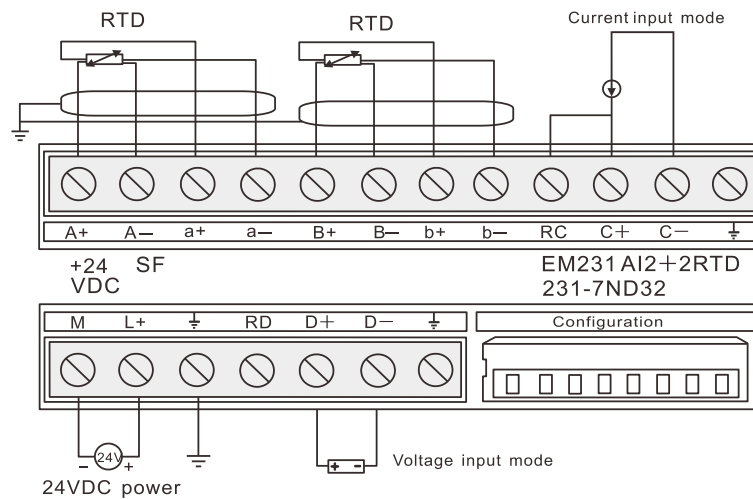
【Acquire input value】

- You can read 2 bytes of input data for each channel in a user program or

programmatic monitoring software such as MagicWorks PLC or MicroWin.

- NTC full range reading is -400~1200 (R25=10K B=3950), corresponding temperature range: -40.0~120.0°C; Or-400~1500 (R25=10K B=3435), corresponding temperature range: -40.0~150.0°C; PT100 full range reading is -500~2000, the corresponding temperature range: -50.0~200.0°C; The return reading in excess range or off line is determined by the direction of the calibration configuration switch (-32768 or 32767).
- Voltage/current full-scale reading is -32000 ~ 32000, readings out of range until 32767/32765 is still the effective.
- Readings is 32766 for all no user power channels.

Terminal connection



4.5.3.2 CTS7 231-7NF32 specifications

Table 4-5-14 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 temperature input modules 8AI×Pt100/NTC, isolated 16 accuracy	CTS7 231-7NF32

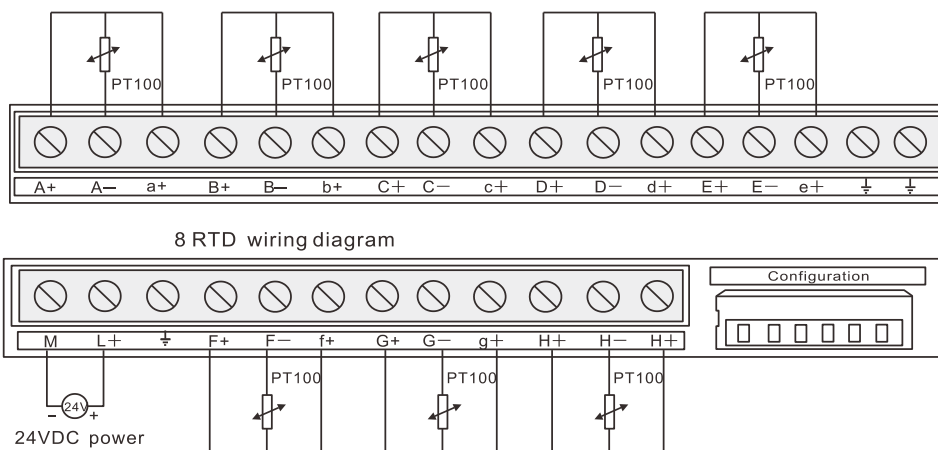
Table 4-5-15 EM231 hybrid input modules specifications

Connect to	CTSC-200 CPU 224/226/PSC266 S7-224/226
(W×H×D)	71 x 80 x 62mm
LED indicator lights	24VDC, ON: power supply normal, OFF: abnormal power supply SF, ON: Abnormal, OFF: Normal, BLINK: wrong input signal
Current dissipation	
+5V DC current	60mA
L+	32.5mA
L+ coil voltage	20.4~28.8V DC

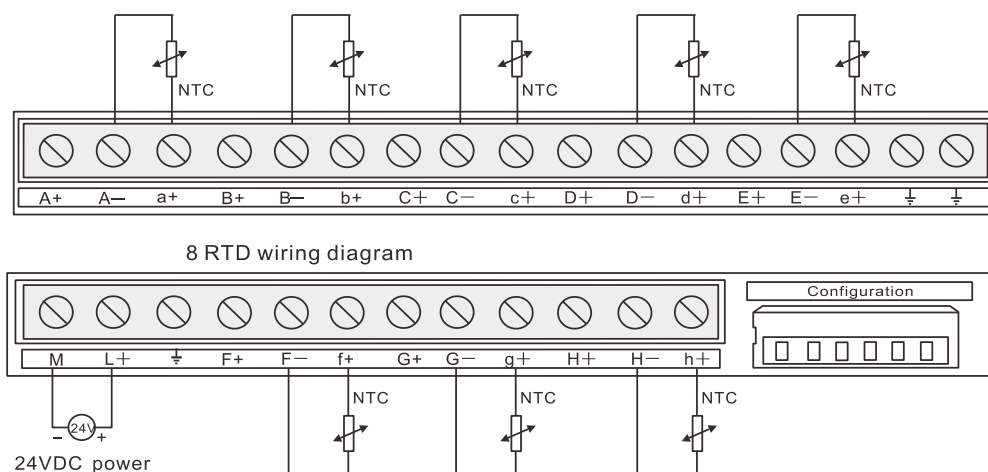
Power	1.8 W
Inputs	
Type	Pt100: $\alpha=3850\text{PPm}/3920\text{PPM}/3850.55\text{PPM}/3916\text{PPM}/3902\text{PM}$ NTC: R25=10k/B=3950 or R25=10k/B=3435
Temperature measuring range	Pt100: $-50^{\circ}\text{C} \sim 800^{\circ}\text{C}$ NTC (R25=10k/B=3950) : $-40^{\circ}\text{C} \sim 120^{\circ}\text{C}$ NTC (R25=10k/B=3435) : $-40^{\circ}\text{C} \sim 150^{\circ}\text{C}$ <Note> Measuring out of range report an error, refer to table 4-5-2.
Counts	8PT100/8NTC
Plug-in I/O	Yes
Power isolated	Yes
Field to logic	500 V AC
Field to 24V DC	500 V AC
24V to logic	500 V AC
CMR	120dB@120VAC
Sampling performance	
Temperature resolution	$0.1^{\circ}\text{C} / 0.1^{\circ}\text{F}$
Measuring principle	Sigma-Delta
Turnover time(all channel)	825 ms
Cable length	100m
Loop resistance	20Ω
Noise suppression	85dB@50/400 Hz
Input resistance	$>1\text{K}\Omega$
Measuring precision	$\pm 0.3^{\circ}\text{C}$
Breaking detection	Pt100 test 3 wires(NTC test 2 wires), test for 3min most

Wiring diagram

Pt100 wiring



NTC wiring



DIP switch specifications

Input type	SW1	SW2	SW3
100Ω Pt 0.003850(Default)	0	0	0
100Ω Pt 0.003920	0	0	1
100Ω Pt 0.00385055	0	1	0
100Ω Pt 0.003916	0	1	1
100Ω Pt 0.00302	1	0	0
NTC R25=10K B=3950	1	0	1
NTC R25=10K B=3435	1	1	0
Disable	1	1	1

SW4	Calibration direction	SW5	Measuring unit	SW6	Back up
0	Positive(+3276.7 ℃)	0	℃	0	No effect
1	Negative(-3276.8 ℃)	1	℉	1	No effect

Software Configuration

For EM231 8AI×16 bits, reading in VW instead of AIW. The relative location of the module is different, so is the corresponding address. Calculation formula:

$$x (VWx) = \text{Slot no.} \times 64 + \text{Input channel} \times 2$$

The slot number of the address quick reference table corresponds to the installation location of the module. The first expansion module close to the CPU has the slot number of 0, the second expansion module has the slot number of 1, and so on. There are 8 input channels, from A to G, and the corresponding number is 0 to 7.

Table 4-5-12 Quick reference of address based on formula above:

VWxx	channel 0	channel 1	channel 2	channel 3	channel 4	channel 5	channel 6	channel 7
slot 0	VW0	VW2	VW4	VW6	VW8	VW10	VW12	VW14
slot 1	VW64	VW66	VW68	VW70	VW72	VW74	VW76	VW78
slot 2	VW128	VW130	VW132	VW134	VW136	VW138	VW140	VW142
slot 3	VW192	VW194	VW196	VW198	VW200	VW202	VW204	VW206
slot 4	VW256	VW258	VW260	VW262	VW264	VW266	VW268	VW270
slot 5	VW320	VW322	VW324	VW326	VW328	VW330	VW332	VW334
slot 6	VW384	VW386	VW388	VW390	VW392	VW394	VW396	VW398

**Notice**

Since the address assigned by COTRUST's TD2X text display panel and SIEMENS 'TD200 text display panel is fixed at VW0, if you need to use both TD2X or TD200 and EM231 8TC modules in your system, the EM231 8TC module cannot be installed in the first position (Slot 0), otherwise it will not work properly.

4.6 PID Control Modules Specifications

Table 4-6-1 PID control modules Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 thermocouple PID modules, 4AI×TC PID	CTS7 231-7TD32
TrustPLC CTSC-200 EM231 thermocouple PID modules, 8AI×TC PID	CTS7 231-7TF32
TrustPLC CTSC-200 EM231 8 current input, 0-20mA/4-20mA, isolated 16 bits accuracy, with intelligent PID	CTS7 231-7HF32

4.6.1 PID Control Modules Specifications

【Specifications】

- Bus, power supply, isolation between channels, high reliability, strong anti-jamming capability.
- 16-bit sampling precision, hardware filtering technology, more accurate and stable measurement.
- Power supply with reverse connection protection and surge absorption function, suitable for harsh industrial environments
- Integrated advanced fuzzy logic control algorithm, do not occupy CPU resources, without programming to realize accurate temperature control, good dynamic performance.
- PID output can be PWM or analog, bipolar output, be able to heat and cool.

【Usage】

- Use insulation thermocouple to obtain ideal anti-jamming and improve reliability.
- Signal lines should be shielded wire and single-ended grounded.
- The earthing terminal of modules should be on ground under good grounding system, otherwise not grounded
- Short unused channels to eliminate the break line fault alarm.

Table 4-6-2 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 thermocouple PID modules, 4AI×TC PID	CTS7 231-7TD32
TrustPLC CTSC-200 EM231 thermocouple PID modules, 8AI×TC PID	CTS7 231-7TF32

Table 4-6-3 7TD/7TF technical specifications

Types	EM231 4AI×TC PID		EM231 8AI×TC PID
Features			
(W×H×D)	71×80×62mm		
LED indicator lights	24V, ON: Power supply normal, OFF: Abnormal power supply SF, ON: Abnormal, OFF: Normal, BLINK: Wrong input signal		
Current dissipation			
+5V DC current	54 mA		
L+	34 mA	39 mA	
L+ coil voltage	20.4~28.8V DC		
Power	1.8W		
Inputs			
Type	Suspension thermocouple		
	K type thermocouple		
Counts	4	8	
Isolation			
Field to logic Field to 24V DC 24V to logic	500V AC 500V AC 500V AC		
Common mode input range(input channel to input channel)	120V AC		
CMR	>120dB@120V AC		
Input resolution			
Temperature	0.1℃/0.1℉		
Voltage	15 bits+sign bit		
Measuring principle	Sigma-Delta		
Turnover time(all channel)	405ms	825ms	
Cable length to sensor	Maximum 100 m		
Loop resistance	Maximum 100Ω		
Noise suppression	85dB@ 50Hz/60Hz/400Hz		
Data word format	Voltage: -27648~+27648		
Input impedance	>1MΩ		
Maximum input voltage	30V DC		
Resolution	15 bits+sign bit		
Input filter attenuation	-3dB@ 21kHz		
Basic error	0.1% FS(voltage)		
Repeatability	0.05% FS		
CJ error	±1.5℃		
Diagnosis program	LED: EXTf, SF		
PID features			
PID arithmetic	PID+FUZZY parameter self-adjustment		
Sampling time	1 s		

Minimum Output pulse width	10ms
PID type	P, PI, PD, PID
PID output type	Analog or PWM pulse width control
PID output polarity	Bipolar or unipolar

PID address and parameter configuration

• PID address formula

Address name	Formula	Note
PID parameter address	$A = (2048 + S * 256) + 16 * C$	S is the slot where modules are (range : 0~6) C is channel number, 231-7TF is 0~7, 231-7TD is 0~3
PID positive pulse output address	$X = (2048 + S * 256) + 12$	
PID negative pulse output address	$Y = (2048 + S * 256) + 13$	

• PID parameter output (modules to CPU)

Items	Address	Range	Actual corresponding value
Actual temperature	VW A	-2000~13000	-200~1300℃
Status Word	VW A+2		
PID analog output	VW A+4	-32000~32000	

• PID parameter input (CPU to modules)

Items	Address	Range	Actual corresponding
Setting temperature	VW A+128	-2000~13000	-200~1300℃
Control byte		When VB A+130 bit= 0	When VB A+130 bit= 1
	V(A+130).0	PID do not run	PID run
	V(A+130).1	Ti works always, Kp do not auto-adjust	Ti separated, Kp auto-adjust
	V(A+130).2	PID unipolar output, 0~32000	PID bipolar output, -32000~32000, able to heat and cool
	V(A+130).3	Not use	
	V(A+130).4	Integral works	Integral does not work
	V(A+130).5	Differential works	Differential does not work
	V(A+130).6	Actual temperature filter	Actual temperature do not filter
PID pulse output cycle setting	VW A+132	1~255	1~255 s
Kp	VW A+134	0~9999	0~999.9
Ti	VW A+136	0~3600	0~3600 s
Td	VW A+138	0~3600	0~3600 s

Positive pulse output address:

Pulse output for channel 0	V X.0
Pulse output for channel 1	V X.1
Pulse output for channel 2	V X.2
Pulse output for channel 3	V X.3
Pulse output for channel 4	V X.4
Pulse output for channel 5	V X.5
Pulse output for channel 6	V X.6
Pulse output for channel 7	V X.7

Negative pulse output address:

Pulse output for channel 0	V Y.0
Pulse output for channel 1	V Y.1
Pulse output for channel 2	V Y.2
Pulse output for channel 3	V Y.3
Pulse output for channel 4	V Y.4
Pulse output for channel 5	V Y.5
Pulse output for channel 6	V Y.6
Pulse output for channel 7	V Y.7

Examples:

Calculate the last address of PID loop on 231-7TF second expansion modules, first calculate address: S=1,C=7	
Address A	$A = 2048 + 1 * 256 + 16 * 7 = 2416$
Address X	$X = 2048 + 1 * 256 + 12 = 2316$
Address Y	$Y = 2048 + 1 * 256 + 13 = 2317$
Then set or read parameter items according to following illustration:	
VW2544	//setting temperature
VB2546	// Control byte (parameter self-adjustment, bipolar output)
VW2548	// pulse output cycle
VW2550	//Kp
VW2552	//Ti(s)
VW2554	//Td(s)
VW2416	// Actual temperature
VW2418	//Status byte
VW2420	//PID analog output
V2316.7	// Positive pulse output
V2317.7	// Negative pulse output

To ensure normally use for PID modules, don't use PID V memory block when editing other POU. You can call the specified EM231 PID parameter configuration library.

- ☐ Occupied address in slot 0: VW2048 to VW2298
- ☐ Occupied address in slot 1: VW2304 to VW2554
- ☐ Occupied address in slot 2: VW2560 to VW2810

- ❑ Occupied address in slot 3: VW2816 to VW3066
- ❑ Occupied address in slot 4: VW3072 to VW3322
- ❑ Occupied address in slot 5: VW3328 to VW3578
- ❑ Occupied address in slot 6: VW3584 to VW3834

LADDER figure of parameter configuration:

NOTE

The parameter address of PID module occupy CPU V storage area.
To ensure PID module normally use, please do not use CPU V storage area the PID Module occupied when you write other program.

The address module occupied in slot 0: VW2048 to VW2098

The address module occupied in slot 1: VW2304 to VW2554

The address module occupied in slot 2: VW2560 to VW2810

The address module occupied in slot 3: VW2816 to VW3066

The address module occupied in slot 4: VW3072 to VW3322

The address module occupied in slot 5: VW3328 to VW3578

The address module occupied in slot 6: VW3584 to VW3834

Network 1

NOTE

set parameter of first EM231 PID module loop(channel 0) of first expansion module(slot 0):

Assuming that PID parameters need to be changed after debugging of this loop, PIDSetting is called to set parameters of this loop

No need to calculate PID parameter address, just input the slot and channel number where input loop is, and enable Run to run the loop.

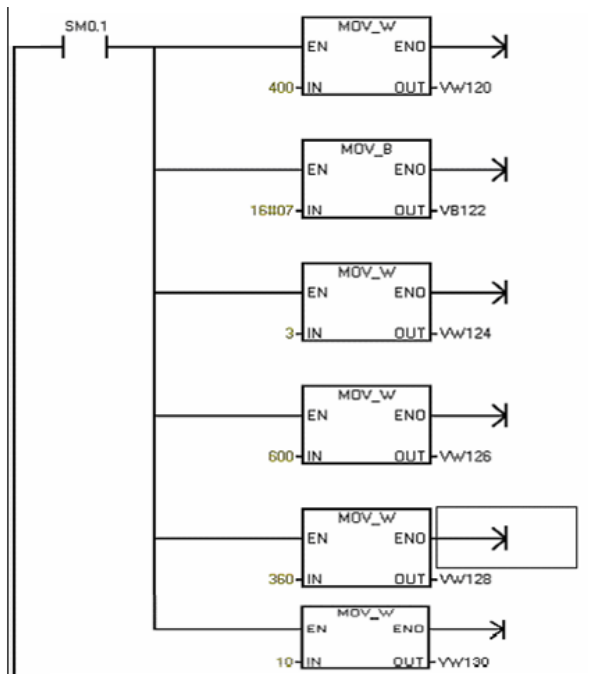
This case including network 1 and 2, network 1 for PID parameter initialization, and network 2 for PID running.

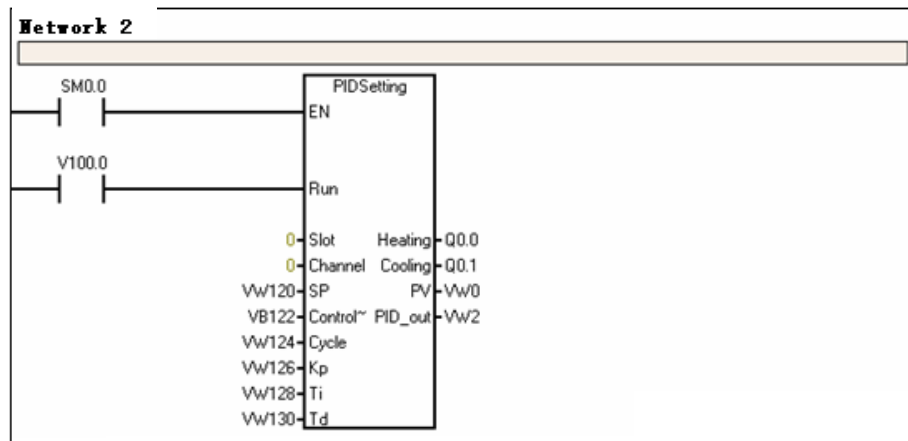
Q0.0 is positive pulse output, Q0.1 is negative pulse output, VW0 is actual temperature, VW2 is PID analog output.

you can also modify PID setting parameter to use other address, corresponding address in this case is:

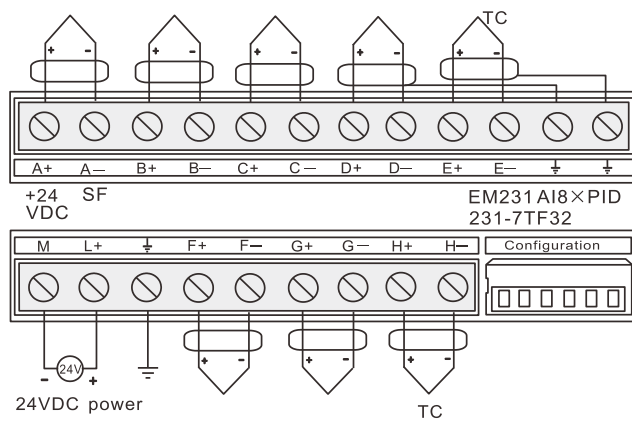
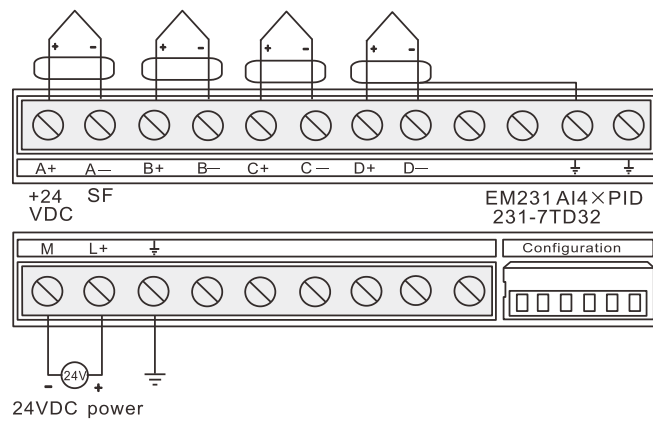
set temperature--VW120, control Word--VB122, pulse output cycle--VW124, proportion coefficient--VW126

integral time--VW128, differential time--VW130





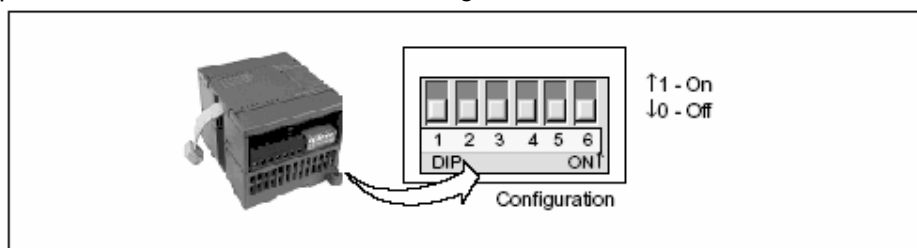
Terminal connection



DIP switch configuration

EM231 thermocouple modules supports J/K thermocouple. The module provides a 6-bit DIP switch for selecting measurement unit, cold end compensation and open circuit fault direction. As for whether to conduct wire breaking detection, EM231 thermocouple forces the wire breaking detection just as the EM231 8TC.

The 6-bit DIP switch is located below the module, as shown in the figure below. SW1 ~ SW2 was not used, use SW3 to select the thermocouple type, use SW4 to select the direction of wire breakage detection, use SW5 to select the measuring unit, and use SW6 to select whether to conduct cold end compensation. Power off the PLC power supply and then power on to make the DIP switch setting work.



Setting items	Switch position	Setting
Select thermocouple	SW3	0: J type 1: K type
Direction of broken line detection	SW4	0: positive(+3276.7°) 1: negative(-3276.8°)
Measuring unit	SW5	0: °C 1: °F
CJC	SW6	0: yes 1: no

Range of thermocouple measuring

The temperature range(°C) and accuracy of multiple thermocouple

Data word (1 digit bit=0.1°C)		TypeJ	TypeK	TypeT	TypeE	TypeR,S	TypeN	±80mV	
Decimalism	Hexadecimal								
32767	7FFF	>1200.0°C	>1372.0°C	>400.0°C	>1000.0°C	>1768.0°C	>1300.0°C	>94.071mV	OF
↑	↑							↑	↑
32511	7EFF							97.071mV	
⋮	⋮								
27649	6C01							80.0029mV	OR
27648	6C00							80mV	
⋮	⋮								
17680	4510					1768.0°C			
⋮	⋮								
13720	3598		1372.0°C over range						
⋮	⋮								
13000	32C8		1300.0°C				1300.0°C		NR
⋮	⋮								
12000	2EE0	1200.0°C							
⋮	⋮								
10000	2710				1000.0°C				
⋮	⋮								
4000	0FA0			-400.0°C		400.0°C			
⋮	⋮								
1	0001	0.1°C	0.1°C	0.1°C	0.1°C	0.1°C	0.1°C	0.0029mV	
0	0000	0.0°C	0.0°C	0.0°C	0.0°C	0.0°C	0.0°C	0.0mV	
-1	FFFF	-0.1°C	-0.1°C	-0.1°C	-0.1°C	-0.1°C	-0.1°C	-0.0029mV	
⋮	⋮								
-500	FE0C					Below the range -50.0°C			
-1500	FA24	-150.0°C							
⋮	⋮								
-2000	F830	Below the range	-200.0°C						
⋮	⋮								
-2100	F7CC	-210.0°C	Below the range						
⋮	⋮								
-2550	F60A			-255.0°C	-255.0°C				
⋮	⋮								
-2700	F574		Below the range	Below the range	Below the range				
⋮	⋮								
-27648	9400							-80mV	
-27649	93FF							-80.0029mV	
⋮	⋮								
-32512	8100							-94.071mV	UR
#	#								
-32768	8000	<-210.0°C	<-270.0°C	<-270.0°C	<-270.0°C	<-50.0°C	<-270.0°C	<-94.07mV	UF
Full-range precision		S0.1%	S0.3%	S0.6%	S0.1%	S0.6%	S0.1%	S0.1%	
Precision		S1.5°C	S1.7°C	S1.4°C	S1.3°C	S3.7°C	S1.6°C	S0.10°C	
Cold end error		S1.5°C	S1.5°C	S1.5°C	S1.5°C	S1.5°C	S1.5°C	NA	
*OF=Upper overflow OR=Over range NR=rated range VR=Below the range UF=Lower overflow									
↑Indicates that all analog quantities greater than this value but less than the break threshold are reported as upper overflow values, 32767 (10x7FFF)									
↓Indicates that all analog quantities less than this value but greater than the break threshold are reported as lower overflow values, -32768 (0x8000)									

4.6.1.1 Using CTSC-200 PID Modules Control Library

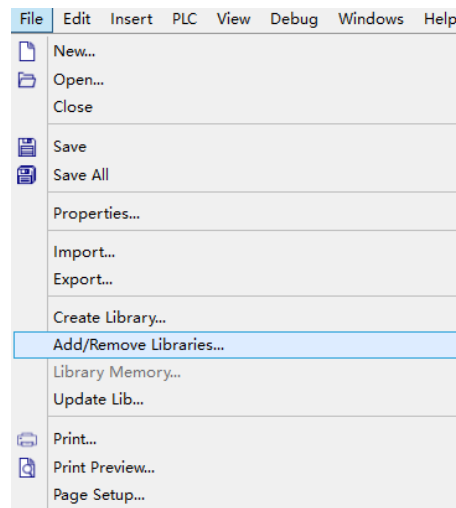
Function introduction

PID_setting function block is specially provided for the PID module of CTSC-200 as a library function which mainly aimed at intelligent PID module of temperature control. Without complex programming, you only need to call and set some simple parameters to use, and the temperature control is accurate.

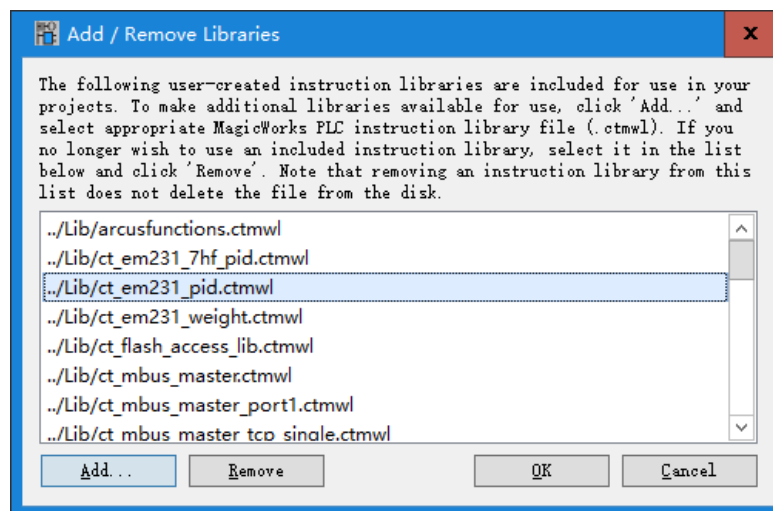
Installation

- Add library file

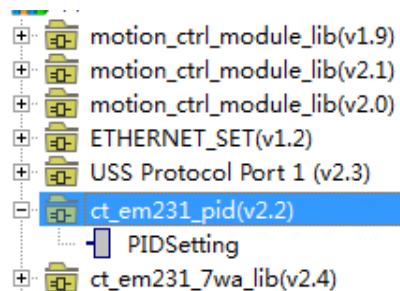
Use "File" -> "Add/Remove Libraries", find "ct_em231_pid.ctmwl" as following.



Find “ct_em231_pid.ctmwl” where you save it, then click “Add”.

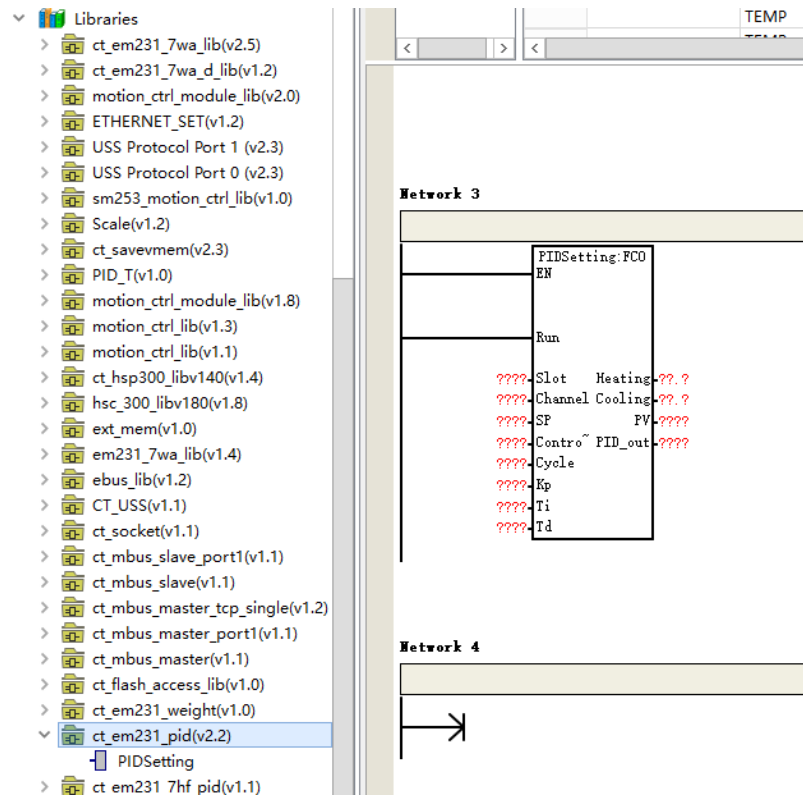


You can see library named “PID_setting” under catalogue tree “Libraries” after successful installation:



- Call PID library

Click “Internet” where you want to add function block, then double click “PIDSetting”, it will show in “Internet”:



PID library functions

Table 4-6-4 parameter address specifications

parameter address	illustration	Type	Value range	Note
Run	Run	Bit	0 or 1	
Slot	Slot from 0	Word, constant or variable	0~6	
Channel	Channel No.	Word, constant or variable	0~7	
SP	Setting value	Word, constant or variable	-32767~32767 0~65535	
CTRL Byte	Control byte, for PID run	Constant or variable		General control byte: 1 16#03(only for heating with self-adoptve 2 16#07(both heating and cooling with self-adoptve)
Cycle	Pulse output cycle	Word, constant or variable	1~255	Unit: s
Kp	Kp	Word, integer, constant or variable		No self-adoptve for constant
Ti	Ti	Word, integer, constant or	1~3600	Unit: s No self-adoptve for constant

		variable		
Td	Td	Word, integer, constant or variable	0~3600	Unit: s No self-adoptive for constant
Heating	Output	Bit		
Cooling	Output	Bit		
PV	Measurement(feedback)	Word, variable	-32767~32767 0~65535	
PID_out	PID analog output	Word, integer, variable	Only for heating output range: 0~32000, range with cooling: -32000~32000	

Table 4-6-5 Control bit address

Control bit	Setting	Note
0	0	PID stop
	1	PID run
1	0	Ti constantly work, Kp no self-adoptive
	1	Ti separated, Kp self-adoptive
2	0	PID unipolar
	1	PID bipolar
3	0	Retained
	1	Retained
4	0	Ti constantly work
	1	Ti do not work
5	0	Td constantly work
	1	Td do not work
6	0	Retained
7	1	Retained

Application example

System illustration:

This example sets the parameters of the first PID loop (channel 0) of EM231 PID module on the first expansion module (slot 0). Call PIDSetting to set loop parameters, no need to calculate the PID parameter address, just input the slot number and channel number to Run the loop.

Q0.0 is positive pulse output, Q0.1 is negative pulse output

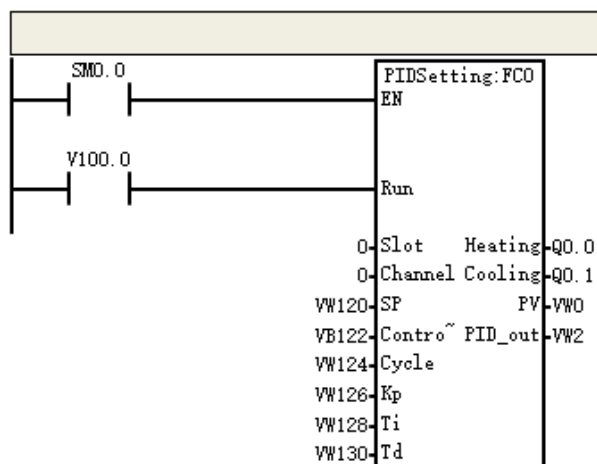
VW0 is actual temperature, VW2 is analog output PID

Modify PID parameters to use another address, setting temperature: VW120; control word : VB122;

Pulse output cycle: VW124; Kp: VW126; Ti: VW128; Td: VW130

Application program

Network 3



4.6.2 Current-mode PID Modules Specifications

TrustPLC ® is special for temperature control application of PID temperature control extension module with built-in PID temperature control algorithm, complex closed-loop temperature control without programming. Faster and more effective because of reducing CPU computing burden.

Table 4-6-6 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 8 current input, 0-20mA/4-20mA, isolated 16 bits accuracy, with intelligent PID	CTS7 231-7HF32

Table 4-6-7 231-7HF32 specifications

General	
Specification	EM231, 8AI×PID
Order No.	CTS7 231-7HF32
Connect to	CTSC-200 CPU224/226/PSC266 S7-224/226
Features	
(W×H×D)	120.5 x 80 x 62mm
Weight	210 g
Insert I / O terminal	No
Current dissipation	
+5V DC current	30 mA
L+	37 mA
Power	1.8 W
Inputs	
Counts	8, analog
Range / resistance	Current input
Maximum voltage input	30 V DC

Isolation	Yes
Field to logic	500 V AC
Field to 24V DC	500 V AC
24V to logic	500 V AC
Update time	825 ms(all channels)
Measuring principle	SIGMA-DELTA
Resolution	15 bit + sign bit
Temperature	0.1 °C / 0.1 °F
Noise suppression	85 dB
Noise frequency	50 / 60 / 400 Hz
CMV	120 V AC
CMR	120 dB @120 V AC
Basic error	0.1% FS(current)
Reappearance	0.05% FS
Diagnose program	LED : EXTTF, SF
Cable length	100 m —sensor
Maximum loop resistance	10 Ω
PID specifications	
PID arithmetic	PID+FUZZY parameter self-adoptive
Sampling time	1s
Minimum output pulse	10ms
PID type	P, PI, PD, PID
PID output type	Analog or PWM pulse width control
PID output polarity	Bipolar or unipolar

PID address and parameter configuration

- PID address formula

Address name	Formula	Note
PID parameter address	$A=(2048+S*256)+16*C$	S is the slot where modules are(range : 0~6) C is channel number, 231-7TF is 0~7, 231-7TD is 0~3
PID positive pulse output address	$X=(2048+S*256)+12$	
PID negative pulse output address	$Y=(2048+S*256)+13$	

- PID parameter output (modules to CPU)

Items	Address	Range	Actual corresponding value
Actual temperature	VW A	-2000~13000	-200~1300°C
Status Word	VW A+2		
PID analog output	VW A+4	-32000~32000	

- PID parameter input (CPU to modules)

Items	Address	Range	Actual corresponding
Setting temperature	VW A+128	-2000~13000	-200~1300°C
Control byte		When VB A+130 bit= 0	When VB A+130 bit= 1
	V(A+130).0	PID do not run	PID run
	V(A+130).1	Ti works always, Kp do not auto-adjust	Ti separated, Kp auto-adjust
	V(A+130).2	PID unipolar output, 0~32000	PID bipolar output, -32000~32000, able to heat and cool
	V(A+130).3	Not use	
	V(A+130).4	Integral works	Integral does not work
	V(A+130).5	Differential works	Differential does not work
	V(A+130).6	Actual temperature filter	Actual temperature do not filter
PID pulse output cycle setting	VW A+132	1~255	1~255 s
Kp	VW A+134	0~9999	0~999.9
Ti	VW A+136	0~3600	0~3600 s
Td	VW A+138	0~3600	0~3600 s

Positive pulse output address: $X=(2048+S*256)+12$

Pulse output for channel 0	V X.0
Pulse output for channel 1	V X.1
Pulse output for channel 2	V X.2
Pulse output for channel 3	V X.3
Pulse output for channel 4	V X.4
Pulse output for channel 5	V X.5
Pulse output for channel 6	V X.6
Pulse output for channel 7	V X.7

Negative pulse output address: $Y=(2048+S*256)+13$

Pulse output for channel 0	V Y.0
Pulse output for channel 1	V Y.1
Pulse output for channel 2	V Y.2
Pulse output for channel 3	V Y.3
Pulse output for channel 4	V Y.4
Pulse output for channel 5	V Y.5
Pulse output for channel 6	V Y.6
Pulse output for channel 7	V Y.7

Examples:

Calculate the last address of PID loop on 231-7TF second expansion modules, first calculate address: S=1,C=7

Address A	$A=2048 + 1 * 256 + 16 * 7 = 2416$
-----------	------------------------------------

Address X	$X=2048 + 1 * 256 + 12 = 2316$
Address Y	$Y=2048 + 1 * 256 + 13 = 2317$
Then set or read parameter items according to following illustration:	
VW2544	//setting temperature
VB2546	// Control byte (parameter self-adjustment, bipolar output)
VW2548	// pulse output cycle
VW2550	//Kp
VW2552	//Ti(s)
VW2554	//Td(s)
VW2416	// Actual temperature
VW2418	//Status byte
VW2420	//PID analog output
V2316.7	// Positive pulse output
V2317.7	// Negative pulse output

To ensure normally use for PID modules, don't use PID V memory block when editing other POU. You can call the specified EM231 PID parameter configuration library.

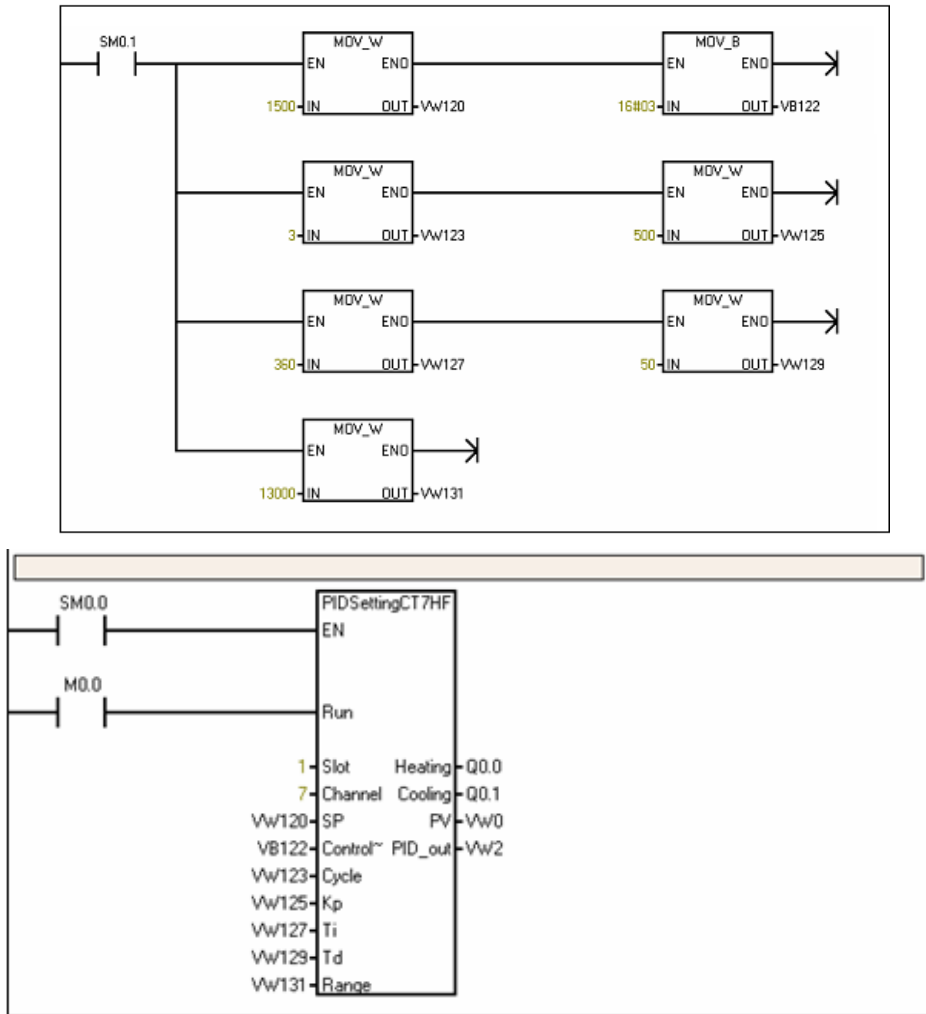
- ☐ Occupied address in slot 0: VW2048 to VW2298
- ☐ Occupied address in slot 1: VW2304 to VW2554
- ☐ Occupied address in slot 2: VW2560 to VW2810
- ☐ Occupied address in slot 3: VW2816 to VW3066
- ☐ Occupied address in slot 4: VW3072 to VW3322
- ☐ Occupied address in slot 5: VW3328 to VW3578
- ☐ Occupied address in slot 6: VW3584 to VW3834

LADDER figure of parameter configuration:

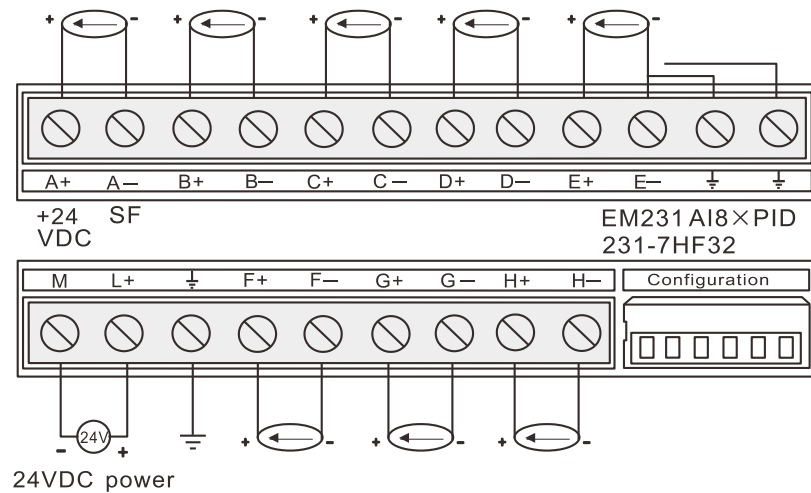
NOTE
 The parameter address of PID module occupy CPU V storage area.
 To ensure PID module normally use, please do not use CPU V storage area the PID Module occupied when you write other program.
 The address module occupied in slot 0: VW2048 to VW2300
 The address module occupied in slot 1: VW2304 to VW2556
 The address module occupied in slot 2: VW2560 to VW2812
 The address module occupied in slot 3: VW2816 to VW3068
 The address module occupied in slot 4: VW3072 to VW3324
 The address module occupied in slot 5: VW3328 to VW3580
 The address module occupied in slot 6: VW3584 to VW3836

Network 1

NOTE
 set parameter of first EM231-7HF PID module loop(channel 0) of first expansion module(slot 0):
 Assuming that PID parameters need to be changed after debugging of this loop, PIDSettingCT7HF is called to set parameters of this loop(you need to add em231-7hf_pid library first).
 no need to calculate PID parameter address, just input the slot and channel number where input loop is, and enable Run to run the loop.
 this case including network 1 and 2, network 1 for PID parameter initialization, and network 2 for PID running.
 Q0.0 is positive pulse output, Q0.1 is negative pulse output, VW0 is actual temperature, VW2 is PID analog output.
 you can also modify PID setting parameter to use other address, corresponding address in this case is:
 set temperature--VW120, control Word--VB122, pulse output cycle--VW123, proportion coefficient--VW125
 integral time--VW127, differential time--VW129, temperature sensor range--VW131



Terminal connection



DIP switch configuration

SW1	SW2	SW3	SW4	SW5	SW6	Range	Resolution
ON	ON	OFF	OFF	OFF	OFF	0~20mA	0.1℃/F
ON	ON	ON	OFF	OFF	OFF	4~20mA (default)	0.1℃/F

4.7 Communication Modules Specifications

Table 4-7-1 Order No.

Description	Order No.
TrustPLC CTSC-200 EM277A DP slave port modules, 12Mbps, isolated RS485	CTS7 277-0AA32
TrustPLC CTSC-200 EM277B PROFIBUS DP slave modules, 1.5Mbps, isolated RS485	CTS7 277-0AB32

4.7.1 EM277A DP Slave Port Modules Specifications

Table 4-7-2 EM277A DP Order No.

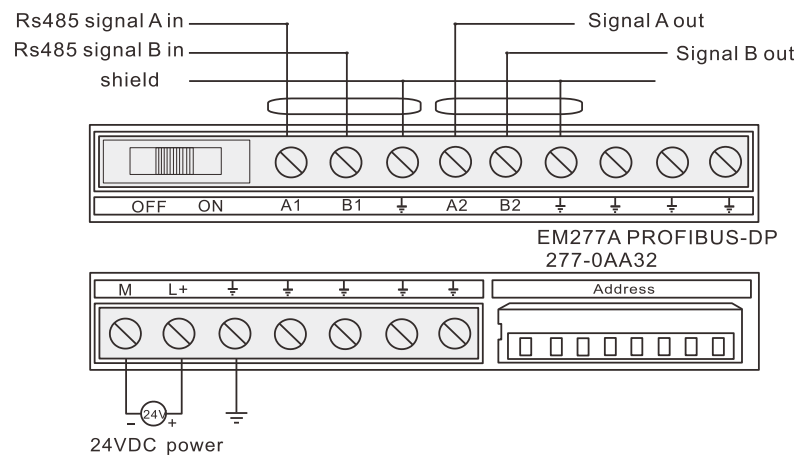
Description	Order No.
TrustPLC CTSC-200 EM277B PROFIBUS DP slave modules, 1.5Mbps, isolated RS485	CTS7 277-0AA32

Table 4-7-3 EM277A modules specifications

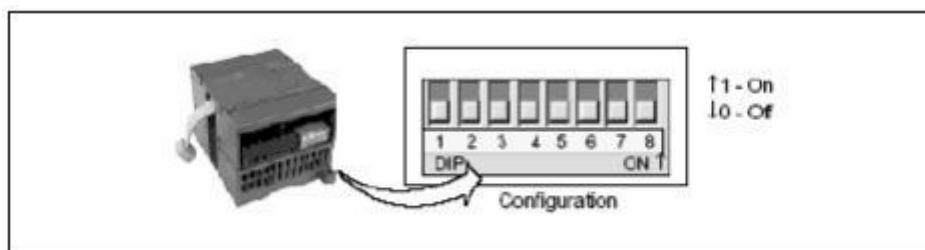
Features				
(W×H×D)	71 × 80 × 62mm			
Ports	1			
Electric port	isolated RS485			
I/O	0			
Expansion modules	Maximum 6			
PROFIBUS-DP/ MPI baud rate(auto-setting)	9.6, 19.2, 45.45, 93.75, 187.5 and 500K; 1, 1.5, 3.6 and 12M			
Protocol	PROFIBUS-DP slave and MPI slave			
LED	OFF	Red	Red blink	green
DX mode	No data exchange	--	--	Data exchange
DP error	No error	No data exchange	Parametric /configuration error	--
Power	No 24VDC	--	--	24VDC
CPU fault	Modules stable	Modules fail	--	--
Cable length(maximum)				
max 93.75K baud	1200 m			
187.5K baud	1000 m			
500K baud	400 m			
1 ~ 1.5 Mbaud	200 m			
3 ~ 12 Mbaud	100 m			
Internet performance				
Station address	0 to 126(set by DIP switch)			
Maximum station	32			

for each segment	
Maximum station for each net	126, maximum 125 EM277 station
MPI connection	6 in all, 2 retained(1 for PG, 1 for OP)
24VDC requirement	
Voltage range	20.4 to 28.8VDC(2 type or sensor power of PLC)
Power	2.3W
Maximum current (communication activation)	70mA
Ripple noise (< 10Mhz)	<1V peak value

Terminal connection



Address switch configuration



DIP SW1-8, in binary form, SW1 is LSB, SW8 is MSB(must be OFF), ON of SW1-SW7 means “1”, OFF means “0”. Formula:

$$\text{Address} = \text{SW1} \times 2^0 + \text{SW2} \times 2^1 + \text{SW3} \times 2^2 + \text{SW4} \times 2^3 + \text{SW5} \times 2^4 + \text{SW6} \times 2^5 + \text{SW7} \times 2^6$$

Table 4-7-4 address switch configuration reference

Address	SW1~SW8	Address	SW1~SW8	Address	SW1~SW8
0	0000 0000	43	1101 0100	86	0110 1010
1	1000 0000	44	0011 0100	87	1110 1010
2	0100 0000	45	1011 0100	88	0001 1010

3	1100 0000	46	0111 0100	89	1001 1010
4	0010 0000	47	1111 0100	90	0101 1010
5	1010 0000	48	0000 1100	91	1101 1010
6	0110 0000	49	1000 1100	92	0011 1010
7	1110 0000	50	0100 1100	93	1011 1010
8	0001 0000	51	1100 1100	94	0111 1010
9	1001 0000	52	0010 1100	95	1111 1010
10	0101 0000	53	1010 1100	96	0000 0110
11	1101 0000	54	0110 1100	97	1000 0110
12	0011 0000	55	1110 1100	98	0100 0110
13	1011 0000	56	0001 1100	99	1100 0110
14	0111 0000	57	1001 1100	100	0010 0110
15	1111 0000	58	0101 1100	101	1010 0110
16	0000 1000	59	1101 1100	102	0110 0110
17	1000 1000	60	0011 1100	103	1110 0110
18	0100 1000	61	1011 1100	104	0001 0110
19	1100 1000	62	0111 1100	105	1001 0110
20	0010 1000	63	1111 1100	106	0101 0110
21	1010 1000	64	0000 0010	107	1101 0110
22	0110 1000	65	1000 0010	108	0011 0110
23	1110 1000	66	0100 0010	109	1011 0110
24	0001 1000	67	1100 0010	110	0111 0110
25	1001 1000	68	0010 0010	111	1111 0110
26	0101 1000	69	1010 0010	112	0000 1110
27	1101 1000	70	0110 0010	113	1000 1110
28	0011 1000	71	1110 0010	114	0100 1110
29	1011 1000	72	0001 0010	115	1100 1110
30	0111 1000	73	1001 0010	116	0010 1110
31	1111 1000	74	0101 0010	117	1010 1110
32	0000 0100	75	1101 0010	118	0110 1110
33	1000 0100	76	0011 0010	119	1110 1110
34	0100 0100	77	1011 0010	120	0001 1110
35	1100 0100	78	0111 0010	121	1001 1110
36	0010 0100	79	1111 0010	122	0101 1110
37	1010 0100	80	0000 1010	123	1101 1110
38	0110 0100	81	1000 1010	124	0011 1110
39	1110 0100	82	0100 1010	125	1011 1110
40	0001 0100	83	1100 1010	126	0111 1110
41	1001 0100	84	0010 1010		
42	0101 0100	85	1010 1010		

Terminal resistance setting

Terminal resistance	Meaning
ON	Last node in Internet
OFF	Not last node in Internet

4.7.2 EM277B DP Slave Modules Specifications

Table 4-7-5 EM277B DP Order No.

Description	Order No.
TrustPLC CTSC-200 EM277B PROFIBUS DP slave modules, 1.5Mbps, isolated RS485	CTS7 277-0AB32

Table 4-7-6 EM277B electrical and environmental specifications

Run temperature	0 ~ 55℃
Store temperature	-25 ~ 70 ℃
Air mobility	Flowing Air in facility up and down 50.8 mm
Humidity	5 ~ 95%(no condensation)
Resistance to electrical interference	Pulse 50ns, repeat frequency 5kHz, 2,000 V peak voltage
Shake resistance	Frequency 10~57Hz, range0.1mm, frequency 57~150Hz, acceleration 1.0g, 10 times in 3D
Shock resistance	15g, constantly 11ms, 3 times shock in 3D
High voltage insulation	DC2200V for terminal to ground. I/O give AC1500V to other ports, last 1min
Grounding	3 type grounding(multi-branch single-ground)
Environment	Dust filter and no corruption
Fall	100mm, 4 times fall, unpacked
Free-falling	1m, 5 times, package

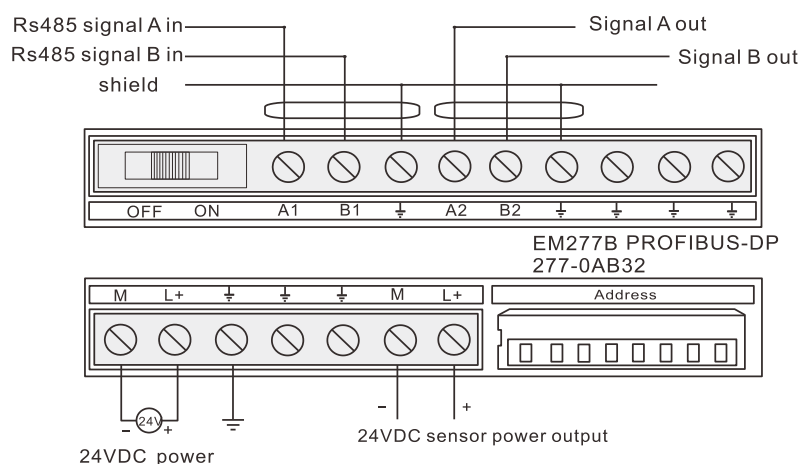
3 types grounding: grounding separately for each device

Table 4-7-7 EM277B modules specifications

General	
Order no.	277-0AB32
(W×H×D)	71 × 80 × 62mm
Ports	1
Electric port	RS-485 isolated (500VAC)
PROFIBUS-DP baud rate(auto-setting)	9.6, 19.2, 45.45, 93.75, 187.5 and 500 K baud; 1, 1.5Mbaud
Protocol	PROFIBUS-DP V0 slave facility
IO	-
Expansion IO	
Maximum expansion modules	6

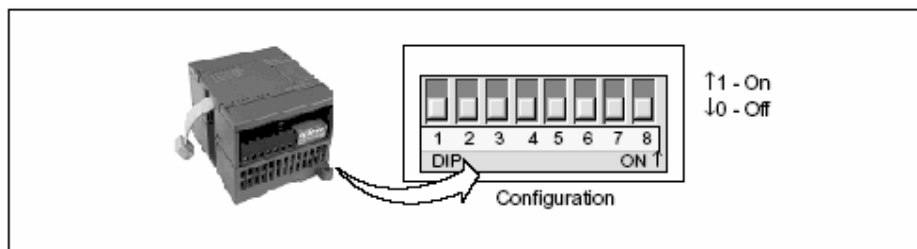
Maximum intelligent modules	-
Maximum I/O(map)	256 (128 In/128 Out)
LED indicator lights	
ON(green)	Light on after power on
SF(red)	Lights on when wrong in EM 277B self-check(including address setting out of range), or error in expansion I/O modules
BF(red)	Lights on when no DP exchange with master station
Maximum cable length	
Maximum 93.75Kbaud	1200m
187.5Kbaud	1000m
500Kbaud	400m
1 to1.5Mbaud	200m
Internet performance	
Station address	0 to 126(set by DIP switch)
Maximum station for each segment	32
Maximum station for each net	126, 125 EM 277B stations at most
24VDC requirement	
Voltage range	20.4 to 28.8 VDC(2 type or sensor power of PLC)
Isolation	-
Power	8W
Output current for sensor (24VDC)	
Voltage range	20.4 to 28.8 VDC(2 type or sensor power of PLC)
Maximum current	400mA
Output current for expansion modules (5VDC)	
Maximum current	660mA

Terminal connection



Set PROFIBUS address

Set PROFIBUS address through DIP switch. After changing the PROFIBUS address, EM 277B needs to be cut off and reenergized.



DIP SW1-8, in binary form, SW1 is LSB, SW8 is MSB(must be OFF), ON of SW1-SW7 means “1”, OFF means “0”. Formula:

$$\text{Address} = \text{SW1} \times 2^0 + \text{SW2} \times 2^1 + \text{SW3} \times 2^2 + \text{SW4} \times 2^3 + \text{SW5} \times 2^4 + \text{SW6} \times 2^5 + \text{SW7} \times 2^6$$

Table 4-7-8 address switch configuration reference

Address	SW1~SW8	Address	SW1~SW8	Address	SW1~SW8
0	0000 0000	43	1101 0100	86	0110 1010
1	1000 0000	44	0011 0100	87	1110 1010
2	0100 0000	45	1011 0100	88	0001 1010
3	1100 0000	46	0111 0100	89	1001 1010
4	0010 0000	47	1111 0100	90	0101 1010
5	1010 0000	48	0000 1100	91	1101 1010
6	0110 0000	49	1000 1100	92	0011 1010
7	1110 0000	50	0100 1100	93	1011 1010
8	0001 0000	51	1100 1100	94	0111 1010
9	1001 0000	52	0010 1100	95	1111 1010
10	0101 0000	53	1010 1100	96	0000 0110
11	1101 0000	54	0110 1100	97	1000 0110
12	0011 0000	55	1110 1100	98	0100 0110
13	1011 0000	56	0001 1100	99	1100 0110
14	0111 0000	57	1001 1100	100	0010 0110
15	1111 0000	58	0101 1100	101	1010 0110
16	0000 1000	59	1101 1100	102	0110 0110
17	1000 1000	60	0011 1100	103	1110 0110
18	0100 1000	61	1011 1100	104	0001 0110
19	1100 1000	62	0111 1100	105	1001 0110
20	0010 1000	63	1111 1100	106	0101 0110
21	1010 1000	64	0000 0010	107	1101 0110
22	0110 1000	65	1000 0010	108	0011 0110
23	1110 1000	66	0100 0010	109	1011 0110
24	0001 1000	67	1100 0010	110	0111 0110
25	1001 1000	68	0010 0010	111	1111 0110
26	0101 1000	69	1010 0010	112	0000 1110

27	1101 1000	70	0110 0010	113	1000 1110
28	0011 1000	71	1110 0010	114	0100 1110
29	1011 1000	72	0001 0010	115	1100 1110
30	0111 1000	73	1001 0010	116	0010 1110
31	1111 1000	74	0101 0010	117	1010 1110
32	0000 0100	75	1101 0010	118	0110 1110
33	1000 0100	76	0011 0010	119	1110 1110
34	0100 0100	77	1011 0010	120	0001 1110
35	1100 0100	78	0111 0010	121	1001 1110
36	0010 0100	79	1111 0010	122	0101 1110
37	1010 0100	80	0000 1010	123	1101 1110
38	0110 0100	81	1000 1010	124	0011 1110
39	1110 0100	82	0100 1010	125	1011 1110
40	0001 0100	83	1100 1010	126	0111 1110
41	1001 0100	84	0010 1010		
42	0101 0100	85	1010 1010		

Terminal resistance setting

Terminal resistance	Explanation
ON	Last node in Internet
OFF	Not last node in Internet

4.7.3 EM277C Slave Modules Specifications

Table 4-7-9 Order No.

Description	Order No.
TrustPLC CTSC-200 EM277C CAN slave modules, with 8DI/6DO	CTS7 277-0AC32

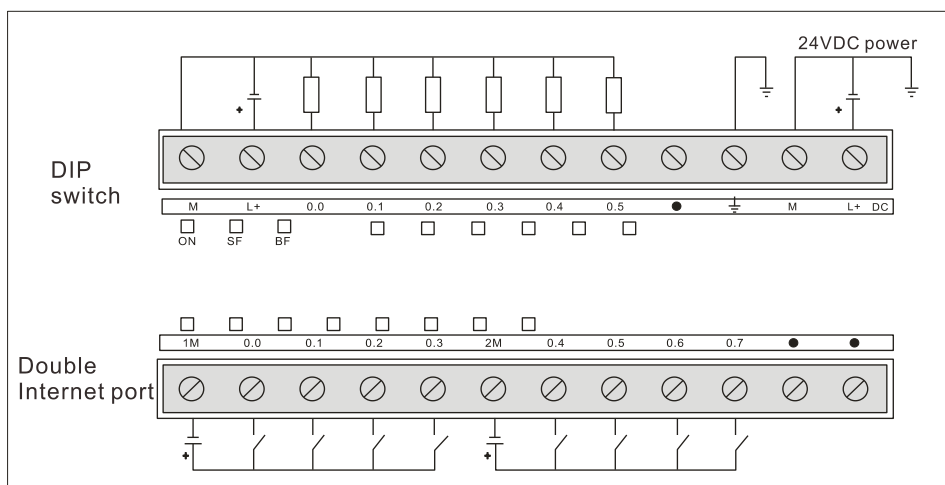
Table4-7-10 EM277C specifications

General								
(W×H×D)	90×80×62mm							
Power	5W							
Supply	20.4V to 28.8V							
Rated supply	24VDC							
Communication								
Transmission speed (kbit/s)	1000	800	500	250	125	50	20	
Maximum length (m)	25	50	100	250	500	1000	2500	
Digital input								
General	24 VDC							
Type	drain/source (IEC 1 source)							
Rated voltage	Typical, 24 VDC for 4mA							
Maximum constant voltage	30 VDC							

Transient voltage		35 VDC last for 0.5 s
Logic 1(minimum)		14 VDC in 2.5mA
Logic 0(maximum)		5VDC in1mA
Input delay		6.4 ms(minimum 6.3 ms)
2-wire Bero Leakage current		1mA
Isolation(field to logic)		Yes
Optic(current)		500 VAC, 1min
Isolation group		Refer wiring diagram
Counts simultaneous		8 (all 55°C)
Maximum cable length	unshielded	500m
	Shielded	300m
Digital output		
General		+24VDC transistor output
Type		Solid –MOSFET(source)
Rated voltage		24 VDC
Range		20.4~28.8 VDC
Logic 1(minimum)		Maximum 0.1 VDC (10K Ω)
Logic 0(maximum)		Minimum 20 VDC
Output current		
Signal"1"(maximum)		0.75A
Group		1
Counts each group		6
Simultaneous ON counts		6
Current each group		4.5A
Lights power		5W
Contact resistance		0.3 Ω , Maximum 0.6 Ω
Leakage current		Maximum 10
Maximum surge current		8A, 100ms
Isolation		Opto-isolation, 500 VAC, last in 1min
Isolation each group		Yes
Output delay		
OFF-ON		Maximum 15 μ s
ON-OFF		Maximum 130 μ s
Output in parallel		Yes, only for same output group
Maximum cable length		
Unshielded		500m
Shielded		300m

Terminal connection

EM 277C wiring diagram



4.7.4 EM277PN Slave Modules Specifications

Table 4-7-11 Order No.

Description	Order No.
TrustPLC CTSC-200 EM277PN PROFINET slave modules, 100Mbps, 8 expansion modules, do not support EM253, weighing modules and communication modules	CTS7277-0PN32

Table 4-7-12 EM277-PN environmental specifications

Environmental condition	
Temperature	-40°C ~+70°C
Humidity	10%~95% no condensation
Electromagnetic compatibility	
Electrostatic discharge IEC61000-4-2	Contact discharge: ±4KV Air discharge: ±8KV
Electrical fast transient pulse train IEC61000-4-4	Power line: 2KV, 5KHz Signal line: 2KV, 5KHz (I/O coupling clamp) 1KV, 5KHz (communications coupling clamp)
Surge IEC61000-4-5	Power line: 2KV(asymmetry), 1KV(symmetry)
Radiofrequency electromagnetic radiation IEC61000-4-3	80MHz~1GHz, 10V/m, 80%AM (1KHz)
Radiofrequency field inductively conducts interference IEC61000-4-6	0.15MHz~80MHz, 10V/m, 80%AM (1KHz)
DC power input port short - term interruption and voltage variation	Interruption time: 10ms

IEC61000-4-29	
Run in high temperature IEC60068-2	65°C, stay time: 24h, temperature change rate≤1°C/min, recovery time≥1h
Run in low temperature IEC60068-2	-20°C, stay time: 24h, temperature change rate≤1°C/min, recovery time≥1h
Start in high temperature IEC60068-2	65°C, 2h, not boot 3 times after temperature stable
Start in low temperature IEC60068-2	-20°C, 2h not boot 3 times after temperature stable
Cycle operation both high and low temperature IEC60068-2	-10°C~60°C stay time:3h temperature change rate 1°C/min, 5 loop, recovery time 1h
Store in high temperature IEC60068-2	75°C, stay time: 72h temperature change rate≤1°C/min, recovery time≥1h
Store in low temperature IEC60068-2	-40°C, stay time: 72h temperature change rate≤1°C/min, recovery time≥1h
Hot and cold shock IEC60068-2	-40°C~75°C, stay time: 3h, temperature change time: 2min, 3 loops
Stable high humidity IEC60068-2	Store temperature: 40°C, stay time: 48h, store humidity: 93%
Alternate damp and hot IEC60068-2	25°C~55°C, humidity: 95%, single loop: 24h, 2 loops
Sine shake(bare) IEC60068-2	5~150Hz, 0.05G ² /Hz 150Hz~500Hz, -3dB/oct, 1h/axis, X, Y, Z 3 axes in all
Mechanical shake(bare) IEC60068-2	15G, 11ms pulse, 3 times/direction
Fall	1m, 10 times, package

Table4-7-13 EM277-PN modules specifications

Features	
(W×H×D)	71×80×62 mm
Current dissipation	
Rated input voltage	24V DC
Input voltage range	20.4V~28.8V DC
Input current	0.8A
Bus voltage	+5V DC
Bus current	1.1A
Isolation	External power isolated with system power
Power protection	Supply power offer reverse connect protection and surge absorption
LED indicator lights	
24VDC(green)	ON = 24VDC supply normal
	OFF = no 24VDC supply
SF(red)	ON = expansion I/O bus wrong or PROFINET modules wrong
	OFF = no error

BF(red)	ON = PROFINET bus communication error(not connect to interchanger, no network)
	Blinking = configuration inconsistent
	OFF = no error
MT(yellow) Maintenance lights	ON = lost simultaneous
	OFF = no error
P1R/P2R port (green)	ON = connect to interchanger /PN bus
	OFF = no connect to interchanger /PN bus
P1R/P2R port (yellow)	ON = data send-receive connect to interchanger /PN bus
	OFF = no data send-receive connect to interchanger /PN bus

Table4-7-14 EM277-PN modules specifications

Expansion I/O	
Maximum modules number for each slave	8(digital modules, analog modules, PID control modules and temperature modules)
Protocol type	CTSC-200 PLC costume 4MHZ bus protocol
Maximum configuration of slave for each I/O	Analog IO maximum configuration :64AI/32AQ Digital IO maximum configuration :256DI/256DQ
Topological structure	Support star shape, tree structure, line structure and loop structure
PROFINET communication port	
Communication port	1 double RJ45
Data transmit speed	Ethernet: 10Mbps
	PROFINET: 100Mbps, duplex
Ethernet service	ping, arp, network diagnosis(SNMP)/MIB-2, LLDP
Send cycle	250us~4ms
Third PROFINET master	Support
Maximum communication distance	100m (100BASE-TX)
Topological structure	Support star shape, tree structure, line structure and loop structure
Isolation	Isolated communication port
Hardware configuration	
Import file type	PROFINET GSD file and XML format
Third PN master station	Add CTS7 277-0PN32 to support 8 slot expansion
	digital modules, analog modules, PID control modules and temperature modules

4.8 Weighing Modules Specifications

Table 4-8-1 Order No.

Description	Order No.
TrustPLC CTSC-200 EM231 weighing modules, 2 sensor signal input, sampling frequency 20HZ(single channel)/10HZ(two-	CTS7 231-7WB32

channel), 6VDC excitation power supply output, isolated 16 accuracy	
TrustPLC CTSC-200 EM231 weighing modules, 1 sensor signal input, sampling frequency 50Hz, 6VDC excitation power supply output, isolated 16 accuracy	CTS7 231-7WA32

Table 4-8-2 E231 weighing modules specifications

Type	CTS7 231-7WA32	CTS7 231-7WB32
LED indicator lights	24VDC, ON: power supply normal, OFF: abnormal power supply SF, ON: Abnormal, OFF: Normal, BLINK: sensor break	
Current dissipation		
+5V DC current	<140mA	
L+	<100mA	
L+ coil voltage	20.4—28.8VDC	
Power	5W	
Inputs		
Type	Deformeter, 4 or 6 wiring method	
Range	1mV/V 2mV/V 4mV/V	0~1mV/V 0~2mV/V 0~3mV/V 0~4mV/V
Counts	1AI	2AI
Isolation		
Field to logic	500VAC	
Field to 24V DC	500VAC	
24V to logic	500VAC	
CMV range(input channel to input channel)	0	
CMR	>120dB@120VAC	
Input resolution		
Temperature coefficient zero	±0.1uV/K	
Measuring principle	Sigma-Delta	
Cable length to sensor	Maximum 500m	
Noise resistance	85db@50Hz/60Hz	
Data Word format	Voltage: 0 to +32000	
Resolution	16 bits	
Basic error	0.05%Fs	
Repeatability	0.1%Fs	

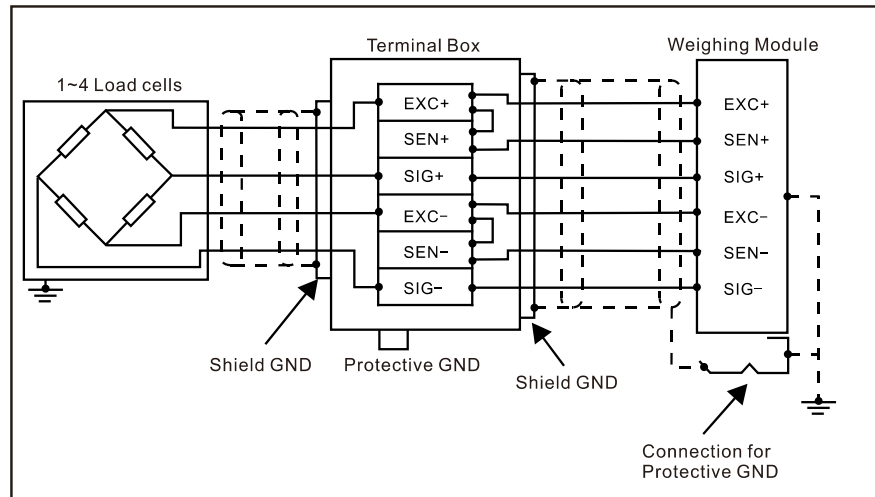
Terminal connection

The weighing module is a high-precision measurement module that must reliably measure low signal levels (about 1.5 μV). Therefore, proper assembly and cable connections are essential for trouble-free operation

Notes for connecting weighing sensors:

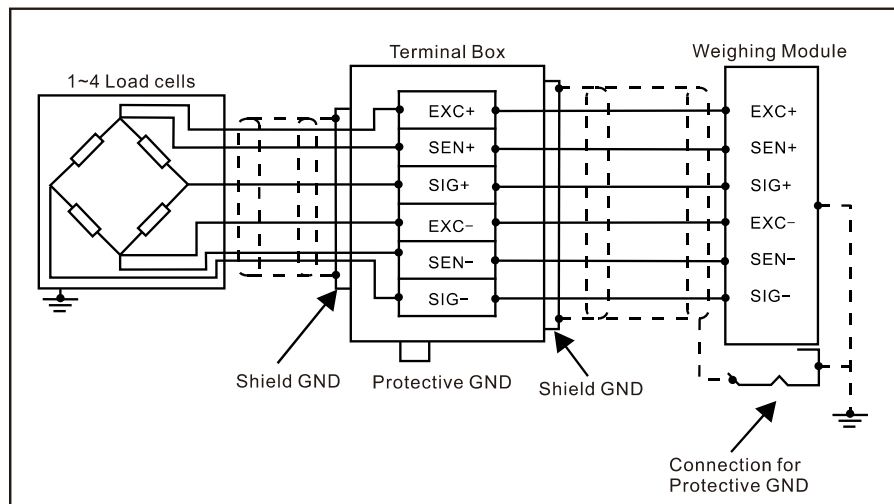
- 1) A junction box must be used for more than one sensor connection (the weighing sensor must be in parallel). As well as when the distance from the weighing sensor to the module or to the junction box is greater than the available cable length of the weighing sensor. Special extension box.
- 2) The cable shielding should always be connected to the cable gland of the junction box or extension box.
- 3) Use shielded and twisted pair cables on specified lines
 - sensor line SEN+/SEN-;
 - measuring voltage line SIG+/SIG-;
 - supply voltage line EXC+/EXC-;
- 4) The shielded wire must be connected to the shielded connector clip

1 Four-wire connection



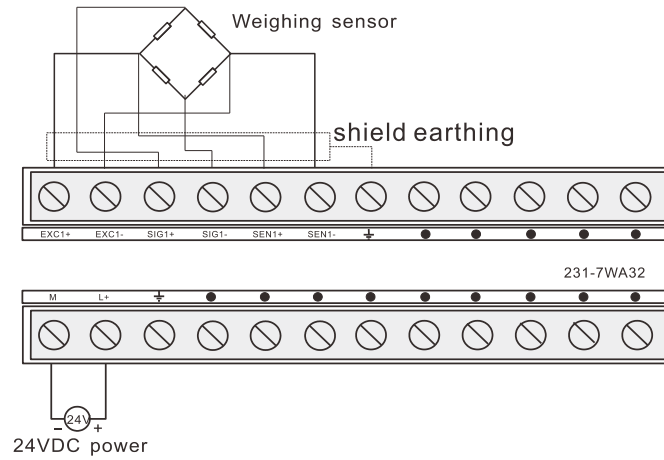
Short EXC+ and SEN+, EXC- and SEN- in junction box when connecting the four-wire sensor, otherwise the weighing module may not work normally.

2 Six-wire connection

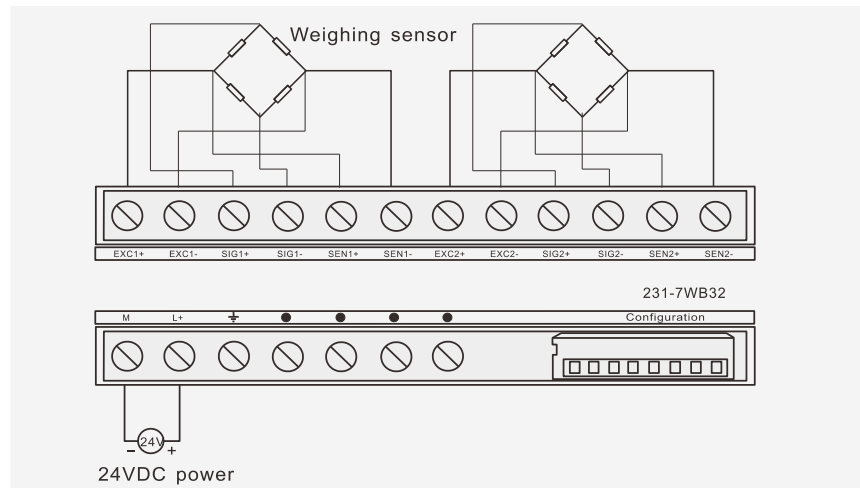


3 Modules wiring diagram

EM231-7WA weighing modules



EM231-7WB weighing modules



4 DIP switch setting

Switch 1, 2, 3, 4, 5, 6	Setting	Function
SW1, 2	01	Select 1mV/V sensor
	00	Select 2mV/V sensor(default)
	10	Select 3mV/V sensor
	11	Select 4mV/V sensor
SW3	0	Select double channel work mode (default)
	1	Select single channel work mode
SW4	0	Select filter type 1(default)
	1	Select filter type 2
SW5, 6, 7, 8	00	Retained
Note: 1)Default: 000000 2)Debug according to filter mode, set by software		

4.9 Motion Control Modules Specifications

Table 4-9-1 digital input specifications

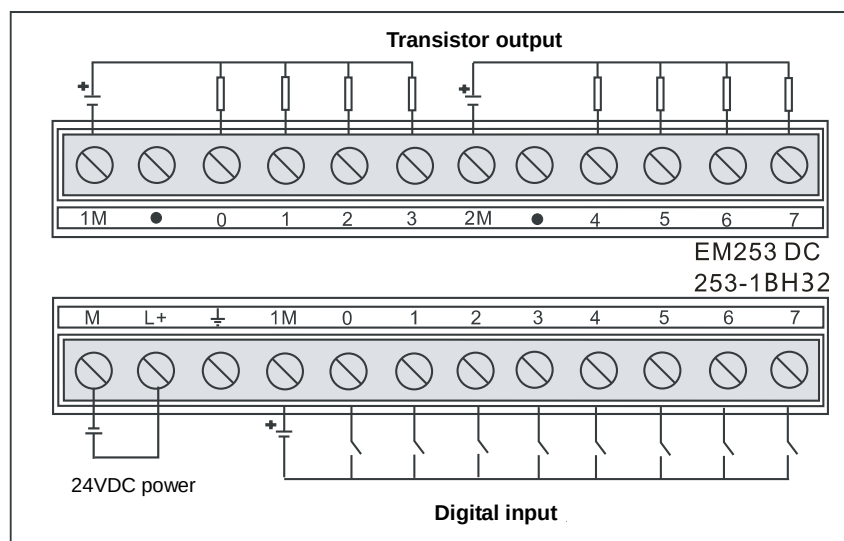
General		24VDC input
Counts		8 inputs
Type		Drain /source (IEC type 1/drain)
Rated voltage		24 VDC on typical 5mA
Maximum constant voltage		30 VDC
Surge voltage		35 VDC last for 0.5 s
Logic 1(minimum)		15.6 VDC in 2.72mA (I0.0, I0.1, I0.2, I0.4, I0.5, I0.6) 12.8VDC in 2.55mA (I0.3, I0.7)
Logic 0(maximum)		15.4VDC in 2.69mA (I0.0, I0.1, I0.2, I0.4, I0.5, I0.6) 12.6 VDC in 2.51mA (I0.3, I0.7)
Input delay		< 1.1us (I0.0, I0.1, I0.2, I0.4, I0.5, I0.6) < 1ms (I0.3, I0.7)
2-wire Bero Leakage current		1mA
Isolation(field to logic)		Yes
Optic(current)		500 VAC, 1min
Isolation group		Refer to diagram
High speed input rate		200KHz(unipolar&bipolar) (I0.0, I0.4)
HSC logic 1=16~26 VDC		200KHz (A/B phase) (I0.0 and I0.1, I0.4 and I0.5) (Input waveform duty cycle 40%~60%)
Simultaneous input		All
Maximum cable length	Unshielded	500 m standard input, 50m HSC input(use shield twisted pair and ground shielding)
	Shielded	300m standard input
Maximum turning frequency of HSC		50KHz

Table 4-9-2 digital output specifications

General		Transistor output
Counts		8 output
Type		Solid - MOSFET(drain)
Rated voltage		24 VDC
Output voltage range		5 to 28.8 VDC
Surge voltage		8A last for 100ms
Logic 1(minimum)		0.5V
Logic 0(maximum)		VCC-0.5V
Maximum rated current		0.5A
Maximum common port rated current		2.0A
Maximum leakage current		10 μ A
Maximum Lights power		3.5W
Induction clamp voltage		L+ - 48 VDC, 1W power

On resistance(junction)	0.3 Ω typical (maximum 0.6 Ω)
Isolation(field to logic) Optic(electrical, field to logic) Logic circuit to junction Resistance(Logic circuit to junction) Isolation group	500 VAC, 1 min --- --- Refer to diagram
Maximum delay OFF to ON ON to OFF	0.2 μ s (Q0.0, Q0.1, Q0.2, Q0.3), 50 μ s (Q0.4, Q0.5, Q0.6, Q0.7) 0.2 μ s (Q0.0, Q0.1, Q0.2, Q0.3), 130 μ s (Q0.4, Q0.5, Q0.6, Q0.7) Note: The execution of the DO output instruction has a certain delay due to communication transmission: If the CPU is followed by single EM253 motion control module, the delay from enable to execution of the output instruction is about 780us. If there are up to 5 EM253 motion control modules hanging behind the CPU, the delay from enable to execution of the output instruction is about 930us.
Maximum pulse frequency	200KHz (Q0.0, Q0.2)
Mechanical life cycle	/
Contactor life	/
Output simultaneous	Output all in 55° C
Output parallel	Only simultaneous
Maximum cable length Shield Unshielded	500m 150m

Terminal connection



EM253 module wiring diagram

Table 4-9-3 I/O defined function

IO.0	IO.1	IO.2	IO.3		IO.4	IO.5	IO.6	IO.7
MC253_ HSC0 pulse input	MC253_ HSC0ex ternal direction signal	MC253_ _HSC0 reset signal	Motion axis 0 stop signal		MC253_ HSC1 pulse input	MC253_ HSC1 external direction signal	MC253_H SC1 reset signal	Motion axis 1 stop signal
Q0.0	Q0.1	Q0.2	Q0.3		Q0.4	Q0.5	Q0.6	Q0.7
Axis 0 pulse output	Axis 0 direction signal	Axis 1 pulse output	Axis 1 direction signal		Common output port			

4.10 BD Expansion Modules Specifications and Installation

Table 4-10-1 Order No.

Description	Order No.
EBH AMS-03 analog input&output expansion board, 2*12 bits accuracy voltage input, 1*12 bits accuracy voltage/current output	CTH2 AMS-03S1-EB
EBH-AMS-06 analog input&output expansion board, 4*12 bits accuracy voltage input, 2*12 bits accuracy voltage output	CTH2 AMS-06S1-EB
EBH-AMS-06 analog input&output expansion board, 4*12 bits accuracy voltage input, 2 *12 bits accuracy current output	CTH2 AMS-06S2-EB
EBH CAN bus communications expansion board, 1Mbps, opto-isolated	CTH2 CAN-01S1-EB
EBH PWM 04 charging guide expansion board, double-channel voltage input, single-channel VPW PWM output	CTH2 PWM-04S1-EB
1 road RS485 communication expansion board	CTH2 485-01S1-EB

4.10.1 Analog Expansion Modules Specifications

Table 4-10-2 Analog expansion modules specifications

Features	EBH-AMS-03	EBH-AMS-06S1	EBH-AMS-06S2
(W×H×D)	24×14.4×68.2mm		
BUS supply voltage	+5VDC		
BUS supply current	60mA		60~250mA
LED			
Supply lights	ON = 5VDC supply nomad, OFF = No 5VDC supply		
SF light	Calibration indicator lights, blink: calibration failed		
Input specifications			
Counts	2	4	

Type	Single voltage input		
Range	±10V		
DataWord format	-32000~+32000		
DC input resistance	>100KΩ		
Maximum input voltage	30VDC		
Resolution	11bit + 1 sign bit		
Isolation	No		
Precision			
Worst (0-55 °)	±2.5% full-scale		
Typical (25 °)	±1.0% full-scale		
Repeatability	±0.05% full-scale		
A/D switching time	125ms		
Transmission principle	Σ-Δ		
Step response	Maximum 250ms		
Noise resistance	Typical: -20dB@50Hz		
Output specifications			
Counts	1	2	
Range	Voltage: 0-10V	Voltage: 0-10V	--
	Current: 0-20mA	--	Current: 0-20mA
Data Word format	0 to +32000		
Resolution	12 bits		
LSB	Voltage: 2.50mV		--
	Current: 5.00μA	--	Current: 5.00μA
Isolation	No		
Precision			
Worst (0-55 °)	Voltage:±2% full-scale		--
	Current: ±3% full-scale	--	Current:±3% full-scale
Worst (0-55 °)	Voltage:±1% full-scale		--
	Current: ±1% full-scale	--	Current: ±1% full-scale
Stable time			
• With impedance load	0.1ms		0.2 ms
• With capacitive load			3.3 ms
• With inductive load			0.5 ms (1mH)
Maximum output drive	≥5000Ω		≤500Ω

Terminal definition

1. The structure and terminal diagrams of the AMS-03 analog I/O expansion board are shown below. The following table lists the definitions of each terminal bit. Please strictly follow the definition of terminal bit to install and connect the expansion board.

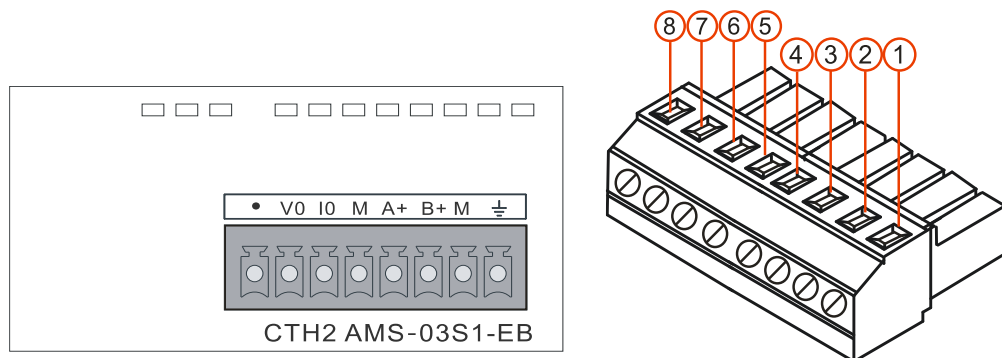


Table 4-10-3 AMS-03 terminal definition

Terminal bit	Signal definition	Terminal bit	Signal definition
1	Vacant	5	Analog input A+
2	Voltage output VO	6	Analog input B+
3	Current output IO	7	GND-M
4	GND-M	8	Ground

2. The structure and terminal diagrams of the AMS-06 analog I/O expansion board are shown below. The following table lists the definitions of each terminal bit. Please strictly follow the definition of terminal bit to install and connect the expansion board.

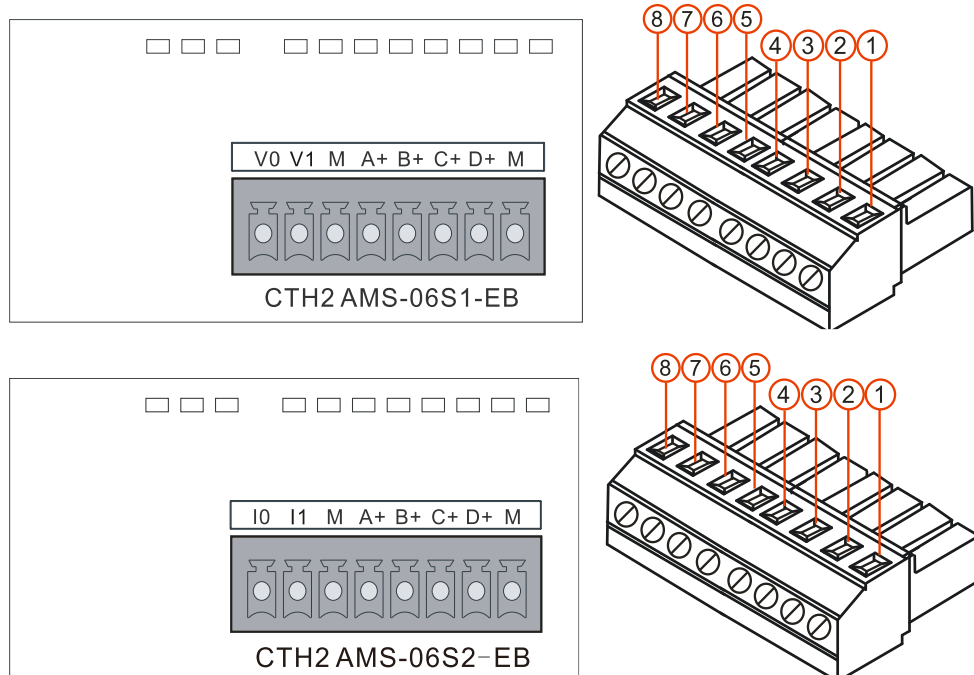


Table4-10-4 AMS-06 terminal definition

Terminal bit	Signal definition	Terminal bit	Signal definition
1	Output V0/I0	5	Analog input B+
2	Output V1/I1	6	Analog input C+
3	GND-M	7	Analog input D+
4	Analog input A+	8	GND-M

<Note> A+, B+, C+, D+ is the voltage input terminal, both positive voltage and negative voltage. External wiring need to connect terminal V1 to grounding terminal No.3. Connect analog input D+ to ground terminal 8. M is common terminal, do not connect the output terminal with M incase causing short circuit in the output.

4.10.2 CAN Expansion Modules Specifications

Table 4-10-5 CAN-01 specifications

Features							
(W×H×D)	137 x 96 x 62mm						
Connect to	CPU H224X/H226XL						
Power	9W						
CANopen communication function							
Ports	1(8 Pin)						
Transmit speed(kbps)	1000	800	500	250	125	50	20
Maximum length(m)	25	50	100	250	500	1000	2500
Maximum station	127						
Station address range	1-127						
Maximum slave number	32						
Maximum digital visits		Bytes		Memory starting address			
	Input	64		IB16			
	Output	64		QB16			
Maximum analog visits		Channels		Memory starting address			
	Input	8		AIW64			
	Output	8		AQW64			

The structure and terminal diagrams of the CAN-01 analog I/O expansion board are shown below. The following table lists the definitions of each terminal bit. Please strictly follow the definition of terminal bit to install and connect the expansion board.

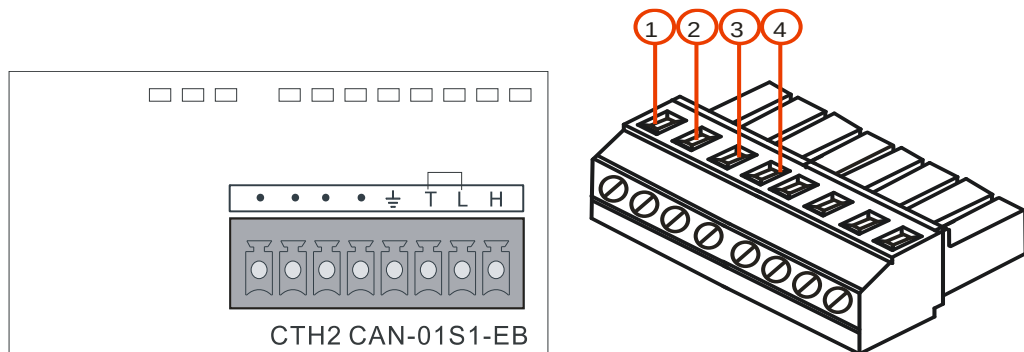


Table4-10-6 CAN-01 terminal definition

Terminal bit	Signal definition
1	CAN_H
2	CAN_L
3	Connect 2 with 3 when matching terminator
4	GND

4.10.3 RS485 Expansion Modules Specifications

Terminal definition

The structure and terminal diagrams of the RS485 analog I/O expansion board are shown below. The following table lists the definitions of each terminal bit. Please strictly follow the definition of terminal bit to install and connect the expansion board.

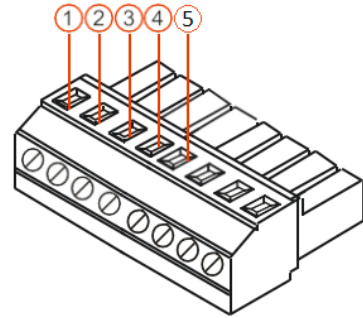
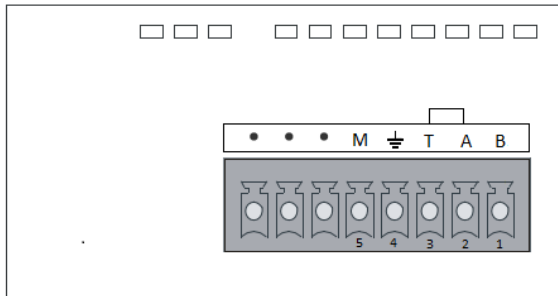


Table 4-10-7 RS485 terminal definition

Terminal bit	Signal definition
1	RS485 signal B
2	RS485 signal A
3	Terminal resistance
4	Equipment ground
5	RS485 signal GND

Note: Connect 2 with 3 when matching terminator

4.10.4 BD Expansion Board Installation

Installation steps:

1. Ensure that both CPU and connected equipment have been cut off, and lay the CPU master control module flat.
2. Remove the cover plate along the left side of the expansion board
3. Insert the expansion board module into the socket slot to ensure that the pin fits the socket tightly
4. Insert the right flange of the expansion plate cover plate into the side socket of the expansion plate slot on the main control module;
5. Press down the left side of the expansion board cover obliquely and put it back on the main control module.



**Note**

- V5 upgrade series CPU H224X and H226XL support CAN-01 main station expansion board
- The CPU CAN be equipped with CAN-01 main station expansion board directly, but the analog I/O expansion board can only be used by calling the special expansion board instruction library. For specific usage, please refer to the user manual of CTH200 series programmable logic controller. Download address: <http://www.co-trust.com>
- Do not plug expansion board with power on.

4.11 Display Panel Specifications

Table 4-11-1 Order No.

Description	Order No.
Copanel TD2X, LCD displayer, 2 row, 20 ASCII characters, RS485, IP65	CTS6 D02-MH010
Copanel TD4X, LCD displayer, 4 row, 96 ASCII characters, RS485, IP65	CTS6 D04-MH010
Copanel TD4S, LCD displayer, 4 row, 96 ASCII characters, RS485, IP65	CTS6 D4S-MH010

4.11.1 TD2X Specifications

Table 4-11-2 TD2X specifications

Type	TD2X
Order No.	CTS6 D02-MH010
Connect to	TrustPLC CTSC-100 and 200 series CPU SIMATIC S7-200 series CPU
Dimension	
Front panel (W×H)	148×76mm
Tapping (W×H×D)	128×68×28mm
Display	
Type	High grade STN display panel
Characters/row	Small font(16×16 raster font) 2×10 Chinese
Character height(mm)	5
Backlight type	Secure low voltage LED lights
Keyboard	
Type	Coating keyboard
Function/system	4/5
DC 24V direct /DP direct	--
Interface features	
Communication ports	1 RS485 PPI/MPI ports, support run in power on Baud: 9.6Kbps, 19.2Kbps and 187.5Kbps

	Not isolated
Supply port	1*2 pin external supply ports
Software features	
Screen protection	LCD backlight delay off controllable, delay backlight service life
Defined screen picture	64
Defined warning	80
Keyboard tone	--
Password protection	Use in permission
Function	
Information display	√
Information font	√
Setting input	√
Configuration software	MagicWorks PLC and Step-7 MicroWin V4.0 of Siemens

Tips: “√” means usable;

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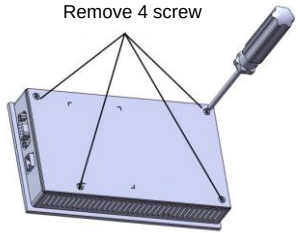
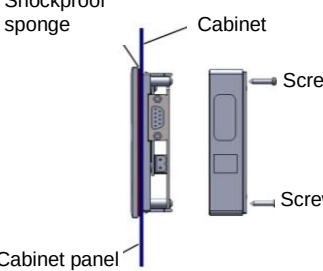
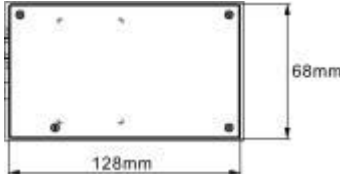
Standard configuration

TD 2X is a small compact text interactive device, a full range of products to provide all CTSC-100 PLC interconnect components.

Table 4-11-3 Configuration Table

Items	Description
TXT display	The text display area is a backlit LCD with a resolution of 160X32 pixels; this area shows information from CTSC-100/200 CPU.
Buffer washer	Buffer washer and mattress are used for severe environment TD 2X installation.
Communication port	Communication port is a standard 9 pin D-SUB, which can connect TD2X to CTSC-100/200 CPU.
Power	When TD/CPU cable is not used, TD2X can be powered directly through the interface and external power plug
External power	When TD/CPU cable is not used, TD2X can be powered directly through the interface and power plug
TD/CPU cable	TD/CPU cables allow TD 2X to connect with CTSC-100/200 CPU directly, cables offer communication and power connection to TD 2X (2 ports of cable are symmetric.)
User label	Pull-out allowed, reload user-defined label.
Button	9 buttons of TD 2X, 5 for predefined function, the functions are context-sensitive, 4 for user-defined.
Angle mat	Use to fix TD 2X to a surface occasion.

Installation dimension

 Remove 4 screw	 Shockproof sponge Cabinet Cabinet panel Screw Screw	 68mm 128mm
Open the shell of TD2X		Trepanning a hole sized 68mm×128mm in surface, install TD2X

4.11.2 TD4X Specifications

Table 4-11-4 TD4X specifications

Items	TD4X
Order No.	CTS6 D04-MH010
Connect to	TrustPLC CTSC-100 and 200 series CPU SIMATIC S7-200 series CPU
Display	
Type	High grade STN display panel
Characters/row	Small font(16×16 raster font) 4×12 Chinese Big font(24×24 raster font) 2×8 Chinese
Display effect	Fashionable high brightness with white characters in blue background.
Backlight type	Secure low voltage LED lights
Dimension	
Front panel (W×H)	192×64 mm
Tapping (W×H×D)	163.5×93.5×31 mm
Keyboard	
Type	Coating keyboard
Function/system	8/7
DC 24V direct /DP direct	--
Interface features	
Communication ports	1 RS485 PPI/MPI communications ports, support run in power on Baud: 9.6Kbps, 19.2Kbps and 187.5Kbps available; Not isolated
Supply port	1*2 pin external supply ports
Software features	
Screen protection	LCD backlight delay off controllable, delay backlight service life
Defined screen picture	64
Defined warning	80
Keyboard tone	Press with Keyboard tone, strength confirmation
Password protection	Use with permission

Supportive language set(for different language)	9 language set: Latin 1/bold, Latin 2/bold, CH(simplify), Cyrillic, Arabic, Baltic, Greek, Hebrew, Turkish
Function	
Information display	√
Information font	√
Setting input	√
Configuration software	MagicWorks PLC and Step-7 MicroWin V4.0 SP5 or above

Tips: “√” means usable;

SIEMENS is trademark of SIEMENS AG

Standard configuration

TD 4X is a small compact text interactive device, a full range of products to provide all CTSC-100 PLC interconnect components.

Table 4-11-5 Configuration Table

Items	Description
TXT display	The text display area is a backlit LCD with a resolution of 160X32 pixels; this area shows information from CTSC-100/200 CPU.
Buffer washer	Buffer washer and mattress are used for severe environment TD 2X installation.
Communication port	Communication port is a standard 9 pin D-SUB, which can connect TD4X to CTSC-100/200 CPU.
Power	When TD/CPU cable is not used, TD4X can be powered directly through the interface and external power plug
External power	When TD/CPU cable is not used, TD4X can be powered directly through the interface and power plug
TD/CPU cable	TD/CPU cables allow TD 4Xconnect with CTSC-100/200 CPU directly, cables offer communication and power connection to TD 4X (two ports of cable are symmetric.)
Button	15 buttons of TD 4X, 7 for predefined function, the functions are context-sensitive, 8 for user-defined.

Paste the panel

Paste panel to mainframe before TD4X installation

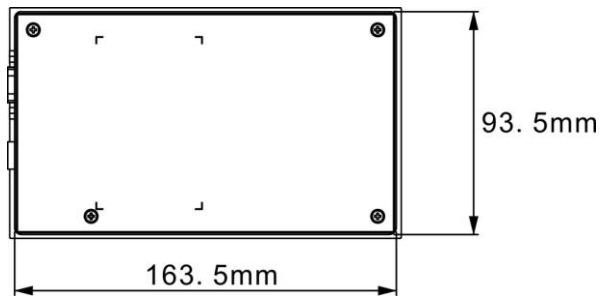
Steps	Diagram	Illustration
1		Tear off protection paper on the left side, revealing sticky paper

2		Center the panel to align the TD4X border, first pressing the left side
3		Then carefully pick up the unglued part of the panel and tear off the remaining protective paper
4		Finally, press the panel from left to right on the keyboard

Installation dimension

1) Cabinet opening

Hole size: 163.5 mm × 93.5 mm



2) TD4X installation preparation

Take down the front cover from TD4X shell as diagram 1

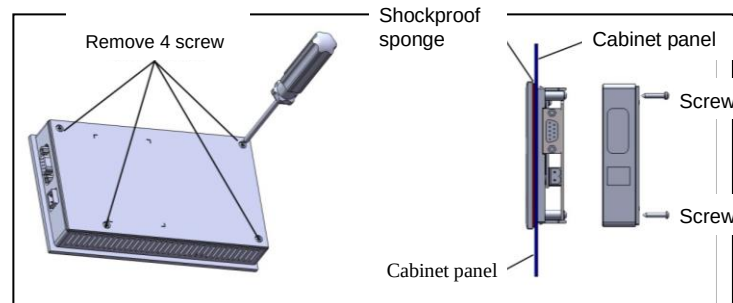


Figure 1

Figure 2

3) Install TD4X

① take down gasket from protection mat.

② put gasket to front panel of TD4X (as diagram 2)

- ③ Load the front panel into the holes already made on the mounting panel
- ④ Screw back panel from back panel to TD4X front panel. Be sure to tighten the screws

Communication cables

Two TD4X communication cables, one is with supply cable, the other is without cable.

Random distribution of communication cable is with power cable, communication port metal shell earth.

With supply cables: P2(0V), P3(D+), P7(+24V), P8(D-)

Without supply cables: P3(D+), P8(D-)

4.11.3 TD4S Specifications

Table 4-11-6 TD4S specifications

Items	TD4S
Order No.	CTS6 D4S-MH010
Connected PLC	TrustPLC CTSC-100 and 200 CPU/SIMATIC S7-200 CPU Note: TD4S only be used for one-one connection of CTSC-100/200 and S7-200 CPU
Features	
Dimension(W×H×D) Front panel Trepanning	174×102 mm 163.5×93.5×31 mm
Protection Dust&water Reliability	Front panel: IP65 Back cover: IP20 ESD: ±4KV(connection discharge), ±8KV(Air discharge) Surge protection: 2KV Power reverse protection
Temperature For working For transport and storage	0℃~+60℃ -30℃~+80℃
LCD features	
Type	STN
Resolution(row×line)	192×64
Dot pitch(row×line)	0.48mm×0.48mm
Characters/row	Small font(16×16 raster font) 4×12 Chinese Big font(24×24 raster font) 2×8 Chinese
Display effect	Fashionable high brightness with white characters in blue background.
Backlight type	Secure low voltage LED lights
Keyboard	
Keyboard type	Coating keyboard
Button type	Coating switch, can be clicked a million times
Default function	8 【F1, F2, F3, F4, F5, F6, F7 and F8, use shift to F16】

button	
Default system button	7 【ENTER, ESC, SHIFT, ↑, ↓, ← and →】
User-defined button	All 15 buttons can be defined by users.
Interface features	
Communication ports	1 RS485 PPI/MPI communications ports, support run in power on Baud: 9.6Kbps, 19.2Kbps available; Not isolated
Supply port	1*2 pin external supply ports
Software features	
Screen protection	LCD backlight delay off controllable, delay backlight service life
Defined screen picture	64
Defined warning	80
Keyboard tone	Press with Keyboard tone, strength confirmation
Password protection	Use with permission
System option menu	Enforced I/O Password protection Set time&date CPU status Switch language(if configured) Change CPU operation mode Clean keypad Edit CPU memory Program memory box
Supportive language set(for different language)	9 language set: Latin 1/bold, Latin 2/bold, CH(simplify), Cyrillic, Arabic, Baltic, Greek, Hebrew, Turkish
Functions	
Information display	√
Information font	√
Setting input	√
Configuration software	MagicWorks PLC and Step-7 MicroWin V4.0 SP5 or above
Power	
Working voltage	9V DC~28V DC
Working current	80mA@24V DC(typical)
Isolation	No isolation, reverse protection circuit

Tips: “√” means usable;

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Standard configuration

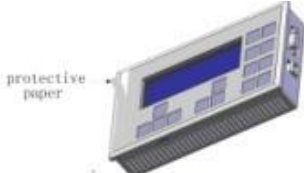
TD 4S is a small compact text interactive device, a full range of products to provide all CTSC-200 PLC interconnect components.

Table 4-11-7 configuration Table

Items	Description
TXT display	The text display area is a backlit LCD with a resolution of 160X32 pixels; this area shows information from CTSC-200 CPU.
Buffer washer	Buffer washer and mattress are used for severe environment TD 4S installation.
Communication port	When TD/CPU cable is not used, TD4S can be powered directly through the interface and power plug
Button	15 buttons of TD 4X, 7 for predefined function, the functions are context-sensitive, 8 for user-defined.
Magnet ring	Strengthen ant jamming capability of communication/power cables.

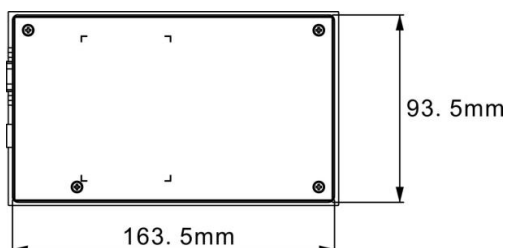
Paste the panel

Paste panel to mainframe before TD4S installation

Steps	Diagram	Illustration
1		Tear off protection paper on the left side, revealing sticky paper
2		Center the panel to align the TD4S border, first pressing the left side
3		Then carefully pick up the unglued part of the panel and tear off the remaining protective paper
4		Finally, press the panel from left to right on the keyboard

Installation dimension

1) Cabinet opening



2) TD4S installation preparation

Take down the front cover from TD4S shell as figure 1

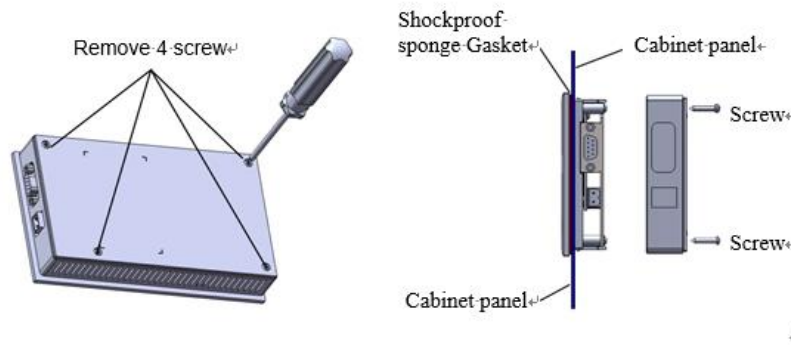


Figure 1

Figure 2

3) Install TD4S

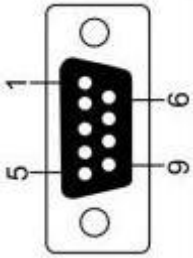
- ① Take down gasket from protection mat.
- ② Put gasket to front panel of TD4S (as diagram 2)
- ③ Load the front panel into the holes already made on the mounting panel
- ④ Screw back panel from back panel to TD4X front panel. Be sure to tighten the screws

There is no communication cable distributed by TD4S, but a magnetic ring is distributed randomly. It is suggested to attach the magnetic ring to the power line or communication cable to increase the anti-interference ability of TD4S under server environment. Use the communication cable with or without power supply, connect metal shell of the communication port to the ground. Communication cables with power supply: P2 (0V), P3 (D+), P7 (+24V), P8 (D-); Communication cable without power supply: P3 (D+), P8 (D-). You can also purchase communication cables with power supply from our company, Order No.: CTS7 291-TDC01

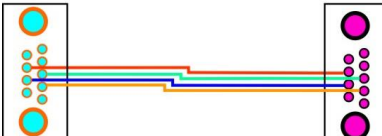
Magnetic ring installation

Steps	Diagram	Illustration
1		First cut out the 2 buttons of the magnetic ring
2		Put the power wire or the communication cables into magnetic ring
3		Fasten the buttons

Communication ports and PIN signal

DB9 female	DB9 female PIN signal	
	PIN	Signal definition
	1	Vacant
	2	Power ground
	3	Data+
	4	Vacant
	5	Vacant
	6	Vacant
	7	Power input(24V DC)
	8	Data-
	9	Vacant

Communication cable connection

Male	Connection
Without power supply(3-8)	
With power supply(3-8, 2-7)	

5 Network

5.1 CTSC-200 Network Communication Conception

【Select network communication port】

Select the communication network supported by CTSC-200 in the 'set PG/PC interface' properties dialog box. The selected network will be used as an interface. MagicWorks PLC can access the communication network interface default to 'PC/PPI Cable'. Specific steps are as follows:

- 1) Double click icon or click 'Set PG/PC Interface' in communication setting.
- 2) Select MagicWorks PLC interface parameters(default PC/PPI Cable(PPI))

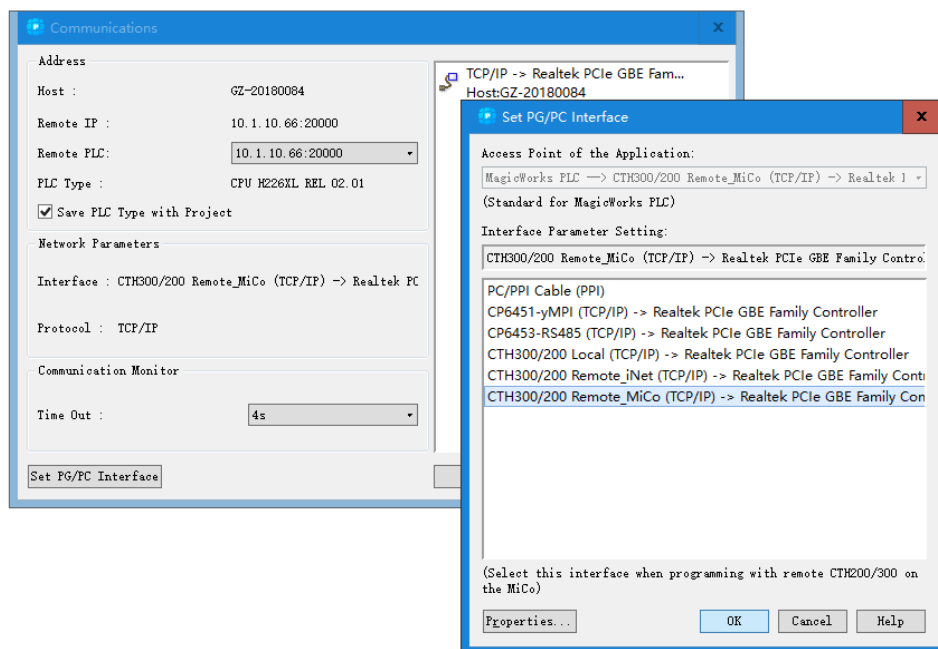


Figure 5-1 MagicWorks PLC communication port

【Set MagicWorks PLC baud rate and station address】

The speed data travels over the network is baud rate. The units are usually Kbps or Mbps. Devices that communicate in the same network must be configured at the same baud rate. Therefore, the highest baud rate of a network depends on the device with the lowest baud rate connected to the network.

Table5-1 CTSC-200 supported baud rate

Network	Baud rate
Standard network	9.6K to 187.5K
Use EM277	9.6K to 12M
Freeport	1200 to 115.2K

A unique station address is specified for each device in the network, which ensures that data is sent to or from the correct device. CTSC-200 supports network addresses from 0 to 126. If a CTSC-200 has two ports, each port has a network address.

Table5-2 CTSC-200 default station address

CTSC-200 facility	Default address
MagicWorks PLC	0
HMI (TD2X,TD4X or TD4S)	1
CTSC-200 CPU	2

MagicWorks PLC must configure baud rate and station address. The baud rate must be the same with other facilities in the network and the station address must be unique. No need to change default station address 0 of MagicWorks PLC. Change the MagicWorks PLC station address when there is another programming kit in the network.

As figure 5-2 shows, configure MagicWorks PLC baud rate and station address, steps as follows:

- 1) Double click icon in communication setting.
- 2) Click property button in 'Set PG/PC Interface'.
- 3) Select station address and transmission rate for MagicWorks PLC.

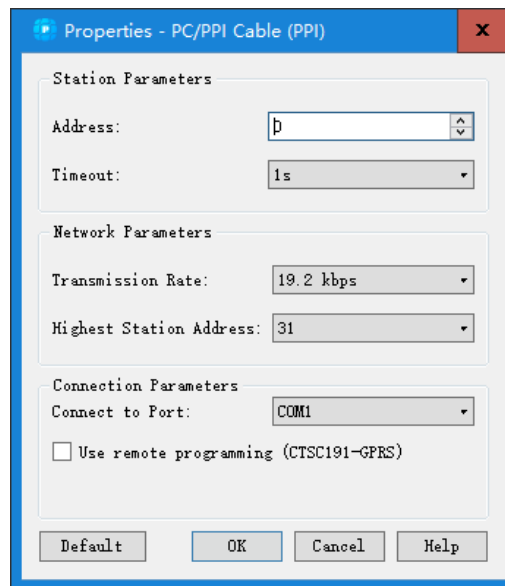


Figure 5-2 MagicWorks PLC communication parameter setting

【Set CTSC-200 baud rate and station address】

CTSC-200 must configure baud rate and station address, stored in system block. Download system block into CTSC-200 after setting parameters. Default baud rate is 9.6kbps, default station address is 2 for each CTSC-200 communication port.

As figure 5-3 shows, configure MagicWorks PLC baud rate and station address, steps as follows:

- 1) Select station address for CTSC-200.

- 2) Select baud rate for CTSC-200.
- 3) Download system block to CTSC-200.

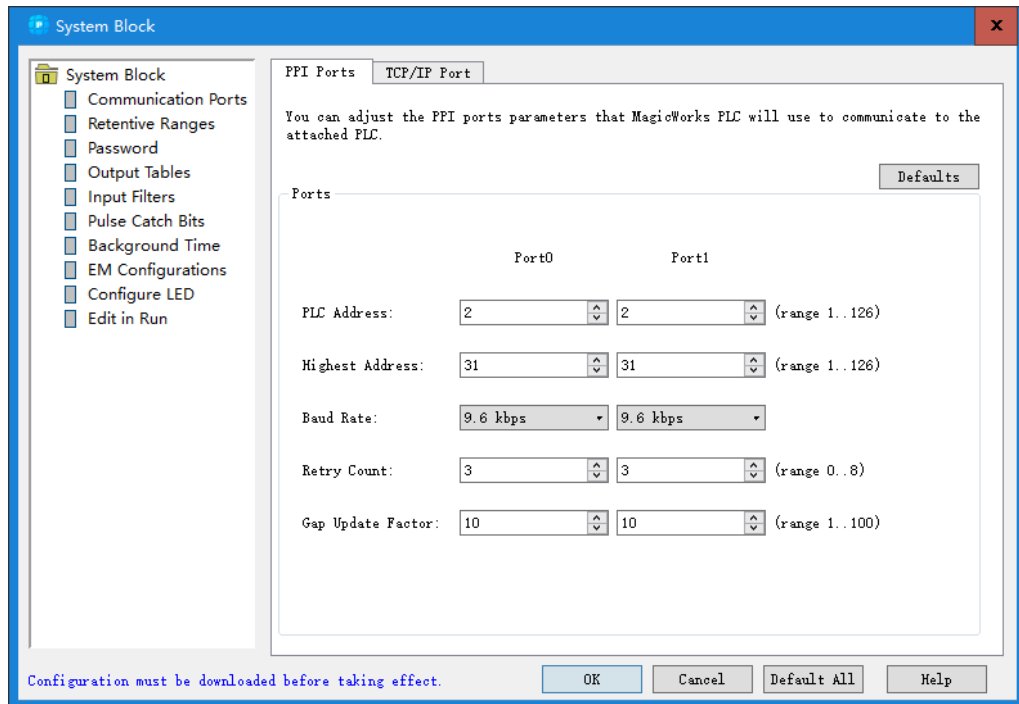


Figure 5-3 CTSC-200 communication parameter setting

【Searching CTSC-200 CPU in network】

CTSC-200 can be found and identified in the network. When searching for CTSC-200, look for networks at specific baud rates. The search begins with the currently selected baud rate.

- 1) Double click "Double-Click on the Refresh" for searching.
- 2) Select appropriate baud rate.

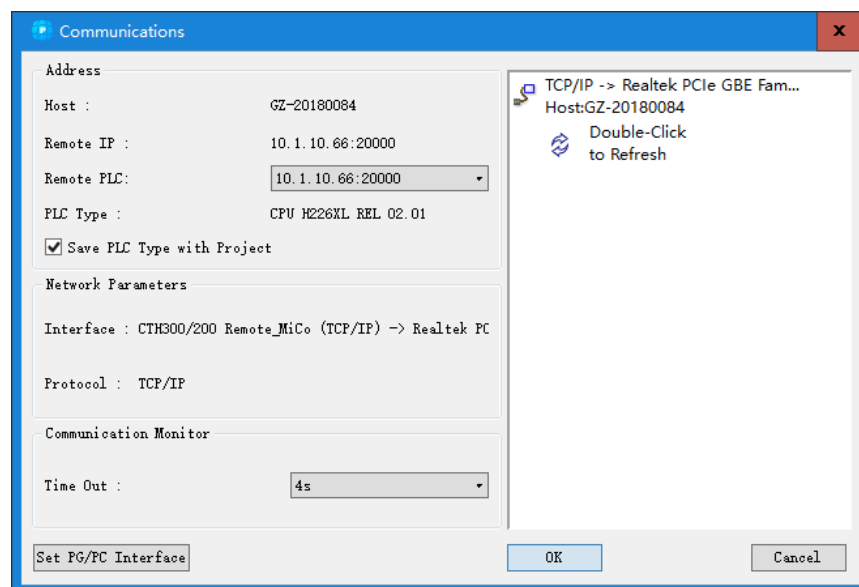


Figure 5-4 Searching CTSC-200

5.2 Communication Protocol and Networking Mode

Trust PLC CTSC-200 series PLC support PPI, MPI, Freeport, Ethernet and DP.

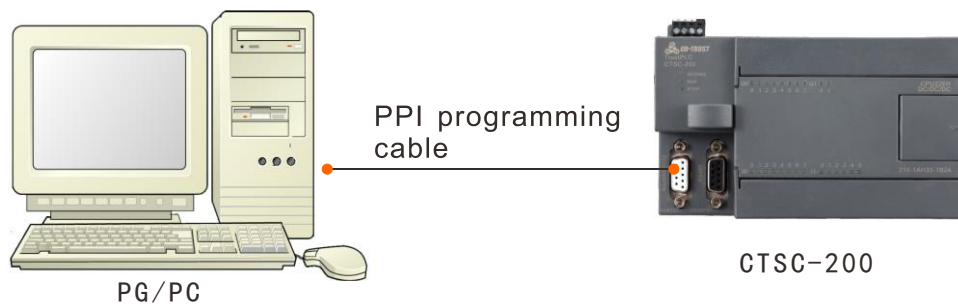
5.2.1 PPI Communication

PPI is a master-slave protocol in which the master sends a request to the slave station and the slave station responds. The slave station wait when master station does not send a request.

TrustPLC CTSC -200 supports PPI protocol and can be connected to the Internet through normal two-core shielded twisted pair cables. Three baud rates are supported: 9.6kBit/s, 19.2kBit/s, and 187.5kBit/s. The integrated programming port on the TrustPLC CTSC-200 is also the PPI communication network interface. Using PPI protocol for communication is simple and convenient, use only NETR and NETW for data signal transmission, no additional configuration module or software. PPI communication network is a token-passing network, which can be composed of up to 31 TrustPLC CTSC-200 series PLC, Copanel series HMI, Siemens OP/TP panel or upper computer inserted MPI card.

Single master station PPI network

The picture below give an example of one master station with one slave station, programming station PG\PC as master station while CTSC-200 as slave station.



Picture below Copanel HMI as master station while CTSC-200 as slave station.

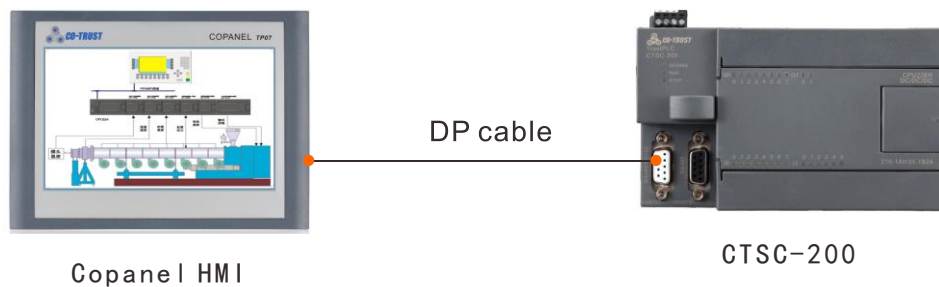


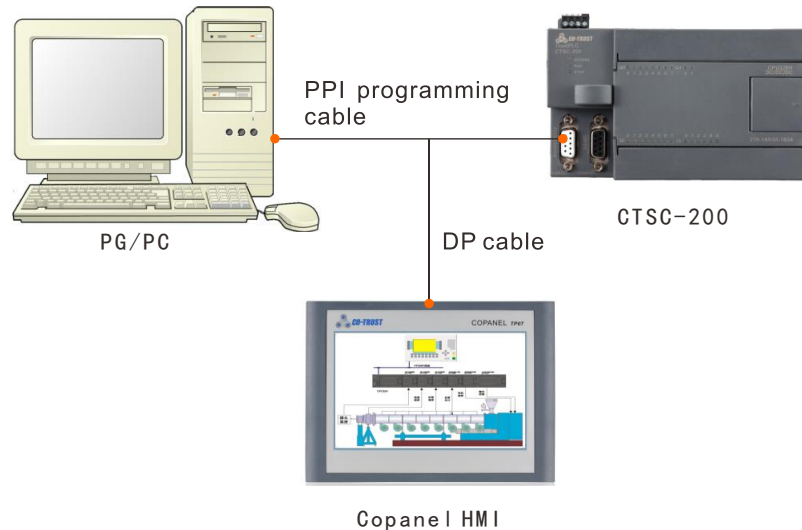
Figure 5-5 Single master station PPI network

Note: recommend PLC programming cables of COTRUST as PPI programming cables (Order No.: CTS7 191-USB30)

Please use appropriate communication cables for different HMI.

Multi-master station PPI network

The picture below give an example of 2 master station with one slave station, programming station PG\PC as master station while CTSC-200 as slave station.



The picture below give an example of multi-master station with multi-slave station, programming station PG\PC as master station while CTSC-200 as slave station.



Figure 5-6 Multi-master station PPI network

The figure below shows the network example of multiple-CPU interconnection. Multiple CTSC-200 CPUs are connected via the third-party DP cable, and the network communication between multiple CPUs is realized through the network NETR/NETW. In the network communication process, each CTSC-200 CPU is both the master and slave station. CPU communicate via the PPI port.

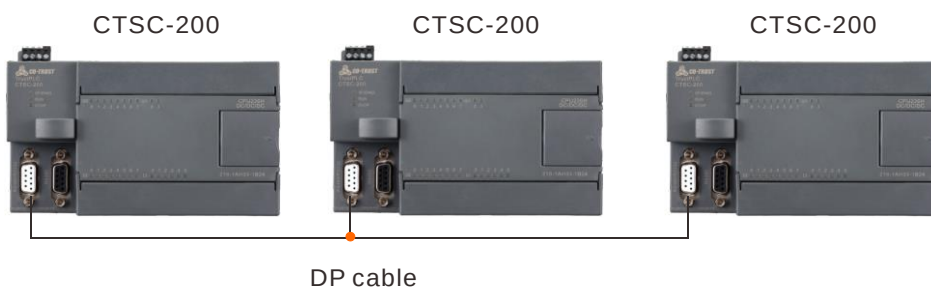


Figure 5-7 Multi-CPU connection network

Note: recommend to use third DP cables.

5.2.2 MPI Communication

MPI supports master-slave and master-master communication.

TrustPLC CTSC -200 supports PPI protocol and can be connected to the Internet through normal two-core shielded twisted pair cables. Three baud rates are supported: 9.6kBit/s, 19.2kBit/s, and 187.5kBit/s. It can communicate with SIMATIC S7-300/S7-400 CPU. TrustPLC CTSC-200CPU only can be slave station in MPI network.

As picture shows S7-300 and Copanel HMI is master station in MPI network while CTSC-200 is slave station. S7-300 PLC read and write CTSC-200 data via XGET and XPUT instruction.

Multi master station MPI network

The picture below give an example of multi-master station with one slave station, programming station PG/PC S7-300 and HMI is master station while CTSC-200 as slave station.

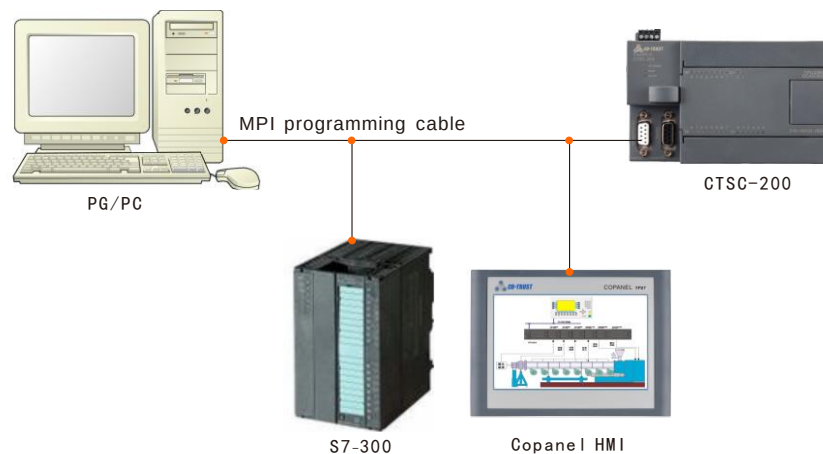


Figure 5-8 Multi-master station MPI network

5.2.3 Freeport Communication

The implementation of the free port communication function makes TrustPLC CTSC-200 can communicate with any third-party device, that is, TrustPLC CTSC-200 can be user-defined. Third party devices such as data acquisition modules, controllers, printers or bar code readers, frequency converters, modems and upper PCS, etc.

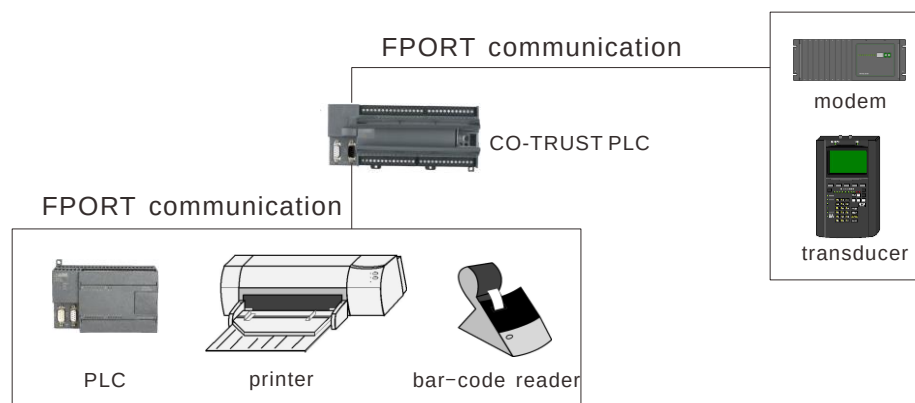


Figure 5-9 Freeport communication mode

5.2.4 DP Communication

Through the DP slave station module, the CPU can be connected to the PROFIBUS field bus. As a slave station, it can exchange data quickly and in real time and run independently with high reliability.

5.2.5 Ethernet Communication

CPU224I and CPU226I can communicate via Ethernet in MagicWorks PLC. Taking CPU224I as an example, Ethernet communication is described as follows:

CPU224I Ethernet port connect to the PLC industrial Ethernet communications processor, accomplish TCP/IP to MPI protocol conversion. This series of PLC support MiCo remote monitoring system for communication through the server, MagicWorks PLC connect remote PLC through MiCo, for programming and monitoring remote CTH200 PLC in the network. For more detailed information, please visit: <http://www.co-trust.com/>

Communication architecture and typical network structure are shown as follows:

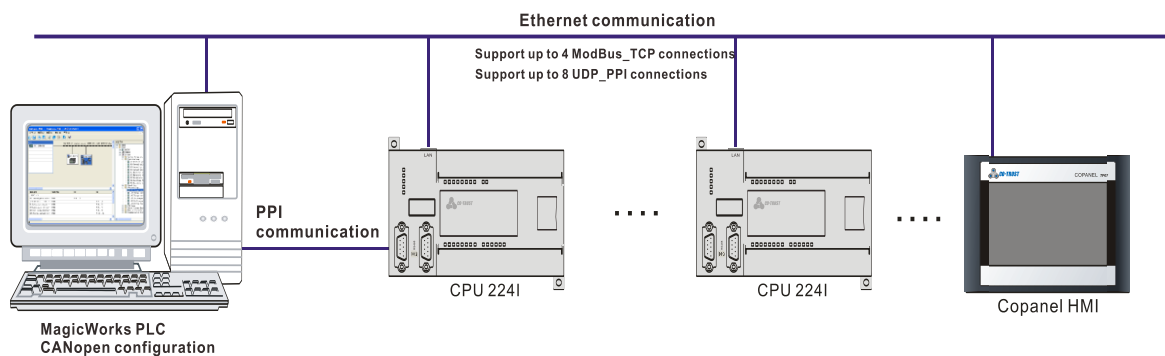


Figure 5-10 CTSC-200 Ethernet communication framework

- Baud rate: 10/100Mbps self-adaption
- Protocol type: UDP/TCP protocol, support Ethernet programming and communications
- Communicate via PC
- Most connections for one station: CPU224I supports 8, 4 for UDP_PPI and 4 for Modbus_TCP; CPU226I supports 8 UDP_PPI and 16 Modbus_TCP.
- Support Ethernet UDPNETR, UDPNETW and Ethernet TCP_MBUS_MSG

5.3 Communication Port

Table 5-3 CTSC-200 communication port

Name	Order No.	Description
CPU 224+ transistor	CTS7 214-1AD33-0X24	2 ports. PORT0: RS485(from JE3363, no RS232); FPORT: RS232/RS485
CPU 224+	CTS7	2 ports. PORT0: RS485(from JF3078, no RS232);

Relay	214-1BD33-0X24	FPORT: RS232/RS485
CPU 224+ V5(upgrade) transistor	CTS7 214-1AD35-0x24	2 ports. PORT0: RS485, PPI; FPORT: RS485, Freeport
CPU 224+ V5(upgrade) Relay	CTS7 214-1BD35-0x24	
CPU 224E transistor	CTS7 214-1AE33-0X24	2 ports. PORT0: RS485, PPI; FPORT: RS485, Freeport
CPU 224E Relay	CTS7 214-1BE33-0X24	
CPU 224I transistor	CTS7 214-1AD41-0X24	3 ports. PORT0: RS485, PPI; FPORT: RS485, Freeport LAN: RJ45
CPU 224I Relay	CTS7 214-1BD41-0X24	
CPU 226I transistor	CTS7 216-2AD41-0X40	3 ports. PORT0 and PORT1: PPI/ Freeport LAN: RJ45
CPU 226I Relay	CTS7 216-2BD41-0X40	
CPU 226M transistor	CTS7 216-1AF33-0X24	3 ports. FPORT0/FPORT1: RS485, Freeport; PORT0: RS232/RS485, PPI
	CTS7 216-1AD33-0X24	3 ports. PORT0: RS485, PPI; PORT1: RS232/RS485, PPI/ Freeport; FPORT0: RS485, Freeport
CPU 226M Relay	CTS7 216-1BF33-0X24	3 ports. FPORT0/FPORT1: RS485, Freeport; PORT0: RS232/RS485, PPI
	CTS7 216-1BD33-0X24	3 ports. PORT0: RS485, PPI; PORT1: RS232/RS485, PPI/ Freeport; FPORT0: RS485, Freeport
CPU 226M V5(upgrade) transistor	CTS7 216-1BD35-0x24	2 ports. PORT0/PORT1: RS485, PPI/ Freeport After adding RS485 expansion board: PORT0: RS485, PPI PORT1: RS485 , PPI/ Freeport FPORT0: RS485, Freeport
CPU 226M V5(upgrade) Relay	CTS7 216-1AD35-0x24	
CPU 226L transistor	CTS7 216-2AF33-0X40	3 ports. FPORT0/FPORT1: RS485, Freeport; PORT0: RS232/RS485, PPI
	CTS7 216-2AD33-0X40	3 ports. PORT0: RS485, PPI; PORT1: RS232/RS485, PPI/ Freeport; FPORT0: RS485, Freeport
CPU 226L Relay	CTS7 216-2BF33-0X40	3 ports. FPORT0/FPORT1: RS485, Freeport; PORT0: RS232/RS485, PPI
	CTS7 216-2BD33-0X40	3 ports. PORT0: RS485, PPI; PORT1: RS232/RS485, PPI/ Freeport; FPORT0: RS485, Freeport
CPU 226L	CTS7	2 ports. PORT0 and PORT1: RS485, PPI/

V5(upgrade) transistor	216-2BD35-0x40	Freeport After adding RS485 expansion board:
CPU 226L V5(upgrade) Relay	CTS7 216-2AD35-0x40	PORT0: RS485, PPI PORT1: RS485 , PPI/ Freeport FPORT0: RS485, Freeport
CPU 226H Motion control	CTS7 216-1AH33-0A24	3 ports. FPORT0/FPORT1: RS485, Freeport; PORT0: RS232/RS485, PPI
	CTS7 216-1AH33-1A24	
	CTS7 216-1AH33-0B24	3 ports. FPORT0/FPORT1: RS485, Freeport; PORT0: RS485 , PPI
	CTS7 216-1AH33-1B24	
	CTS7 216-1AH33-0X24	
	CTS7 216-1AH34-0B24	3 ports. PORT0: RS485, PPI; PORT1: RS485 , PPI/ Freeport; FPORT0: RS485, Freeport
	CTS7 216-1AH34-1B24	
	CTS7 216-1AH34-0X24	
	CTS7 216-1AH34-2B24	3 ports. PORT0/PORT1: RS485 , PPI/ Freeport; FPORT0: LAN, RJ45
CPU 226H V5(upgrade) transistor	CTS7 216-1AH35-0B24	2 ports. PORT0 and PORT1: RS485 , PPI □/ Freeport
	CTS7 216-1AH35-2B24	After adding RS485 expansion board: PORT0: RS485, PPI PORT1: RS485 , PPI/ Freeport FPORT0: RS485, Freeport After adding CAN expansion board: PORT0: RS485, PPI/Freeport PORT1: RS485 , PPI/Freeport CAN communication port
PCS266	CTSC 266-3AT33-2424	3 ports. PORT0/PORT1: RS485, PPI; FPORT: RS485, Freeport

**Note**

- All PPI ports support PPI protocol and MPI protocol; all FPORT support Freeport protocol and Modbus communication protocol.
- CPU224+ transistor is with RS232 from JD3117 to JD3136 PPI port.
- PPI/FPORT: PPI protocol and Freeport protocol can be switchover.
- Ethernet port support TCP/UDP protocol.

5.3.1 CTSC-200 CPU Communication Ports

※ One PPI port and one FPORT(CPU224+)



Item	Order no.	Description
CPU 224+ transistor	CTS7 214-1AD33-0X24	2 ports: PORT0: RS485(from JE3363, no RS232), only support PPI protocol
CPU 224+ relay	CTS7 214-1BD33-0X24	FPORT: RS485/RS232 (from JF3078, no RS232), support FPORT and Modbus protocol

Table 5-4 CPU224+ communication PIN definition

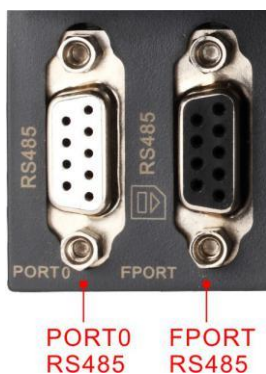
Connector	PIN	PORT0 (RS485)	FPORT (RS232/RS485)
	1	Rack earth	
	2	24V ground	RXD (RS232)
	3	RS485+ signal B	
	4	Send application	CTS (RS232)
	5	Logic ground	
	6	+5 V, 100Ω series resistor	
	7	+24V	RTS (RS232)
	8	RS485- signal A	
	9	10 bit protocol (input)	TXD (RS232)
	shell	Rack earth	



Note

The RS485 signal of some instruments is defined as A positive, B negative. Positive to positive and negative to negative when wiring (universal).

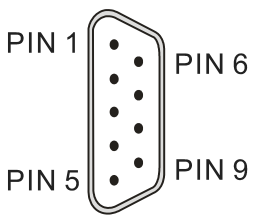
※ One PPI port and one FPORT(CPU224E/ 224+ V5(upgrade))



Item	Order no.	Description
CPU 224E Transistor	CTS7 214-1AE33-0X24	2 ports: PORT0: RS485, only support PPI protocol FPORT: RS485, support FPORT and Modbus protocol
CPU 224E Relay	CTS7 214-1BE33-0X24	
CPU224+V5(upgrade) Transistor	CTS7 214-1AD35-0x24	
CPU224+V5(upgrade) Relay	CTS7 214-1BD35-0x24	

Table 5-5 CPU224E/224+ V5(upgrade) PIN definition

Connector	PIN	PORT0 (RS485)	FPORT (RS485)
-----------	-----	---------------	---------------

	1	Rack earth
	2	24V earth
	3	RS485+ signal B
	4	Send application
	5	Logic ground
	6	+5 V, 100Ω series resistor
	7	+24V
	8	RS485- signal A
	9	10 bit protocol (input)
	Shell	Rack earth

※ 2个PPI/FPORT(CPU226M V5(upgrade))



Item	Order No.	Description
CPU226M V5(upgrade) transistor	CTS7 216-1BD35-0x24	2 ports PORT0: RS485/FPORT, support protocol switching, PPI, FPORT, Modbus protocol PORT1: RS485/FPORT, support protocol switching, PPI, FPORT, Modbus protocol
CPU226M V5(upgrade) Relay	CTS7 216-1AD35-0x24	After adding RS485 expansion board: PORT0: RS485, PPI PORT1: RS485 , PPI/FPROT FPORT0: RS485, FPROT

2个PPI/FPORT(226L V5(upgrade))



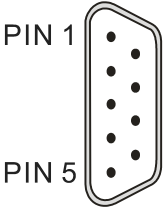
CPU226L V5(upgrade) transistor	CTS7 216-2BD35-0x40	2 ports PORT0: RS485/FPORT, support protocol switching, PPI, FPORT, Modbus protocol PORT1: RS485/FPORT, support protocol switching, PPI, FPORT, Modbus protocol
CPU226L V5(upgrade) Relay	CTS7 216-2AD35-0x40	After adding RS485 expansion board: PORT0: RS485, PPI PORT1: RS485, PPI/ FPORT FPORT0: RS485, FPORT

※ 2 PPI/FPORT(226H V5(upgrade))



CPU226L V5(upgrade) transistor	CTS7 216-2BD35-0x40	2 ports PORT0: RS485/FPORT, support protocol switching, PPI, FPORT, Modbus protocol PORT1: RS485/FPORT, support protocol switching, PPI, FPORT, Modbus protocol
CPU226L V5(upgrade) Relay	CTS7 216-2AD35-0x40	After adding RS485 expansion board: PORT0(RS485, PPI port) PORT1(RS485, PPI/FPORT) FPORT0(RS485, FPORT) After adding CAN expansion board: PORT0(RS485, PPI/ FPORT) PORT1(RS485, PPI/ FPORT) CAN port

Table 5-6 CPU226H V5(upgrade)/226M V5(upgrade)/226L V5(upgrade) PIN definition

Connector	PIN	PORT1/FPORT1 (RS485)	PORT0/FPORT0 (RS485)
	1	Rack earth	
	2	24V earth	
	3	RS485 signal B	
	4	Send application	
	5	+5V earth	
	6	+5V, 100 Ω series resistor	
	7	+24V	
	8	RS485 signal A	
	9	NC	
	Shell	Rack earth	

※ One Ethernet, one PPI and one FPORT(CPU224I)



Item	Order No.	Description
CPU 224I Transistor	CTS7 214-1AD41-0X24	3 ports PORT0: RS485, support PPI FPORT: RS485, support
CPU 224I Relay	CTS7 214-1BD41-0X24	FPORT and Modbus protocol LAN: RJ45, Modbus/TCP, PPI/UDP

※ One Ethernet, two PPI/ FPORT (CPU226I)



Item	Order No.	Description
CPU 226I transistor	CTS7 216-2AD41-0X40	3 ports PORT0: RS485/FPORT, support protocol switching, PPI, FPORT, Modbus protocol PORT1: RS485/FPORT,
CPU 226I Relay	CTS7 216-2BD41-0X40	support protocol switching, PPI, FPORT, Modbus protocol LAN: RJ45, Modbus/TCP, PPI/UDP

Table 5-7 CPU224I/226I PIN definition

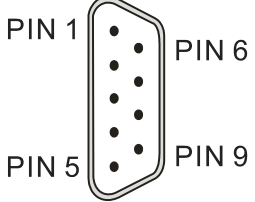
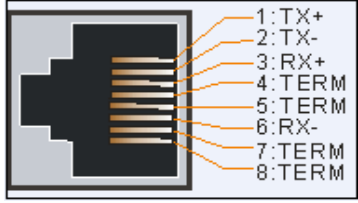
Connector	PIN	PORT0 (RS485)	FPORT (RS485)
	1	Rack earth	
	2	24V earth	
	3	RS485 signal B	
	4	Send application	
	5	Logic earth	
	6	+5 V, 100Ω series resistor	
	7	+24V	
	8	RS485- signal A	
	9	10 bit protocol (input)	
	Shell	Rack earth	

Table 5-8 CPU224I Ethernet PIN definition

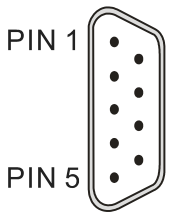
RJ45 communication port	Bit	Signal	Signal definition
	1	TX+	Data send positive
	2	TX-	Data send negative
	3	RX+	Data receive positive
	4	TERM	--
	5	TERM	--
	6	RX-	Data receive negative
	7	TERM	--
	8	TERM	--

※ 2 PPI and one FPORT(PSC266)



Item	Order No.	Description
PCS266	CTSC 266-3AT33-2424	3 ports PORT0: RS485, support PPI PORT1: RS485, support PPI, FPORT, Modbus protocol FPORT: RS485, support FPORT
CPU226H motion control	CTS7 216-1AH34-0B24	
	CTS7 216-1AH34-1B24	
	CTS7 216-1AH34-0X24	

Table5-9 PSC266 PIN definition (2 PPI, 1 FPORT)

Connector	PIN	PORT1(RS485)	PORT0 (RS485)	FPORT
	1	Rack earth		Rack earth
	2	-	Logic earth	RS485+ signal B
	3	RS485+ signal B		RS485- signal A
	4	Send application		4: short connect 3 with 4 when you need to match termination
	5	Logic earth		
	6	+5V, 100Ω		
	7	+24V		
	8	RS485-signal A		
	9	-	10 bit protocol (input)	
	shell	Rack earth		

※ One PPI, one PPI/FPORT, one FPORT(CPU226M/CPU226L)



Item	Order No.	Description
CPU 226M Transistor	CTS7 216-1AD33-0X24	3 ports PORT0: RS485, support PPI PORT1: RS232/RS485, support PPI/FPORT, Modbus protocol FPORT: RS485, support FPORT, Modbus protocol
CPU 226M Relay	CTS7 216-1BD33-0X24	
CPU 226L Relay	CTS7 216-2BD33-0X40	
CPU 226L Transistor	CTS7 216-2AD33-0X40	

Table5-10 CPU226M/226L PIN definition(2 PPI, 1 FPORT)

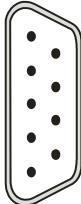
Connector	PIN	PORT1 (RS485/RS232)	PORT0 (RS485)	FPORT
PIN 1 PIN 5  PIN 6 PIN 9	1	Rack earth		Rack earth
	2	RS232 signal RXD	Logic earth	RS485+ signal B
	3	RS485+ signal B		RS485- signal A
	4	Send application		4: short connect 3 with 4 when need to match termination (except CPU226L)
	5	Logic earth		
	6	+5V, 100Ω		
	7	-/+24V	+24V	
	8	RS485- signal A		
	9	RS232 signal TXD	10 bit protocol (input)	
	shell	Rack earth		

※ One PPI, two FPORT(CPU226M/CPU226L/CPU226H)



Item	Order No.	Description
CPU 226M transistor	CTS7 216-1AF33-0X24	3 ports PORT0: RS232/RS485, support PPI FPORT0/FPORT1: RS485, support PPI/FPORT, Modbus protocol
CPU 226M Relay	CTS7 216-1BF33-0X24	
CPU 226L transistor	CTS7 216-2AF33-0X40	
CPU 226L Relay	CTS7 216-2BF33-0X40	
CPU 226H Motion control	CTS7 216-1AH33-0A24	
	CTS7 216-1AH33-1A24	

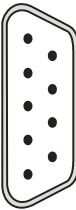
Table 5-11 CPU226M/CPU226L/CPU226H PIN definition(1 PPI, 2 FPORT)

Connector	PIN	PORT0 (RS485/RS232)	FPORT0 (RS485)	FPORT1
PIN 1 PIN 6 PIN 5 PIN 9 	1	Rack earth		1: Rack earth
	2	RS232 signal RXD	Logic earth	2: RS485 signal B
	3	RS485 signal B		3: RS485 signal A
	4	Send application		4: short connect 3 with 4 when need to match termination(except CPU226L)
	5	Logic earth		
	6	+5V, 100Ω		
	7	-/+24V	+24V	
	8	RS485 signal A		
	9	RS232 signal TXD	10 bit protocol (input)	
	Shell	Rack earth		



Item	Order No.	Description
CPU 226H motion control	CTS7 216-1AH33-0B24	3 ports PORT0: RS485 (from 0349, no RS232) , support PPI FPORT0/FPORT1: RS485, support FPORT, Modbus
	CTS7 216-1AH33-1B24	3 ports PORT0: RS485(from KC0160, no RS232), support PPI FPORT0/FPORT1: RS485, support FPORT, Modbus
	CTS7 216-1AH33-0X24	3 ports PORT0: RS485 (from 0126, no RS232) , support PPI FPORT0/FPORT1: RS485, support FPORT, Modbus

Table 5-12 CPU226H PIN definition (1 PPI, 2 FPORT)

Connector	PIN	PORT0(RS485)	FPORT0 (RS485)	FPORT1
	1	Rack earth		1: Rack earth
	2	Logic earth		2: RS485 signal B
	3	RS485 signal B		3: RS485 signal A
	4	Send application		
	5	Logic earth		
	6	+5V, 100Ω		
	7	+24V		
	8	RS485 signal A		
	9	-	10 bit protocol (input)	
	Shell	Rack earth		

**Note**

Do not use RS-232A communication cables on 33 version CPU224+ and CPU226M/L/H or the RS-232 port, or it may be burnout.

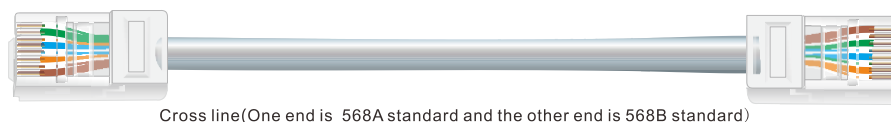
5.3.2 Make Standard Cable

Use standard network cable for CTSC-200 communication. Standard network cable has two forms: parallel line and cross line. The ends of parallel lines are standard 568A or 568B; the cross line is standard 568A at one end while 568B at the other end. The standard line sequence color of 568A/568B shows as below:

568B standard line sequence: white and orange, orange, white and green, blue, white and blue, green, white and brown, brown.

568A standard line sequence: white and green, green, white and orange, blue, white and blue, orange, white and brown, brown.

How to make standard cable:

**Note**

CPU 224I does not support adaptive crossover. Therefore, select crossover line as standard cable when directly connected to PLC for ModBus_TCP communication.

5.4 Network Connections

The protocol RS232 and RS485 network port of TrustPLC CTSC-200 CPU supports:

Table5-13 CTSC-200 interface

Type	Work mode	Supported protocol	Usage
RS232	Half-duplex	PPI, MPI, FPORT (note1)	Connect other CPU or third facilities, no need for RS485/RS232 converter when connect to RS232 facility.
RS485	Half-duplex	PPI, MPI, FPORT, Modbus (note 1)	Connect other CPU or third facilities
RJ45	Half-duplex	UDP/TCP protocol	Connect other CPU or third facilities

【note 1】 ModBus protocol requires the support of STL library, and PLC as slave station. Make the PLC as main station through the free port programming.

**Note**

The two ends of the cable and connector are connected tightly and fixed firmly. When wiring, do not lay the communication cable together with other strong electromagnetic field cables. When the communication cable is long or the communication cable needs to pass through the environment with electrical noise, adopt the shielded double-stranded cable to make communication cable. One end of the cable shield should be grounded.

5.4.1 Connect with IPC

When connect CTSC-200 CPU with IPC serial port, choose RS485/RS232 cable or Siemens PPI cable.

Note: An isolator should be added between PLC and PC (on PC side).

RS232 cable

Wiring as picture below when you need to connect RS232 cable with IPC.

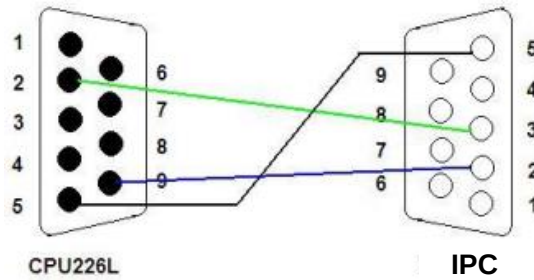


Figure 5-10 RS232 wiring diagram of CPU226L

RS485 cable

How to make RS485 cable:

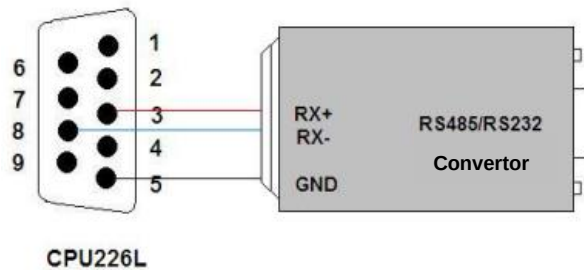


Figure 5-11 RS485 wiring diagram

PPI cable

Recommend to use COTRUST PLC communication cable (Order No.: CTS7 191-USB30), refer to Appendix [O Programming Cable](#). CTSC-200 is also compatible with Siemens PPI communication cables of. For specific usage, please refer to instructions of PPI.



Note

When using the Siemens PPI cable, one end of the PPI cable plug (RS485) can only be connected to the CTSC-200 CPU PORT0's 9-hole serial port, connect to the PORT1 and FPORT ports will not be able to communicate

5.4.2 Connect with Third Facility

Making cables on the basis of interface type when connecting with third facilities.

Facility with RS232 interface

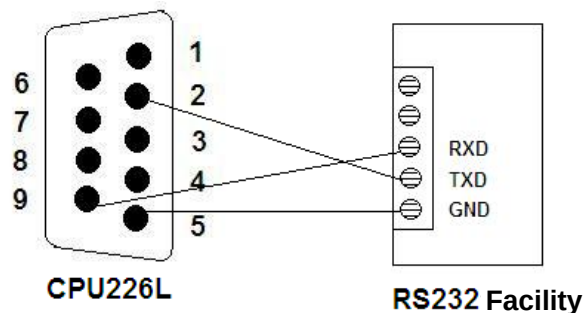


Figure 5-12 CPU224+, CPU226M/L/H connect with RS232 facility

Facility with RS485 interface

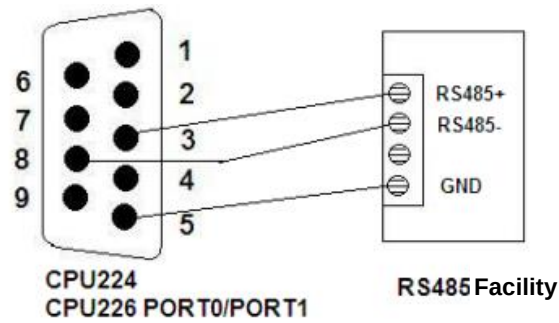


Figure 5-13 CPU224E/224+/V5 /224I/226L/226M/226H connect with RS485 facility

5.5 Modbus TCP Communication Example

This section guide you through an application example that will familiarize you with the Modbus TCP communication capabilities of CTSC-200.

In Modbus TCP communication, as slave station, the CPU communication is independent of the whole big loop cycle. Controlled by the user program.

CTSC-200 as the slave station of Modbus TCP to realize communication without any other configuration.

When CTSC-200 is used as the Modbus TCP master station to communicate with other slave stations, the Modbus TCP wizard or Modbus TCP library file is used to configure the master station communication. The default port number is 502. If you need to change the IP address, refer to Chapter [2 Getting Started](#).



Note

Refer to “*MagicWorks PLC user manual*” for usage instruction of Modbus TCP wizard and single R&W instructions library file. More info refer to: www.co-trust.com

5.5.1 The Sample Components

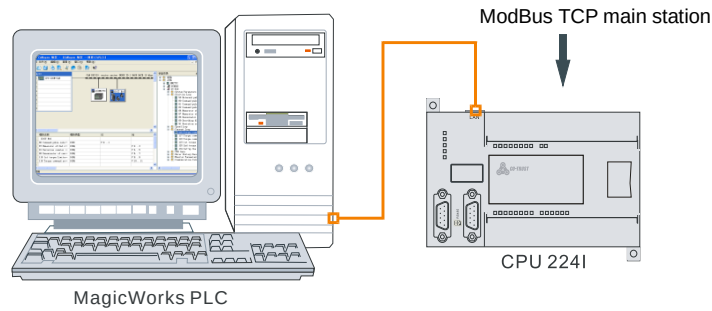
Table5-14 Modbus TCP communication sample components

Component	Function
Program facility PG/PC	MagicWorks PLC(V2.08 and above), able to configure, edit and debug on CTSC-200 PLC
CTSC-200 main-control module	2 CPU224I, one as Modbus main station, the other as Modbus slave station, Modbus TCP communication through EtherNET interface
Standard cable	Connect CPU224I with programming facility Connect CPU224I(Modbus main station) with CPU224I(Modbus slave station)

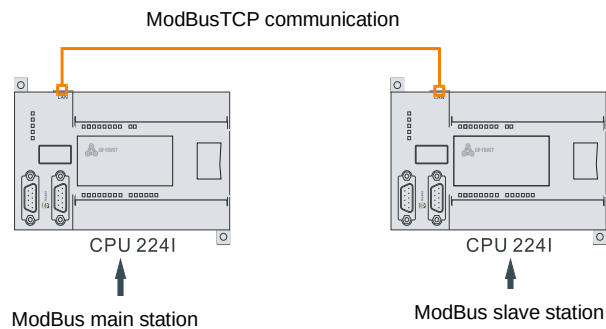
5.5.2 Network Connections

Use standard cable to connect program facility with CPU224I, then programming on

COTRUST offered facility CT_Modbus_TCP library to Modbus main station (CPU224I):



Use Standard cable to connect Modbus master station and slave station. The Modbus master station will perform a write operation to write the data in the specified address to the Modbus slave station. Then, the data in the corresponding address will be read from the Modbus slave station to realize Modbus TCP communication



Note

CPU 224I does not support adaptive crossover. Therefore, select crossover line as standard cable when directly connected to PLC for ModBus_TCP communication.

5.5.3 Operation Steps

Step 1: Connection

Use standard cable to connect PC with Modbus main station.

Step 2: Set communication

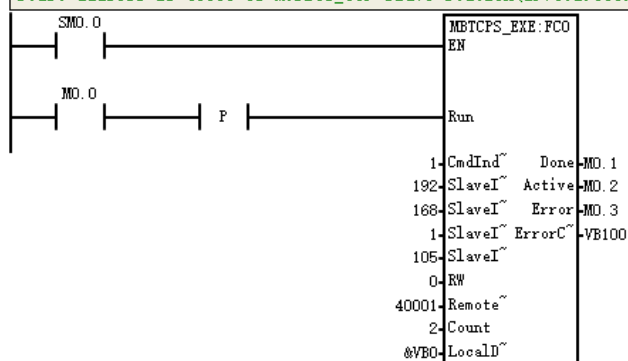
Build a new project in MagicWorks PLC, add CPU224I station in the project, and please refer to Chapter [2.2 Communication Settings](#).

Step 3: program Modbus main station (CPU224I)

1) Program Modbus_TCP main station in MagicWorks PLC program block (use Modbus TCP library or Modbus TCP wizard to communicate, use Modbus TCP single read and write instruction library to program in this example) , the following is the “MBTCPS_EXE” of “Ct_Mbus_master_tcp_single” :

Network 1

Note: MODBUS_TCP main station write 120 data(start from VB0) into a memory that the start address is 40001 of MODBUS_TCP slave station(IP: 192.168.1.100)



<Note> To configure more read/write operation, use MagicWorks PLC Modbus TCP wizard, and refer to *MagicWorks PLC user's manual* for detailed information.

2) Edit and download program to Modbus main station after programming.

Table below are the "MBTCPS_EXE" library instruction definition

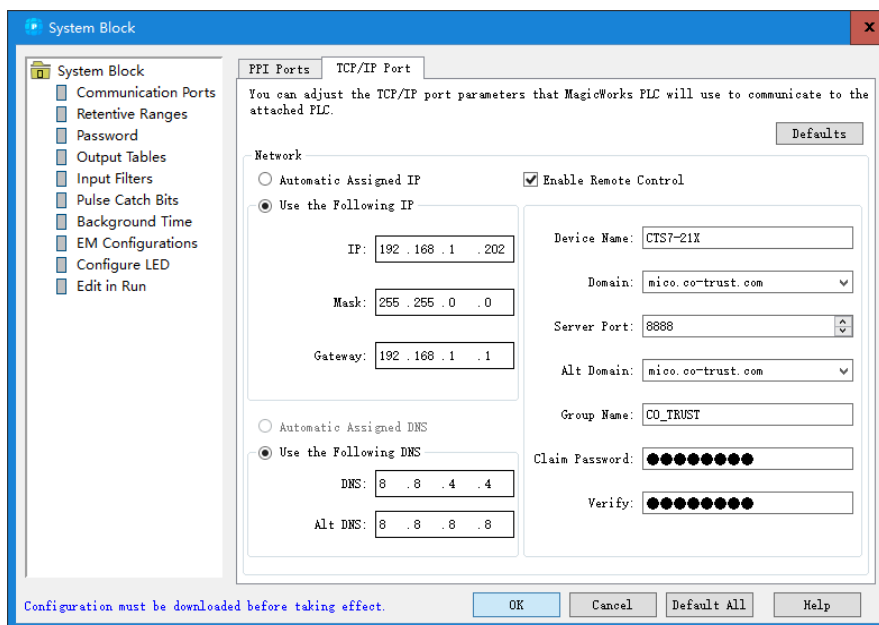
Symbol	Variable type	Data type	Note
RUN	IN	BOOL	Use edge trigger when communication start bit
CmdIndex	IN	BYTE	Call MBTCPS_EXE number, do not repeat range 1~255
SlaveIP0	IN	BYTE	First byte of slave station, for example, the first byte is 192 for 192.168.1.202
SlaveIP1	IN	BYTE	Second byte of slave station
SlaveIP2	IN	BYTE	Third byte of slave station
SlaveIP3	IN	BYTE	Fourth byte of slave station
RW	IN	BYTE	R/W symbol, read = 0; write = 1
Remote Addr	IN	DWORD	Remote data address(such as 40001)
Count	IN	WORD	R/W units, valid range: 1~120 Word or 1-1920 bits
Local DataPtr	IN	DWORD	Local data pointer(such as &VB0)
	IN		
	IN_OUT		
Done	OUT	BOOL	Completion bit, 0=unfinished or unstart; 1=operation finished
Active	OUT	BOOL	Activation bit, 0=inactive or unfinished;; 1=activated but unfinished
Error	OUT	BOOL	Wrong bit, 0=no wrong, 1=wrong
ErrorCode	OUT	BYTE	Wrong code, valid when the completion bit is 1, table below is detailed information

ErrorCode output wrong code definition:

Wrong code	explanation
0	No error
1	Connections reach maximum
2	Building connections
3	Receive time-out, slave station no response
4	Wrong request parameter
5	Instructions not enabled
6	Busy connections(activate multi MBTCPS_EXE)

Step 4: Configure Modbus TCP slave station

When configuring Modbus TCP slave station, the slave station IP must correspond to the IP set by the master station programming. Open the system block in MagicWorks PLC to modify the IP address as shown below. In addition, when ModBus TCP communication is carried out between PLC, the slave station port number should be more than 1024, which is the system default.



Step 5: Debug

Use standard cable to connect Modbus master station and slave station after the above steps are performed sequentially, read 120 data starting from 40001 of Modbus slave station through the status table of MagicWorks PLC. If the content of this segment is consistent with the data in the corresponding memory of Modbus master station, the communication succeed.

5.5.4 ModBus TCP Mapping from Slave Station

ModBus addresses are typically 5 - or 6-character values containing data types and offsets. The first one or two characters determine the data type, and the next few characters are a corresponding value that matches the current data type. ModBus-TCP master stations map this address to the correct function. Taking CTSC-200 cpus that support Ethernet communication (such as CTS7 214-1bd41-0x24) as an example,

ModBus-TCP slave station support the following addresses:

TCP protocol corresponding slave station address mapping

ModBus slave address	CTSC-200 address
000001	Q0.0
000002	Q0.1
000003	Q0.2
...	...
000127	Q15.6
000128	Q15.7
010001	I0.0
010002	I0.1
010003	I0.2
...	...
010127	I15.6
010128	I15.7
030001	AIW0
030002	AIW2
030003	AIW4
...	...
030032	AIW62
040001	VW0
040002	VW0+2
...	...
04xxxx	VW0+2 x (xxxx-1)

5.6 UDP PPI Communication Sample

This section guides you through building an application example that will familiarize you with the CTSC-200 UDP PPI communication (EtherNET communication port).

In UDP PPI communication, CPU acts as slave station, and the communication is independent of the whole large loop cycle, i.e. As the main station, the CPU is controlled by the user program.

The CTSC-200 can be used as both master or slave station to communicate with other devices on the LAN using the network read/write instruction UDP_NETR/UDP_NETW or the NETW/NETR instruction wizard. Meanwhile, HMI can also be used as UDP_PPI master station and CTSC-200 as slave station for communication.

5.6.1 The Sample Components

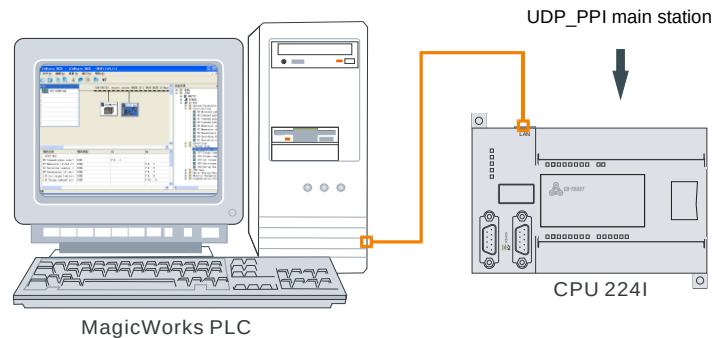
Table5-15 UDP PPI communication sample components

Component	Function
Program device PG\PC	MagicWorks PLC(V2.08 and above), able to configure, edit and debug on CTSC-200 PLC
CTSC-200 main-control modules	2 CPU224I, one as UDP_PPI main station, the other as UDP_PPI slave station, UDP_PPI communication through EtherNET interface

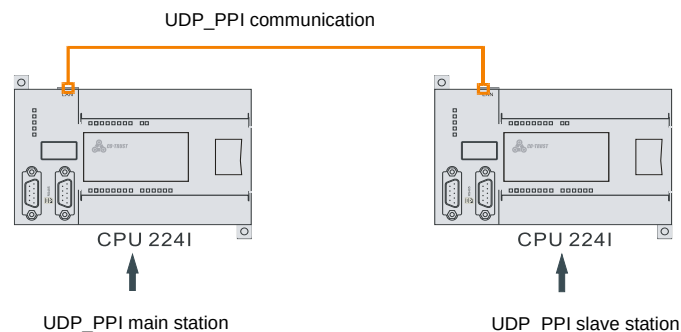
Standard cable	• Connect CPU224I with program device • Connect CPU224I(UDP_PPI main station) with CPU224I (UDP_PPI slave station)
----------------	---

5.6.2 Network Connections

Use standard cable to connect program facility with CPU224I, then programming to UDP_PPI main station (CPU224I):



Use Standard cable to connect UDP_PPI master station and slave station. The UDP_PPI master station will perform a write operation to write the data in the specified address to the UDP_PPI slave station. Then, the data in the corresponding address will be read from the UDP_PPI slave station to realize UDP_PPI communication



Note

CPU 224I does not support adaptive crossover. Therefore, select crossover line as standard cable when directly connected to PLC for ModBus_TCP communication.

5.6.3 Operation Steps

Step 1: Connection

Use standard cable to connect PC with UDP_PPI main station.

Step 2: Set communication

Build a new project in MagicWorks PLC, add CTSC-200 station in the project, and refer to Chapter [2.2 Communication Settings](#).

Step 3: program UDP_PPI main station (CPU224I)

Take UDP network R/W as example, you can read and write the UDP_PPI master station according to one of the two methods.

1) Use UDP network R/W instruction UDP_NETR/UDP_NETW to program

UDP_NETR/UDP_NETW parameter:

D	A	E	0	Wrong code	0
First byte of IP address					1
Second byte of IP address					2
Third byte of IP address					3
Fourth byte of IP address					4
Port number high byte					5
Port number low byte					6
First byte point to remote station pointer <I, Q, M, V, DB>(4bytes in all)					7
Second byte point to remote station pointer					8
Third byte point to remote station pointer					9
Fourth byte point to remote station pointer					10
Data length					11
Data byte 0					12
Data byte 1					13
...					...
Data byte 199					211

D: accomplish (function finished), 0=unfinished, 1=finish

A: current (function in), 0=not current, 1=current

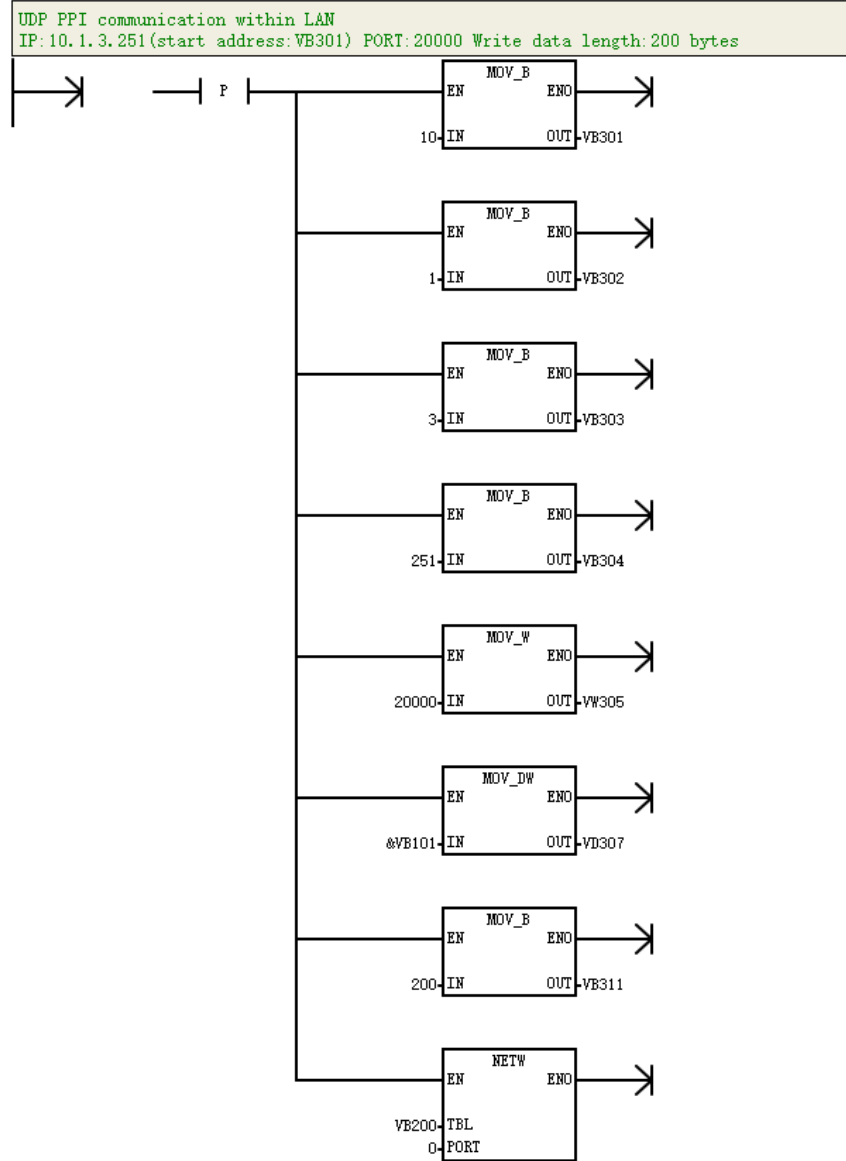
E: wrong, 0=no error, 1=error

Error code

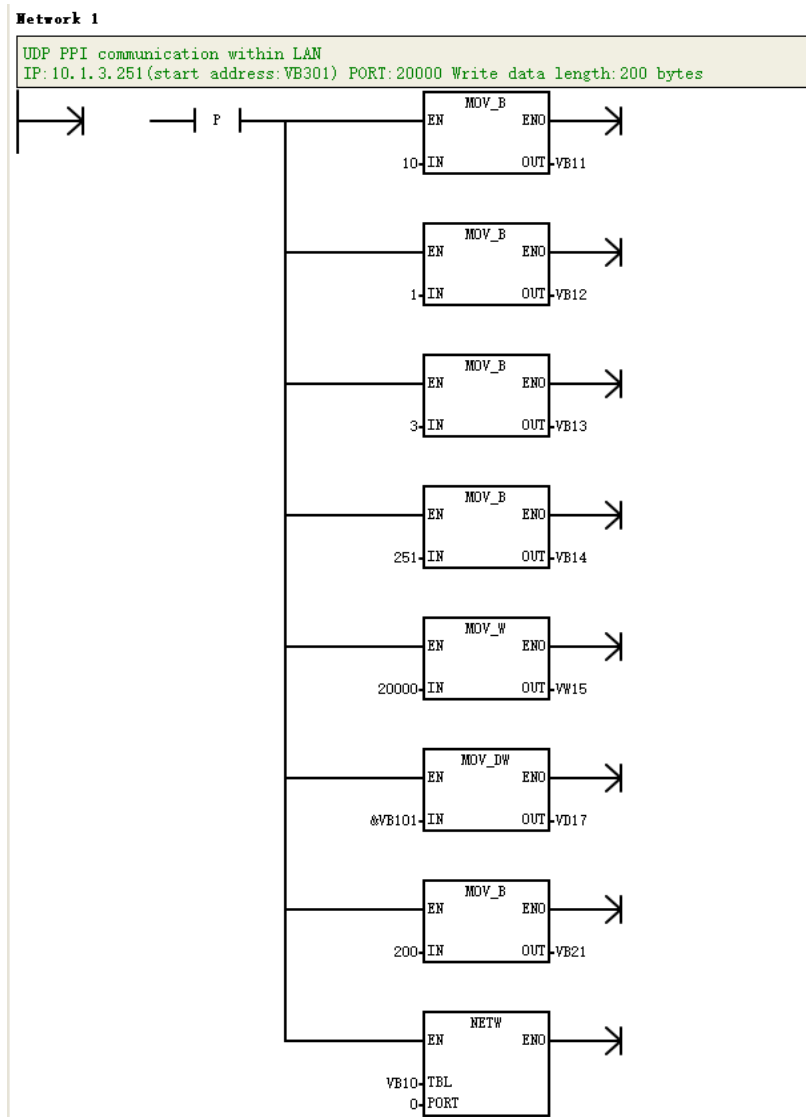
Code	Illustration
1	No error
2	Time-out error; remote station no response
3	Receive error; frame verify error in response
4	Offline error; repeat station address or hardware failure
5	Queue overflow error; over 8 UDP_NETR/UDP_NETW activated
6	Illegal parameter; UDP_NETR/UDP_NETW table involve illegal or invalid value
7	No resource; busy remote station(uploading or downloading sequence)
8	Layer 7 error; violate application protocol
9	Information error; data address error or data length wrong

Network 1: Write 200 byte data of UDP_PPI main station (start address: VB312) into UDP_PPI slave station (IP: 10.1.3.251, start address: VB101)

Network 1

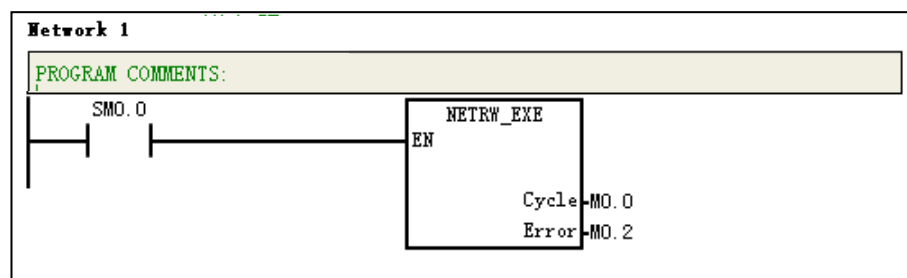


Network2: Read 200 byte data of UDP_PPI slave station(start address: VB312) into UDP_PPI main station(IP: 10.1.3.251, start address:VB101)



2) Program UDP_PPI main station through NETR/W instruction wizard

Expand CTSC-200 in the project tree, select the wizard in the sub options, and then double-click the NETR/W wizard in the right working area to open the wizard dialog box. Call the FC in the OB1 network after configuration as shown below:



3) Finish program according to 1) or 2) , download program into UDP_PPI main station device(must call by SM0.0)

Step 4: UDP_PPI master station communicate with UDP_PPI slave station

- 1) Power UDP_PPI master and slave station on
- 2) Use standard cable to connect UDP_PPI master station and slave station for UDP PPI communication.

5.6.4 UDP PPI Mapping from Slave Station

UDP_PPI communication use normal network R/W, multiple register can be configured when perform read and write, mapping from slave station is directly map.

5.7 CANopen Communication Sample

CPU 226M-CAN use EM277C slave station modules to realize CANopen communication, this section shows how to conduct CANopen connection, configuration and diagnosis. Specific usage of CANopen communication refer to *CANopen user manual*, download address: www.co-trust.com

5.7.1 The Sample Components

Table 5-16 sample components

Component	Function
PG/PC with MagicWorks PLC	MagicWorks PLC V1.73 or above
One PLC program cable	Connect program station with master station(Order No.: CTS7 191-USB30)
One CANopen master station system	CPU 226M-CAN
One slave station device	EM 277C
Bus cable	Recommend CTSC-200 port convert module
3 EM connect to EM277C	2 CTSC-200 digital modules and 1 analog module

Sample components:**PG/PC**

Ensure the PG/PC has connect with CANopen master station (by PPI). MagicWorks PLC V1.73 or above.

PLC program cables

Connect PG/PC program station with CANopen master station CPU, download network hardware program and monitor data.

CANopen master station

In this example, using the CPU226M-CAN with CAN communication port as CANopen master station.

CANopen bus cable

CANopen bus cable use screened cable or twisted pair, recommend to use CTSC-200 communication convert module.

EM277C

In this example, use EM277C with CAN double communication port and baud DIP switch as CANopen slave station.

Expansion modules

Table below lists CTSC-200 series expansion modules use for EM 277C. Sample in this section select 3 expansion modules: digital modules EM 223 8DI/8DO, EM221 8DI and analog module EM 231 AI2*16.

Table 5-17 EM277C supported IO modules

Module	Order No.
Digital modules	
TrustPLC CTSC-200 EM221 8DI	CTS7 221-1BF32
TrustPLC CTSC-200 EM221 16DI	CTS7 221-1BH32
TrustPLC CTSC-200 EM221 32DI	CTS7 221-1BL32
TrustPLC CTSC-200 EM221 32DI compact	CTS7 221-2BL32
TrustPLC CTSC-200 EM222 8DO transistor output	CTS7 222-1BF32
TrustPLC CTSC-200 EM222 8DO relay output	CTS7 222-1HF32
TrustPLC CTSC-200 EM222 16DO transistor output	CTS7 222-1BH32
TrustPLC CTSC-200 EM222 16DO relay output	CTS7 222-1HH32
TrustPLC CTSC-200 EM222 32DO transistor output	CTS7 222-1BL32
TrustPLC CTSC-200 EM223 4DI/4DO transistor output	CTS7 223-1BF32
TrustPLC CTSC-200 EM223 4DI/4DO relay output	CTS7 223-1HF32
TrustPLC CTSC-200 EM223 8DI/8DO transistor output	CTS7 223-1BH32
TrustPLC CTSC-200 EM223 8DI/8DO relay output	CTS7 223-1PH32
TrustPLC CTSC-200 EM223 16DI/16DO transistor output	CTS7 223-1BL32
TrustPLC CTSC-200 EM223 16DI/16DO relay output	CTS7 223-1PL32
Analog modules	
TrustPLC CTSC-200 EM231 high-speed and high-precision analog input module 2AI×16BIT, integrated 10VDC power output, sampling speed 1000HZ, repeatability error less than 0.02%	CTS7 231-7HB32
TrustPLC CTSC-200 EM231 thermal resistance input module, 2AI×RTD,16BIT isolated	CTS7 231-7PB32
TrustPLC CTSC-200 EM231 4-channels analog input modules, 4AI×12BIT precision, current or voltage input, opto-isolator	CTS7 231-0HC32
TrustPLC CTSC-200 EM231 high-speed and high-precision analog input module 4AI×16BIT, integrated 10VDC power output, sampling speed 500HZ, repeatability error less than 0.05%	CTS7 231-7HC32
TrustPLC CTSC-200 EM231 thermal resistance input module, 4AI×RTD,16BIT isolated	CTS7 231-7PC32
TrustPLC CTSC-200 EM231 thermal resistance input module, 4AI×TC, J/K/R/S/T/E/N, 16BIT isolated	CTS7 231-7PD32
TrustPLC CTSC-200 EM231 NTC input modules 2 NTC or PT100, 2 voltage or current input, isolated 16-precision	CTS7 231-7ND32

TrustPLC CTSC-200 EM231 thermal resistance input module, 8AI×TC, J/K/R/S/T/E/N, 16BIT isolated	CTS7 231-7PF32
TrustPLC CTSC-200 EM231 high-precision analog input module, 8 inputs, 16BIT, voltage input, opto-isolator	CTS7 231-0HF32
TrustPLC CTSC-200 EM231 high-precision analog current input module, 8 inputs, 16BIT, current input, opto-isolator	CTS7 231-1HF32
TrustPLC CTSC-200 EM231 analog input module, 8AI×Pt100/NTC, 16BIT isolated	CTS7 231-7NF32
TrustPLC CTSC-200 EM232 analog output module 2-output, current and voltage, 12BIT, opto-isolator	CTS7 232-0HB32
TrustPLC CTSC-200 EM232 analog output module 4-output, current and voltage, 12BIT, opto-isolator	CTS7 232-0HF32
TrustPLC CTSC-200 EM235 analog input output module, 4DI/1DO, 12BIT, current and voltage, opto-isolator	CTS7 235-0KD32

5.7.2 Network Connections

CANopen network connection example:

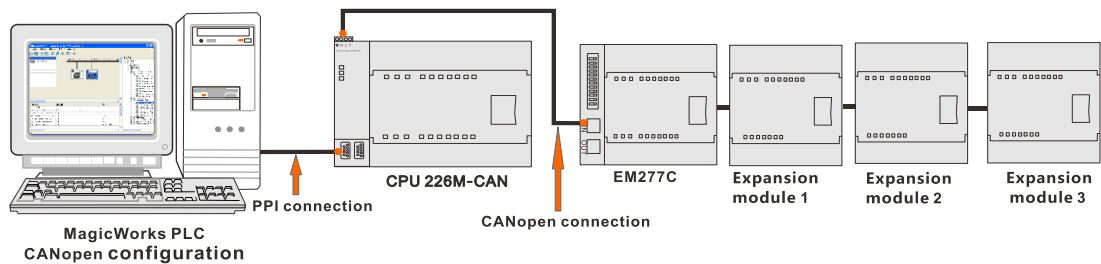


Figure 5-14 CANopen network example 1

CANopen bus wiring method:

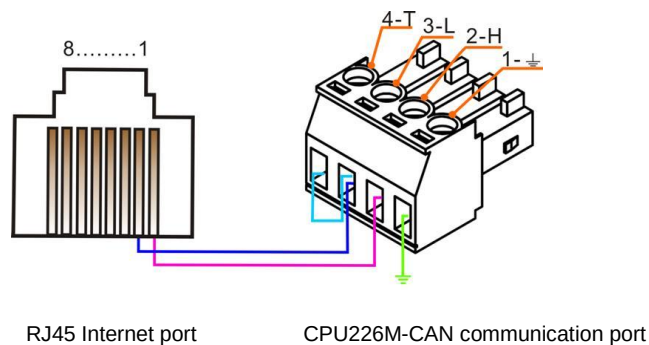


Figure 5-15 CANopen bus connection

Table 5-18 EM 277C double access connector

Number	Signal name	Description
1	CAN_H	CAN send signal+
2	CAN_L	CAN send signal -
3	---	---
4	---	---
5	---	---
6	---	---
7	---	---
8	---	---

Table5-19 CPU 226M-CAN ports PIN definition

Connector	PIN	Description
4-hole terminal (CPU226)	1	Shielding ground
	2	CAN_H
	3	CAN_L
	4	Terminal resistance(short with 3# means terminal resistance)

**Note**

Bus cable use line shielded or twisted pair, recommend CTSC-200 communication port convert module. For detailed information, refer to *CANopen user manual*, download address: www.co-trust.com.

Wiring method when using CTSC-200 communication port convert module:

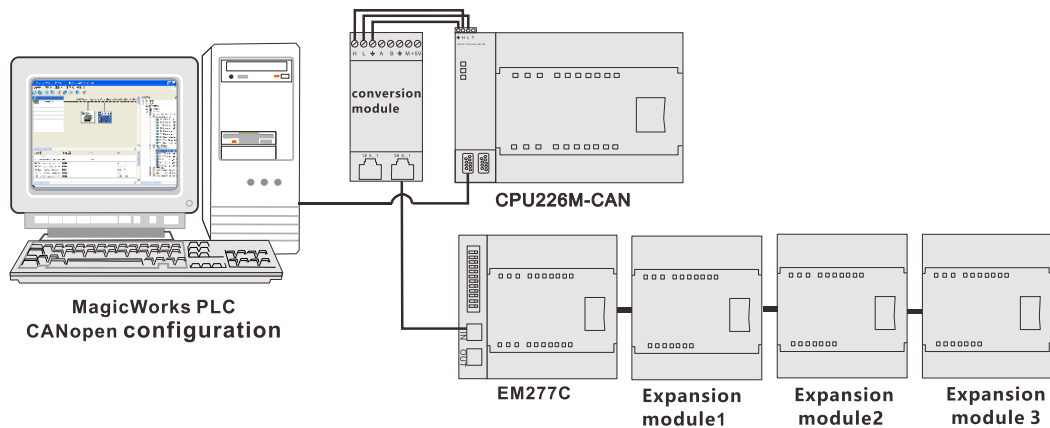


Figure 5-16 CANopen network example 2

**Note**

The connection length between CPU226M-CAN and communication port convert module should be short (recommend to hang on the same guide rail).

Wiring method of communication port convert module and CPU 226M-CAN:

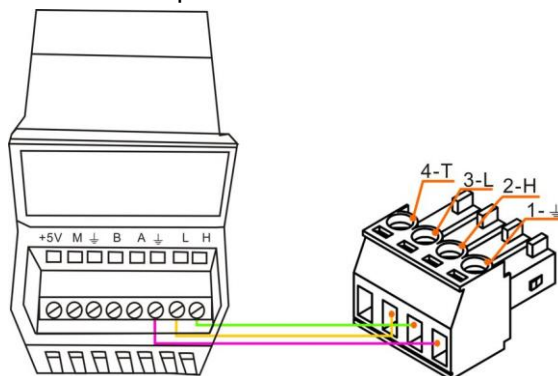


Figure 5-17 Communication port convert module and CPU 226M-CAN connection

**Note**

Use shielded twisted pair to connect communication port convert module with CPU 226M-CAN.

CAN communication cable creation and connection method:

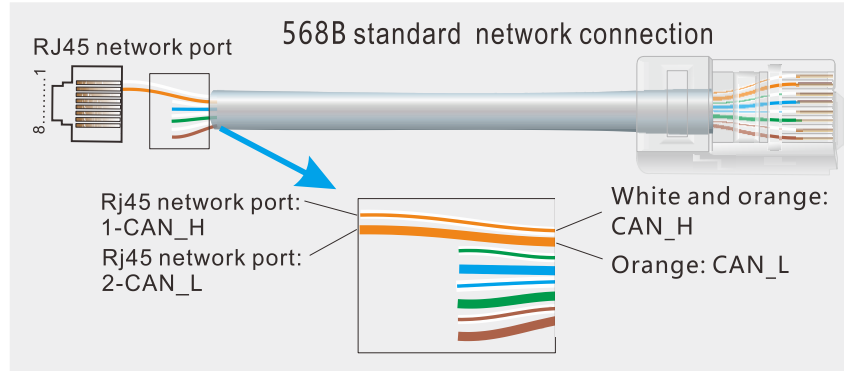


Figure 5-18 CAN communication cable creation and connection method

5.7.3 Operation Steps

1. Open EasyCAN Config interface

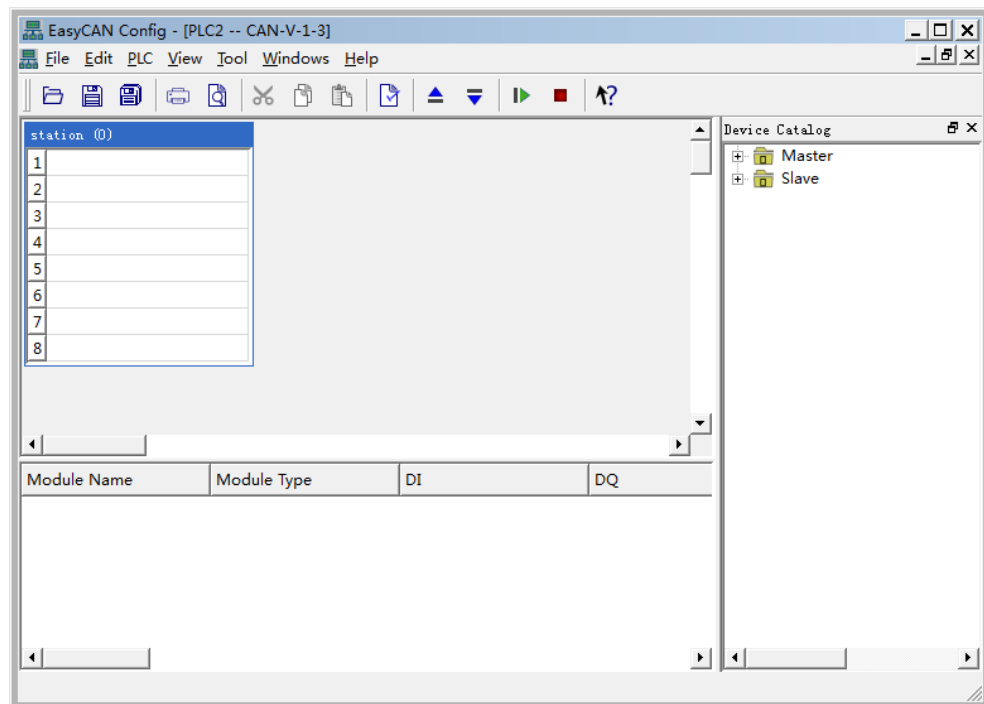


Figure 5-19 EasyCAN Config page

2. CANopen bus editor is as below, select corresponding CPU226XLS in device directory tree master station, and drag it to the "station 1".

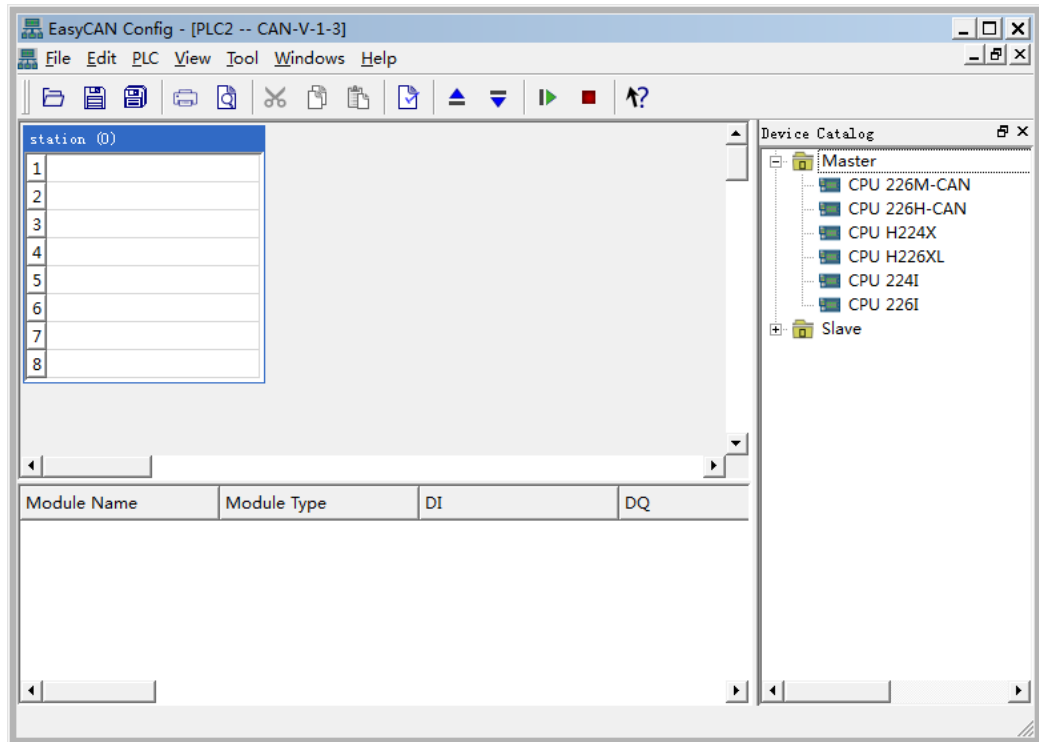


Figure 5-20 Add master station

Then double click CPU226XLS of slot 1 shows master station information interface:

Set communication speed to 1000kbps, select CPU as master station, select in the node ID drop-down box

3 Click "Use CAN Bus"

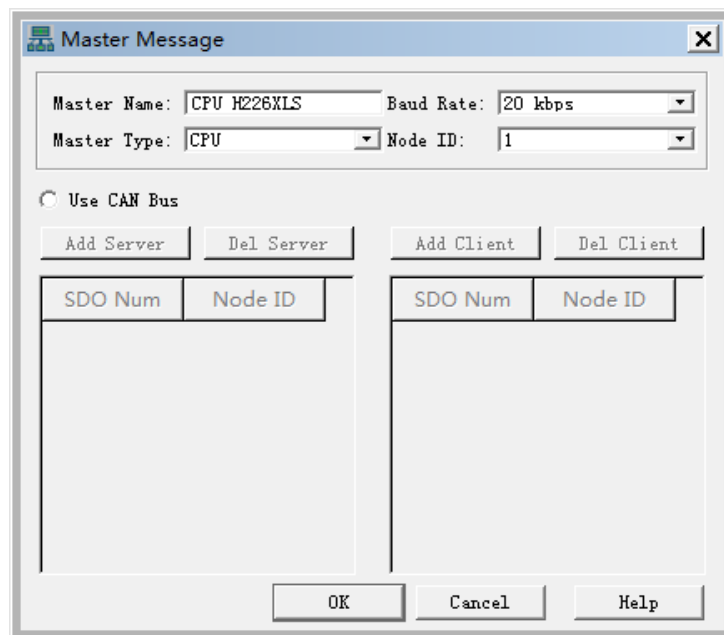


Figure 5-21 Set master station information

Select SM 277C under the slave station of the device directory tree and drag it into the CAN BUS area configuration interface. The system will automatically connect the slave

station with CAN BUS. Relevant information of slave station modules will appear in the list of slave station modules information.

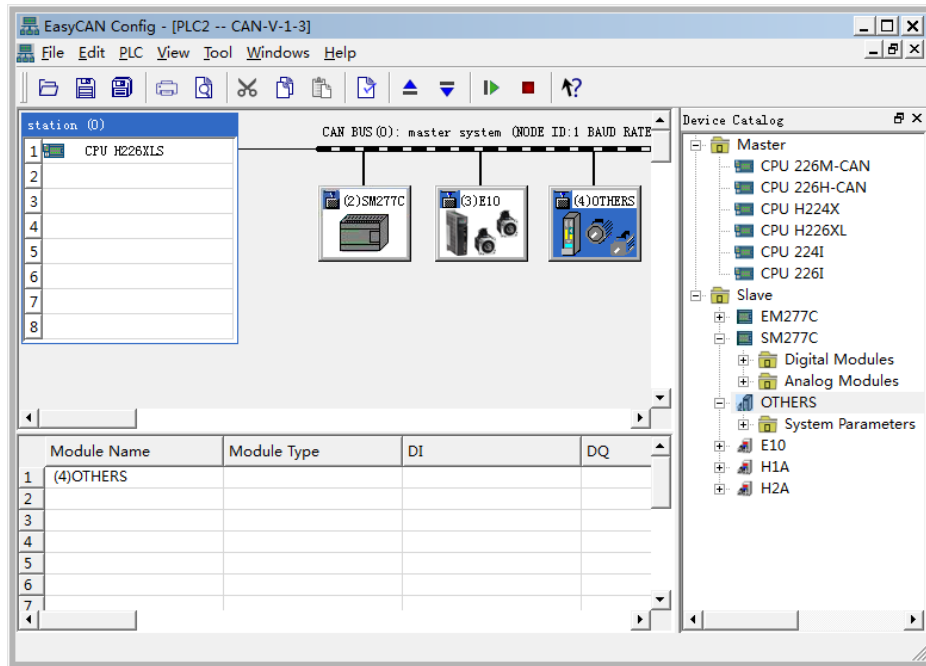


Figure 5-22 Add slave station

Then double-click icon under CAN BUS connection:

Open node ID drag-down box, select 4(set node ID different from master station), click to ensure

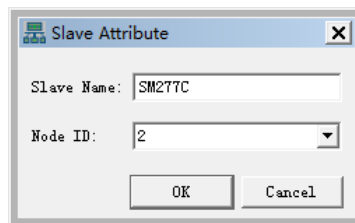


Figure 5-23 Set slave station information

4 Open the digital module under the slave station node in the device directory tree, select SM 223 8DI/8DO and SM 221 8DI, then open the analog module node, select SM231 AI2*16BIT, double-click the selected module or drag it into the module information list, the module will be added to the current slave station. The system will arrange its corresponding I/Q address after the module is inserted, as shown in the figure:

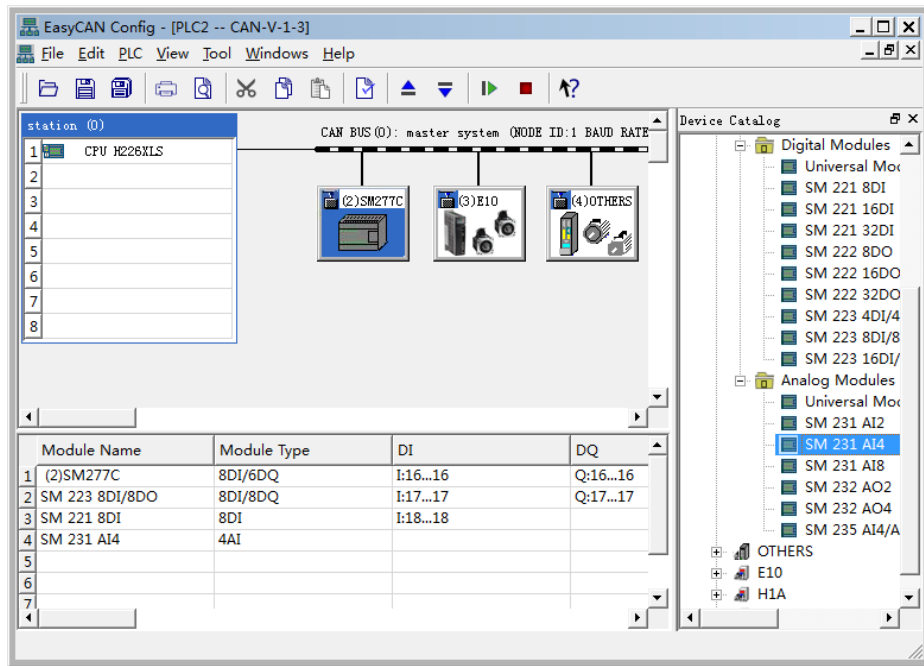


Figure 5-24 Add modules

5 Connection and configuration of hardware

- 1) Communication cables for connecting CPU 226 XLS and PC (USB to USB, RS485 port to RS485 port).
- 2) Use communication cable to connect CAN port of CPU 226 XLS (2, 3 PIN) and SM277C (1, 2 PIN).
- 3) Set DIP switch (communication speed 1000kbps, node ID 4), toggle SM277C DIP switch as reference table below:



Figure 5-25 SM277C DIP switch

DIP7-DIP1 (node address): binary representation, DIP7 is high, DIP1 is low.

DIP10-DIP8 (baud rate): binary representation, all are high.



Note

0 is global address, forbidden when use.

Table 5-20 switch reference table

DIP10	DIP9	DIP8	Communication speed (kbit/s)	Maximum communication distance (m)
OFF	OFF	ON	20	2500
OFF	ON	OFF	50	1000
OFF	ON	ON	125	500
ON	OFF	OFF	250	250
ON	OFF	ON	500	100
ON	ON	OFF	800	50
ON	ON	ON	1000	25

DIP11 (terminal resistance): Dial to “ON” when facility located in end of network, otherwise dial to “OFF”. DIP12: reserved.

1) Put digital module SM 223 8DI/8DO, SM 221 8DI, and analog module SM231 AI2*16BIT to the expansion port of SM277C.

2) Make sure the wiring OK after connection.

3) Power on facilities.

6 Save current project after CANopen configuration, then compile it in CANopen configuration interface and download to PLC.

7 CPU XLS expand 64Byte I /64Byte Q / 128Words AI / 128 Words AQ for CANopen mapping address. The I/Q address of CANopen configuration modules show as below, you can monitor all the input and output in the status table.

	Module Name	Module Type	DI	DQ
1	(2)SM277C	8DI/6DQ	I:16...16	Q:16...16
2	SM 223 8DI/8DO	8DI/8DQ	I:17...17	Q:17...17
3	SM 221 8DI	8DI	I:18...18	
4	SM 231 AI4*12Bit/16Bit	4AI		
5				
6				

Figure 5-26 Slave station modules information

Monitor the I/O of SM277C, SM221 8DI, SM223 8DI/8DO and SM231 AI2*16 in status table after configuration.

	Address	Format	Current Value	
1	Q16.0	Bit	#1	6 Outputs for SM277C(Q16.0~Q16.5)
2	Q16.1	Bit	#1	
3	Q16.2	Bit	#1	
4	Q16.3	Bit	#1	
5	Q16.4	Bit	#1	
6	Q16.5	Bit	#1	
7	Q17.0	Bit	#0	8 Outputs for SM223 8DI/8DO(Q17.0~Q17.7)
8	Q17.1	Bit	#1	
9	Q17.2	Bit	#1	
10	Q17.3	Bit	#0	
11	Q17.4	Bit	#0	
12	Q17.5	Bit	#1	
13	Q17.6	Bit	#1	
14	Q17.7	Bit	#1	
15	Q18.0	Bit	#0	8 Inputs for SM231 8DI(I18.0~I18.7)
16	Q18.1	Bit	#0	
17	Q18.2	Bit	#0	
18	Q18.3	Bit	#0	
19	Q18.4	Bit	#0	
20	Q18.5	Bit	#0	
21	Q18.6	Bit	#0	
22	Q18.7	Bit	#0	
23	AIW64	Signed	12768	4 Inputs for SM231 4AI(AIW64~70)
24	AIW66	Signed	12768	
25	AIW68	Signed	12768	
26	AIW70	Signed	12768	

Figure 5-27 Status monitor table

5.7.4 Diagnosis

EM 277C/E10 and MagicWorks PLC SMB status word can be used to diagnose CANopen master-slave network.

Diagnose by hardware

EM277C LED

Power on EM 277C, the POWER LED indicating "ON" (green) will light up.

If the BF and SF LED lights remain off, then EM 277C is functioning normally.

If the BF or SF LED lights up or flickers, there is an error in hardware configuration or wiring. If the operation of the main system is trouble-free, the cause of the error can be referred to the following table:

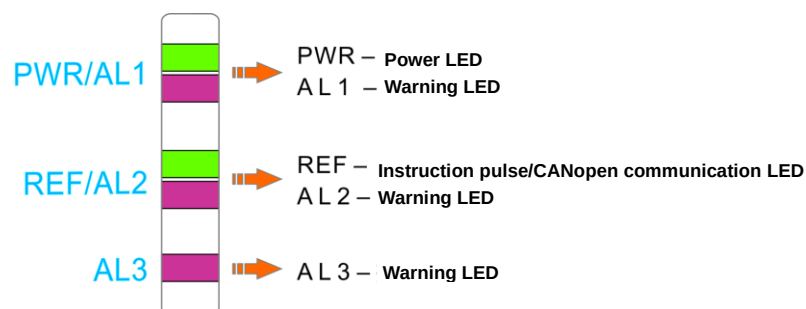
Table5-21 EM 277C LED function

LED	On	Off	Flicker	Note
ON	Power on	No power	---	Power indicator
SF	IO expansion module no error	IO expansion module no error	---	System error
BF	No CAN network	No CAN network	Inconsistent configuration	Bus error

E10 CANopen communication indicator light

If the slave station connect to E10 servo, the E10 CANopen communication indicator light keep off before the communication connection. The indicator light on if the communication succeed, indicating that the CANopen communication is normal., the CANopen communication indicator light turn off if the CANopen communication is abnormal.

E10 CANopen communication indicator light:



Diagnose by MagicWorks PLC

CPU226M-CAN is allocated with 100 bytes SM. Users can get error information by viewing status bytes. The meaning of CAN site status word is shown in the following table:

Table5-22 CAN station status table

	Byte	Address distribution	Status value illustration
SMB (status)	100	SMB550: CAN communication status of master station(CPU)	0x00: Initialize 0x01: Disconnect 0x04: Stop 0x05: Run 0x7f: Pre-run 0xff: Configuration data error
		SMB551-SMB582: CAN communication status of 1~32 slave station(Node ID from small to large)	0x00: Initialize 0x01: Disconnect 0x04: Stop 0x05: Run 0x7f: Pre-run 0x7: Configuration parameter error
		SMB583-SMB614: Run warning of 1~32 slave station (Node ID from small to large)	bit7: IO information configuration compare with 277C reality: 0: No error 1: Error Bit6-Bit0: 277C IO expansion module diagnostic information bit0 indicate the first module, bit6 indicate seventh module: 0: No warning 1: Warning (Above is the definition of 277C slave station different from others)
		SMB615-SMB649	Reserved

**Note**

SMB550-SMB649 is read-only memory, user cannot write

5.8 Application of Analog Expansion Board

CPU224I//224XP/226I support optional analog expansion boards (CPU224XP comes with a 2AI/1AQ analog expansion board). Please refer to Chapter [4.10.1 Analog Expansion Modules Specifications](#) for detailed specifications, wiring and terminal definitions.

Select the access mode of analog expansion board: call special extension instruction library or directly access SM area.

CPU224XP comes with an expansion board 2AI/1AQ access, AIW/AQW area.

- **Access SM area**

The SMW116~SMW126 in CPU are mapping area in expansion board. First map AI,

SMW116 allocate 4 words, then map AQ and SMW124 allocate 2 words.

2AI/1AQ: SMW116 corresponding to AIW0; SMW118 to AIW2; SMW124 to AQW0

Function	Map access address	Function	Map access address
A+input	SMW116	Module type	SMB114
B+input	SMW118	Module status	SMB115
VO output	SMW124		

4AI/2AQ: SMW116 corresponding to AIW0; SMW118 to AIW2; SMW120 to AIW4;
SMW122 to AIW6; SMW124 to AQW0; SMW126 to AQW2

Function	Map access address	Function	Map access address
A+ input	SMW116	V0/I0 output	SMW124
B+ input	SMW118	V1/I1 output	SMW126
C+ input	SMW120	Module type	SMB114
D+ input	SMW122	Module status	SMB115

SMB115 module status definition

Name	Function	Value
Module type	Modules	0x1E: 4AI/2AQ 0x19: 2AI/1AQ 0x20: CAN expansion board
	No modules	0x00
Module status	Run normally	0x00
	Communication error	0x01
	Calibration failure	0x02
	Access error	0xFF

5.9 Fault Diagnosis

When system fail, CPU does following:

- 1) Enter STOP mode
- 2) SF/DIAG(red)LED indicator light and STOP indicator light on
- 3) Disconnect output

Check the following conditions first:

- 1) If the CTSC-200 master control module power normal
- 2) If the screw and connector of CTSC-200 master control module and I/O expansion modules loose
- 3) Make sure the wiring OK after connection.
- 4) Please check communication setting, such as modify baud rate, connection port or IP to research again if you haven't find PLC.

You can also read diagnostic information by MagicWorks PLC or the LED indicator lights.

5.9.1 Diagnose by MagicWorks PLC

Fault status continue to error clear, diagnostic information read: open MagicWorks PLC software - double click to open the project interface - select the menu item "PLC" - "information" to view error information.

Table 5-23 diagnosis function

Supported event type		Specific event code and description	
		Code	Description
CPU error	nonfatal	0x00	No error
		0x01	Enabled HSC before execute HDEF (Reserved)
		0x02	Input interruption distribution clash, HSC already been distributed(Reserved)
		0x03	An input allocation conflict to an HSC that has been allocated to an input interrupt or other HSC (Reserved)
		0x04	Attempt to execute ENI, DISI or HDEF instructions in interrupt routine program
		0x05	Attempt to execute a second HSC/PLS with the same number before completing the first HSC/PLS, has an interrupt routine conflict with the HSC/PLS in the main program (Reserved)
		0x06	Indirect addressing error
		0x07	TODW(daily write time) or TODR(daily read time) data error
		0x08	Exceeded the maximum number of nested layers for user subroutines
		0x09	Execute XMT/RCV instruction in COM 0 at the same time.
		0x0A	Attempt to redefine by executing another HDEF instruction for the same HSCHSC(Reserved)
		0x0B	Execute XMT/RCV instruction in COM 1 at the same time.
		0x0C	Reserved
		0x0D	Reserved
		0x0E	Reserved
		0x0F	Invalid numeric value encountered in comparison contact instruction
		0x10	Reserved
		0x11	Reserved
		0x12	Reserved
		0x13	Illegal PID loop table
		0x80	Program is too large, the CPU cannot generate executable code; Please reduce the program size
		0x81	Reserved
		0x82	Illegal instruction; Check the instruction mnemonic 0
		0x83	Lack MEND, or instructions are not allowed in main program; add MEND or remove wrong instructions
		0x85	Lack FOR; add FOR or delete NEXT
		0x86	Lack NEXT; add NEXT or delete FOR

	0x87	Lack label (LBL, INT, subprogram) ; add proper label
	0x88	Lack RET or no instructions allowed in subroutine; Add RET or remove incorrect instructions at the end of a subroutine
	0x89	Lack RETI or no instruction allowed in the interrupt routine. Add RETI or remove incorrect instruction at the end of the interrupt routine
	0x8B	From SCR illegal JMP or from SCR illegal JMP
	0x8C	Repetitive label (LBL, INT, SBR) ; rename one label
	0x8D	Illegal label(LBL, INT, SBR); verify allowed label number is not exceed
	0x90	Illegal parameter; verify allowed parameters
	0x91	Error range(including address information); check operand range
	0x92	Instruction count field error (including count information); Verify maximum count
	0x93	Over FOR/NEXT nested layer number
	0x94	Error writing range to nonvolatile memory with address information
	0x95	Lack LSCR(load SCR)
	0x96	Lack SCRE(SCR end) or no instructions before SCRE
	0x97	User program contains EU/ED with or without numbers
	0x98	Try to perform runtime editing in a program with no EU/ED instructions
	0x99	Too many implicit program segments
	0x9A	Attempt to switch to FPORT mode in user interrupt
	0x9B	Illegal index(String operation in which a starting position value is specified)
CPU fatal error	0x00	No fatal error
	0x01	Reserved
	0x02	Reserved
	0x03	Scan watchdog timeout error
	0x04	Reserved
	0x05	Reserved
	0x06	Reserved
	0x07	Reserved
	0x08	Reserved
	0x09	Reserved
	0x0A	Reserved
	0x0B	Reserved
	0x0C	Reserved
	0x0D	Reserved
	0x0E	Reserved
	0x0F	Reserved
	0x10	Internal software error

	0x11	Comparison contactor indirection addressing error
	0x12	Invalid floating point value for comparison node
	0x13	Reserved
	0x14	Comparing node range error
Monitor record every scan cycle		
Modules diagnosis event Reference table 5-16	0x00	Modules no error
	0x01	Modules busy
	0x02	Modules time-out no response
	0x03	Modules type unmatched
	0x04	Modules version unmatched
	0x05	Software error
	0x06	Modules wait symbol timeout
	0x07	Bus response error
	0x08	Bus CRC calibration error
	0x10	Memory offset out of range
	0x11	Modules unready
	0x12	Modules configuration error
	0x13	Modules do not support the instruction
	0x15	Module internal diagnosis
	0x16	No power in module

Table5-24 special memory diagnostics

SMB	Error description
SMB8	Module 1 maker register
SMB9	Module 1 error register
SMB10	Module 2 maker register
SMB11	Module 2 error register
SMB12	Module 3 maker register
SMB13	Module 3 error register
SMB14	Module 4 maker register
SMB15	Module 4 error register
SMB16	Module 5 maker register
SMB17	Module 5 error register
SMB18	Module 6 maker register
SMB19	Module 6 error register
SMB20	Module 7 maker register
SMB21	Module 7 error register

5.9.2 Diagnose by CTSC-200 Main-control Modules

Table 5-25 CTSC-200 LED status indicator light

Indicator light	Color	Description
SF/DIAG light	Red 	On: System fault, off: no error
RMC light (CPU 224I)	Green 	On: CPU communicate with remote server(Ethernet communication port configure properly) Off: Failed or disabled to communicate with the remote server (enable and disable communication with the server via the DIP switch)
RUN light	Green 	On: System run, off: System stop
STOP light	Orange 	On: System stop, off: System run
LINK/ACT light (CPU 224I)	Green 	On: Connected, flicker: Transmission, off: Disconnect

Note: STOP light and SF light will on simultaneously when the facility completely off electricity.

6 Power Budget

CTSC -200 series CPU has an internal power supply for the unit and the expansion modules. It also provides 5VDC logic power supply, which can be used for power supply of any expansion in the system. Pay close attention to the system configuration to ensure that the CPU can provide the 5V power for the selected expansion modules. If the configuration requires more power than the CPU provides, some expansion modules must be removed or select CPU that can provide more power. See Chapter [3 Installation](#) for detailed information on CTSC-200CPU's 5VDC logic budget and expansion module 5VDC power requirements.

CTSC-200 series CPU also provide 24VDC sensor power supply (except PSC266), which can provide 24VDC for input, relay coil on expansion module or other power supply. If the required power exceeds budget, an external 24VDC power supply must be added. For specific CTSC-200 CPU 24VDC sensor power budget, refer to Chapter [6.3 24VDC Power](#). If you need an external 24VDC power supply, make sure that the power supply is not connected in parallel to the sensor power supply of the CTSC -200 series CPU. To improve electrical noise protection, please connect to the common terminal (M) of different power sources.

6.1 Power Requirement

CPU provides 5VDC and 24VDC power supply itself:

Each CPU has an external 24VDC power supply, which can provide 24VDC for input and relay coil of the expansion module.

When expansion unit connect, CPU provides 5VDC power. If the 5VDC requirement of the extension unit exceeds the CPU rating power output, the extension unit must be unloaded until the demand is within the predetermined power supply.

Chapter 4 provides information on CPU power budget and expansion module power requirements.



Note

If 5VDC power exceeds CPU power budget, it may unable to connect maximum modules the CPU allowed.



Notice

An external 24VDC power supply connected in parallel to CTSC-200DC sensor power supply can cause a conflict because each power supply seeks to establish its own potential output. This conflict will shorten the service life of one or two power supplies or cause an immediate failure, then uncertain operation of the PLC system. The

unpredictable operation may cause personal injury and equipment damage.

CTSC-200 CPU's 24VDC sensor and any external power supply should be supplied at different points. Allow separate connections to the utility power.

6.2 5VDC Power

Table 6-1 5VDC power consumption table

CPU and DP module	5VDC current offered for expansion IO (mA)	Expansion module	5VDC current (mA)
CPU224E CPU224+ CPU224I CPU224XP CPU226M CPU226H CPU226M-CAN CPU226H-CAN EM277B EM277C	660	EM221 DI8×DC24V	40
		EM221 DI16×DC24V	65
		EM221 DI32×DC24V	111
		EM221 DI32×DC24V high density	110
		EM222 DO8×DC24V	73
		EM222 DO8×relay	56
		EM222 DO16×DC24V	126
		EM222 DO16×relay	80
		EM222 DO32×DC24V	250
		EM223 DI4/DO4×DC24V	54
		EM223 DI4/DO4×DC24V/ relay	49
		EM223 DI8/DO8×DC24V	63
CPU226I CPU226L PSC266	1000	EM223 DI8/DO8×DC24V/ relay	79
		EM223 DI16/DO16×DC24V	173
		EM223 DI16/DO16×DC24V/ relay	138
		EM231 AI2×16-bits	40
		EM231 AI4×12-bits	34
		EM231 AI4×16-bits	40
		EM231 AI8×16-bits (voltage)	44
		EM231 AI8×16-bits (current)	25
		EM231 AI8×12-bits	50
		EM231 AI4×thermocouple	35
EM277PN	1100	EM231 AI2×RTD/ EM231 AI4×RTD	35
		EM231 AI2+RTD2	87
CPU224E/	660	EM232 AQ2×12-bits	24

CPU 224+/224+V5(upgrade)/ CPU224I/224 XP/CPU226M EM277B		EM232 AQ4×12-bits	31
		EM235 AI4/AQ1×12-bits	34
		EM231 AI4 PID(thermocouple)	54
CPU226I/L/H PSC266	1000	EM231 AI8 PID(thermocouple)	54
		EM231 AI8 PID(current)	30
		EM231 8AI×Pt100/NTC	54
		EM253 motion control module	90
		EM231-7WA	140
		EM231-7WB	140
		EM277 A	70

Calculate max I/O configuration of CTSC-200 system according to the above table, refer to Chapter [6.4 Power Calculation Example](#).

6.3 24VDC Power

Table 6-3 24VDC power consumption table

24VDC power supply (mA)		Expansion module 24VDC current consumption (mA)	
CPU224E CPU224+ CPU224I CPU224XP CPU226M CPU226H CPU226M-CAN CPU226H-CAN	280	EM231 AI4×12-bits	17
		EM231 AI2×16-bits	40
		EM231 AI4×16-bits	35
		EM231 AI8×16-bits s (voltage)	31
		EM231 AI8×16-bits (current)	31
		EM231 AI8×12-bits	50
		EM231 AI4×thermocouple	30
		EM231 AI8×thermocouple	30
		EM231 AI2×RTD	34
		EM231 AI4×RTD	37
		EM232 AQ2×12-bits	61
CPU226L CPU226I SM277B	400	EM232 AQ4×12-bits	112
		EM235 AI4/AQ1×12-bits	48
		EM231 AI4 PID(thermocouple)	34
		EM231 AI8 PID(thermocouple)	39
		EM231 AI8 PID(current)	39
		EM231 8AI×Pt100/NTC	39
		EM231 AI2+RTD2	60
		EM253 motion control module	80
		EM231-7WA	100
		EM231-7WB	100

		EM277A	70
		TD2X	40
		TD4X	45
		TD4S	50

6.4 Power Calculation Example

Use following design methods to determine how many power configurations can be provided by the ontology unit power supply.

Table 6-4 shows power calculation of CTSC-200 PLC, CTSC-200 including:

- CTSC-200CPU224+ AC/ DC/relay
- Each CTSC 223-1PH32 in EM223 with 8DC inputs/8 relay outputs
- Each CTSC 221-1BF32 in EM221 with 8DC inputs
- The device has 62 inputs and 42 outputs

In this example, the CPU provides enough 5VDC current for the expansion module, but not enough 24VDC current for all input and expansion relay coils from the sensor. I/O requires 536mA while CTSC-200 CPU only provides 280mA. The device requires at least 256mA of 24VDC additional power to operate all contained 24VDC inputs and outputs.

Table6-4 power budget table

CPU power budget	5VDC	24VDC
Current CPU224+ offered	660 mA	280 mA
-		
System requirement	5VDC	24VDC
CPU 224+, 14 inputs		14* 4 mA= 56 mA
4 CTS7 223-1PH32, 5V current	4*80 mA= 320mA	
2 CTS7 221-1BF32, 5V current	2*30 mA= 60 mA	
4 CTS7 223-1PH32, 8 inputs each		4*8*4 mA= 128 mA
4 CTS7 223-1PH32, 8 relay coil each		4*8*9 mA= 288 mA
2 CTS7 221-1BF32, 8 inputs each		2*8*4 mA= 64 mA
Total demand power consumption	380 mA	536 mA
=		
Current balance budget	5VDC	24 VDC
Total current balance	280 mA	-256 mA

Appendix

CTSC-200 series CPU support all the library files in appendix.

A Using CT-MODBUS Master and Slave Station

A.1 Introduction of CT_MODBUS library

4 libraries in all, PORT0 master&slave station, PORT1 master&slave station.

CT_MODBUS function block is a set of embedded and easy MODBUS protocol library provided by Siemens MODBUS function block, which occupies a large amount of CPU program and data space. Integrated in the CPU, it does not occupy user data space, provided to the user as a set of library functions.



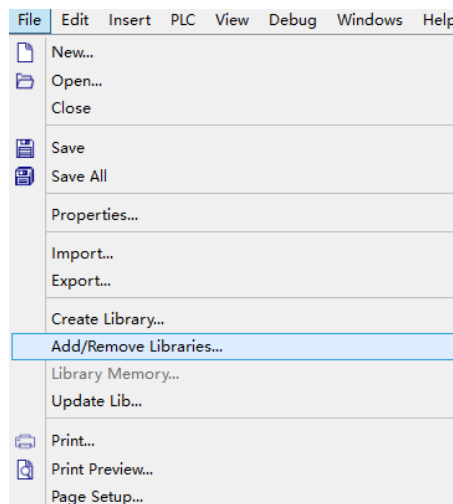
Note

Library download address: www.co-trust.com

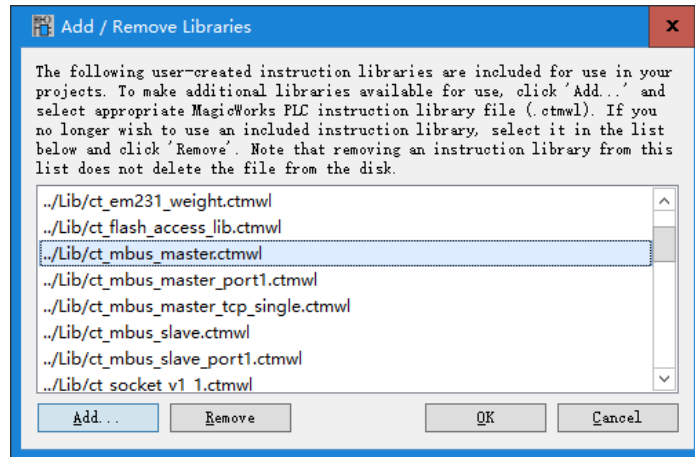
A.2 Software Installation

【Add library files】

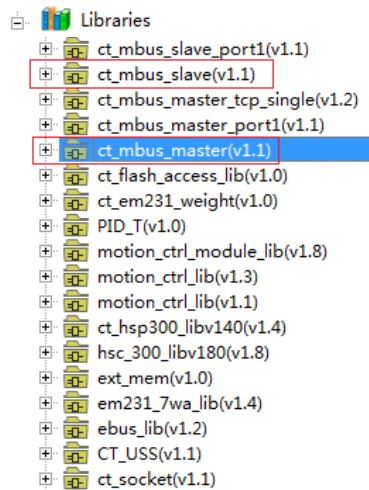
Find “ct_mbus_master.ctmwl” and “ct_mbus_slave.ctmwl” in “File” –“Add/Remove library” as picture below:



Find files where you save “ct_mbus_master.ctmwl” and “ct_mbus_slave.ctmwl”, click “Add”.

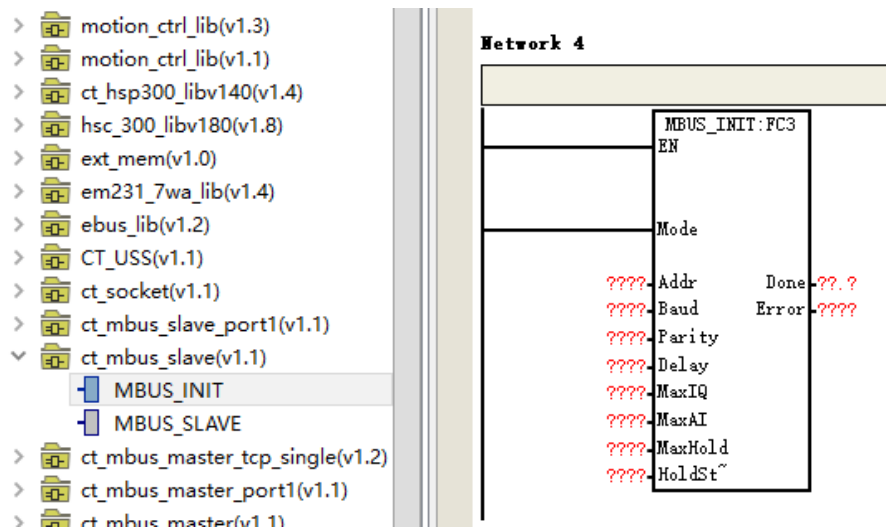


You can find CT_MBUS_MASTER and CT_MBUS_SLAVE library under “libraries” of directory tree after installation:



【call CT_MODBUS library】

Click “Network” you need to add function block, double click “MBUS_INIT”, “MBUS_SLAVE”, “MBUS_CTRL”, “MBUS_MASTER” in “libraries”, the corresponding function block will shows in “Network”:



A.3 CT_MODBUS Library Function Explanation

【Modbus address】

Modbus addresses are typically 5 - or 6-character values containing data type and offset. The first one or two characters determine the data type, and the last four characters match the data type. The Modbus master station maps this address to correct function.

Modbus slave station instruction supports the following addresses: 00001 to 00128 are the actual output, corresponding to Q0.0--Q15.7; 10001 to 10128 are actual inputs corresponding to I0.0 -- I15.7; 30001 to 30032 are analog input registers corresponding to AIW0 to AIW62; 40001 to 4XXXX is the holding register, corresponding to V memory area.

All Modbus addresses are numbered from 1. The following table shows the correspondence between Modbus addresses and CTSC-200 addresses. The Modbus slave protocol allows you to limit the number of inputs, outputs, analog inputs, and holding registers (V zones) that Modbus master can access.

Modbus address	CTSC-200 address
000001	Q0.0
000002	Q0.1
000003	Q0.2
.....	
000127	Q15.6
000128	Q15.7
010001	I0.0
010002	I0.1
010003	I0.2
.....	
010127	I15.6
010128	I15.7
030001	AIW0
030002	AIW2
030003	AIW4
.....	
030032	AIW62
040001	HoldStart
040002	HoldStart+2
040003	HoldStart+4
.....	
04xxxx	HoldStart+2 x (xxxx-1)

【Using Modbus slave station protocol instructions】

※ CT_MODBUS slave station protocol instructions occupy CTSC-200CPU resource

1) According to different Modbus protocol libraries, the FPORT 0 or 1 is used as the slave station protocol. When Port 0 or Port 1 is used as Modbus protocol communication, it cannot be used for any other purpose, including communication with MagicWorks PLC or SETP7-Micro/WIN, FPORT communication. The MBUS_INIT controls Port is set to Modbus is still PPI.

2) All SM corresponding to selected FPORT

3) Occupy 92 byte program space

※ Steps use Modbus slave station protocol in CTSC-200 program

1) MBUS_INIT can be used to initialize or modify Modbus communication parameters when there are only one loop cycle. When you insert the MBUS_INIT, several hidden subroutines and interrupt service routines are automatically added to your program.

2) Use only one MBUS_SLAVE instruction in your program. It execute in each cycle and serves all requests received.

3) Connect CTSC 200CPU with Modbus master station by communication cable.

※ All functions Modbus slave station protocol instruction support

Modbus slave protocol instruction supports Modbus RTU protocol. It uses the FPORT function of S7-200, and supports most common Modbus functions. Support following Modbus functions:

Function	Description
1	Read single/multiple coil (actual output) status. 1 return any digital outputs ON/OFF status(Q)
2	Read single/multiple contactor (actual input) states. 2 returns the on/off status for any number of input points(I)
3	Read single/multiple hold registers. 3 returns the contents of V memory, keep the register in Modbus as the word-type, and up to 120 words in a request.
4	Read single/multiple input registers. 4 returns simulated input values.
5	Write single coil (actual output). 5 sets the actual output point which is not mandatory to the specified value. User program can override the values written by the Modbus request.
6	Write a single hold register. 6 writes a single hold register value to the S7-200 V memory.
15	Write multiple coils (actual output). 15 writes multiple actual output values to the Q image region of S7-200. The starting output point must be the beginning of a byte (such as Q0.0 or Q2.0), and the number of outputs to write is a multiple of 8. This is the limit of Modbus slave station protocol instruction. These points are not mandatory, user program can override the values written by the Modbus request.
16	Write multiple hold registers. 16 write multiple save registers to S7-200 V zone. Up to 120 words can be written in a request.

※ MBUS_INIT instruction

The MBUS_INIT directive is used to enable and initialize or disable Modbus communication. The MBUS_INIT directive must execute error-free before using the MBUS_SLAVE directive. Before proceeding to the next instruction, the MBUS_INIT instruction must finish executing and set Done bit immediately. The MBUS_INIT directive should be executed only once per communication state change. Therefore, EN input should use edge detection elements to trigger with pulses, or only execute during the first cycle.

Parameter address	Illustration	Type	Value range	Note
Mode	Select communication protocol: Input 1 defines Port as the Modbus protocol and enables it, and 0 defines Port as the PPI and disables Modbus protocol.	Bit		
Addr	Set site address	Byte	1~247	
Baud	Set baud rate	DWord	1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200	
Parity	Set parity	Byte	0—No parity 1—Odd parity 2—Even parity	All settings use one stop bit.
Delay	Extends standard Modbus information end timeout condition by increasing the specified number of ms for the standard Modbus information timeout.	Int	0~32767	Unit: ms
MaxIQ	Set available I and Q.	Int	0~128 0 means forbid I/O read and write	Recommend MaxIQ 128, allow to ask all IO of S7-200.
MaxAI	Sets the number of word input registers (AI).	Int	0~32 0 means forbid read analog input.	Recommend MaxAI value: - CPU221 0 - CPU222 16 - CPU224, CPU226 and CPU224XP are 32.
MaxHold	Sets the number of	Int	0~32767	Unit: Word

	available V storage word hold registers.			
HoldStart	Sets the starting address of the available holding register in the V storage area.	DWord	Pointer point to V memory	
Done	When the MBUS INIT instruction completes, the Done output is on.	Bit		
Error	Error output byte contains the result of the execution of the instruction.	Byte		

※ MBUS_SLAVE instruction

MBUS_SLAVE instruction is used to service requests from the Modbus master station and must be executed every cycle in order to check and respond to Modbus requests. When EN input switch on, it execute each cycle without input parameters.

Parameter illustration:

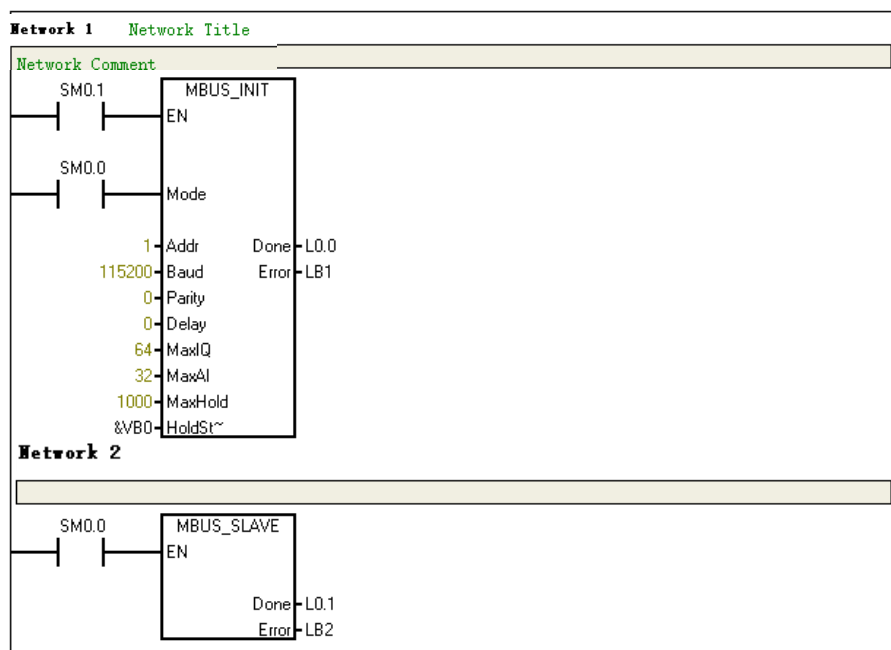
Parameter address	Illustration	Type	Value range	Note
Done	When MBUS_SLAVE respond to Modbus request Done output on. If no service request, Done output disconnect.	Bit		
Error	Error output byte contains the result of the execution of the instruction.	Byte	Error code as following	Valid only when Done is on. Error code will not change if Done is off.

Error code table:

0	No error
1	Memory area range error
2	Illegal baud rate or parity
3	Illegal slave station address
4	Illegal value of Modbus parameter
5	Holding register and Modbus slave station symbol address overlap
6	Receive parity error
7	Receive CRC error
8	Illegal function request/no supported functions
9	Illegal memory area address in request
10	Slave station function disable

※ Modbus slave station protocol instruction example

LAD program below established a no-parity slave station, address 1 and baud rate 115200:



※ MBUS_INIT parameter configuration illustration

Addr	Set slave station address 1
Baud	Set baud rate 115200
Parity	Set parity no-parity
Delay	Set delay time 0ms
MaxIQ	Set maxIQ 4 I/O(000001-0000064 and 010001-010064)
MaxAI	Set maxAI 32 analog inputs(030001-030032)
MaxHold	Set maxhold number of V area holding register(word as unit)
StartHold	Starting address of S7-200 V area holding register that Modbus master station can visit (such as &VB0).

【use Modbus master station protocol instructions】

※ Modbus master station protocol instructions occupy CTSC-200CPU resource

1) According to different Modbus protocol libraries, the FPORT 0 or 1 is used as the slave station protocol. When Port 0 or Port 1 is used as Modbus protocol communication, it cannot be used for any other purpose, including communication with MagicWorks PLC or SETP7-Micro/WIN, FPORT communication. The MBUS_INIT controls Port is set to Modbus is still PPI.

2) All SM corresponding to selected FPORT

3) Occupy 119 byte program space

※ MBUS_CTRL instruction

Use SM0.0 to call MBUS_CTRL instruction to initialize master station and enable it.

Parameter illustration:

Parameter address	Illustration	Type	Value range	Note
Mode	Set communication mode: 1 enable Modbus protocol, and 0 recovery system to PPI protocol.	Bit		
Baud	Set baud rate	DWord	11200, 2400, 4800, 9600, 19200, 38400, 57600, 115200	
Parity	Set parity	Byte	0—No parity 1—Odd parity 2—Even parity	All settings use one stop bit.
Timeout	Time master station wait for slave station, ms as unit.	Int	1~32767	Typical value 1000ms.
Done	Completion bit, automatically set 1 after initialization.	Bit		
Error	Initialization error code	Byte	0—no error 1—illegal parity selection 2—illegal baud-rate selection 3—illegal mode selection	Valid only when Done bit is 1.

※ MBUS_MSG instruction

Use SM0.0 to call Modbus RTU master station R/W subroutine MBUS_MSG, send a Modbus request after First connected. One R/W function (MBUS_MSG) enable at same time.

Parameter illustration:

Parameter address	Illustration	Type	Value range	Note
First	R/W request bit	Bit		Use pulse trigger for each new R/W request.
Slave	Set slave station address	Byte	1~247	
RW	Operation order	Byte	0~read 1~write	
Addr	Select data type of R/W	DWord	00000 to xxxx—on-off output 10000 to 1xxxx--on-off input 30000 to 3xxxx—	

			analog input 40000 to 4xxxx— holding register	
Count	Communication data number (number of word or bit)	Int		Maximum R/W of Modbus master station MBUS_MSG instruction is 120 words.
DataPtr	Data pointer, for read instruction, put data back into this data area; for write instruction, the data to be written is placed in this data area	DWord		
Done	Completion bit	Bit		
Error	Error code		Error code as following	Valid only when Done bit is 1.

Error code:

0	No error
1	Response parity error
2	unused
3	Receive timeout(slave station nonresponse)
4	Request parameter error
5	Modbus/FPORT disable
6	Modbus is busy for other request
7	Response error(response is not request operation)
8	Response CRC parity error
101	Slave station do not support request
102	Slave station do not support data address
103	Slave station do not support this data type
104	Slave station device fault
105	Slave station received information, but response delayed
106	Slave station is busy and refuse the information
107	Slave station refuse the information
108	Slave station register parity error

B Using TrustPLC CPU Expansion Program Space

B.1 Function Introduction

The dynamic library function block is a special function provided by CTSC-200 CPU to expand the user programming program space and increase program privacy. It is an independent program block which is downloaded to PLC in advance and compiled to the program when the application is downloaded.

B.2 Instructions

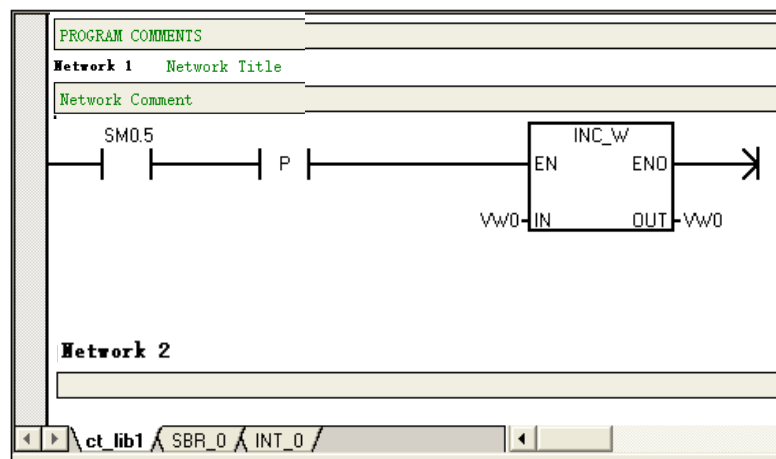
【Usable range of DLL】

CTSC-200CPU can load up to two dynamic libraries (" ct_lib1 "and" ct_lib2 "). The dynamic libraries different CPUs support and their sizes are shown in below table:

CPU type	ct_lib1	ct_lib2
CPU224E/224+V5(upgrade)/224+/224I	Allow 4K	Nonsupport
CPU224XP/CPU226M/226L/226H/226I/226M V5(upgrade)/226H V5(upgrade)/226L V5(upgrade) PSC266	Allow 24K	Allow 24K

【Create DLL】

In the project, create all the program blocks used as dynamic library, name it as ct_lib1 or ct_lib2, download to the PLC, in the PLC will generate library function including all the subroutine blocks in the project of the dynamic connection library.



【Download DLL】

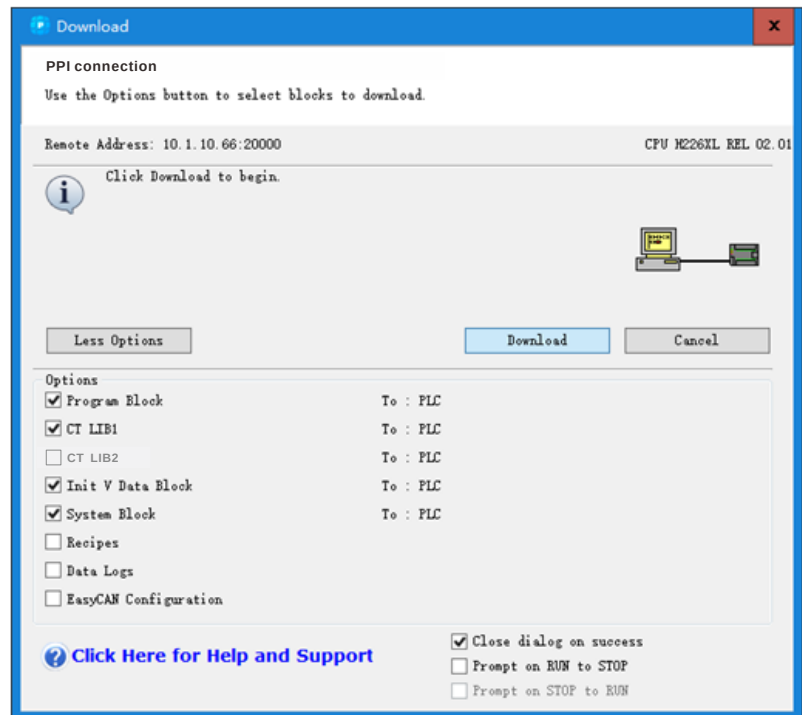
Download dynamic library program that edited well to the PLC (as shown in the figure below), in the PLC will generate a library function including all the subroutines in the project block dynamic connection library.

Every time download a new dynamic library, the PLC original library and program block is completely cleared. A dynamic library named ct_lib1 is generated in the PLC.



Note

Make sure only to download program block when download DLL.



Download another dynamic library named ct_lib2 to PLC the same way.

【Use DLL】

First create the empty blocks the same name with PLC subroutine of dynamic libraries, such as ct_lib1 or ct_lib2, then call it in project, after the program downloaded to the PLC, the empty block in the CPU will be replaced with the original download same name library functions, the corresponding dynamic database program execute when CPU run.



Note

Load the dynamic library before you load the program

【Clear DLL】

When downloading the new dynamic library, the original dynamic library in the PLC is cleared. Download a main program name ct_lib1 or ct_lib2 empty block project into PLC, the corresponding dynamic library is completely cleared.

C Using TrustPLC Expand 100K Data Block Library

C.1 Function Introduction

Extension data space is addressable 100K data storage out of CTSC - 200 CPU existed storage area, its keep features same with V memory data space, keep the data by super capacitor when CPU power off, maximum data keep time up to 100 hours, the data storage capacity is 100 k bytes. COTRUST has provided library instructions for this data

access. Users can add the library provided by COTRUST to MicroWin, access this data space through the read-write instructions provided in the library, and realize data exchange between extended data space and other data space.



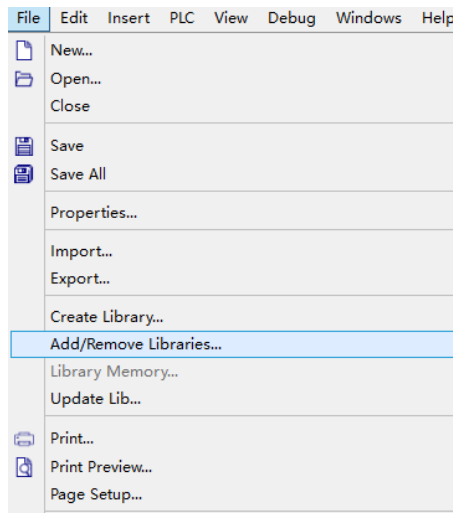
Note

Library download address: www.co-trust.com

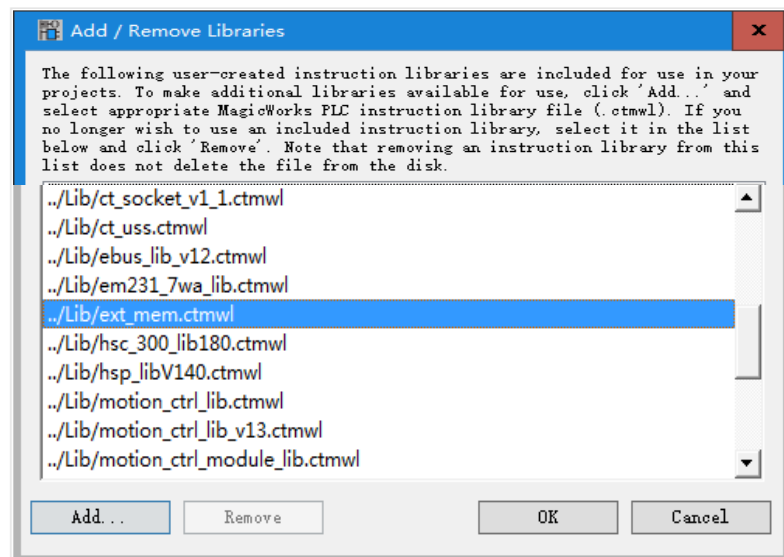
C.2 Software Installation

【Add library file】

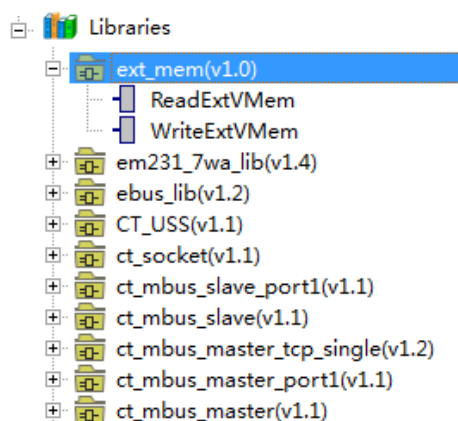
“File” -> “Add/delete library”, find “ext_mem.ctmwl” as shows below



Find file where you save ext_mem.ctmwl, click “Add”

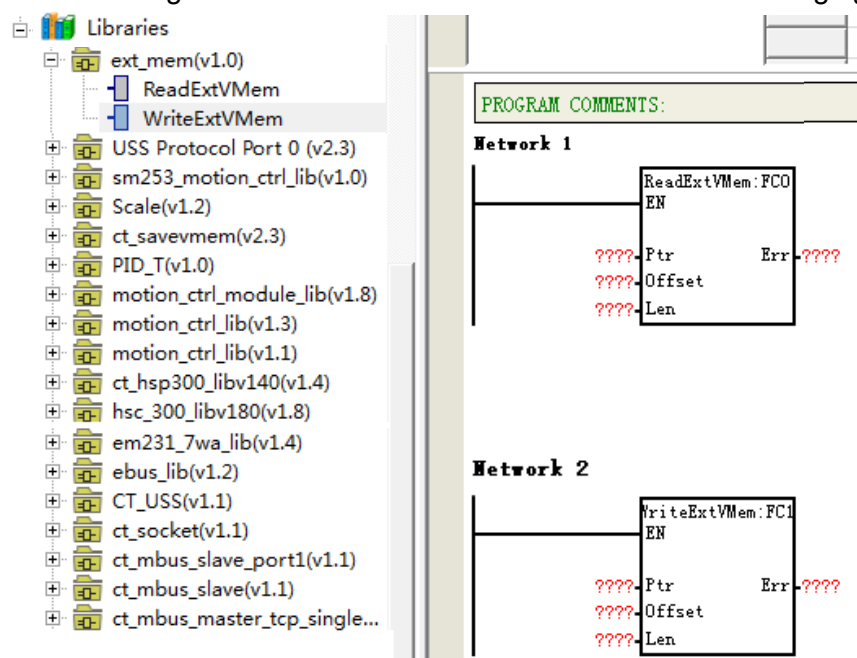


You can see new ext_mem library under directory tree “Libraries” after installation.



【Call ext_mem library】

Click “Network” to add a function block, then double-click “ReadExtVMem” and “WriteExtVMem” to get the function block in “Network”. As the following figure:



C.3 ext_mem library Function Explanation

【Use ReadExtVMem to read data from expansion data space】

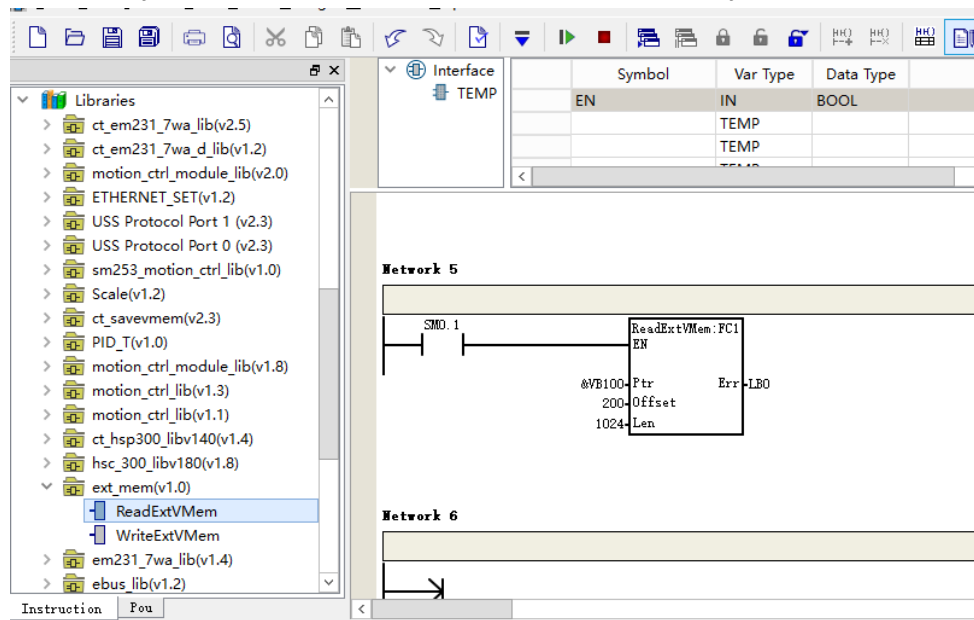
Parameter address	Illustration	Type	Value range	Note
Ptr	Memory address pointer to read	DWord		Such as: &VB0,&IB0
Offset	Read starting offset address in expansion memory	DWord	0-102399	
Len	Read memory length(byte)	DWord		
Err	Return value indicate if it read successfully	BYTE		0 means read successfully others fail.

【Use WriteExtVMem to write data to expansion data space】

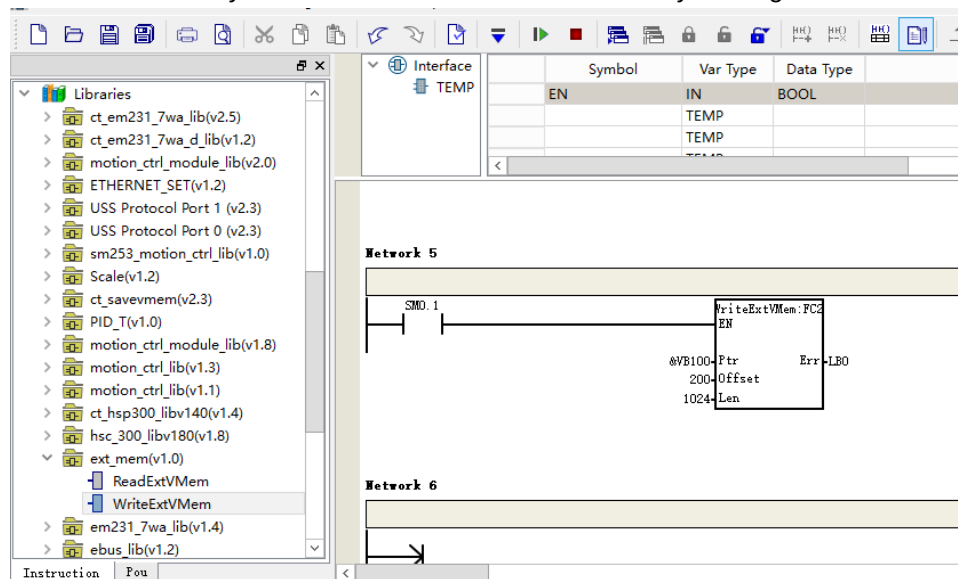
Parameter address	Illustration	Type	Value range	Note
Ptr	Write memory length(byte)	DWORD		Such as: &VB0,&IB0
Offset	Return value indicate if it write successfully	DWORD	0-102399	
Len	Write memory length(byte)	DWORD		
Err	Return value indicate if it write successfully	BYTE		00 means write successfully others fail.

C.4 Application Example

1 Read 1024 bytes from offset 200 to VB100 in extended memory



2 Write the first 1024 bytes of VB100 to the extended memory starting at offset 200



D Using TrustPLC PID Control Library

D.1 Using CPU Embedded PID_T Library

【Introduction】

PID_T function block integrate inside the CPU, does not occupy user data space, offered to user as a library function. It mainly aims at intelligent temperature control PID function, with self-tuning and self-adaptive function. No complex programming, but only need to call and set some simple parameters to use, and accurate temperature control.



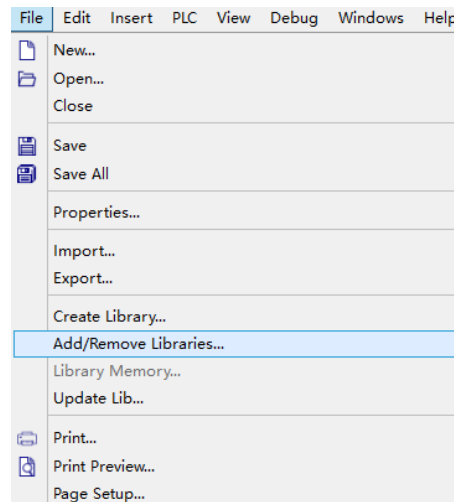
Note

Library download address: www.co-trust.com

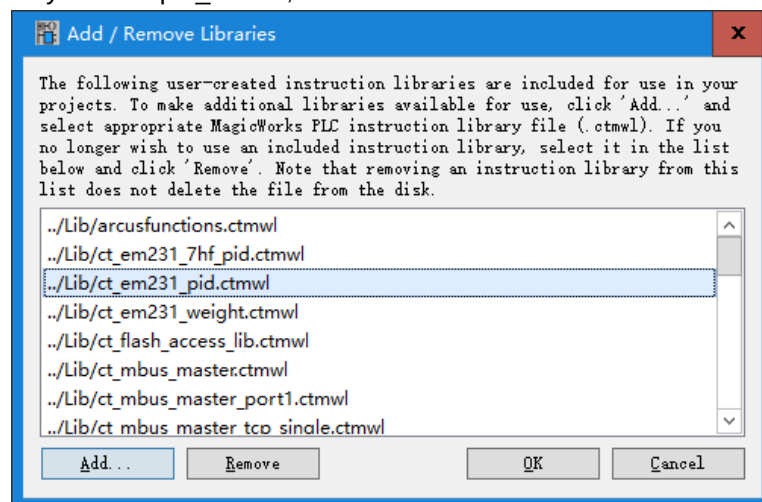
【Software Installation】

※ Add library file

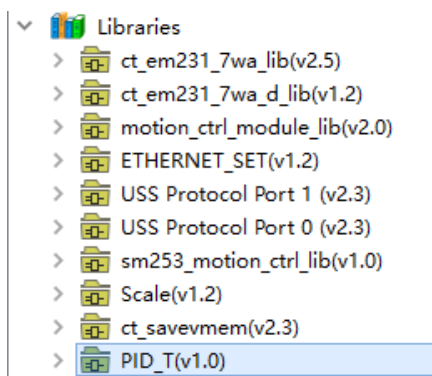
“File” -> “Add/delete library”, find “pid_t.ctmwl” , as the following figure



Find file where you save pid_t.ctmwl, click “Add”

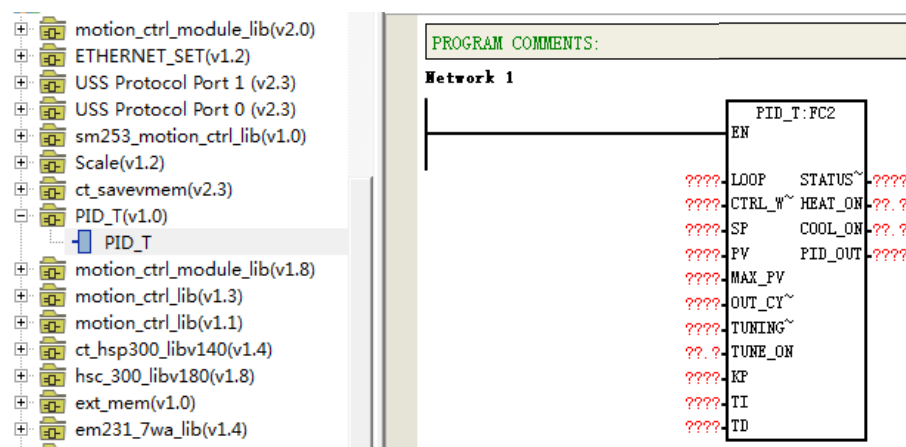


You can see new PID_T library under directory tree “Libraries” after installation.



※ Call PID_T library

Click “network” to add a function block, then double-click “PID_T” to get the function block in “network”. As the following figure:



【PID_T library function illustration】

Address parameter illustration

Parameter address	Illustration	Type	Value range	Note
LOOP	Belong to which PID, no repeat. Start from 0.	Word, constant or variable	0-63	Control the loop ID
CTRL_WORD	Control Word, control PID run.	Word, constant or variable		Common control Word: 1) 16#03(Only heating output with adaptive function) 2) 16#07(Only cold output with adaptive function)
SP	Setting value	Word, constant or variable	-32768-32767	Unit: 0.1℃

PV	Measurement(feedback value)	Word, variable	-32768-32767	Unit: 0.1°C
MAX_PV	Maximum measurement	Word, constant or variable	-32768-32767	Unit: 0.1°C
OUT_CYCLE	Pulse output cycle	Word, constant or variable	1-255	Unit: s
TUNING_K	Self-tuning coefficient	DWord, float	0.5-2.0	0.5: The system control is required small overshoot 1.0: Normal response 2.0: The system control is required to response fast and large overshoot
TUNING_ON	Start self-tuning	Bit, variable		Automatic reset after self-tuning
Kp	Kp	Word, variable		If the word variable is assigned constant, self-tuning function cannot be performed
Ti	Ti	Word, variable	1-3600	Unit: s If the word variable is assigned constant, self-tuning function cannot be performed
Td	Td	Word, variable	0-3600	Unit: s If the word variable is assigned constant, self-tuning function cannot be performed
STATUS_WORD	Status Word	Word, variable		Run and warning status
HEAT_ON	Heat-output	Bit		
COOL_ON	Cool-output	Bit		
PID_OUT	PID analog output	Word, constant or variable	Only define heat output: 0-32000. Cool output: -32000-32000	

【control Word and status Word bit address】

Control word bit address definition:

Bit	Setting	Note
0	0	PID stop
	1	PID run
1	0	Integral always work, Kp do not self-tuning
	1	Integral separate Kp self-tuning
2	0	PID unipolar output
	1	PID bipolar output
3	0	Reserved
	1	Reserved
4	0	Integral works
	1	Integral does not work
5	0	Differential works
	1	Differential does not work
6		Reserved
7		Reserved

Status word bit address definition:

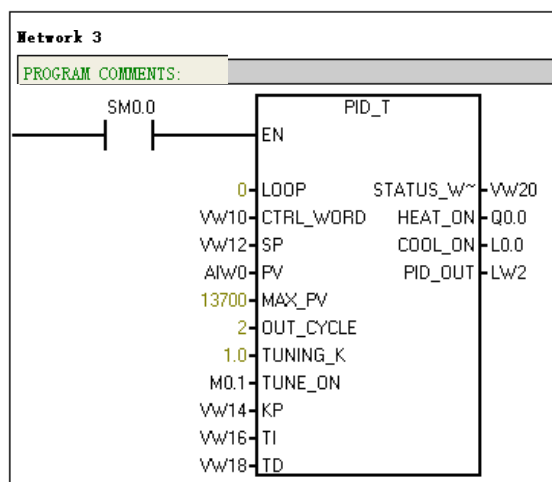
Bit	Setting	Note
0	0	No disconnection error
	1	Disconnection error
1	0	Not implemented self-tuning
	1	Self-tuning
2	0	No self-tuning error
	1	Self-tuning error
3	0	No heating
	1	Heating
4	0	No cooling
	1	Cooling
5	0	PID stop
	1	PID run
6		Reserved
7		Reserved

【example】

Take a temperature zone as example, and the system and I/O configuration table is as follows:

System configuration	CPU226L+231-7PD32	Temperature acquisition using four - way thermocouple module	
Control request	1 only heating-output, no cooling-output 2 request self-tuning 3 K thermocouple		
I/O distribution			
Q0.0	heating-output		
AIW0	Temperature input	K thermocouple	
M0.0	PID run/stop bit		
M1.0	Self-tuning start bit		

Application program



Illustration

PID_T parameter illustration

Parameter	Address or value	Illustration	Note
LOOP	0	0 for first loop	
CTRL_WORD	VW10		
SP	VW12		
PV	AIW0		
MAX_PV	13700	Maximum input is 13700 for K thermocouple	
OUT_CYCLE	2	2s, pulse output cycle	
TUNING_K	1.0		
TUNING_ON	M0.1	Set from 1, reset after setting.	
Kp	VW14	Kp, setting value will automatically write to this variable after setting, users can adjust themselves.	
Ti	VW16	Ti, setting value will automatically write to this variable after setting, users can adjust themselves.	
Td	VW18	Td, setting value will automatically write to this variable after setting, users can adjust themselves.	
STATUS_WORD	VW20	Status Word	
HEAT_ON	Q0.0	Heating output	
COOL_ON	L0.0	Use a local variable for unused	
PID_OUT	LW2	Use a local variable for unused	

D.2 Using TrustPLC Thermocouple PID Control Module Library

【Function introduction】

The "PID_setting" function library is specially designed to provide parameter for the thermocouple PID expansion module (e.g., EM231-7TD, EM231-7TF). Integrate PID algorithm inside the thermocouple PID module. No complex programming, but only needs to call "PID_setting" library to set some simple parameters to use module, and the

temperature control is accurate.

Library download address: www.co-trust.com



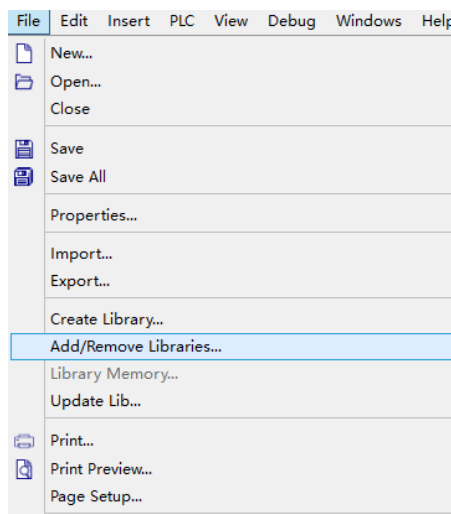
Note

- Apply to CTS7 231-7TD32, CTS7 231-7TF32 modules;
- Use EM231-7TD32 and EM231-7TF32 will occupy some V memory area, do not use V memory area when programming
- Siemens CPU222 cannot use the library for data space restriction

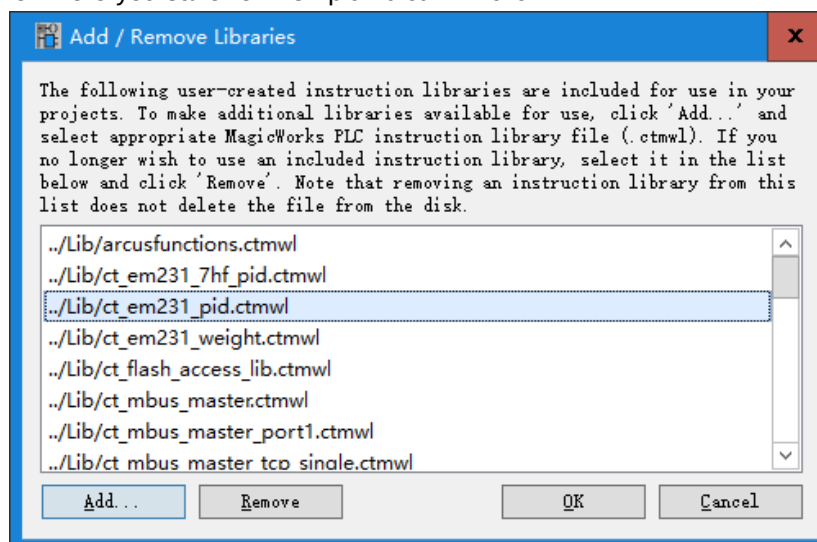
【Software Installation】

※ Add library file

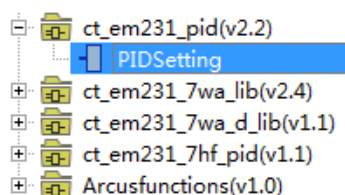
“File” -> “Add/delete library”, find “em231 pid lib.ctmwl” , as the following figure



Find file where you save “em231 pid lib.ctmwl” click “Add”

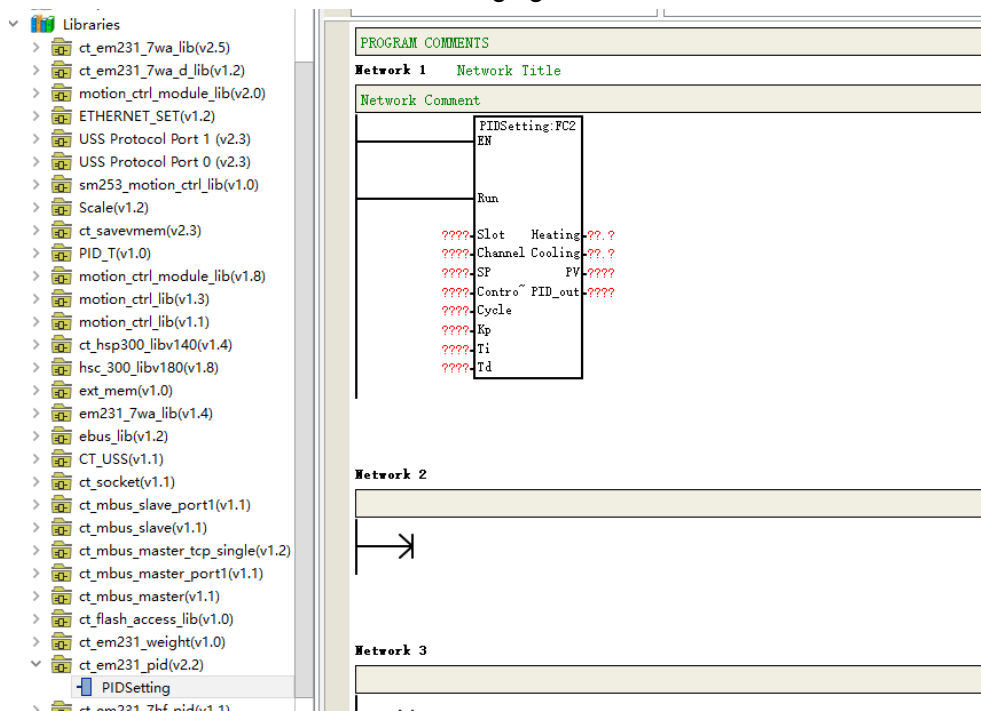


You can see new PID_setting library under directory tree “Add” after installation.



※ Call PID_Setting library

Click “network” to add a function block, then double-click “PID_Setting” to get the function block in “network”. As the following figure:



※ 【PID_setting library function illustration】

Address parameter illustration

Parameter address	Illustration	Type	Value range	Note
Run	Run	Bit	0 or 1	
Slot	Start from slot 0	Word, constant or variable	0~6	
Channel	Channel number	Word, constant or variable	0~7	
SP	Setting value	Word, constant or variable	-2000~32767	Unit: 0.1℃
CTRLByte	Control byte, control PID run	constant or variable		Common control Word: 1) 16#03(Only heating output with adaptive function) 2) 16#07(Only cold output with adaptive function)

Cycle	Pulse output cycle	Word, constant or variable	1~255	Unit: s
Kp	Kp	Word, int, constant or variable		
Ti	Ti	Word, int, constant or variable	1~3600	Unit: s
Td	Td	Word, int, constant or variable	0~3600	Unit: s
Heating	Heating-output	Bit		
Cooling	Cooling-output	Bit		
PV	Measurement(feedback value)	Word, variable	-2000~32767	Unit: 0.1℃
PID_out	PID analog output	Word, int, variable	Only define heat output: 0-32000. Cool output: -32000~32000	

※ Control word bit address

Control word bit address definition:

Bit	Setting	Note
0	0	PID stop
	1	PID run
1	0	Integral always work, Kp do not self-tuning
	1	Integral separate Kp self-tuning
2	0	PID unipolar output
	1	PID bipolar output
3	0	Reserved
	1	Reserved
4	0	Integral works
	1	Integral does not work
5	0	Differential works
	1	Differential does not work
6		Reserved
7		Reserved

※ PID address and parameter configuration

• PID address calculation formula

Address name	Formula	Note
PID parameter address	$A = (2048 + S * 256) + 16 * C$	S is slot where module is(range: 0~6) C is channel, 231-7HF is 0~7
PID positive pulse output address	$X = (2048 + S * 256) + 12$	
PID negative pulse output address	$Y = (2048 + S * 256) + 13$	

• PID parameter output (modules to CPU)

Items	Address	Value set range	Actual value
Actual temperature	VW A	0~13000	0~1300°
Status Word	VW A+2		
PID analog output	VW A+4	-32000~32000	

• PID parameter input(CPU to modules)

Items	Address	Value set range	Actual value
Setting temperature	VW A+128	0~13000	0~1300
Control byte		When VB A+130 bit = 0	
	V(A+130).0	PID stop, no output	PID run
	V(A+130).1	Integral always work, Kp do not self-tuning	Integral separate Kp self-tuning
	V(A+130).2	PID unipolar output, 0~32000	PID bipolar output, -32000~32000, both heat and cool
	V(A+130).3	Unused	
	V(A+130).4	Integral works	
	V(A+130).5	Differential works	
	V(A+130).6	Actual temperature filter, stronger ant jamming capability	Actual temperature do not filter
PID pulse output cycle setting	VW A+132	1~255	
Kp	VW A+134	0~9999	0~999.9
Ti	VW A+136	0~3600	
Td	VW A+138	0~3600	

• Heating pulse output address

Channel 0 pulse output	V X.0
Channel 1 pulse output	V X.1
Channel 2 pulse output	V X.2
Channel 3 pulse output	V X.3
Channel 4 pulse output	V X.4
Channel 5 pulse output	V X.5
Channel 6 pulse output	V X.6
Channel 7 pulse output	V X.7

• Cooling pulse output address

Channel 0 pulse output	V Y.0
Channel 1 pulse output	V Y.1
Channel 2 pulse output	V Y.2
Channel 3 pulse output	V Y.3
Channel 4 pulse output	V Y.4
Channel 5 pulse output	V Y.5
Channel 6 pulse output	V Y.6
Channel 7 pulse output	V Y.7

【Example】

System description

This routine sets the parameters of the first PID loop (channel 0) of EM231-7TD extension module (slot 0). Call PIDSetting to set the loop parameters, no need to calculate the PID parameter address, just input the slot and channel number where the loop is, and then enable Run to run the loop.

Q0.0 is the positive pulse output; Q0.1 is negative pulse output;

VW0 is the actual temperature; VW2 is PID analog output;

Use other addresses to Modify PID setting parameters;

Set temperature: VW120;

Control Word: VB122;

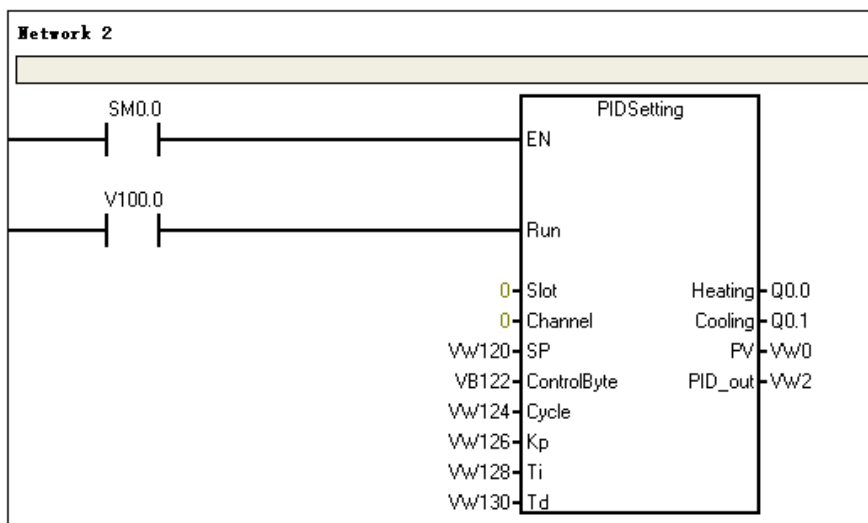
Pulse output cycle: VW124;

Kp: VW126;

Ti: VW128;

Td: VW130;

Application program



**Note**

For PID modules normal use, please do not occupy following PID V storage when programming.

Address that module occupied in slot 0: VW2048 to VW2298

Address that module occupied in slot 1: VW2304 to VW2554

Address that module occupied in slot 2: VW2560 to VW2810

Address that module occupied in slot 3: VW2816 to VW3066

Address that module occupied in slot 4: VW3072 to VW3322

Address that module occupied in slot 5: VW3328 to VW3578

Address that module occupied in slot 6: VW3584 to VW3834

D.3 Using TrustPLC Current Temperature PID Control Module Library

【Introduction】

The "PIDSettingCT7HF" function library is specially designed to provide parameter for the current mode PID expansion module (e.g., EM231-7HF). Integrate PID algorithm inside the PID module. No complex programming, but only needs to call "PIDSettingCT7HF" library to set some simple parameters to use module, and the temperature control is accurate.

Library download address: www.co-trust.com

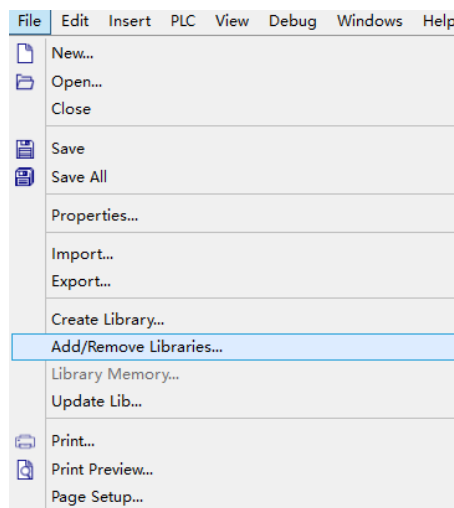
**Note**

- Apply to CTS7 231-7HF32 modules;
- Use EM231-7TD32 and EM231-7TF32 will occupy some V memory area, do not use V memory area when programming
- Siemens CPU222 cannot use the library for data space restriction

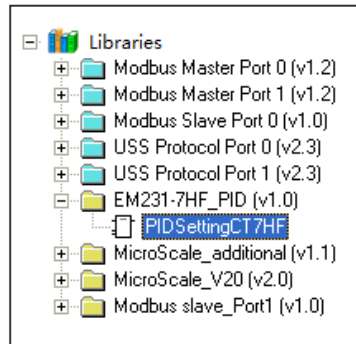
【Software Installation】

※ Add library file

"File" -> "Add/delete library", find "em231-7hf_pid.ctmwl", as the following figure

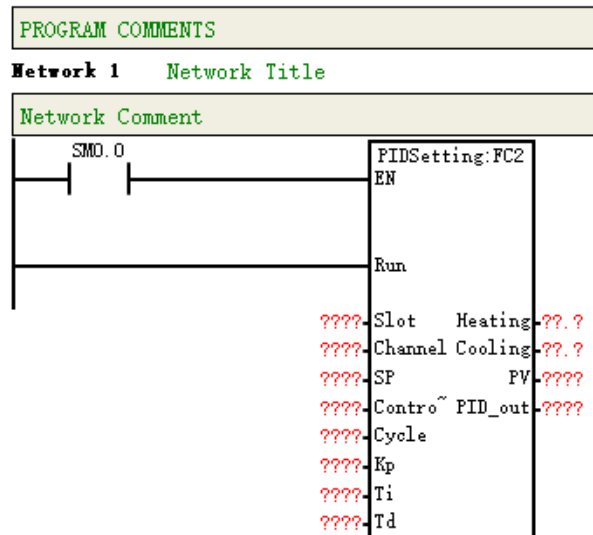


Find file where you save “em231-7hf_pid.ctmwl” click “Add”. You can see new PIDSettingCT7HF library under directory tree “Libraries” after installation.



※ Call PIDSettingCT7HF library

Click “network” to add a function block, then double-click “PIDSettingCT7HF” to get the function block in “network”. As the following figure:



【PIDSettingCT7HF library function illustration】

Address parameter illustration

Parameter address	Illustration	Type	Value range	Note
Run	Run	Bit	0 or 1	
Slot	Start from slot 0	Word, constant or variable	0~6	
Channel	Channel number	Word, constant or variable	0~7	
SP	Setting value	Word, constant or variable	-2000~32767	Unit: 0.1℃
CTRLByte	Control byte, control PID run	constant or variable		Common control Word: 1) 16#03(Only heating output

				with adaptive function) 2) 16#07(Only cold output with adaptive function)
Cycle	Pulse output cycle	Word, constant or variable	1~255	Unit: s
Kp	Kp	Word, int, constant or variable		
Ti	Ti	Word, int, constant or variable	1~3600	Unit: s
Td	Td	Word, int, constant or variable	0~3600	Unit: s
Range	Heating-output	Word, int, constant or variable	0~13000	Unit: 0.1°C
Heating	Cooling-output	Bit		
Cooling	Measurement (feedback value)	Bit		
PV	PID analog output	Word, variable	-2000~32767	Unit: 0.1°C
PID_out	Heating-output	Word, int, variable	Only define heat output: 0-32000. Cool-output: -32000~32000	

※ Control word bit address

Control word bit address definition:

Bit	Setting	Note
0	0	PID stop
	1	PID run
1	0	Integral always work, Kp do not self-tuning
	1	Integral separate Kp self-tuning
2	0	PID unipolar output
	1	PID bipolar output
3	0	Reserved
	1	Reserved
4	0	Integral works
	1	Integral does not work
5	0	Differential works
	1	Differential does not work
6		Reserved
7		Reserved

PID address and parameter configuration

• PID address calculation formula

Address name	Formula	Note
PID parameter address	$A = (2048 + S \times 256) + 16 \times C$	S is slot where module is (range: 0~6) C is channel, 231-7HF is 0~7
PID positive pulse output address	$X = (2048 + S \times 256) + 12$	
PID negative pulse output address	$Y = (2048 + S \times 256) + 13$	

• PID parameter output (modules to CPU)

Items	Address	Value set range	Actual value
Actual temperature	VW A	0~13000	0~1300
Status Word	VW A+2		
PID analog output	VW A+4	-32000~32000	

• PID parameter input (CPU to modules)

Items	Address	Value set range	Actual value
Setting temperature	VW A+128	0~13000	0~1300°
Control byte		When VB A+130 bit = 0	When VB A+130 bit = 1
	V(A+130).0	PID stop, no output	PID run
	V(A+130).1	Integral always work, Kp do not self-tuning	Integral separate Kp self-tuning
	V(A+130).2	PID unipolar output, 0~32000	PID bipolar output, -32000~32000, both heat and cool
	V(A+130).3	Unused	
	V(A+130).4	Integral works	
	V(A+130).5	Differential works	
	V(A+130).6	Actual temperature filter, stronger ant jamming capability	Actual temperature do not filter
PID pulse output cycle setting	VW A+132	1~255	
Kp	VW A+134	0~9999	0~999.9
Ti	VW A+136	0~3600	
Td	VW A+138	0~3600	
Range	VW A+140	0~13000	0~3600℃

• Heating pulse output address

Channel 0 pulse output	V X.0
Channel 1 pulse output	V X.1

Channel 2 pulse output	V X.2
Channel 3 pulse output	V X.3
Channel 4 pulse output	V X.4
Channel 5 pulse output	V X.5
Channel 6 pulse output	V X.6
Channel 7 pulse output	V X.7

• Cooling pulse output address

Channel 0 pulse output	V Y.0
Channel 1 pulse output	V Y.1
Channel 2 pulse output	V Y.2
Channel 3 pulse output	V Y.3
Channel 4 pulse output	V Y.4
Channel 5 pulse output	V Y.5
Channel 6 pulse output	V Y.6
Channel 7 pulse output	V Y.7

【Example】

System description

This routine sets the parameters of the first PID loop (channel 0) of EM231-7HF extension module (slot 0). Call PIDSettingct7HF to set the loop parameters, no need to calculate the PID parameter address, just input the slot and channel number where the loop is, and then enable Run to run the loop.

Q0.0 is the positive pulse output; Q0.1 is negative pulse output;

VW0 is the actual temperature; VW2 is PID analog output;

Use other addresses to Modify PID setting parameters;

Set temperature: VW120;

Control Word: VB122;

Pulse output cycle: VW123;

Kp: VW125;

Ti: VW127;

Td: VW129;

Transmitter 0~20MA corresponding temperature range: VW131

【Example】

System description

This routine sets the parameters of the first PID loop (channel 0) of EM231-7TD extension module (slot 0). Call PIDSetting to set the loop parameters, no need to calculate the PID parameter address, just input the slot and channel number where the loop is, and then

enable Run to run the loop.

Q0.0 is the positive pulse output; Q0.1 is negative pulse output;

VW0 is the actual temperature; VW2 is PID analog output;

Use other addresses to Modify PID setting parameters;

Set temperature: VW120;

Control Word: VB122;

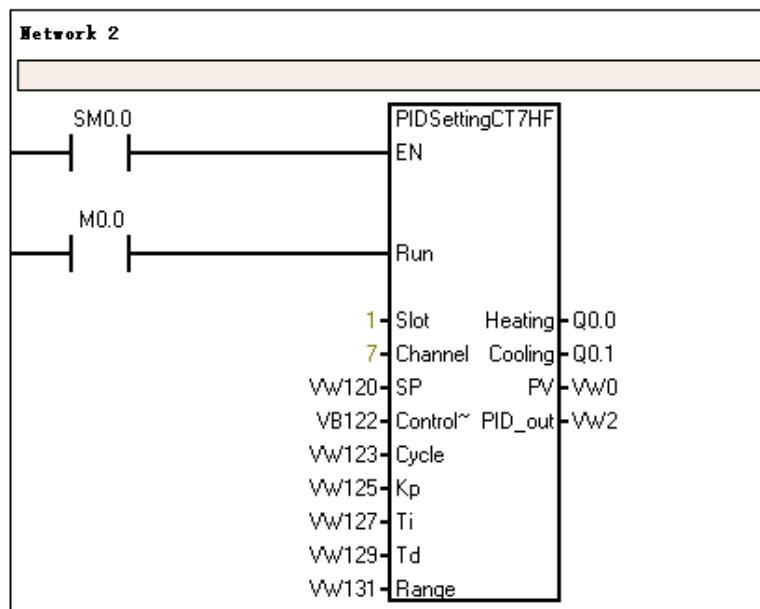
Pulse output cycle: VW124;

Kp: VW125;

Ti: VW127;

Td: VW129;

Application program



Note

For PID modules normal use, please do not occupy following PID V storage when programming.

Address that module occupied in slot 0: VW2048 to VW2300

Address that module occupied in slot 1: VW2304 to VW2556

Address that module occupied in slot 2: VW2560 to VW2812

Address that module occupied in slot 3: VW2816 to VW3068

Address that module occupied in slot 4: VW3072 to VW3324

Address that module occupied in slot 5: VW3328 to VW3580

Address that module occupied in slot 6: VW3584 to VW3836

E Using Motion Control Library “motion_ctrl_lib”

E.1 Introduction

The motion_ctrl_lib library is used for TrustPLC motion control CPU. No complex programming, just call and set some simple parameters to use, this series of CPU has multi-axis independent control function, CPU with interpolation can carry out any two axis linear interpolation and circular interpolation, also support linear acceleration and deceleration control.

Notice

When multiple interpolation instructions are invoked in the user program, only one of them can RUN.

For example, call the two-axis circular arc interpolation movement instruction (MC_CIRCLE_R) etc. When the program is running, only one of the interpolation instructions input parameter RUN can be set as ON.



Note

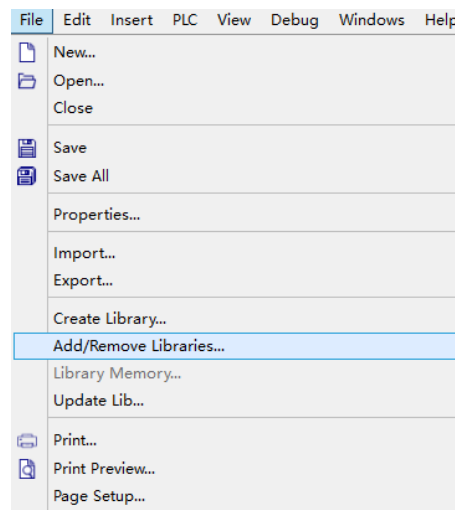
Library download address: www.co-trust.com

CTSC-200 CPU support MicroWIN software program and library.

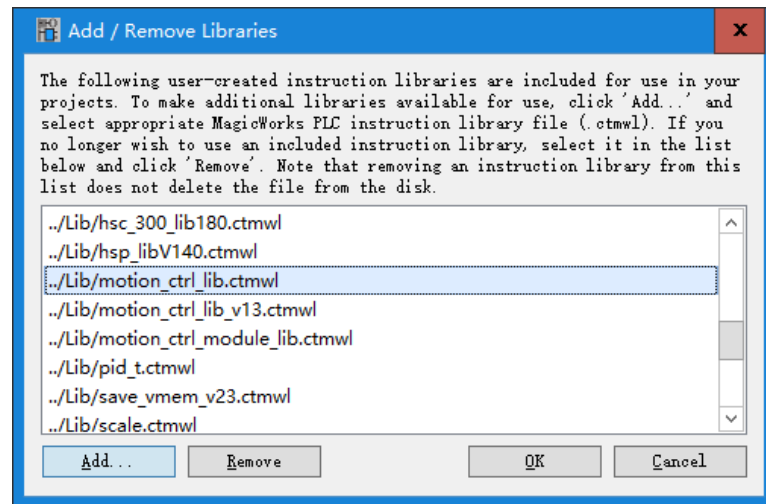
E.2 Software Installation

【Add library file】

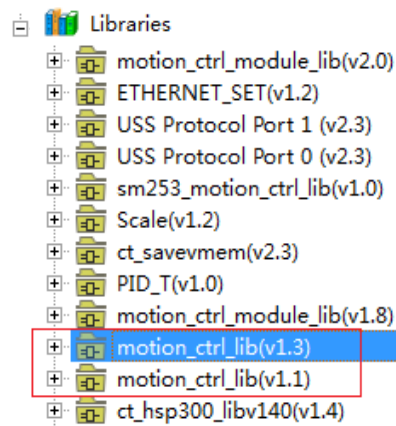
“File” -> “Add/delete library”, find “motion_ctrl_lib.ctmwl”, as the following figure



Find file where you save “motion_ctrl_lib.ctmwl” click “Add”.



You can see new motion_ctrl_lib library under directory tree “Libraries” after installation. V1.1 (at left)/V1.3:



E.3 I/O Correspondence of Motion Axis and CPU

◆ CPU226H

Normal IO	Q0.0	Q0.1	Q0.2	Q0.3	Q0.4	Q0.5	Q0.6	Q0.7	Q1.0	Q1.1
Motion control IO	Pulse _0	Dir _0	Pulse _1	Dir _1	Pulse _2	Dir _2	Pulse _3	Dir _3	Q1.0	Q1.1

◆ CPU224E\CPU226M-CAN\CPU224I\CPU226I

Normal IO	Q0.0	Q0.1	Q0.2	Q0.3	Q0.4 ~ Q1.1
Motion control IO	Pulse _0	Pulse _1	Dir _0	Dir _1	-



Note

- 1) Pulse _0 -----0 axis pulse output; Dir _0 -----0 axis direction output;
Pulse _1 -----1 axis pulse output; Dir _1 -----1 axis direction output;
Pulse _2 -----2 axis pulse output; Dir _2 -----2 axis direction output;
Pulse _3 -----3 axis pulse output; Dir _3 -----3 axis direction output;
- 2) Q0.0 and Q0.1 do not support PTO and PWM HSPSO of program software.
- 3) to avoid conflict between discrete interrupt and HSC, the discrete interrupt has changed from I0.0, I0.1, I0.2, I0.3 to I0.0, I0.1, I1.1, I1.5

E.4 motion_ctrl_lib (V1.3) Instruction List

Table1 CPU 226H library instruction list

Function name	Instruction name	CPU226H type			
		CTS7 216-1AH34-0B24	CTS7 216-1AH34-0X24	CTS7 216-1AH34-1B24	CTS7 216-1AH34-2B24
MC_HOMING	Homing instruction	Support			
MC_INIT_DIR	Configure motor direction instruction	Support			
MC_EXT_RESET_EN	External reset coordinate enable instruction	Support			
MC_EXT_RESET_EN_EXT	External reset coordinate enable instruction II	Support			
MC_SET_POS_ZERO	Software reset instruction	Support			
MC_SET_POS_PV	Set target position instruction	Support			
MC_SET_MAX_ACCELE	Set maximum acceleration instruction	Support			
MC_READ_POS	Read position instruction	Support			
MC_SPEED_CTL	Speed control instruction	Support			
MC_PTP_R	Uniaxial relative motion instruction	Support			
MC_PTP_A	Uniaxial absolute motion instruction	Support			
MC_LINE_R	Two axis linear interpolation relative motion instruction	Nonsupport		Support	
MC_LINE_A	Two axis linear interpolation absolute motion instruction	Nonsupport		Support	
MC_CIRCLE_R	Two axis circular interpolation relative motion instruction	Nonsupport		Support	
MC_CIRCLE_A	Two axis circular interpolation absolute motion instruction	Nonsupport		Support	
MC_3P_CIRCLE_R	Two axis three point draw arc interpolation relative motion instruction	Nonsupport		Support	
MC_3P_CIRCLE	Two axis three	Nonsupport		Support	

_A	point draw arc interpolation absolute motion instruction		
MC_SET_CI_MOD E	Set continuous interpolation function instruction	Nonsupport	Support

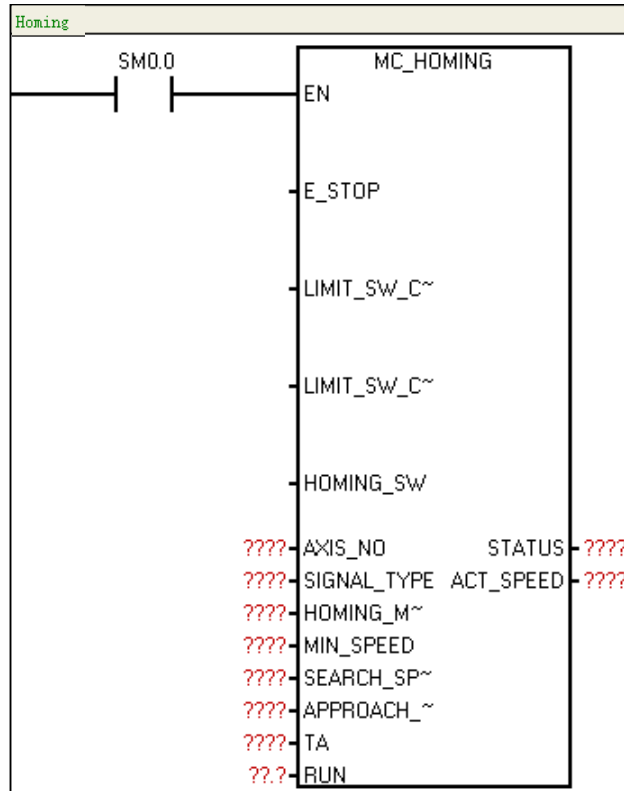
Table 2 Other motion control CPU library instruction

Function name	Instruction name	CPU type	
		CTS7 214-1AE33-0X24/CTS7 216-1AC33-0X24	
		CTS7 214-1AD41-0X24/CTS7 216-2AD41-0X40	
MC_HOMING	Homing instruction	Support	
MC_INIT_DIR	Configure motor direction instruction		
MC_EXT_RESET_EN	External reset coordinate enable instruction		
MC_EXT_RESET_EN_EXT	External reset coordinate enable instruction II		
MC_SET_POS_ZERO	Software reset instruction		
MC_SET_MAX_ACCELERATION	Set maximum acceleration instruction		
MC_READ_POS	Read position instruction		
MC_SPEED_CTL	Speed control instruction		
MC_PTP_RELATIVE	Uniaxial relative motion instruction		
MC_PTP_ABSOLUTE	Uniaxial absolute motion instruction		
MC_SET_POS_POINT	Set target position instruction	Nonsupport	
MC_LINE_RELATIVE	Two axis linear interpolation relative motion instruction		
MC_LINE_ABSOLUTE	Two axis linear interpolation absolute motion instruction		
MC_CIRCLE_RELATIVE	Two axis circular interpolation relative motion instruction		
MC_CIRCLE_ABSOLUTE	Two axis circular interpolation absolute motion instruction		
MC_3P_CIRCLE_RELATIVE	Two axis three point draw arc interpolation relative motion instruction		
MC_3P_CIRCLE_ABSOLUTE	Two axis three point draw arc interpolation absolute motion instruction		
MC_SET_CI_MOD E	Set continuous interpolation function instruction		

E.5 Instructions Specifications

Homing instruction

- ① Function name: MC_HOMING



- ② Function: Find device origin by setting parameters such as homing mode.

The relationship between the axis number and external reset IO signal (such as homing Z pulse):

Axis 0 — I0.2 (HSC0, SM37.0) Axis 1 — I1.0 (HSC1, SM47.0)

Axis 2 — I1.4 (HSC2, SM57.0) Axis 3 — I0.5 (HSC4, SM147.0)

If the homing mode refers to the origin switch (mode 3 or 4), connect origin switch signal (the HOMING_SW parameter of the instruction) to the corresponding point above; otherwise, the origin cannot be found.

- ③Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
LIMIT_SW_CCW	IN	CCW counterclockwise stroke limit input	BOOL	0~1	

LIMIT_SW-CW	IN	CW counterclockwise stroke limit input	BOOL	0~1	
HOMING_SW	IN	Origin signal input	BOOL	0~1	
AXIS_NO	IN	Axis number	BYTE	0~3	Unchangeable in the process
SIGNAL_TYPE	IN	7 6 5 4 3 2 1 0 Signal type: Bit0: Counterclockwise stroke limit input signal type 0—High level 1—Low level Bit1: Clockwise stroke limit input signal type 0—High level 1—Low level Bit2: Origin switch signal type 0—High level 1—Low level	BYTE	0~255	
HOMING_MODE	IN	Homing mode	BYTE	1~14	Please refer to Appendix E.6 Back to Zero Mode Illustration .
MIN_SPEED	IN	Minimum speed Unit: Hz	DWORD	0~200000	1 Pulse output close when speed < 5Hz. 2 Unchangeable in the process 3 Search speed should not be too large, close speed should be as small as possible
SEARCH_SPEED	IN	Origin search speed Unit: Hz	DWORD	0~200000	
APPROACH_SPEED	IN	Origin close speed Unit: Hz	DWORD	0~200000	
TA	IN	Time of ACC/DEC. Unit: ms	DWORD	0~10000	Changeable in the process
RUN	IN	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 when E_STOP=1, RUN

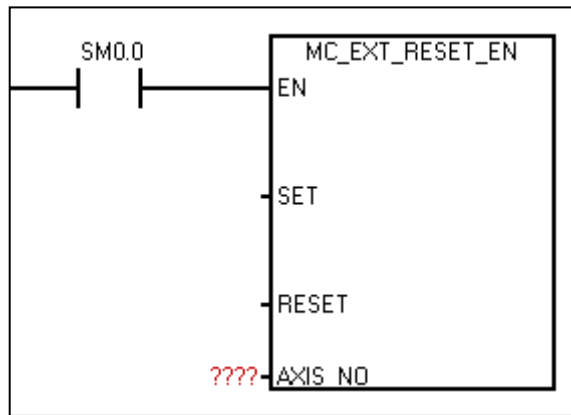
					reset internal.
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done. Bit4: Reserved Bit5: Find origin or not 1—Found origin 0—Not find origin	BYTE	0~255	Bit0: 1 Judge Only to axis parameter configuration error and homing mode out of range. 2 no error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_SPEED	OUT	Current speed	DWORD	0~200000	

④ Explanation

The program detect each input by scanning (the Z phase is not affected by this), so when the switch value changes and process not in time, there may be some delay. If the homing speed (including search and close speed) is too high, the processing delay will be amplified, resulting in inaccurate return.

External reset coordinate enable instruction

① Function name: MC_EXT_RESET_EN



- ② Function: When call this instruction, set whether enable external IO reset absolute coordinate value.

Note: The relationship between the axis number and external reset signal:

Axis 0 —— I0.2 (HSC0, SM37.0)

Axis 1 —— I1.0 (HSC1, SM47.0)

Axis 2 —— I1.4 (HSC2, SM57.0)

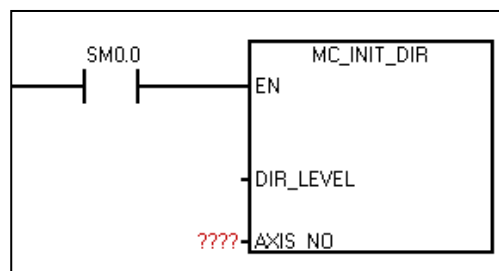
Axis 3 —— I0.5 (HSC4, SM147.0)

- ③ Parameter

Name	I/O	Description	Type	Value range
SET	IN	SET rising edge, set external reset enable, for each call, first reset SET and then set 1	Bool	Range: 0~1
RESET	IN	RESET rising edge, prohibit external reset enable. For each call, first reset and then set 1	Bool	Range: 0~1
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	Range: 0~3

Configure motor direction instruction

- ① Function name: MC_INIT_DIR



- ② Function: Configure motor direction

Note: Execute one time for first scan cycle CPU power on

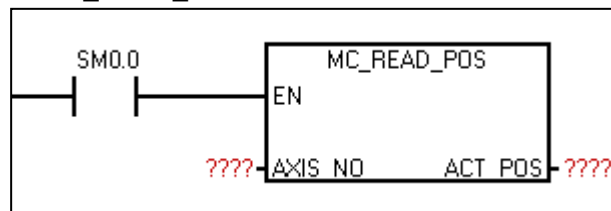
- ③ Parameter

Name	I/O	Description	Type	Value range	Note
DIR_LEVEL	IN	Configure effective level when positive direction signal	Bool	Range: 0~1	Default: 1, Motor

		Set motor forward for corresponding direction axis output "1" when DIR_LEVEL is 1. Set motor reverse for corresponding direction axis output "0" when DIR_LEVEL is 0.			forward when direction axis output is "1"
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	Range: 0~3	

Read position instruction

- ① Function name: MC_READ_POS



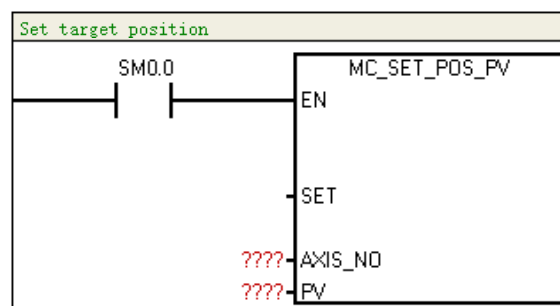
- ② Function: Read the absolute coordinate value of each axis. Once the origin coordinate is set, the value will be calculated according to the relationship between output pulse and direction: positive rotation output one pulse: +1, and reverse rotation output a pulse: -1. What you end up with is an absolute coordinate with the set point as the origin.

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	Range: 0~3	
ACT_POS	OUT	Absolute coordinate of current axis (1 pulse 1 units coordinate)	DInt	-2147483647~+2147483647	This instruction no error and set the axis number correctly.

Set target position instruction

- ① Function name: MC_SET_POS_PV



- ② Function: This instruction is used to write the absolute location of the machine to the module. For example, machine can save current position when the power cut off. And this

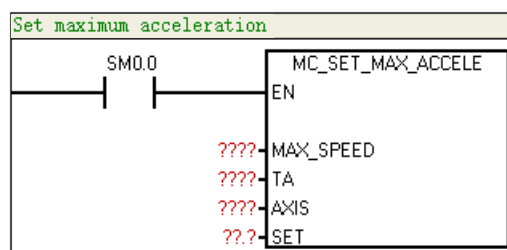
position instead of origin will be written back to the module next time the machine power on, so that the absolute position counting starting point of the module is consistent with the actual starting point of the machine. If this position happens to be the origin, this instruction has the same effect as MC_SET_POS_ZERO.

③ Parameter

Name	I/O	Description	Type	Value range
AXIS_NO	IN	Set axis number	BYTE	0~3
SET	IN	SET rising edge, instruction enable, reset and then set 1 for each call.	BOOL	0~1
PV	IN	Set the target position, points plus or minus. The positive output pulse indicates the positive direction along the X-axis, and the negative pulse number indicates the negative direction along the X-axis.	DINT	-2147483648 ~ +2147483647

Set maximum acceleration instruction

① Function name: MC_SET_MAX_ACCELE



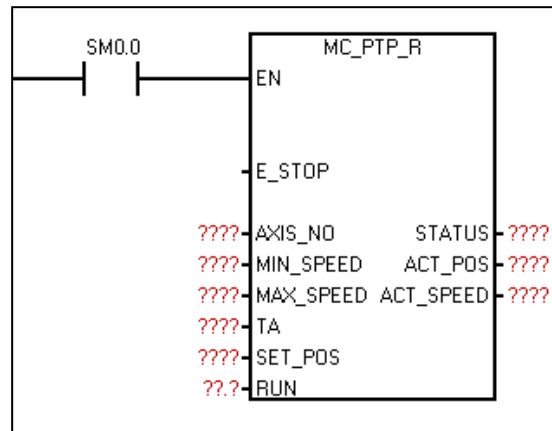
② Function: Set maximum acceleration(= MAX_SPEED/TA)(TA≠0)(no call indicates no setting for acceleration)

③ Parameter

Name	I/O	Description	Type	Value range	Note
MAX_SPEED	IN	Maximum speed Unit: Hz	DWORD	0~200000	Changeable in the process
TA	IN	Time of ACC/DEC, Unit: ms	DWORD	0~10000	Changeable in the process; if TA=0, no settings for acceleration.
AXIS	IN	Set axis number	BYTE	0~3	This instruction no error and set the axis number correctly.
SET	IN	Give SET a rising edge to take effect after above parameter ensure.	BOOL	0~1	

Uniaxial relative motion instruction

① Function name: MC_PTP_R



② Function: Use for single - axis point - to - point control (single - axis fixed - length drive). One call for fixed pulse output, through the maximum and minimum speed and deceleration time setting, the output pulse will gradually accelerate to the largest speed, when the pulse number is ready to run, cut down pulse frequency automatically, to prevent vibration or jammed caused by too much inertia when start or stop the machine.

③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	Bool	0/1	1. Run only when Run =1 and E_Stop =0 2. RUN reset internal when E_STOP = 1
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	0~3	Unchangeable in the process
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	500~200000	1. Set minimum speed less than maximum speed
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	500~200000	2. Changeable in the process
TA	IN	Time of ACC/DEC, Unit: ms	Dword	0~10000	Changeable in the process; if TA=0, no settings for acceleration.
RUN	IN	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete.

					3 when E_STOP=1, RUN reset internal.
STATUS	OUT	76543210 Output status byte: Bit0: parameter configuration error sign 1—Parameter configuration error 0—Parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1 Judge Only to axis parameter configuration error and homing mode out of range. 2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
SET_POS	IN	Output pulse. The positive output pulse indicates the positive direction along the X-axis, and the negative pulse number indicates the negative direction along the X-axis.	DInt	-2147483647 ~ +2147483647	Modifiable during operation. When the new set value is greater than the number of pulses output, the final output pulse will be subject to the new set value. When the new set value is less than the number of output pulses, the pulse output will be stopped immediately
ACT_PO	OUT	Absolute coordinate of	DInt	-2147483647	This instruction no

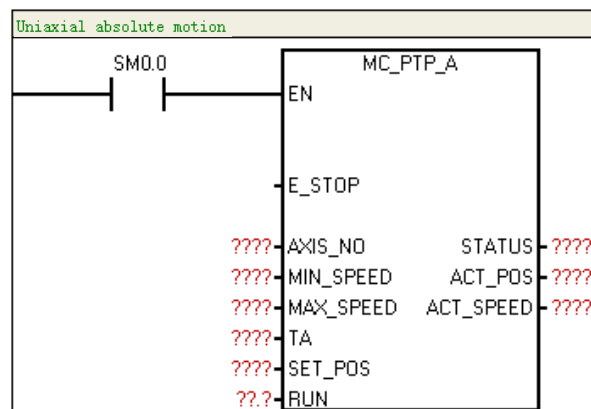
S		current axis (1 pulse 1 units coordinate)		~ +2147483647	error and set the axis number correctly.
ACT_SPEED	OUT	Current run speed	Dword	500~200000	Note: This value may be a little off actual one, at most 5K, related to acceleration time and setting speed

Note 1

Theoretically DCC/DEC time $TA \leq (MAX_SPEED) - (MIN_SPEED)$, such as $TA > (MAX_SPEED) - (MIN_SPEED)$, then default $TA = (MAX_SPEED) - (MIN_SPEED)$ inside the motion control instruction

Uniaxial absolute motion instruction

① Function name: MC_PTP_A



② Function: Use for single - axis point - to - point control (fixed-point instead of fixed - length). One call for fixed pulse output, through the maximum and minimum speed and deceleration time setting, the output pulse will gradually accelerate to the largest speed, when the pulse number is ready to run, cut down pulse frequency automatically, to prevent vibration or jammed caused by too much inertia when start or stop the machine.

③ Parameter

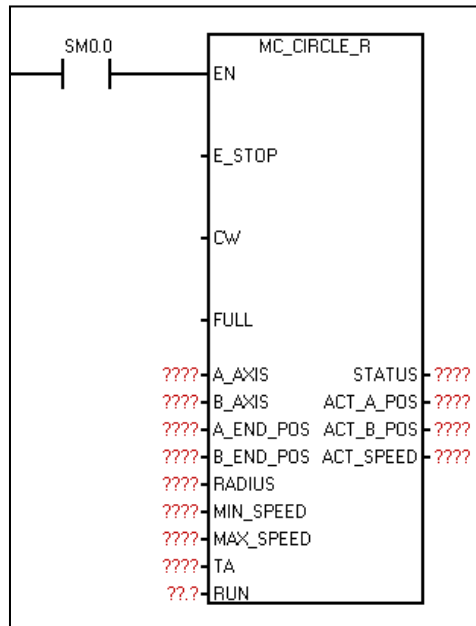
Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	Bool	0/1	1 Run only when Run =1 and E_Stop =0 2 RUN reset internal when E_STOP = 1
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	0~3	Unchangeable in the process
MIN_SPEED	IN	Minimum speed of run or stop.	Dword	0~200000	1 Set minimum speed less than

		Unit: HZ			maximum speed
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	0~200000	2 Changeable in the process 3 Recommend MIN_SPEED ≥ 500. Or the pulse output will complete minimum speed limit (usually 500) at the end of the deceleration. 4 Write MAX_SPEED 0 to realize the soft stop function (the output pulse deceleration stops). Then the enable bit RUN no reset; If write speed back to a large one, output the pulse until it complete.
TA	IN	Time of ACC/DEC, Unit: ms	Dword	0~10000	Changeable in the process; if TA=0, no settings for acceleration.
RUN	IN	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 when E_STOP=1, RUN reset internal.
STATUS	OUT	76543210 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error	BYTE	0~255	Bit0: 1 Judge Only to axis parameter 2 No error reported for other parameters, it will set to the nearest

		0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.			reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
SET_POS	IN	Output pulse. The positive output pulse indicates the positive direction along the X-axis, and the negative pulse number indicates the negative direction along the X-axis.	DInt	-2147483648 ~ +2147483647	Modifiable during operation. When the new set value is greater than the number of pulses output, the final output pulse will be subject to the new set value. When the new set value is less than the number of output pulses, the pulse output will be stopped immediately
ACT_POS	OUT	Absolute coordinate of current axis (1 pulse 1 units coordinate)	DInt	-2147483648 ~ +2147483647	This instruction no error and set the axis number correctly.
ACT_SPEED	OUT	Current run speed	Dword	0~200000	

Two axis circular interpolation relative motion instruction

① Function name: MC_CIRCLE_R



② Function: Circular interpolation between any two axes (set points as relative coordinates).

③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1. Run only when RUN =1 and E_STOP =0. 2. When E_STOP=1, RUN reset internal.
CW	IN	Interpolation of CW or CCW 1: CW 0: CCW	Bool	0/1	Unchangeable in the process
FULL	IN	Full-circle sign 1—full-circle 0—Arc	Bool	0/1	Unchangeable in the process
A_AXIS	IN	Interpolation A axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the A axis maps to. For example, 3 indicates A axis maps to the three axes (Pulse_3 and Dir_3).	Byte	0~3	Unchangeable in the process

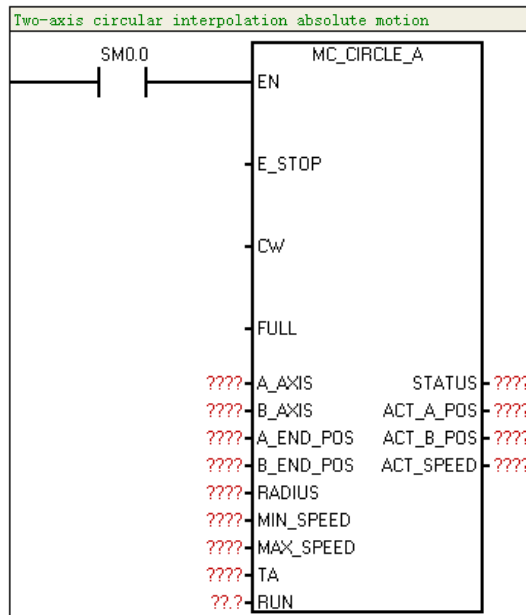
B_AXIS	IN	Interpolation B axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the B axis maps to.	Byte	0~3	Unchangeable in the process
A_END_POS	IN	If FULL = 0, indicates destination coordinate(relative origin) of virtual A-axis If FULL = 1, indicates coordinate (relative starting point) of an arc that is different from the starting point, not the end point.	Dint	-2147483647 ~ +2147483647	1 Default interpolation starting point coordinate(0,0) 2 Unit: pulse, Transform with actual
B_END_POS	IN	If FULL = 0, indicates the coordinates of virtual B axis end point (relative to the starting point). If FULL = 1, indicates the coordinates of the arc at a point different from the starting point, not the ending point.	Dint	-2147483647 ~ +2147483647	1 Interpolation start coordinate default (0,0) 2 Unit: pulse, Transfer with actual displacement Note: Unchangeable in the process
RADIUS	IN	Arc radius 1. There are positive and negative. Positive number: indicates the arc trajectory with radian less than 180°. Negative number: indicates the arc trajectory with a radian greater than 180°. 2. The absolute value of RADIUS represents Arc radius	Dint	$ R < 3 \times 10^6$	1 Unmodifiable in the process 2 Unit: pulse, Transform with actual radius
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	500~200000	Unmodifiable in the process
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	500~200000	Changeable in the process

TA		Time of ACC/DEC, Unit: ms	Dword	0~10000	1 Unmodifiable in the process 2 Calculate acceleration only when startup and TA/TD changes
RUN	IN/OUT	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 when E_STOP=1, RUN reset
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1. Judge Only to axis parameter 2. No error reported for other parameters, it will set to the nearest reasonable value automatically. 3. Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_A _POS	OUT	Current position of A axis(relative coordinate, actual output pulse) Indicates relative coordinate of 0	Dint	-2147483647 ~ +2147483647	

ACT_B_POS	OUT	Current position of B axis(relative coordinate, actual output pulse) Indicates relative coordinate of 1	Dint	-2147483647 ~ +2147483647	
ACT_SPEED	OUT	Current run speed	Dword	500~200000	

Two-axis circular interpolation absolute motion instruction

① Function name: MC_CIRCLE_A



② Function: Circular interpolation between any two axes (set points as absolute coordinates).

③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
CW	IN	Interpolation of CW or CCW 1: CW 0: CCW	Bool	0/1	Unchangeable in the process
FULL	IN	Full-circle sign 1—full-circle	Bool	0/1	Unchangeable in the process

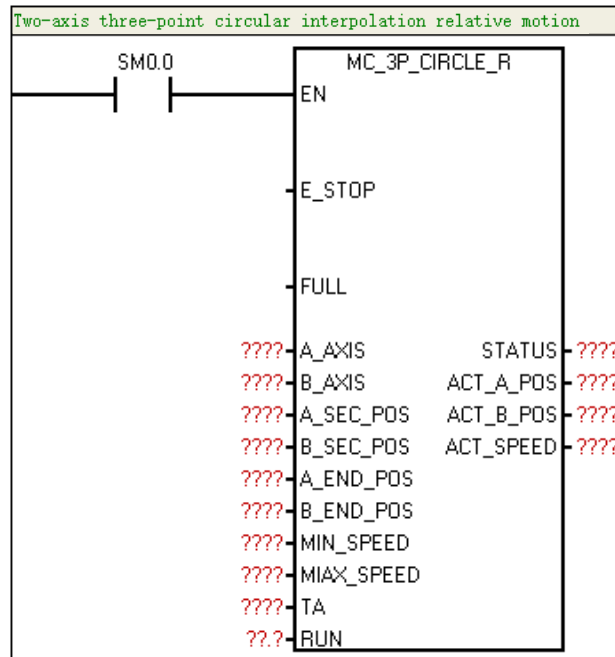
		0—Arc			
A_AXIS	IN	Interpolation A axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the A axis maps to. For example, 3 indicates A axis maps to the three axes (Pulse_3 and Dir_3).	BYTE	0~3	Unchangeable in the process
B_AXIS	IN	Interpolation B axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the B axis maps to.	BYTE	0~3	
A_END_POS	IN	If FULL = 0, indicates destination coordinate(relative origin) of virtual A-axis If FULL = 1, indicates coordinate (relative starting point) of an arc that is different from the starting point, not the end point.	DINT	-2147483648 ~ +2147483647	1 Unmodifiable in the process 2 Unit: pulse, Transform with actual displacement
B_END_POS	IN	If FULL = 0, indicates the coordinates of virtual B axis end point (relative to the starting point). If FULL = 1, indicates the coordinates of the arc at a point different from the starting point, not the ending point.	DINT	-2147483648 ~ +2147483647	
RADIUS	IN	Arc radius 1. There are positive and negative. Positive number: indicates the arc trajectory with radian less than 180°. Negative number:	DINT	$ R < 3 \times 10^6$	1 Unmodifiable in the process 2 Unit: pulse, Transform with actual displacement

		indicates the arc trajectory with a radian greater than 180°. 2. The absolute value of RADIUS represents Arc radius			
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	0~200000	1 Set minimum speed less than maximum speed 2 Changeable in the process 3 Recommend MIN_SPEED ≥500. Or the pulse output will complete minimum speed limit (usually 500) at the end of the deceleration. 4 Write MAX_SPEED 0 to realize the soft stop function (the output pulse deceleration stops). Then the enable bit RUN no reset; If write speed back to a large one, output the pulse until it complete.
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	0~200000	
TA		Time of ACC/DEC, Unit: ms	Dword	0~10000	1 Unmodifiable in the process 2 Calculate acceleration only when startup and TA/TD changes
RUN	IN/OUT	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP

					=0. 2 RUN reset internal when run complete. 3 When E_STOP=1, RUN reset internal.
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1 Judge Only to axis parameter 2 no error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_A_POS	OUT	Current position of A axis(relative coordinate, actual output pulse) Indicates relative coordinate of 0 axis when A axis allocate to 0 axis	DINT	-2147483648 ~ +2147483647	
ACT_B_POS	OUT	Current position of B axis(relative coordinate, actual output pulse) Indicates relative coordinate of 1 axis when B axis allocate to 1 axis	DINT	-2147483648 ~ +2147483647	
ACT_SPEED	OUT	Current run speed	Dword	0~200000	

Two-axis three-point circular interpolation relative motion instruction

① Function name: MC_3P_CIRCLE_R



② Function: Circular interpolation between any two axes (set points as relative coordinates).

③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	Bool	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
FULL	IN	Full-circle sign 1—full-circle 0—Arc	Bool	0/1	Unchangeable in the process
A_AXIS	IN	Interpolation A axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the A axis maps to. For example, 3 indicates A axis maps to the three axes (Pulse_3 and Dir_3).	BYTE	0~3	Unchangeable in the process

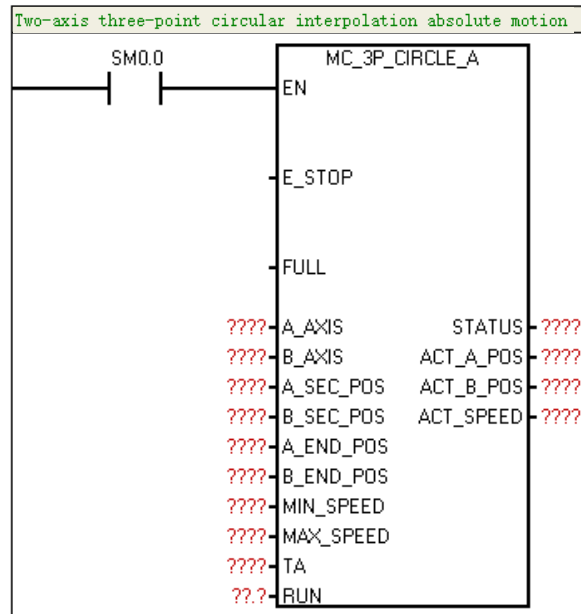
B_AXIS	IN	Interpolation B axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the B axis maps to.	BYTE	0~3	Unchangeable in the process
A_SEC_POS	IN	Determine arc second point of the A axis coordinate (use for certain the second point in the direction of an arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	1 Interpolation starting coordinate is first point of the arc(relative instruction starting coordinate (0, 0)) 2 Unmodifiable in the process 3 Unit: pulse, Transform with actual replacement 4 Interpolation starting coordinate default(0,0)
B_SEC_POS	IN	Determine arc second point of the B axis coordinate (use for certain the second point in the direction of an arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	
A_END_POS	IN	Determine arc third point of axis A coordinate. If FULL = 0, it is also the terminal coordinate. (used for determine arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	
B_END_POS	IN	Determine arc third point of axis B coordinate. If FULL = 0, it is also the terminal coordinate. (used for determine arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	

MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	0~200000	1 Set minimum speed less than maximum speed 2 Changeable in the process 3 Recommend MIN_SPEED ≥500. Or the pulse output will complete minimum speed limit (usually 500) at the end of the deceleration.
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	0~200000	4 Write MAX_SPEED 0 to realize the soft stop function (the output pulse deceleration stops). Then the enable bit RUN no reset; If write speed back to a large one, output the pulse until it complete.
TA		Time of ACC/DEC, Unit: ms	Dword	0~10000	1 Unchangeable in the process 2 Calculate acceleration only when startup and TA/TD changes
RUN	IN/OUT	Run enable bit 1: valid	Bool	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 When E_STOP=1, RUN reset internal.
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte:	BYTE	0~255	Bit0: 1 Judge Only to axis parameter

		Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.			2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_A_POS	OUT	Current position of A axis(relative coordinate, actual output pulse) Indicates relative coordinate of 0 axis when A axis allocate to 0 axis	DINT	-2147483648 ~ +2147483647	
ACT_B_POS	OUT	Current position of B axis(relative coordinate, actual output pulse) Indicates relative coordinate of 1 axis when B axis allocate to 1 axis	DINT	-2147483648 ~ +2147483647	
ACT_SPEED	OUT	Current actual speed	Dword	0~200000	

Two-axis three-point circular interpolation absolute motion instruction

① Function name: MC_3P_CIRCLE_A



② Function: Circular interpolation between any two axes (set points as absolute coordinates).

③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
FULL	IN	Full-circle sign 1—full-circle 0—Arc	Bool	0/1	Unchangeable in the process
A_AXIS	IN	Interpolation A axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the A axis maps to. For example, 3 indicates A axis maps to the three axes (Pulse_3 and Dir_3).	BYTE	0~3	Unchangeable in the process
B_AXIS	IN	Interpolation B axis number. Interpolation requires two	BYTE	0~3	Unchangeable in the process

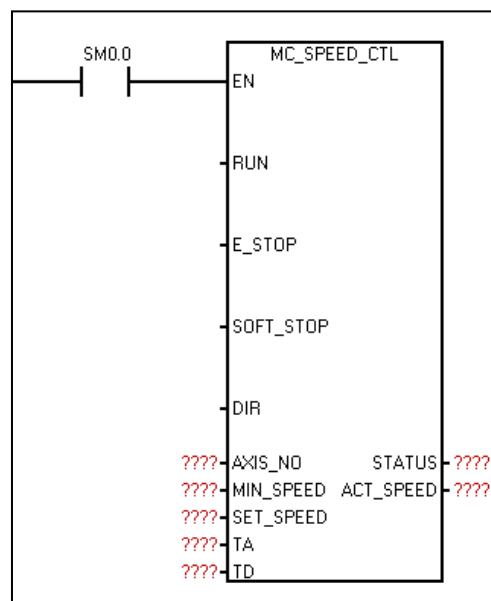
		axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the B axis maps to.			
A_SEC_POS	IN	Determine arc second point of the A axis coordinate (use for certain the second point in the direction of an arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	1. Interpolation starting coordinate is first point of the arc (relative instruction starting coordinate(0,0)) 2. Unmodifiable in the process 3. Unit: pulse 4. Transform with actual replacement
B_SEC_POS	IN	Determine arc second point of the B axis coordinate (use for certain the second point in the direction of an arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	
A_END_POS	IN	Determine arc third point of axis A coordinate. If FULL = 0, it is also the terminal coordinate. (used for determine arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	
B_END_POS	IN	Determine arc third point of axis B coordinate. If FULL = 0, it is also the terminal coordinate. (To determine arc or circle at three points, relative coordinate)	DINT	-2147483648 ~ +2147483647	
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	0~200000	1. Set minimum speed less than maximum speed 2. Changeable in the process 3. Recommend MIN_SPEED ≥500. Or the pulse output will complete minimum speed limit (usually
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	0~200000	

					500) at the end of the deceleration. 4. Write MAX_SPEED 0 to realize the soft stop function (the output pulse deceleration stops). Then the enable bit RUN no reset; if write speed back to a large one, output the pulse until it complete.
TA		Time of ACC/DEC, Unit: ms	Dword	0~10000	1. Unmodifiable in the process 2. Calculate acceleration only when startup and TA/TD changes
RUN	IN/OUT	Run enable bit 1: valid	BOOL	0~1	1. Run only when RUN =1 and E_STOP =0. 2. RUN reset internal when run complete. 3. When E_STOP=1, RUN reset internal.
STATUS	OUT	76543210 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal	BYTE	0~255	Bit0: 1. Judge Only to axis parameter 2. No error reported for other parameters, it will set to the

		Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.			nearest reasonable value automatically. 3. Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_A_POS	OUT	Current position of A axis(relative coordinate, actual output pulse) Indicates relative coordinate of 0 axis when A axis allocate to 0 axis	DINT	-2147483648 ~ +2147483647	
ACT_B_POS	OUT	Current position of B axis(relative coordinate, actual output pulse) Indicates relative coordinate of 1 axis when B axis allocate to 1 axis	DINT	-2147483648 ~ +2147483647	
ACT_SPEED	OUT	Current actual speed	Dword	0~200000	

Speed control instruction

① Function name: MC_SPEED_CTL



- ② Function: Control the frequency of uniaxial output pulse, the output pulse frequency (speed) can be changed any time. When receive the soft stop command, it will automatically slow down to stop. When receive the emergency stop command, the pulse output will stop immediately without any deceleration.

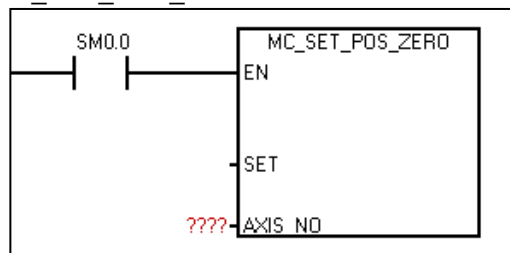
③ Parameter

Name	I/O	Description	Type	Value range	Note
RUN	IN	Run enable bit 1: valid	BOOL	0~1	1. Run only when RUN =1 and E_STOP =0. 2. RUN reset internal when run complete. 3. When E_STOP=1, RUN reset internal.
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1. Run only when RUN =1 and E_STOP =0. 2. When E_STOP=1, RUN reset internal.
SOFT_STOP	IN	Soft stop bit 1: valid 0: invalid Output pulse decelerate to stop when receive the instruction	Bool	0/1	
DIR	IN	Pulse direction bit	Bool	0/1	Unmodifiable in the process
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	0~3	Unmodifiable in the process
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	0~200000	1. Minimum speed should less than setting speed 2. Modifiable in the process
SET_SPEED	IN	Setting speed, output pulse will accelerate or decelerate to this speed before stop command	Dword	0~200000	
TA	IN	Time of ACC Unit: ms	Dword	0~10000	Modifiable in the process; Calculate acceleration only when

					startup and TA/TD changes
TD	IN	Time of ACC, from setting time to minimum time Unit: ms	Dword	0~10000	Unmodifiable in the process
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1 Judge Only to axis parameter 2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_SPEED	OUT	Current actual speed	Dword	0~200000	Current actual speed

Software back to zero instruction

- ① Function name: MC_SET_POS_ZERO



- ② Function: reset absolute coordinate

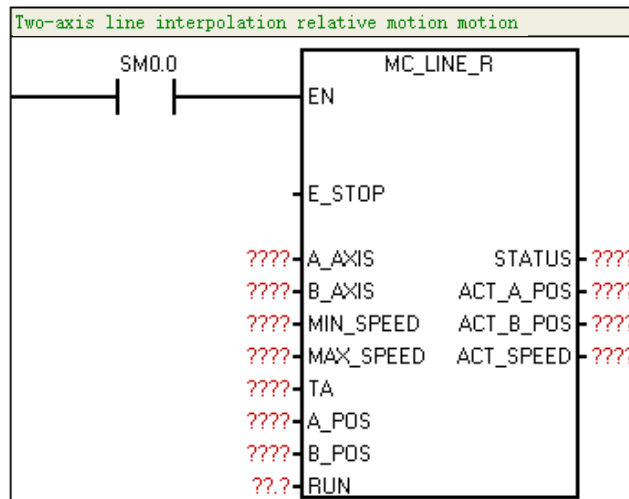
Note: Call the instruction when the machine moves to a certain position, which is equivalent to setting the origin of the axis at that position. Each time you call the "read absolute coordinates" command, you get a coordinate value relative to that point.

③ Parameter

Name	I/O	Description	Type	Value range	Note
SET	IN	Enable reset bit Set absolute coordinate 0 at SET rising edge, set SET bit 0, then set 1 for each call	Bool	Range: 0~1	
AXIS_NO	IN	Set axis number: 0/1/2/3	Byte	Range: 0~3	

Two-axis line interpolation relative motion instruction

① Function name: MC_LINE_R



② Function: line interpolation between any two axes or any plane area (set points as relative coordinates).

③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
A_AXIS	IN	Interpolation A axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual	BYTE	0~3	Unmodifiable in the process

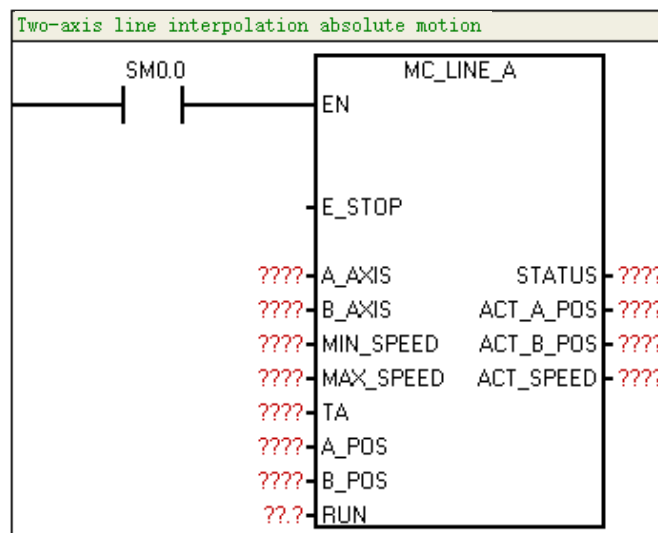
		output. This parameter sets which axis the A axis maps to. For example, 3 indicates A axis maps to the three axes (Pulse_3 and Dir_3).			
B_AXIS	IN	Interpolation B axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the B axis maps to.	BYTE	0~3	
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	0~200000	1 Set minimum speed less than maximum speed 2 Changeable in the process 3 Recommend MIN_SPEED ≥500. Or the pulse output will complete minimum speed limit (usually 500) at the end of the deceleration. 4 Write MAX_SPEED 0 to realize the soft stop function (the output pulse deceleration stops). Then the enable bit RUN no reset; If write speed back to a large one, output the pulse until it complete.
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	0~200000	
TA	IN	Time of ACC Unit: ms	Dword	0~10000	Modifiable in the process Calculate

					acceleration only when startup and TA/TD changes
A_POS	IN	Virtual A axis destination (relative) coordinate.	DINT	-2147483648 ~ +2147483647	1 Unmodifiable in the process 2 Unit: pulse
B_POS	IN	Virtual B axis destination (relative) coordinate.	DINT	-2147483648 ~ +2147483647	
RUN	IN/OUT	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 When E_STOP=1, RUN reset internal.
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—Parameter configuration error 0—Parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1 Judge Only to axis parameter 2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_A_POS	OUT	Current position of A axis(relative coordinate, actual output pulse)	DINT	-2147483648 ~ +2147483647	

		Indicates relative coordinate of 0 axis when A axis allocate to 0 axis		7	
ACT_B_POS	OUT	Current position of B axis(relative coordinate, actual output pulse) Indicates relative coordinate of 1 axis when B axis allocate to 1 axis	DINT	-2147483648 ~ +2147483647	
ACT_SPEED	OUT	Current actual speed	Dword	0~200000	

Two-axis line interpolation absolute motion instruction

① Function name: MC_LINE_A



② Function: line interpolation between any two axes or any plane area (set points as absolute coordinates).

③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: Valid, 0: Invalid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
A_AXIS	IN	Interpolation A axis	BYTE	0~3	Unmodifiable in

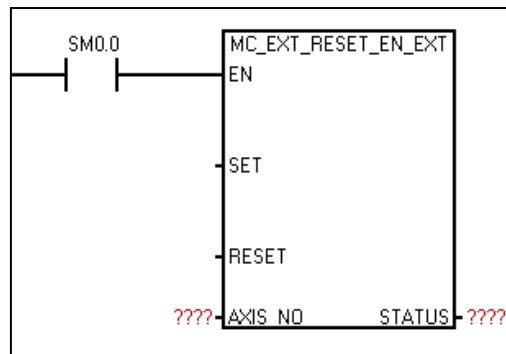
		number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the A axis maps to. For example, 3 indicates A axis maps to the three axes (Pulse_3 and Dir_3).			the process
B_AXIS	IN	Interpolation B axis number. Interpolation requires two axes, the virtual A-axis and B-axis. Map to the 0, 1, 2, 3 axes of the actual output. This parameter sets which axis the B axis maps to.	BYTE	0~3	
MIN_SPEED D	IN	Minimum speed of run or stop. Unit: HZ	Dword	0~200000	1 Set minimum speed less than maximum speed 2 Changeable in the process 3 Recommend MIN_SPEED ≥500. Or the pulse output will complete minimum speed limit (usually 500) at the end of the deceleration.
MAX_SPEED D	IN	Maximum speed of run Unit: HZ	Dword	0~200000	4 Write MAX_SPEED 0 to realize the soft stop function (the output pulse deceleration stops). Then the enable bit RUN no reset; If write speed back to a large one, output the pulse until it complete.

TA	IN	Time of ACC Unit: ms	Dword	0~10000	Modifiable in the process Calculate acceleration only when startup and TA/TD changes
A_POS	IN	Virtual A axis destination (relative) coordinate.	DINT	-2147483648 ~ +2147483647	1 Unmodifiable in the process 2 Unit: pulse
B_POS	IN	Virtual B axis destination (relative) coordinate.	DINT	-2147483648 ~ +2147483647	
RUN	IN/OUT	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 when E_STOP=1, RUN reset internal.
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—Parameter configuration error 0—Parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution	BYTE	0~255	Bit0: 1 Judge Only to axis parameter 2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same

		done.			
ACT_A_POS	OUT	Current position of A axis(relative coordinate, actual output pulse) Indicates relative coordinate of 0 axis when A axis allocate to 0 axis	DINT	-2147483648 ~ +2147483647	
ACT_B_POS	OUT	Current position of B axis(relative coordinate, actual output pulse) Indicates relative coordinate of 1 axis when B axis allocate to 1 axis	DINT	-2147483648 ~ +2147483647	
ACT_SPEED	OUT	Current actual speed	Dword	0~200000	

External reset coordinate enable instruction II

- Function name: MC_EXT_RESET_EN_EXT



- Function: set if enable external IO reset absolute coordinate

Note: correspondence of axis number and external reset signal

Axis 0 —— I0.2 (HSC0, SM37.0)

Axis 1 —— I1.0 (HSC1, SM47.0)

Axis 2 —— I1.4 (HSC2, SM57.0)

Axis 3 —— I0.5 (HSC4, SM147.0)

③ Parameter

Name	I/O	Description	Type	Value range	Note
SET	IN	Enable reset bit Set absolute coordinate 0 at SET rising edge, SET set 0, then set 1 for each call	Bool	0~1	
RESET	IN	RESET rising edge, forbid to enable external reset, RESET then set 0 for each call	Bool	0~1	

AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	0~3	
STATUS	OUT	1: Reset complete 0: Reset uncompleted	Byte	0~1	

④ Explanation

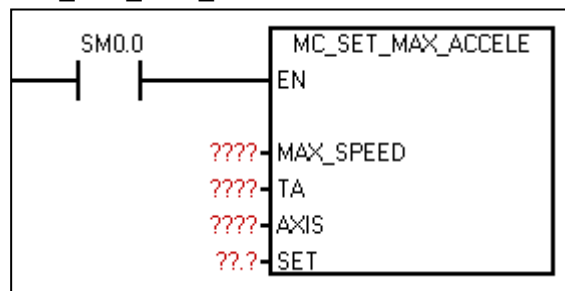
Use 0 axis to call this instruction. After the SET rising edge enables the external reset function and I0.2 detect the "effective reset signal", the system resets the axis 0 absolute coordinate, and also the STATUS setting reset complete. After the RESET rising edge prohibits the external reset function, even if I0.2 detect the "effective reset signal", the system will not reset the 0 axis absolute coordinate, and the STATUS reset instruct is non-reset state.

Note

The so-called "effective reset signal", the reset signal of each axis corresponding to an external IO, and the corresponding register set its effective level. For example, 0 axis corresponds to I0.2, and HSCO control register SM37.0 set the effective reset level of 0 axis. When set to 0, the effective reset signal of 0 axis is in I0.2 high level. When set to 1, the effective reset signal of axis 0 is in I0.2 low level; This setting is only effective if and only if the corresponding high-speed counter (0 axis corresponds to HSC0) is enabled, otherwise (no high-speed counter is enabled) the system default high level is the effective reset signal. If 0 axis, I0.2 high level is the effective reset signal. In the same way for other axes, the corresponding relation of the relevant control of each axis is shown in ② of this section.

Set maximum acceleration instruction

① Function name: MC_SET_MAX_ACCELE



② Function: Set maximum acceleration (= MAX_SPEED/TA; TA≠0), if the instruction isn't called, then the maximum acceleration has not been set.

③ Parameter

Name	I/O	Description	Type	Value range	Note
MAX_SPEED	IN	Maximum speed of long-axis run Unit: HZ	Dword		Modifiable in the process
TA	IN	Time of ACC Unit: ms	Dword	0~10000	Modifiable in the process

					Calculate acceleration only when startup and TA/TD changes
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte		
SET	IN	Enable reset bit Set absolute coordinate 0 at SET rising edge, SET set 0, then set 1 for each call	Bool		

④ Illustration

If set the X-axis parameter TA=0, or the X-axis does not call this instruction, it is considered that the X-axis does not set maximum acceleration; Otherwise, $MAX_ACCELE=MAX_SPEED/TA$ is considered to be set as the X-axis maximum acceleration. The significance of this instruction is:

a. Set an appropriate acceleration to limit the acceleration of each instruction on an axis.

Such as PTP instruction, set AXIS_NO=0, MIN_SPEED=1000, MAX_SPEED=11000, TA=500, then in theory the acceleration of PTP motion is 20HZ/ms ($= (MAX_SPEED-MIN_SPEED)/TA$); if 0 axis call MC_SET_MAX_ACCELE and set maximum acceleration 15HZ/ms, the PTP actual acceleration is 15HZ/ms (the same for MC_LINE_R and MC_CIRCLE_R)

b. An instruction on a certain axis is to obtain the maximum acceleration to run

If the PTP instruction run at the maximum acceleration, call MC_SET_MAX_ACCELE instruction on the coaxes to set the maximum acceleration (the parameter TA of MC_SET_MAX_ACCELE instruction cannot be 0, otherwise cannot obtain the maximum acceleration), and simultaneously set the parameter TA of PTP instruction to 0.

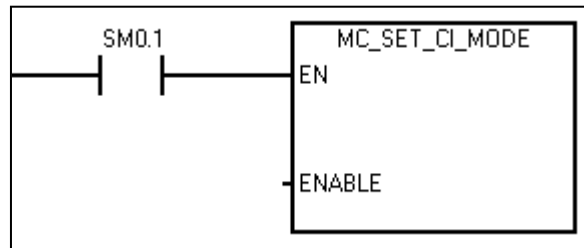
If the maximum acceleration is not set and the PTP instruction TA=0, the alarm parameter fails on the PTP instruction. (Same for MC_LINE_R and MC_CIRCLE_R)

Note: for the biaxial instructions MC_LINE_R and MC_CIRCLE_R, if both set the maximum acceleration, the smaller one will be taken as the maximum acceleration of the biaxial system. If only one axis is set with the maximum acceleration, then take it as the maximum acceleration of the biaxial system. If the maximum acceleration is not set on both axes, the biaxial system has no limitation for acceleration.

Also: after MAX_SPEED, TA, AXIS_NO have been determined, give SET a rising edge, so these parameters are updated and effective.

Set continuous interpolation instruction

- ① Function name: MC_SET_CI_MODE



- ② Function: set if enable continuous interpolation function

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
ENABLE	IN	0: Turn off continuous interpolation 1: Enable continuous interpolation	Bool	0~1	Do not modify in continuous interpolation process

- ④ Illustration

When need continuous interpolation function, call this instruction in PLC program first to activate the continuous interpolation function, and then call each interpolation instruction successively according to the trajectory sequence. Similarly, the RUN bit of each instruction must be activated according to the trajectory sequence. If not call this instruction, default as discontinuous interpolation movement.

Note

Note1: When one axis call the motion control instruction, the function of recovering normal IO can be restored only when the axis is reenergized and reset or the CPU is stopped.

Note2: Calculation of line interpolation instruction(A_POS, B_POS), circular interpolation (A_END_POS, B_END_POS, RADIUS):

POS-----Pulses needed in interpolation instruction;

L -----Absolute value on coordinate actual point, unit: mm;

S -----Lead screw on motion axis, unit: mm;

M----- The subdivision of the stepping driver or the resolution of the servo drive (the number of pulses required for each turn of the motor);

Formula: $POS = L * M \div S$ (unit: pulse number)

Note3: When $TA \neq 0$, $acceleration = (MAX_SPEED - MIN_SPEED) / TA$ (limited to the set maximum acceleration); If $TA = 0$, use instruction MC_SET_MAX_ACCELE to set the maximum acceleration; if it is not set, report the parameter failure. The same for TD. For the biaxial instruction, if both axes set maximum acceleration, take the smaller one as system acceleration.

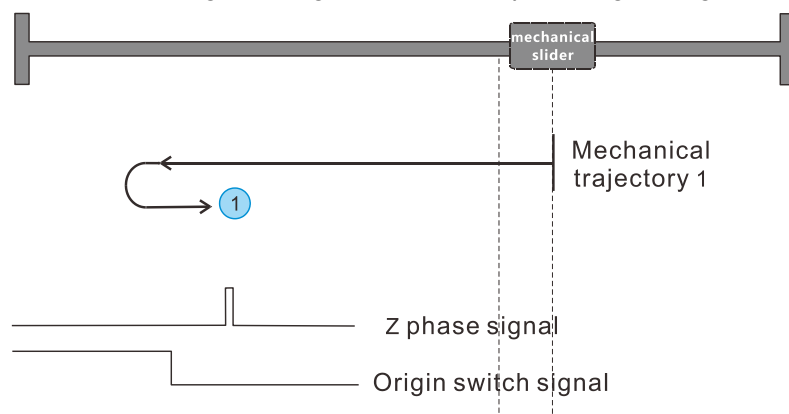
E.6 Back to Zero Mode Illustration

14 origin mode in all, select by your own request to precision and application

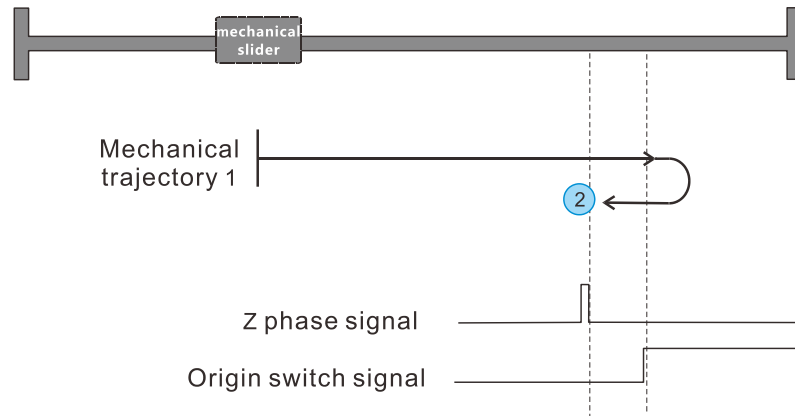
Mode	Definition
1	Refer to negative origin switch and Z phase signal origin mode
2	Refer to positive origin switch and Z phase signal origin mode
3	Only refer to origin mode of negative origin switch
4	Only refer to origin mode of positive origin switch
5	Only refer to origin mode of Z phase signal(negative back to zero)
6	Only refer to origin mode of Z phase signal(positive back to zero)
7	Refer to origin mode of origin switch , Z phase signal and positive limit(adopt Z phase signal of positive origin switch left of left edge)
8	Refer to origin mode of origin switch , Z phase signal and positive limit(adopt Z phase signal of positive origin switch right of left edge)
9	Refer to origin mode of origin switch , Z phase signal and positive limit(adopt Z phase signal of positive origin switch left of right edge)
10	Refer to origin mode of origin switch , Z phase signal and positive limit(adopt Z phase signal of positive origin switch right of right edge)
11	Refer to origin mode of origin switch , Z phase signal and negative limit(adopt Z phase signal of positive origin switch right of right edge)
12	Refer to origin mode of origin switch , Z phase signal and negative limit(adopt Z phase signal of positive origin switch left of right edge)
13	Refer to origin mode of origin switch , Z phase signal and negative limit(adopt Z phase signal of positive origin switch right of left edge)
14	Refer to origin mode of origin switch , Z phase signal and negative limit(adopt Z phase signal of positive origin switch left of left edge)

Regardless of the initial position of the machine, when the equipment (origin switch, positive route limit switch, negative route limit switch) installed, the origin of the equipment sought by the servo is always unique. The vertical line “|” in the schematic diagram of the following modes represents the initial position, and the circle “x” represents the origin position.

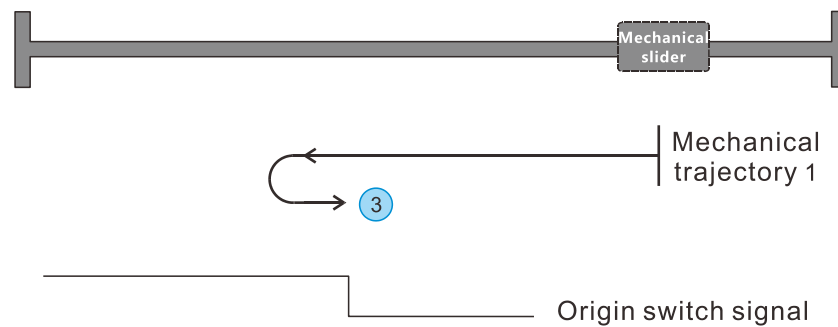
Origin mode 1: Refer to negative origin switch and Z phase signal origin mode



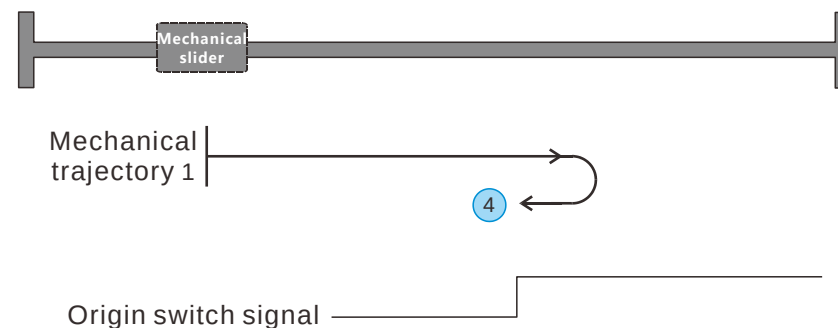
The origin switch located in the negative mechanical direction. The machine moves towards the origin switch, decelerates to stop after detecting the origin switch, then reverses to exit the origin switch, finds the next Z phase signal of the motor and records it as the origin, the motor stops immediately.

Origin mode 2: Refer to positive origin switch and Z phase signal origin mode

The origin switch located in the positive mechanical direction. The machine moves towards the origin switch, decelerates to stop after detecting the origin switch, then reverses to exit the origin switch, finds the next Z phase signal of the motor and records it as the origin, the motor stops immediately.

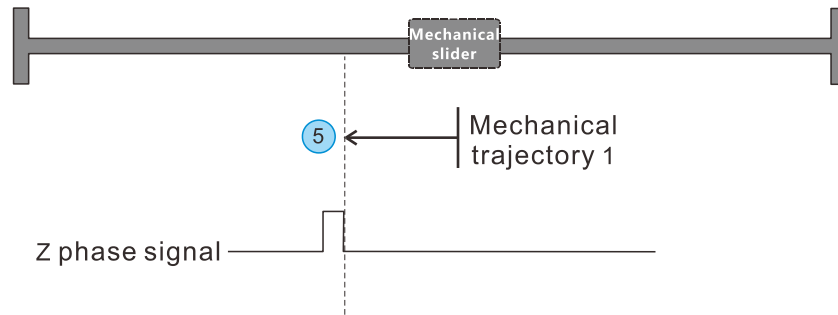
Origin mode 3: Only refer to origin mode of negative origin switch

The origin switch located in the negative mechanical direction. The machine moves towards the origin switch, decelerates to stop after detecting the origin switch, and then reverses to exit the origin switch. Find and record origin switch signal falling edge as the origin, and the motor stops immediately.

Origin mode 4: Only refer to origin mode of positive origin switch

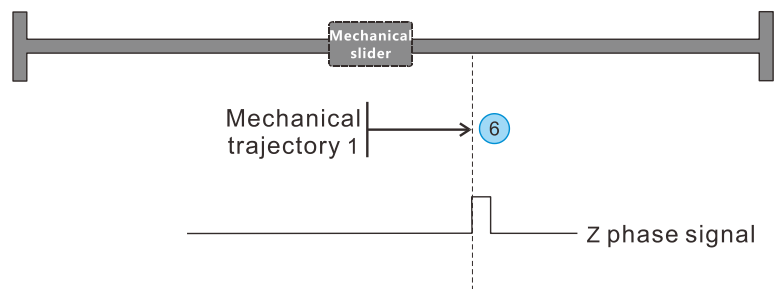
The origin switch located in the positive mechanical direction. The machine moves towards the origin switch, decelerates to stop after detecting the origin switch, and then reverses to exit the origin switch. Find and record origin switch signal falling edge as the origin, and the motor stops immediately.

Origin mode 5: Only refer to origin mode of Z phase signal (negative back to zero)



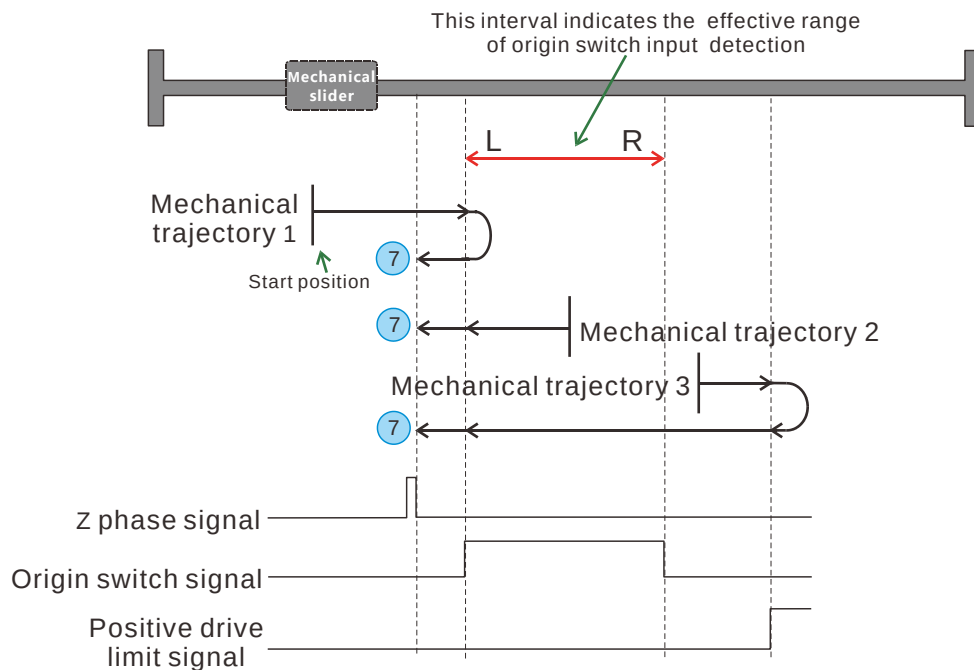
The motor moves along the negative direction from the current position and records it as the origin when the next Z phase signal is found.

Origin mode 6: Only refer to origin mode of Z phase signal (positive back to zero)



The motor moves along the positive direction from the current position and records it as the origin when the next Z phase signal is found.

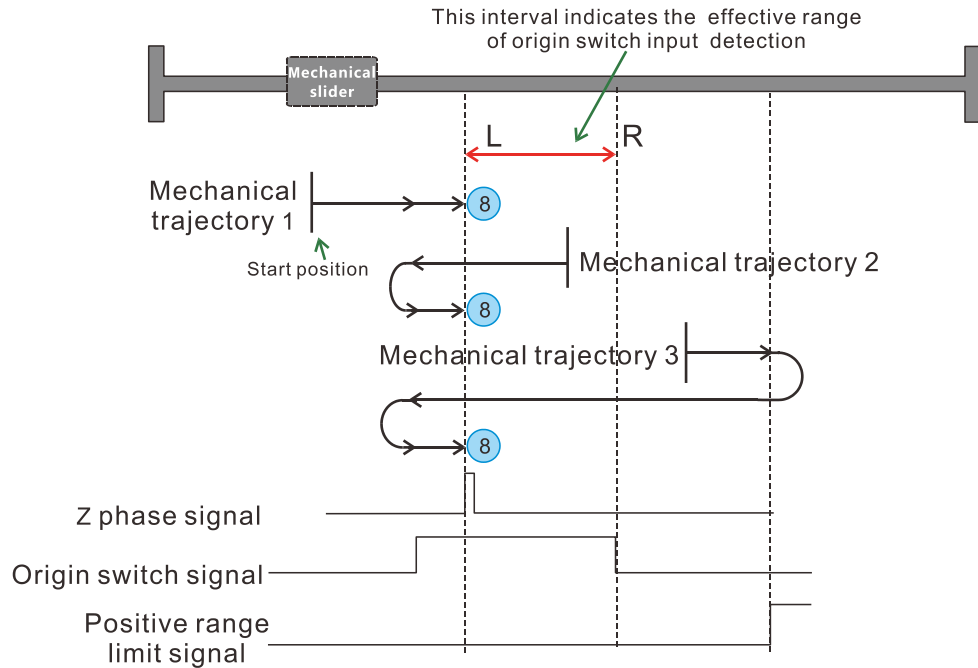
Origin mode 7: Refer to origin mode of origin switch, Z phase signal and positive limit (adopt Z phase signal of positive origin switch left of left edge)



As shown in the figure above, the mechanical slider slides in the positive limit direction (positive direction), and Z in the left position along the left side of the origin switch signal, that is, outside the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the negative direction. When the machine is outside the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (positive direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

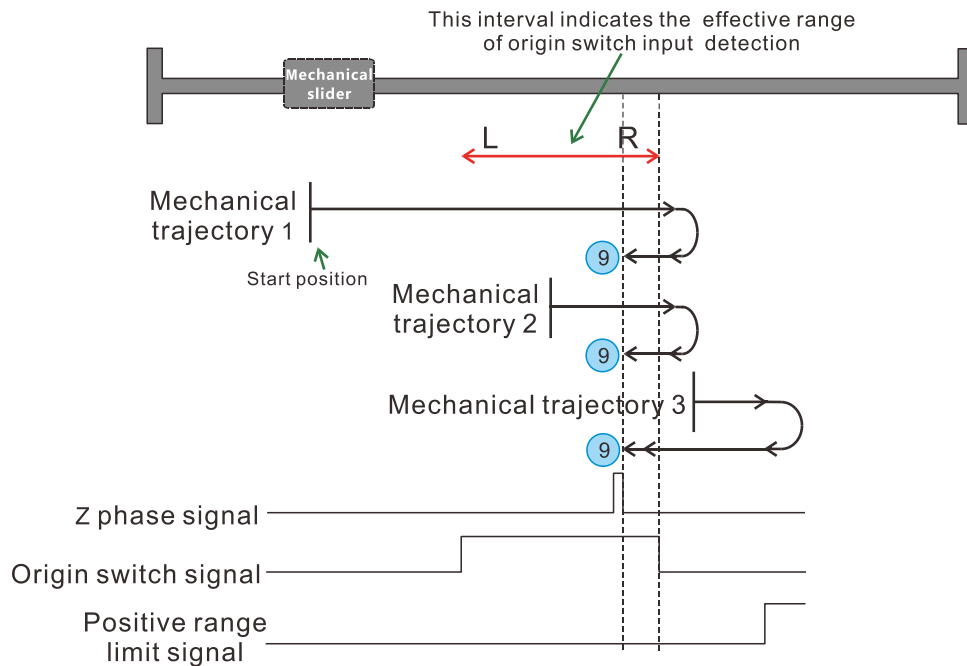
Origin mode 8: Refer to origin mode of origin switch, Z phase signal and positive limit (adopt Z phase signal of positive origin switch right of left edge)



As shown in the figure above, the mechanical slider slides in the positive limit direction (positive direction), and Z in the left position along the right side of the origin switch signal, the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the negative direction. When the machine is outside of the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (positive direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

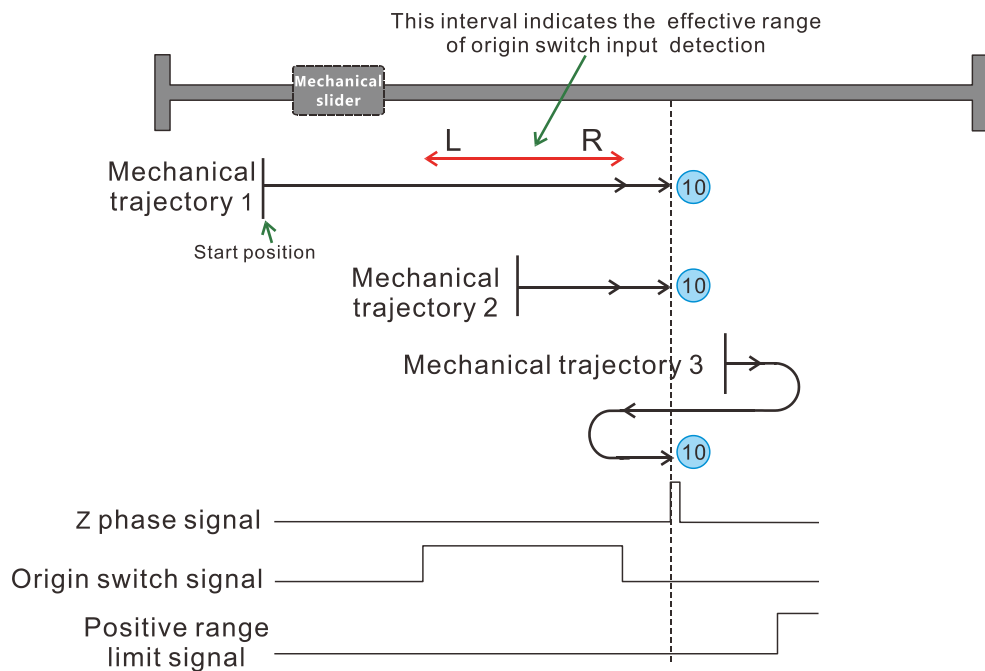
Origin mode 9: Refer to origin mode of origin switch, Z phase signal and positive limit (adopt Z phase signal of positive origin switch left of right edge)



As shown in the figure above, the mechanical slider slides in the positive limit direction (positive direction), and Z in the right position along the left side of the origin switch signal, the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the negative direction. When the machine is outside of the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (positive direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

Origin mode 10: Refer to origin mode of origin switch, Z phase signal and positive limit (adopt Z phase signal of positive origin switch right of right edge)

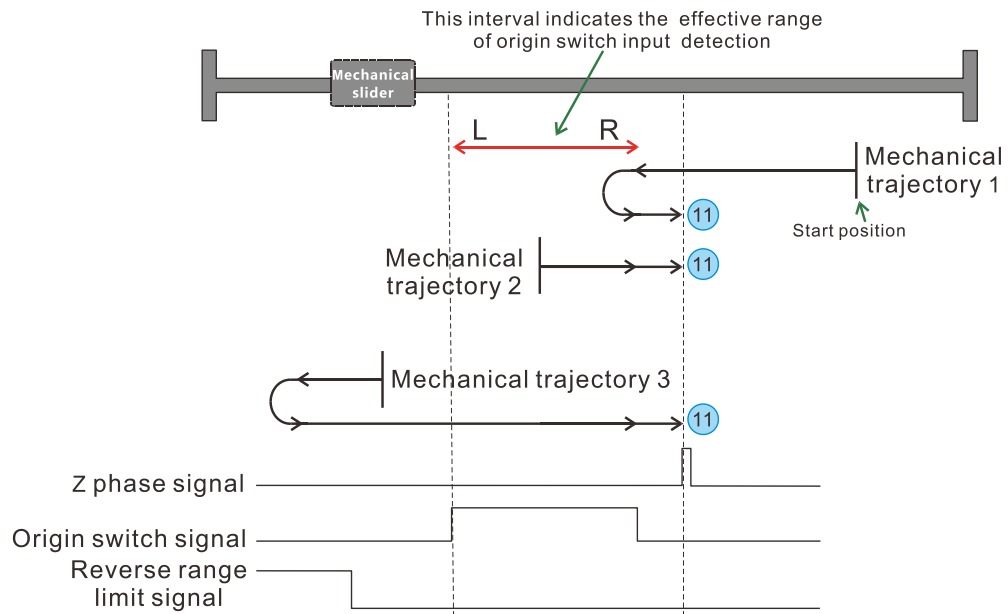


As shown in the figure above, the mechanical slider slides in the positive limit direction

(positive direction), and Z in the right position along the right side of the origin switch signal, the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the negative direction. When the machine is outside of the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (positive direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

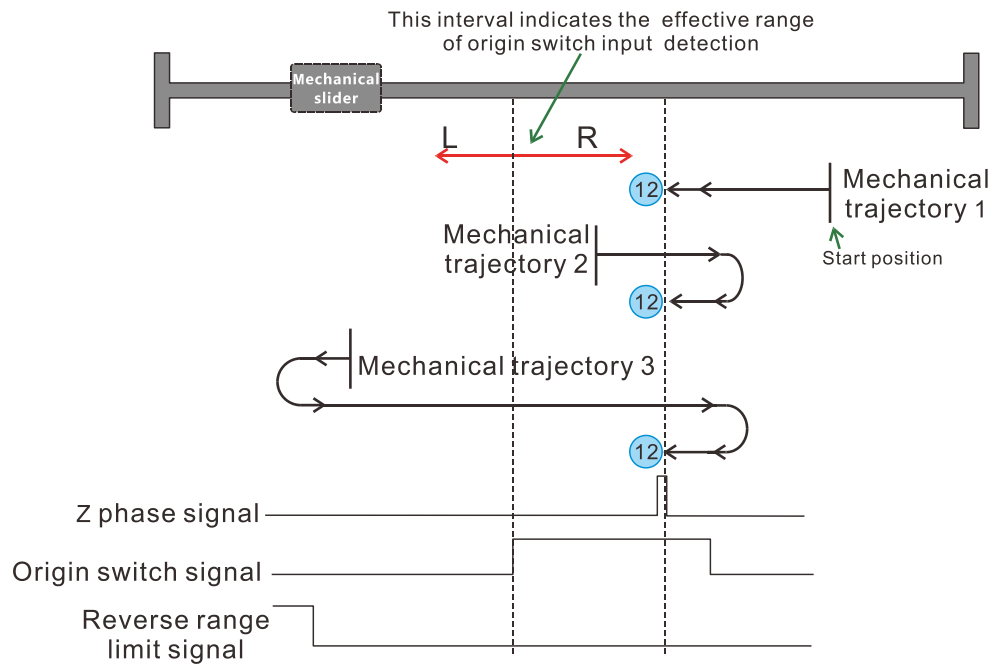
Origin mode 11: Refer to origin mode of origin switch, Z phase signal and negative limit (adopt Z phase signal of positive origin switch right of right edge)



As shown in the figure above, the mechanical slider slides in the negative limit direction (negative direction), and Z in the right position along the right side of the origin switch signal, outside of the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the negative direction. When the machine is outside of the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (positive direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

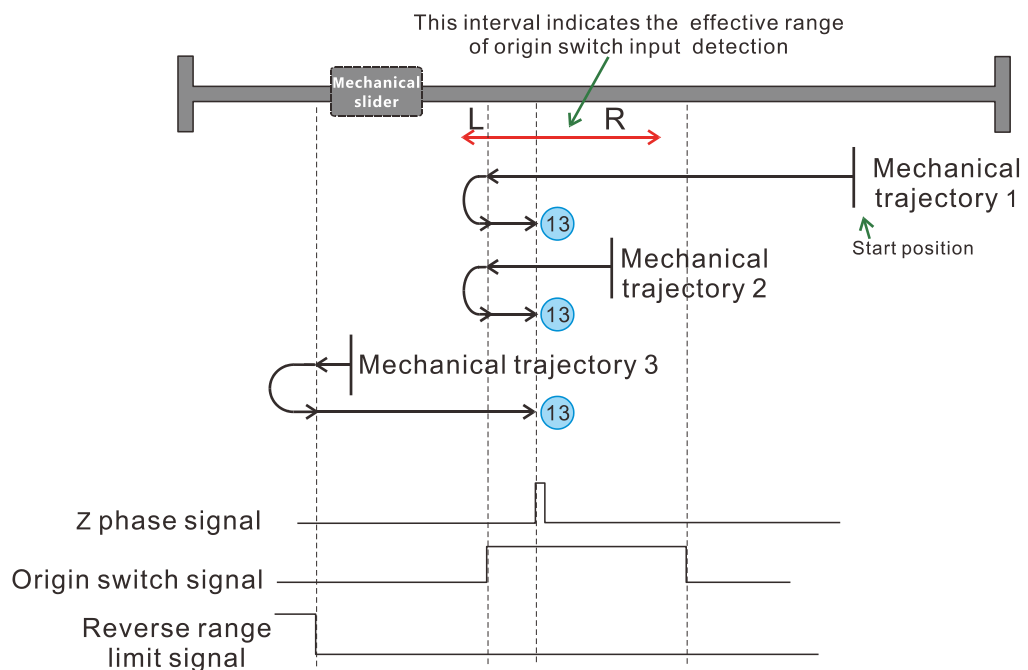
Origin mode 12: Refer to origin mode of origin switch, Z phase signal and negative limit (adopt Z phase signal of positive origin switch left of right edge)



As shown in the figure above, the mechanical slider slides in the negative limit direction (negative direction), and Z in the right position along the right side of the origin switch signal, inside of the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the positive direction. When the machine is outside of the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (negative direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

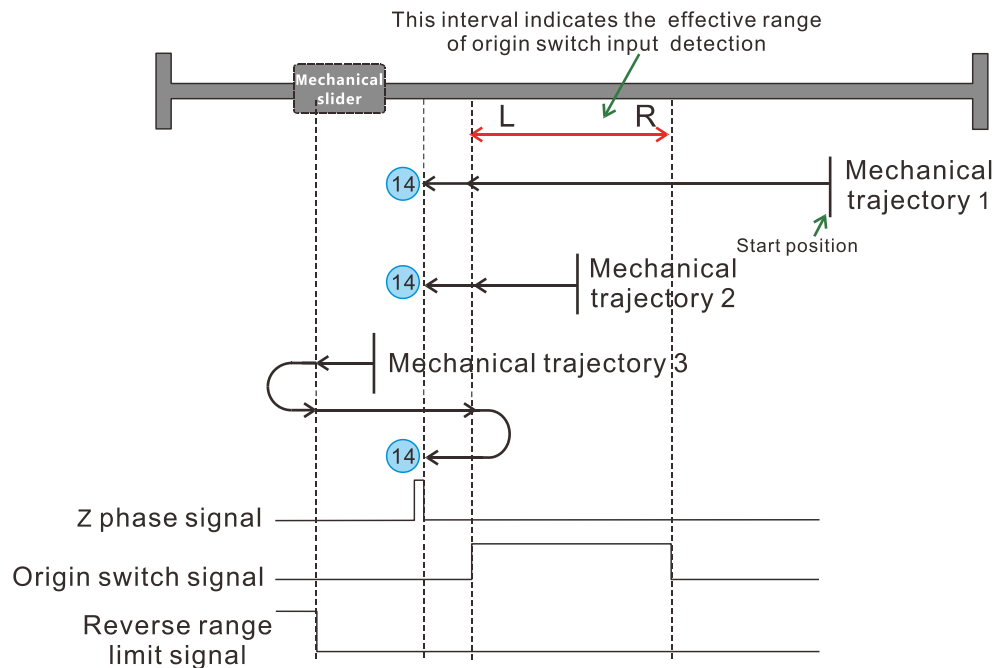
Origin mode 13: Refer to origin mode of origin switch, Z phase signal and negative limit (adopt Z phase signal of positive origin switch right of left edge)



As shown in the figure above, the mechanical slider slides in the negative limit direction (negative direction), and Z in the left position along the right side of the origin switch signal, inside of the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the negative direction. When the machine is outside of the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (negative direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

Origin mode 14: Refer to origin mode of origin switch, Z phase signal and negative limit (adopt Z phase signal of positive origin switch left of left edge)



As shown in the figure above, the mechanical slider slides in the negative limit direction (negative direction), and Z in the left position along the left side of the origin switch signal, inside of the effective range of the origin switch signal.

When the machine is within the origin switch range (mechanical motion trajectory 2), the origin can be found by directly running in the negative direction. When the machine is outside of the origin switch (mechanical motion track 1 and 3), the machine runs in a constant direction (negative direction) toward the limit switch. According to the detected sequence of the origin switch and limit switch, find the movement track and the origin.

E.7 Examples

Case 1) Usage of uniaxial relative motion instruction

PROGRAM COMMENTS

Function: use for uniaxial point to point control (uniaxial fixed length drive) output fixed pulse with one call, by setting the maximum, minimum speed and acceleration time, the output pulse will gradually accelerate to the maximum speed when starting up. When the pulse is about to run out, the pulse frequency will automatically reduce to prevent vibration or jamming caused by too much inertia of the machine when starting or stopping.

M0.0-----emergency stop bit

Axis 0-----Q0.0 is pulse output, Q0.1 is direction output

VD0-----start/stop speed

VD4-----normal speed after acceleration

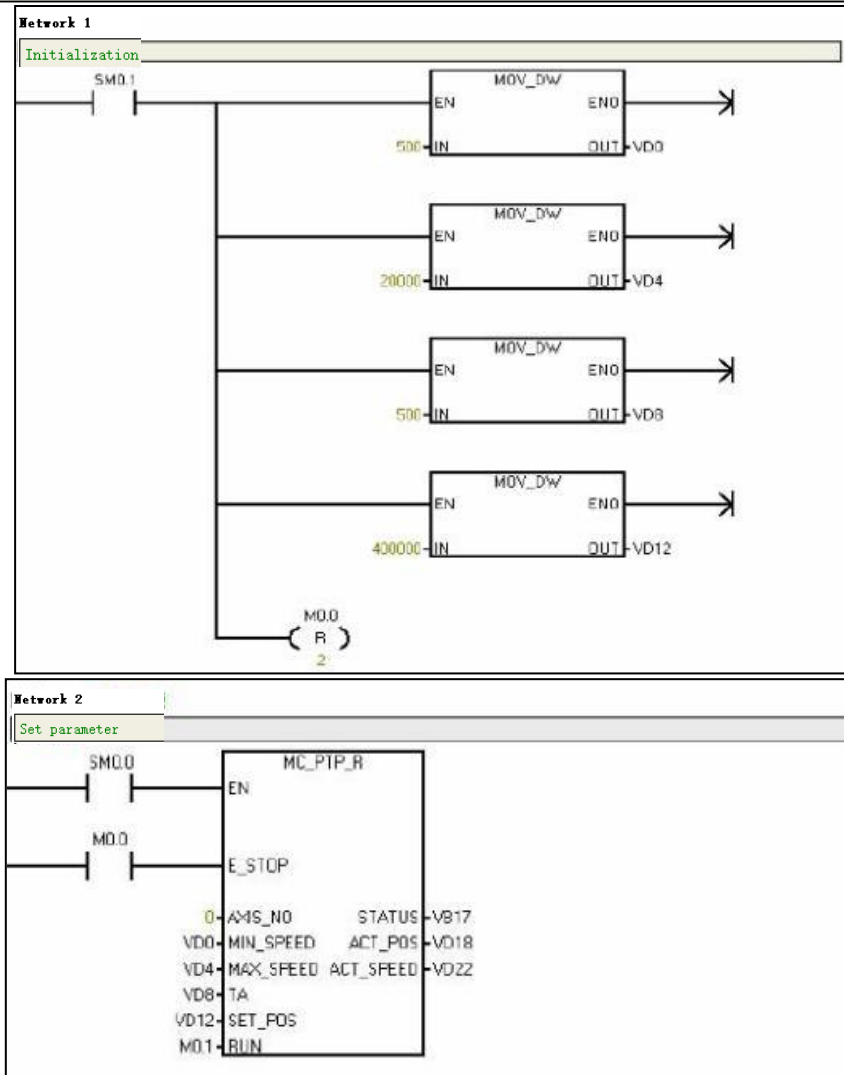
VD8-----acceleration time(ms)

VD12-----pulse number to output

VB17-----output status byte

VD18-----output pulse number

VD22-----current output pulse speed(frequency)



Case 2) Usage of uniaxial speed motion instruction

PROGRAM COMMENTS

Function: control uniaxial output pulse frequency, you can change the frequency anytime(speed).

When receiving the soft stop command, it will automatically reduce to stop. when receiving emergency stop command, it will immediately stop to output pulse without deceleration.

M0.0-----enable run bit

M0.1-----emergency stop bit

M0.2-----soft stop bit

M0.3-----pulse direction bit(0 negative, 1 positive)

Axis 0-----Q0.0 is pulse output, Q0.1 is direction output

VD0-----start/stop speed

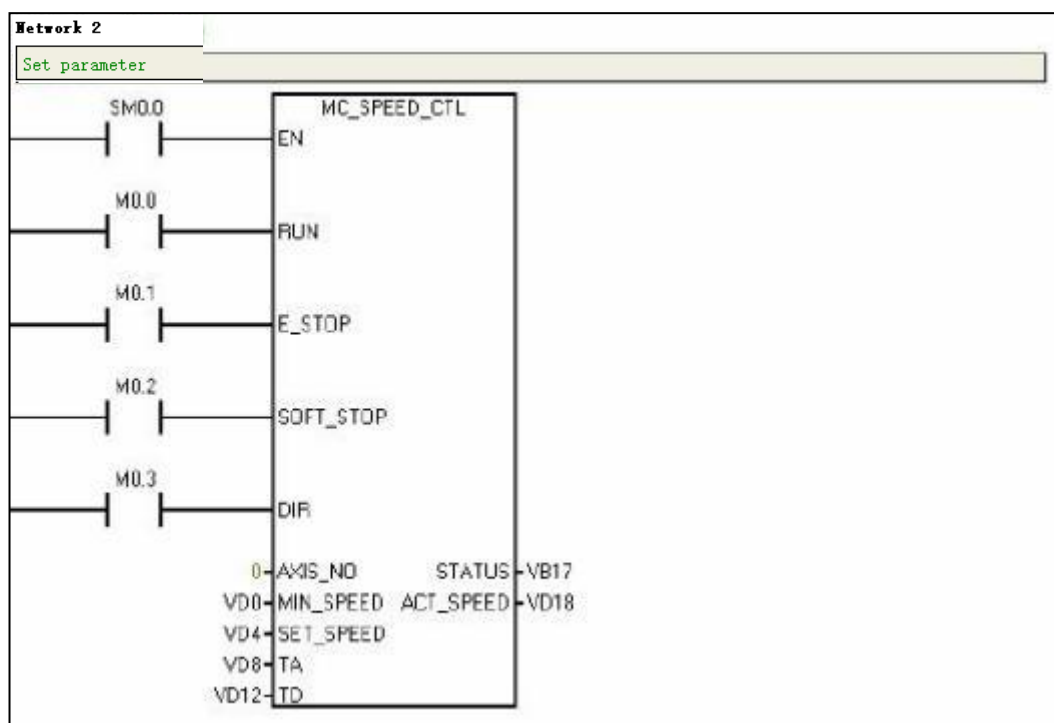
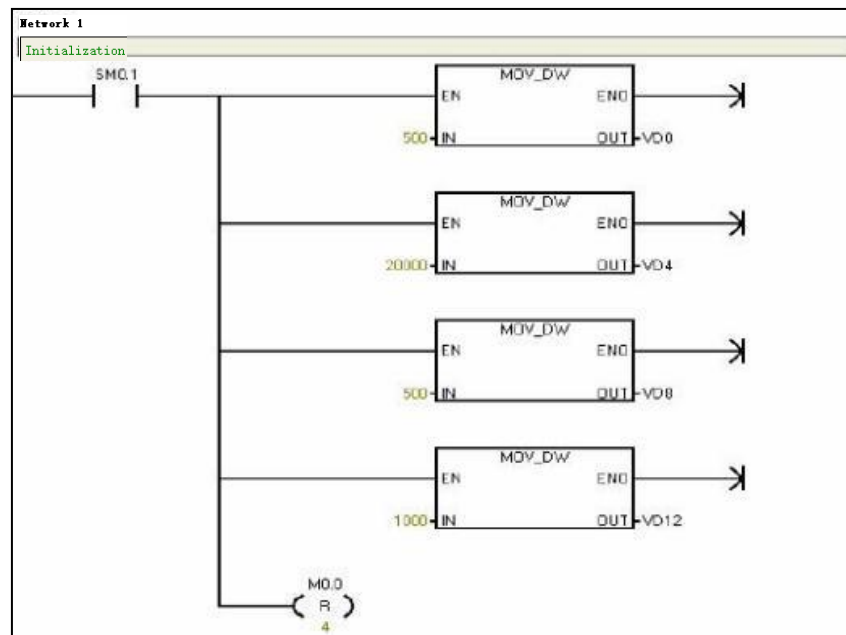
VD4-----normal speed after acceleration

VD8-----acceleration time(ms)

VD12----- deceleration time(ms)

VB17-----output status byte

VD18-----current output pulse speed(frequency)



Case 3) Usage of two-axis line interpolation motion instruction

PROGRAM COMMENTS

Function: line interpolation in any plane area and any two axes.

MO.0-----emergency stop bit

Axis 1-----Q0.2 is pulse output, Q0.3 is direction output

Axis 3-----Q0.5 is pulse output, Q0.7 is direction output

VD0-----long axis start/stop speed

VD4-----normal speed after long axis acceleration

VD8-----ACC/DCC time(ms)

VD12----- virtual A axis terminus (relative) coordinate

VB16-----virtual B axis terminus (relative) coordinate

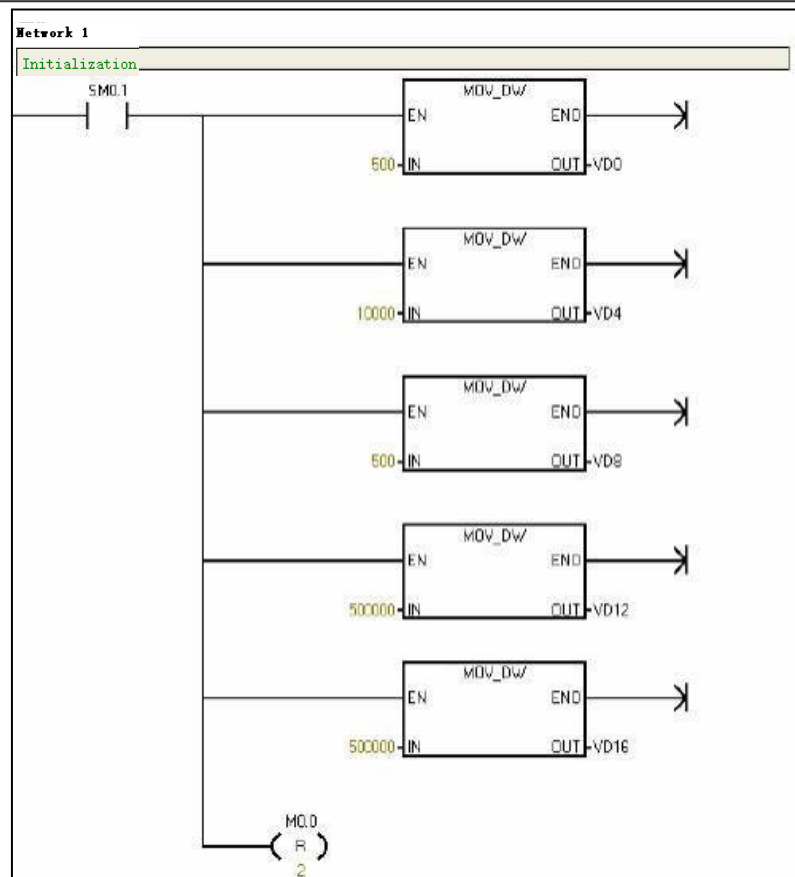
MO.1-----enable run bit

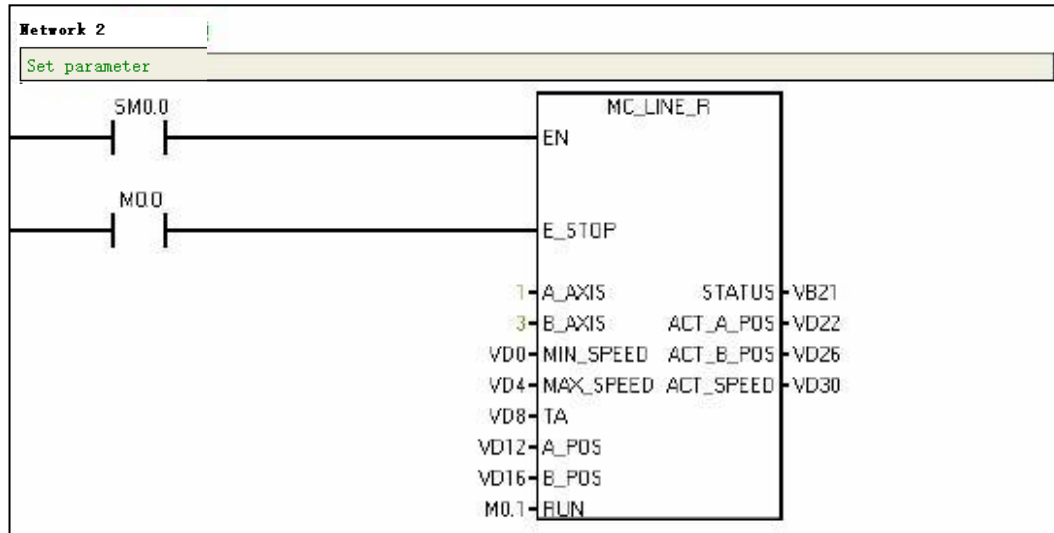
VB21-----output status byte

VD22-----A axis current position(relative coordinate, actual output pulse number for this call)

VD26-----B axis current position(relative coordinate, actual output pulse number for this call)

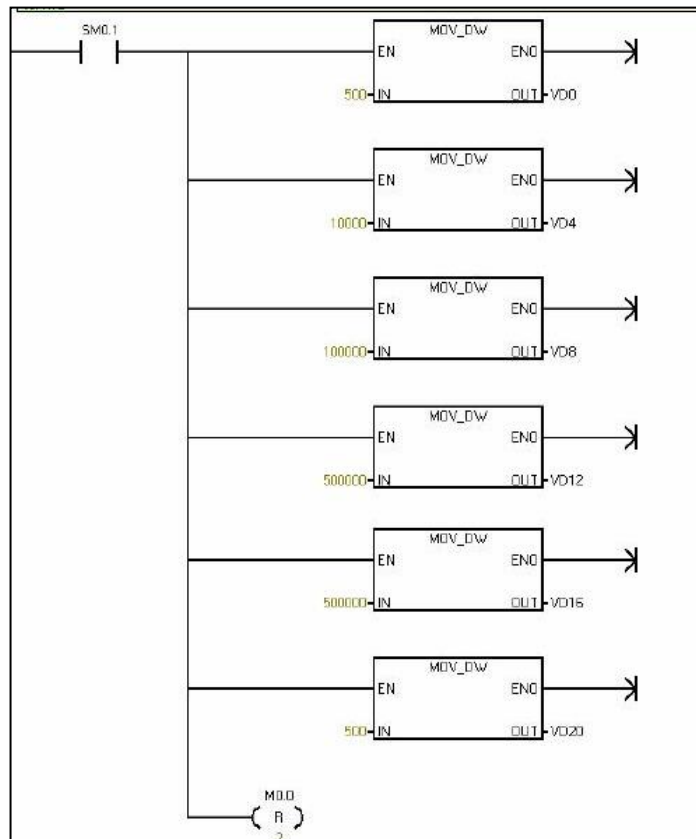
VD30-----current actual speed(frequency)

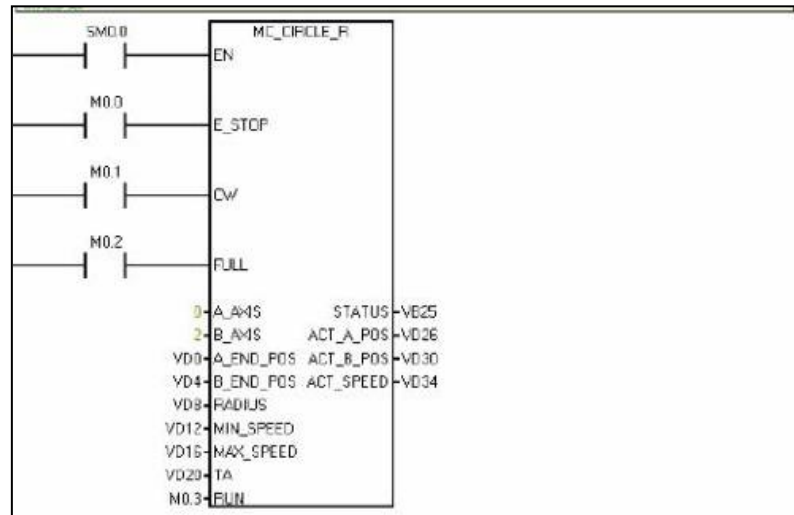




Case 4) Usage of two-axis arc interpolation motion instruction

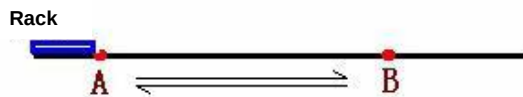
PROGRAM COMMENTS
 Function: circular interpolation in any plane area and any two axes.
 M0.0-----emergency stop bit
 M0.1-----clockwise or counterclockwise interpolation sign bit
 M0.2-----full circle sign bit(1-full circle, 0-arc)
 Axis 0-----Q0.0 is pulse output, Q0.1 is direction output
 Axis 2-----Q0.4 is pulse output, Q0.5 is direction output
 VD0-----long axis start/stop speed
 VD4-----normal speed after long axis acceleration
 VD8-----arc radius
 VD12----- virtual A axis terminus (relative) coordinate
 VB16-----virtual B axis terminus (relative) coordinate
 VD20-----ACC/DCC time(ms)
 M0.3-----enable run bit
 VB25-----output status byte
 VD26-----A axis current position(relative coordinate, actual output pulse number for this call)
 VD30-----B axis current position(relative coordinate, actual output pulse number for this call)
 VD34-----current actual speed(frequency)





Case 5) Usage of mixed motion control instruction

The stepping motor move back and forth from point A to B. subdivide into 1000, the lead screw is 5mm, and the displacement L from A to B is 2000mm.



※ System illustration

In this example, set CPU226H 0 axis parameter of point to point motion mainly by MC_PTP_R.

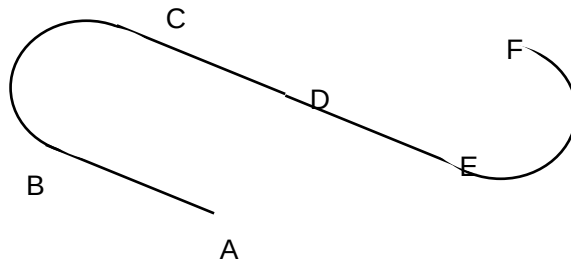
I0.2 is the point A hardware zero reset point (input of the stroke switch quantity, set it mechanical origin)

I2.0 is emergency stop input

Q0.0 is pulse output, Q0.1 is direction output

Case 6) set continuous interpolation instruction

Move in continuous mode (AB → BC → CD), then in discontinuous mode (first line DE, then arc EF)



F Using EM231 Weighing Module Library Function

Weight module converts the voltage signal of weighing sensor to digital AIW, the function of EM231 weight module is to change AIW to actual weight.

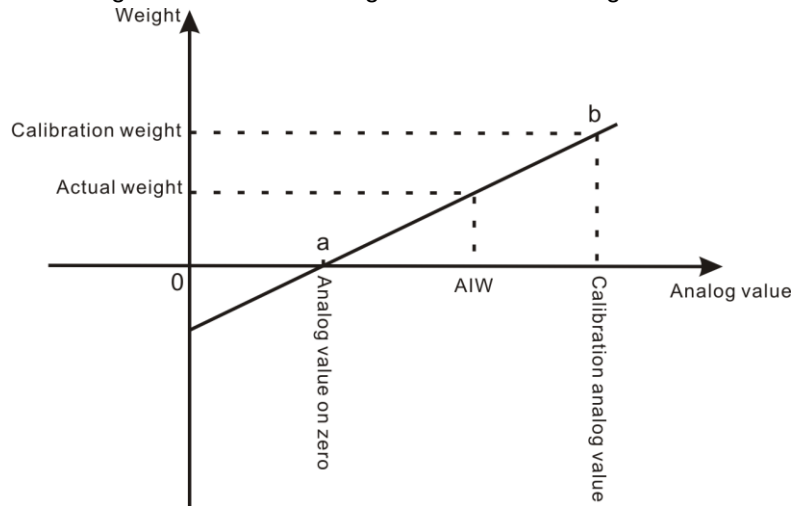


Figure F-1: The principle of measuring weight

You can obtain the linear relationship equation between weight and analog value via point a and b from above diagram, then you can get the actual weight of AIW. So as to get the coordinate value of point a and b, you need adjust zero and calibrate.

Zero Setting and Calibration method:

First suppose the weighting sensor has fixed to the horizontal direction, and the pallet has horizontally fixed on the weighting sensor.

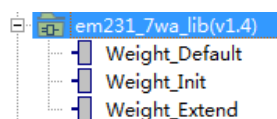
When setting zero, don't put anything on the the pallet, you can get the Zero-analog value while it is stable.

When calibrating, the weights for calibrating are placed to the pallet, you can get the calibration analog value while it is stable.

F.1 Weighing Library Function Illustration

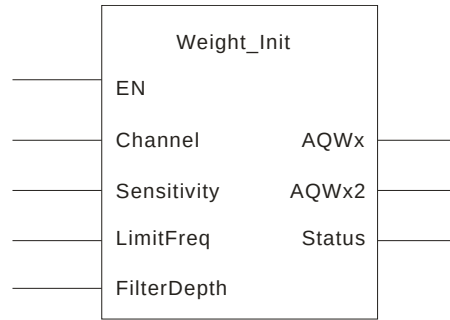
Weighing module correspond to weighing library EM231_7WA_LIB (V1.4), library files download address: www.co-trust.com

Weighing library instructions parameter:



Weighing library configuration

- ① Name: Weight_Init

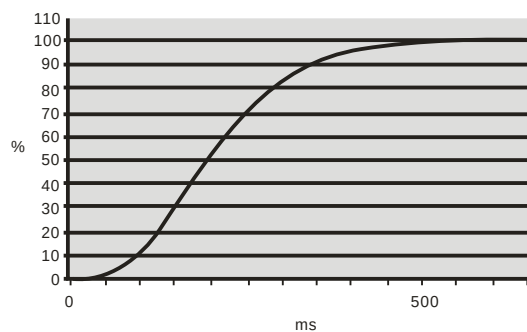


② Function: Configure sensitivity, low pass filter cutoff frequency and average filter depth of sensors which connect to weighing module.

③ Parameter

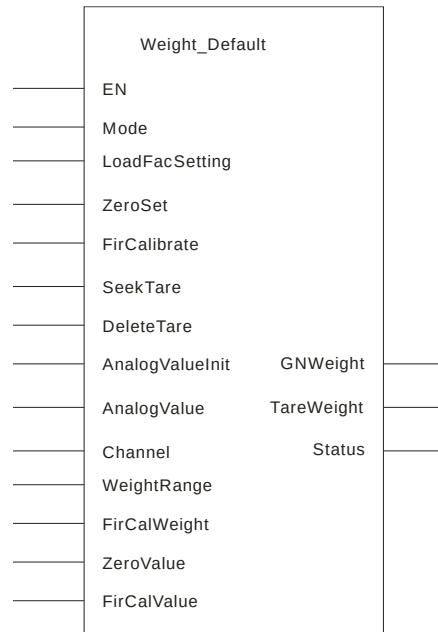
Name	Description	I/O	Type	Value range	Note
Channel	Channel number	IN	BYTE	-	0~6
Sensitivity	Sensor sensitivity	IN/OUT	BYTE	2	1: Eigenvalue range 1mV/V 2: Eigenvalue range 2mV/V 4: Eigenvalue range 4mV/V No other definition
LimitFreq	Low pass filter cutoff frequency	IN/OUT	BYTE	4	3: fg = 5Hz 4: fg = 2Hz 5: fg = 1Hz 6: fg = 0.5Hz 7: fg = 0.2Hz 8: fg = 0.1Hz 9: fg = 0.05Hz No other definition
Filter Depth	Average filter depth	IN/OUT	BYTE	15	0~255, 0 or 1 indicates no average filter
AQWx	First analog output of weighing channel	OUT	WORD	--	Format: "0x53"+"Sensitivity"
AQWx2	Second analog output of weighing channel	OUT	WORD	--	Format: "LimitFreq"+"FilterDepth"
Status	Status byte	OUT	BYTE	--	Bit0: eigenvalue error Bit1: Low pass filter frequency setting error

Figure below shows step response of digital low pass filter when fg=2Hz



Weighing standard library

① Name: Weight_Default



② Function: Calibration, zero set and measurement

③ Parameter

Name	Description	I/O	Type	Value range	Note
Mode	Select mode, 0 is basic mode, 1 is expansion mode	IN	BOOL	--	--
LoadFacSetting	Load factor setting, effective in rising edge	IN	BOOL	--	--
ZeroSet	Set zero point, effective in rising edge	IN	BOOL	--	--
FirCalibrate	First calibration, effective in rising edge	IN	BOOL	--	--
SeekTare	Tare, effective in rising edge	IN	BOOL	--	--
DeleteTare	Delete tare, effective in rising edge	IN	BOOL	--	--
AnalogValueInit	Analog value before filter, first analog output of weighing channel	IN	WORD	--	
AnalogValue	Analog value after filter, second analog output of weighing channel	IN	WORD	--	
Channel	Channel number	IN	BYTE	--	0-6
WeightRange	Maximum weigh range	IN/OUT	INT	2000	
FirCalWeight	Weights for once calibration	IN/OUT	INT	2000	> 5% of range

ZeroValue	Zero sampling value	IN/OUT	WORD	5461	
FirCalValue	Sampling value for once calibration	IN/OUT	WORD	60074	
GNWeight	Rough weight/net weight	OUT	INT	--	
TareWeight	Tare	OUT	INT	--	
Status	Status	OUT	WORD	--	

Load factor setting parameter

Parameter	Factory default	Note
Sensitivity	2	Eigenvalue 2mv/v
LimitFreq	4	Low pass filter frequency 2Hz
FilterDepth	15	
WeightRange	2000	
FirCalWeight	2000	
ZeroValue	5461	
FirCalValue	60074	
TWProcessValue	0	
SecCalWeight	0	
MinWeight	20	20d, d is digital step
Step	1	
StandstillTime	1000	Unit: ms
StandstillRange	10	
TareInput	0	
SecCalValue	0	

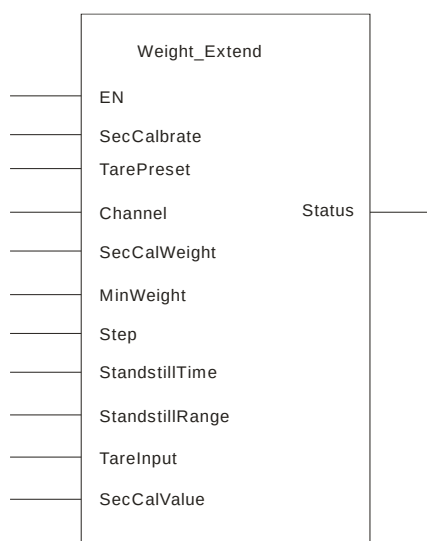
Status note:

Bit	Name	Range	Description
Bit0	Power-down alarm	Basic/expansion mode	0: Module power normal, 1: Module no power. When Bit0=1, sampling value is 0xFFFF
Bit1	Break line alarm	Basic/expansion mode	0: Sensor connection normal, 1: sensor disconnect When Bit1=1, sampling value is 0xFFFE
Bit2	Outrange alarm	Basic mode	1: Rough weight $\geq$ rated weight, sampling value is 0xFFFD
	Max+9e	Expansion mode	1: Rough weight $\geq$ rated weight +9e. e is digital step
Bit3	Tare decided	Basic/expansion mode	1: Tare storage has been occupied(tare process value $\neq$ 0)
Bit4	Tare preset	Expansion mode	1: Preset tare
Bit5	1/4d	Expansion mode	1: Rough weight $\leq \pm 0.25d$, d is digital step
Bit6	Pause	Expansion mode	1: Ensure pause status
Bit7	Zero be found	Basic/expansion mode	1: Find zero already
Bit8	Calibrated	Basic mode	1: Complete once calibration

		Expansion mode	1: Complete first calibration (second calibration weight 0) or complete first and second calibration
Bit9	Low weights	Expansion mode	1: Current weight < minimum weighing range
Bit10	Execute only in pause status	Expansion mode	1: Find zero and tare must be paused in expansion mode
Bit11	Execute only in zero status	Basic/expansion mode	1: Zero setted before first or second calibration
Bit12	Execute only in calibrated status	Basic/expansion mode	1: Calibrated before measure or preset tare
Bit13	Calibrated weight too small	Basic/expansion mode	1: First calibration weight and zero; the weighing difference between second and first calibration cannot be less than 5%FS
Bit14	Illegal tare	Basic/expansion mode	1: Tare cannot be less than 0 or more than maximum weighing range

Weighing extension library

① Weight_Extend



② Function: Realize second calibration, set minimum weight, digital phase step, pause status detection and preset tare.

③ Parameter

Name	Description	I/O	Type	Value range	Note
SecCalibrate	Second calibration, effective rising edge in	IN	BOOL	--	--
TarePreset	Preset tare	IN	BOOL	--	--
Channel	Channel number	IN	BYTE	--	0-6
SecCalWeight	Weight of	IN/OUT	INT	0	0 indicates no second

	second calibration				calibration
MinWeight	Minimum weight	IN/OUT	INT	20	This weight value can only use for calibration records with specified digital steps above the minimum weight. The minimum weight depends on the type of sensor.
Step	Digital phase step	IN/OUT	BYTE	1	Range is 1, 2, 5, 10, 20, no other definition
StandstillTime	Pause time	IN/OUT	INT	1000	Unit: ms
StandstillRange	Pause range	IN/OUT	INT	10	
TareInput	Tare input	IN/OUT	INT	0	Preset tare
SecCalValue	Second calibration analog value	IN/OUT	WORD	0	
Status	Status byte	OUT	BYTE	--	

Status note:

Bit	Name	Note
Bit0	Weight of second calibration is too small	The weighing difference between second and first calibration cannot be less than 5%FS
Bit1	Digital phase step error	Digital phase step can only be 1, 2, 5, 10, 20
Bit2	Pause time error	Pause time must>0
Bit3	Pause range error	Pause range must>0
Bit4	Preset tare outrange	Preseted tare cannot be negative or outrange
Bit5	Reserved	--
Bit6	Pause	Ensure pause status

F.2 Weighing Library Mode Illustration

Weighing module SM231-7WA offer library usage of 2 modes by configure parameter in "Weight_Default" "Mode" to select basic mode or expansion mode.

1) Basic mode (Mode=0)

(1) Call "Weight_Init" and "Weight_Default"2;

(2) Supported functions:

- Set sensor sensitivity, low pass cutoff frequency and filter depth;
- Load factory parameter;
- Set maximum weighing range;
- Set zero;
- First calibration;
- Measure tare;

- Delete tare;
- Read sampling value before and after filter;
- Power-down alarm, line broke alarm, outrange alarm and set zero etc.

2) Expansion mode (Mode=1)

(1) Call "Weight_Init", "Weight_Default" and "Weight_Extend";

(2) Supported functions:

- Supported all functions in basic mode;
- Set minimum weighing range;
- Second calibration;
- Preset tare;
- Set Digital phase (set minimum range scale, show in 1×10^k , 2×10^k or 5×10^k)
- Ensure pause status (set pause status when setting zero and tare)
- Newly added status: Max+9e (GB/T 7724-2008), Preseted tare, 1/4d (GB/T 23111-2008), pause, low weight;

Parameter table

Calling the weighing library requires 322 bytes of V storage address. The parameters definition shows in the following table (take the starting address VB0 as an example):

Name	Address	Illustration	Note
Mode	VB0	Mode	Basic/expansion
Sensitivity	VB1	Sensor sensitivity	
LimitFreq	VB2	Low pass filter frequency	
FilterDepth	VB3	Filter depth	0-255, 0 or 1 indicates no average filter
WeightRange	VW4	Maximum weighing range	
FirCalWeight	VW6	First calibration weight	
SecCalWeight	VW8	Second calibration weight	
TareInput	VW10	Tare input	
MinWeight	VW12	Minimum weighing range	Normally 20d, d is Digital phase step
Step	VB14	Digital phase step	Range: 1, 2, 5, 10, 20
StandstillTime	VW15	Pause time	Unit: ms
StandstillRange	VW17	Pause range	
ZeroValue	VW19	Zero sampling value	
FirCalValue	VW21	Sampling value of first calibration	
SecCalValue	VW23	Sampling value of	

		second calibration	
GWProcessValue	VW25	GrossWeight process value	GrossWeight before integer
NWProcessValue	VW27	NetWeight process value	NetWeight before integer
TWProcessValue	VW29	TareWeight process value	TareWeight before integer
AnalogValueInit	VW31	Sampling value before filter	
AnalogValue	VW33	Sampling value after filter	
GrossWeight	VW35	GrossWeight	
NetWeight	VW37	NetWeight	
TareWeight	VW39	TareWeight	
Status_I	VB41	"Weight_Init" status byte	
Status_D	VW42	"Weight_Default" status byte	
Status_E	VB44	"Weight_Extend" status byte	
InternalVariable1	VB45	Internal variable 1	
Channel 1 parameter table	VB46~VB91	Define reference channel 0	
Channel 2 parameter table	VB92~VB137	Define reference channel 0	
Channel 3 parameter table	VB138~VB183	Define reference channel 0	
Channel 4 parameter table	VB184~VB229	Define reference channel 0	
Channel 5 parameter table	VB230~VB275	Define reference channel 0	
Channel 6 parameter table	VB276~VB321	Define reference channel 0	

G Using EM277A Module

【Main feature】

- Adopt a photoelectric isolation technique, high reliability and strong anti-interference ability.
- Integrated terminal resistance, the bus adopt connection mode, no need dedicated network tap.
- The power supply includes anti - access protection and Surge absorption function, it's suitable for bad industrial environment.

【Terms of use】

- The signal wire should use shielded twisted pair wires, and the double end of the shielded wire should connect to the ground.
- The grounding terminal should connect to the earth wire while the system is under good grounding cases.
- If it's the last station, the terminal resistance should set to the ON position.

PROFIBUS-DP network

【Communication】

PROFIBUS-DP network connect to EM277A DP slave station module by its DP communication port, while EM277A DP slave station module connect to CTSC-200 CPU through serial I/O bus. EM277A use terminal to connect PROFIBUS network without using standard connector.

【Function】

EM277A is a modular slave station device for PROFIBUS DP networks that can connect up to 6 extended modules for digital or analog I/O.

As a DP slave station, the EM 277A module can run on any PROFIBUS baud rate between 9600 and 12M. Also receives a variety of I/O configurations and different amounts of data from the main station and sends data to the main station. The EM 277A can also read and write variable data blocks defined in the CTSC-200 CPU, allows user to exchange any type of data with master station.

MPI network

【Communication】

The EM277A module communicate with other master stations on the same network such as PG/PC programming station, Copanel HMI or S7-300/S7-400CPU as an MPI slave station. Use XGET/XPUT instruction function of S7-300/400 and other main station modules to provide the network communication between MPI main station and CTSC-200. When the EM277A module conducts MPI communication, the MPI master station must use its station address to send information to the CTSC-200 CPU. The MPI information

sent to the EM277A module will be transmitted to the CTSC-200 CPU via EM277A.

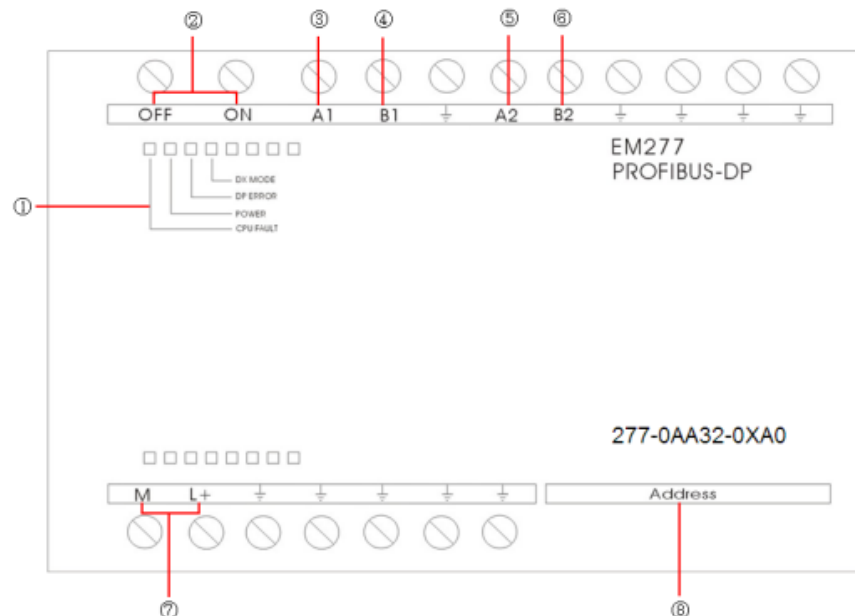
【Function】

EM277A is a slave station device for MPI networks that can connect up to 6 extension modules for digital or analog I/O.

As MPI slave station, the EM 277A module can be run at any MPI baud rate between 9600 and 12M. Also receives a variety of different I/O configurations and different amounts of data from the master station and sends data to the master station, and EM 277A can also read and write variable data blocks defined in the CTSC-200 CPU, allows the user to exchange any type of data with the master station.

Components of EM 277A

Status LED is located on the front of the module, address switch, terminal resistance switch, DP slave interface terminal and user power supply are located on the side of wiring terminal, see as following figure:



- Status LED
- Terminal resistance switch: ON means with terminal resistance, OFF means without terminal resistance.
- Isolated signal A
- Isolated signal B
- Isolated signal A
- Isolated signal B
- User Power supply
- Address switch: Setup by 8 bits dial switch, and expressed in binary form, the available range is 0~126.



Note

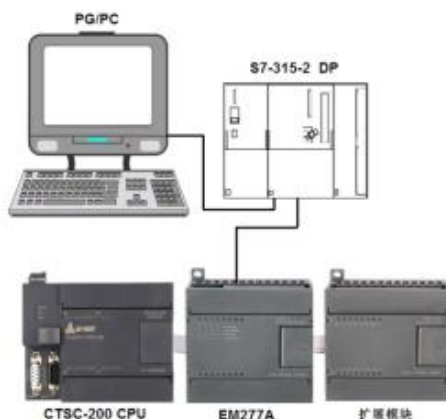
A1/B1 and A2/B2 are both RS-485 electric port, consistent internal electric characters, you can choose to connect one or two group.

PROFIBUS DP communication between S7-300 and EM277A of CTSC-200

PROFIBUS DP communication between S7-300 and EM 277A of CTSC-200 need S7-300 station configuration in STEP7, no need to configure and program the communication in CTSC-200 system, just sort out the data to be communicated and store it in the V storage area corresponding to the hardware I/O address of S7-300 configuration EM 277A slave station.

Write program to call FC1 (DP_SEND) and FC2 (DP_RECV) in OB1, DP master can read and write from the slave station data to achieve S7-300 and CTSC-200 CPU communication. Execute the DP_SEND instruction to output the memory data of CTSC-200 CPU to the EM277A extension module. By performing DP_RECV, input the EM277A extension module data to the memory of CTSC-200 CPU.

Figure G-1 shows a PROFIBUS network with CTSC-200 CPU and EM277A DP slave station module.



- Use S7-300 with CPU 315-2 as the DP master station, and configure with STEP 7 programming software.
- CTSC - 200 CPU is DP slave station of CPU 315-2 DP master station.
- CPU 315-2 DP master station use DP_SEND and DP_RECV of user program to read and write data from CTSC-200.

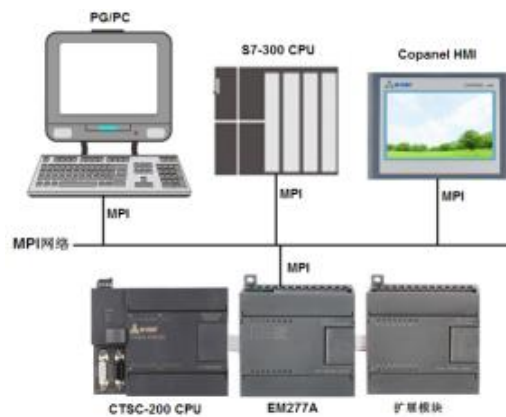
Figure G-1 PROFIBUS network example

Note: To use EM277A as a DP slave, you must set the DP port address that matches in the master group. The slave station address is set by DIP switch of EM277A module.

MPI communication between S7-300 and EM277A of CTSC-200

When CTSC - 200 CPU and S7-300 conduct MPI communication, no need to write any program related to communication in CTSC - 200 PLC, just set the data for exchange into continuous V storage area, and in S7-300 need to call system function X_GET (SFC67) and X_PUT (SFC68) in OB1 (or timer interrupt tissue block OB35) to realize the communications between S7-300 and CTSC - 200 CPU, When call SFC67 and SFC68, the VAR_ADDR parameter is filled in the data address area of CTSC-200, where P# db1.* ** BYTE n corresponds to the data area from VB ** to VB (** +n) in the CTSC-200 CPU V storage area.

Figure G-2 shows an MPI network with CTSC-200 CPU and EM277A DP slave station module.



- Use S7-300 master station module, PG\PC program station and Copanel HMI as MPI master station
- CTSC-200 CPU is a MPI slave station
- Use XGET / XPUT instruction of S7-300/400 master station module to read and write data from CTSC-200.

Figure G-2 MPI network example

Note: Set the slave station address (NEST_ID) the same with dial code address of EM277A module

H Using EM277B Module

EM277B is a modular slave station device for PROFIBUS DP network, which communicates with the DP master station. EM277B can connect up to six extension modules for digital or analog input and output. EM277B USES terminals to connect PROFIBUS network instead of standard connectors, and baud rate is automatically adjusted to be consistent with the main station.

【Main feature】

- Photoelectric isolation technique, strong anti-interference ability, high reliability.
- Power supply with reverse connection protection and surge absorption function, suitable for harsh industrial environments.

【Terms of use】

- Signal lines should be shielded twisted-pair cable, and double side grounding.
- Ground the module earthing terminal under the condition of the system has a good grounding, otherwise not grounded.
- As the last station, termination resistors should be ON position

H.1 Construction of PROFIBUS-DP Network

Distributed I/O systems contain active (master) and passive (slave) nodes that are interconnected through PROFIBUS-DP.

The following figure shows EM 277B in a typical PROFIBUS-DP network configuration:

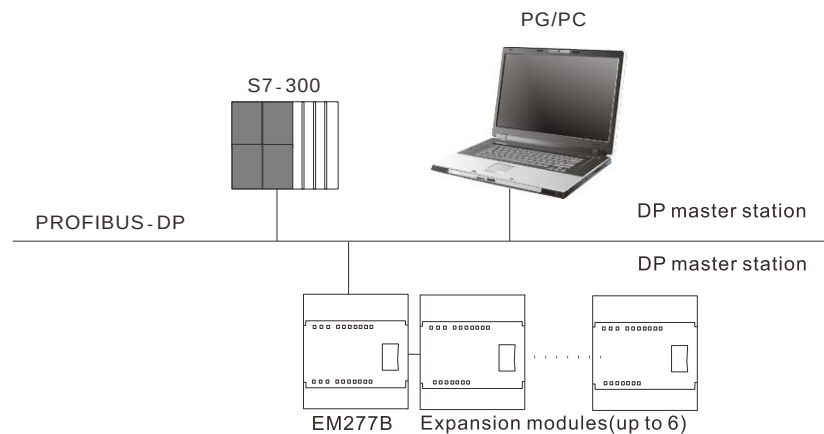


Figure H-1 Construction of PROFIBUS-DP network

H.2 EM 277B Components and Installation Diagram

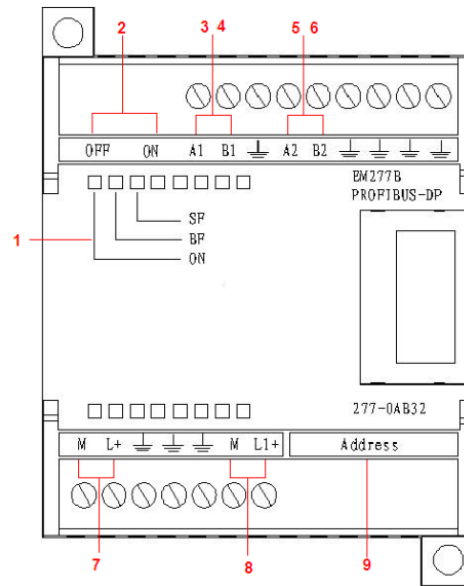


Figure H-2 EM 277B components

- ① Status LED
- ② Terminal resistance option switch: ON indicates with terminal resistance, OFF indicates without terminal resistance
- ③ Isolated signal A (network input)
- ④ Isolated signal B (network input)
- ⑤ Isolated signal A (network cascade)
- ⑥ Isolated signal B (network cascade)
- ⑦ User power
- ⑧ Sensor power
- ⑨ Address switch: set with 8 bit DIP switch, indicates in binary number, effective option range is 1-125

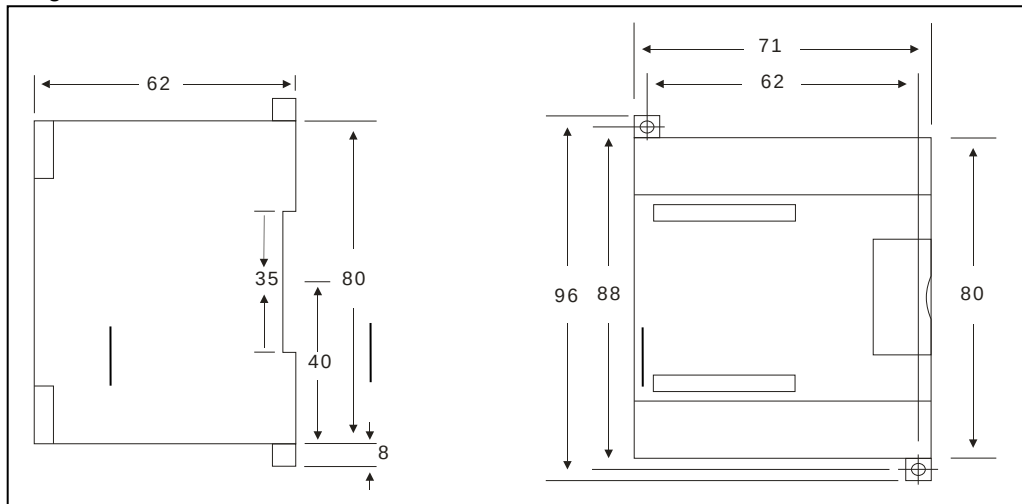


Figure H-3 EM 277B installation diagram

H.3 User Guide

This section introduces how to use EM277B, combined with "CPU312-1AE13 and EM277B communication" example gradually introduced EM277B hardware configuration, user programming and debugging as well as system diagnosis.

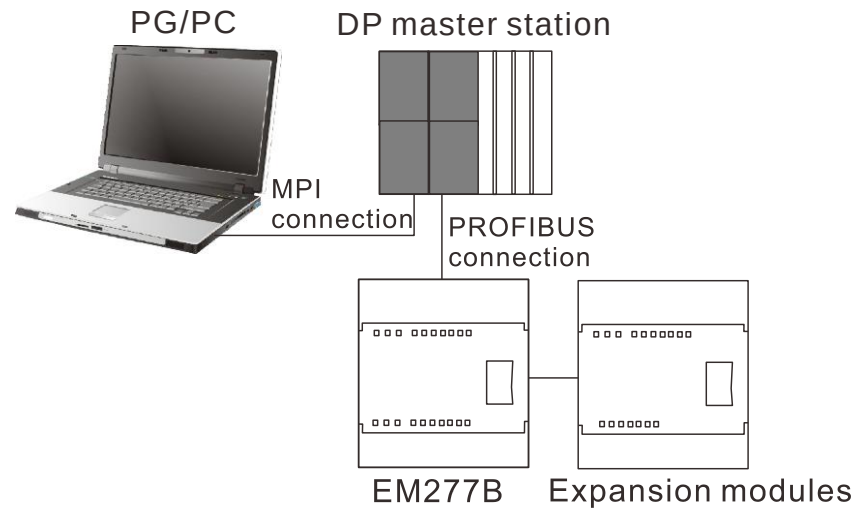


Figure H-4 EM 277B installation construction

Components the example use

Component	Illustration
A PG/PC with STEP7	Version of STEP 7 must support to configure DP master station
MPI program cable	For programming(download network hardware, program and monitor data)
A DP master station system	A CPU312-1AE13, a CP 342-5(master)
CTSC-200 EM 277B	DP slave station device
A PROFIBUS communication cable	PROFIBUS cable must have a standard PROFIBUS connector to connect DP master station
An EM that can connect to EM277B	Such as CTSC-200 EM223-1BL32 module

H.3.1 Hardware Configuration

Steps:

1) Start SIMATIC Manager

Select "File" -> "New", enter project name and select catalogue to save project in dialog box.

2) Create a SIMATIC 300 station

Select station in use (such as: SIMATIC 300 station)

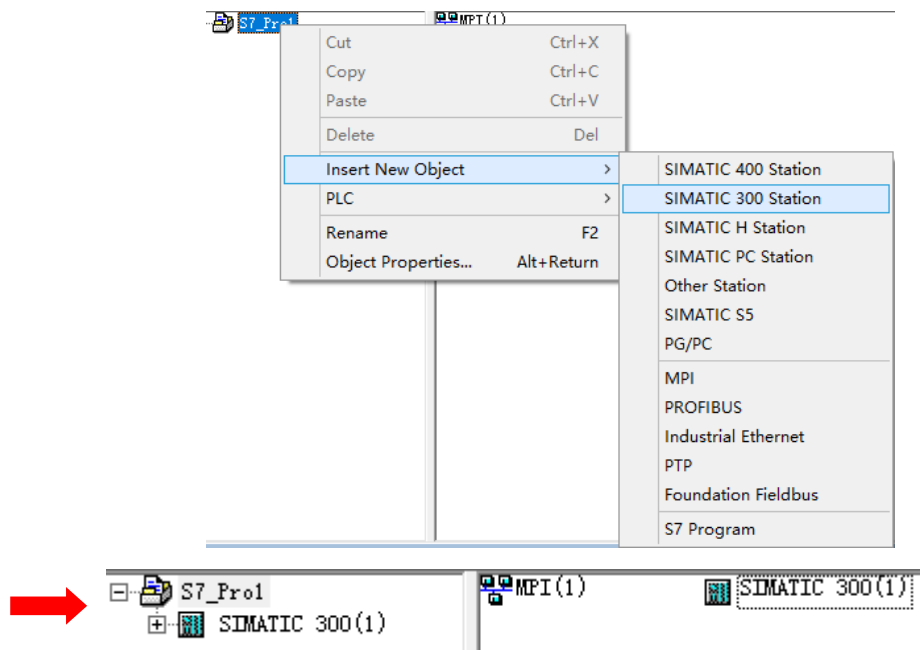


Figure H-5 Insert SIMATIC 300 Station

3) Open HW-Config to configure hardware

Double click the inserted 300 station “SIMATIC 300(1)” in step 2), as shown below:

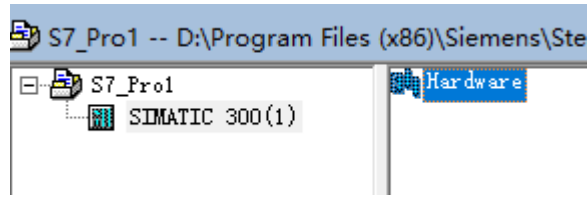


Figure H-6 HW-Config Interface

Double click “Hardware” in picture above to open HW-config interface, set distribute I/O (DP) rack, modules and PROFIBUS connection

4) Add guide rail

First add guide rail for station in use (Rail), use it to install DP master station

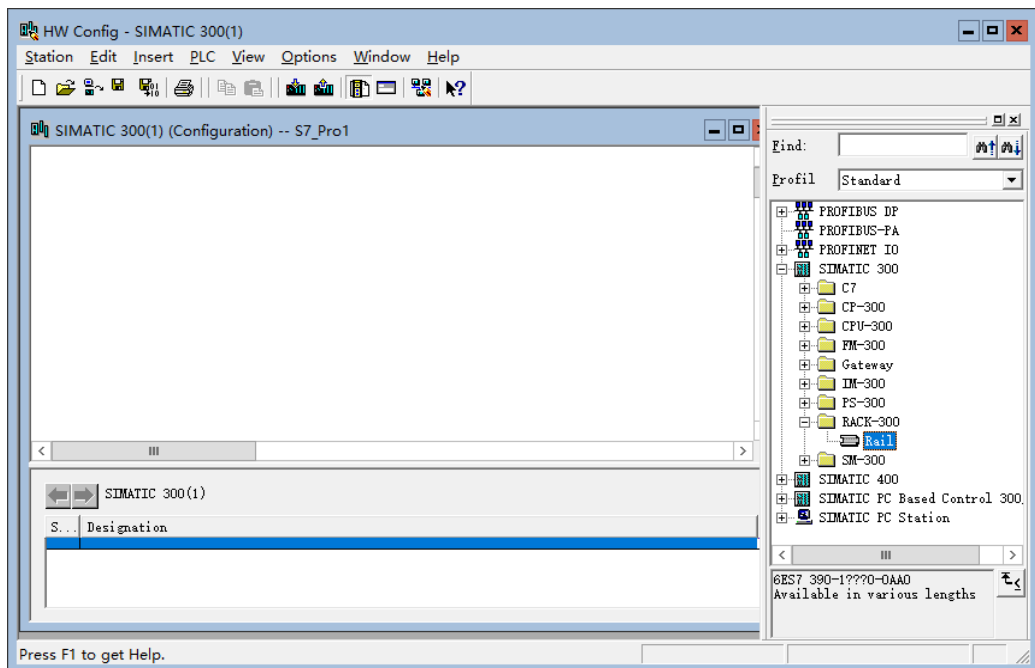


Figure H-7 Add Guide Rail

5) Add Power Supply (selective)

Add power to guide rail. The power can only place in the first slot of the rack.

6) Add DP master station CPU and PROFIBUS connection (CP 342-5)

Drag the CPU of the DP master station actually used in the hardware directory to the second slot of the guide rail or double click to add it, then drag the CP 342-5 actually used in the hardware directory to the fourth slot of the guide rail, as shown in the figure below:

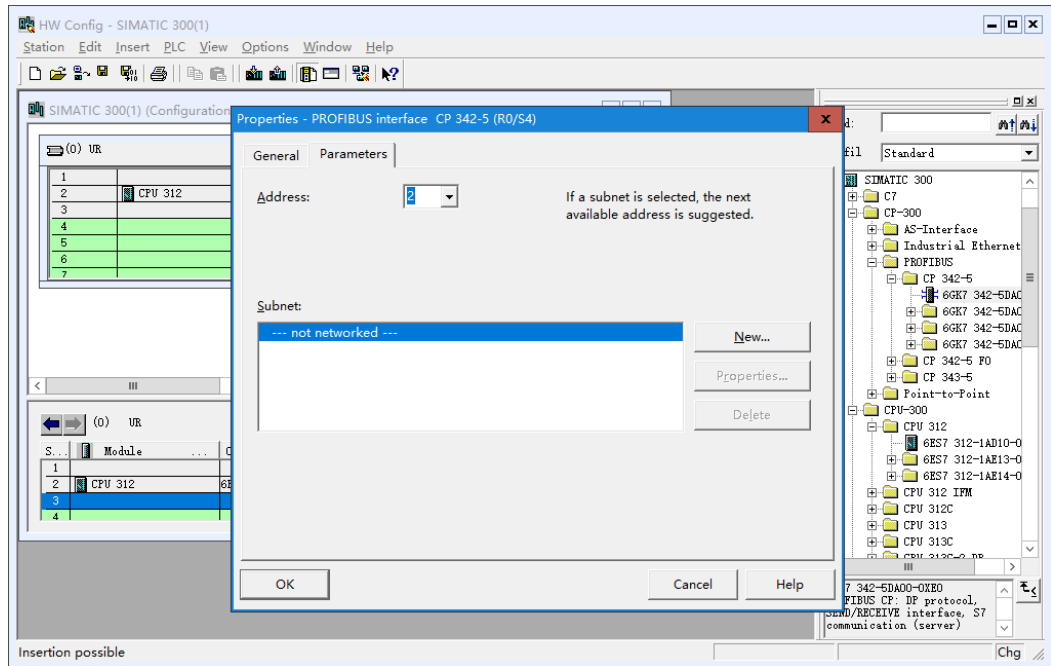


Figure H-8 Add DP Master Station CPU

In the dialog box above, set the PROFIBUS CP 342-5 interface address (as 2), and click the "New" to add a PROFIBUS connection for the DP master station. In the "Properties" dialog box, set the network parameters you need, including the transmission rate (set by 1.5m by default) and the configuration file (select "DP"), as shown below:

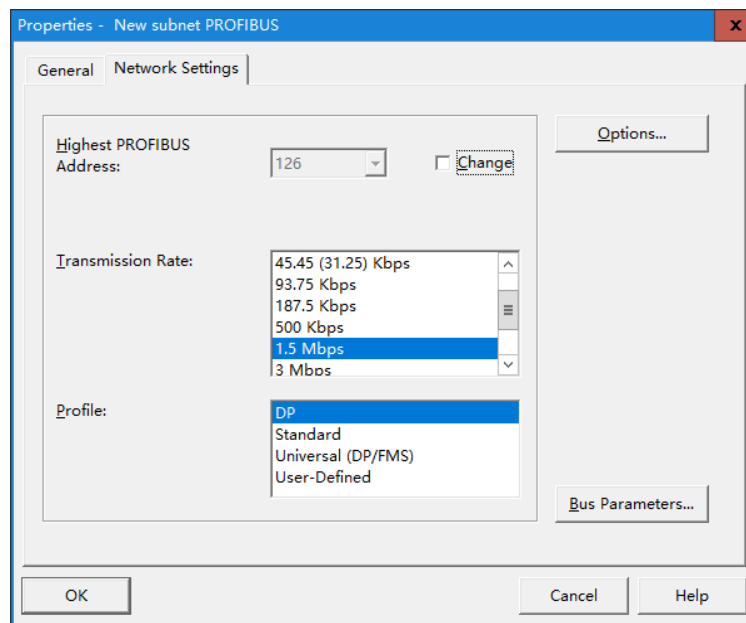


Figure H-9 Master Station Network Setting

Click "OK" to add the CPU and PROFIBUS connections of DP master station.

After confirmation, set the data I/O address and working mode of CP 342-5d. Double-click CP342-5 on the guide rail, and the properties dialog box pops up, as shown below:

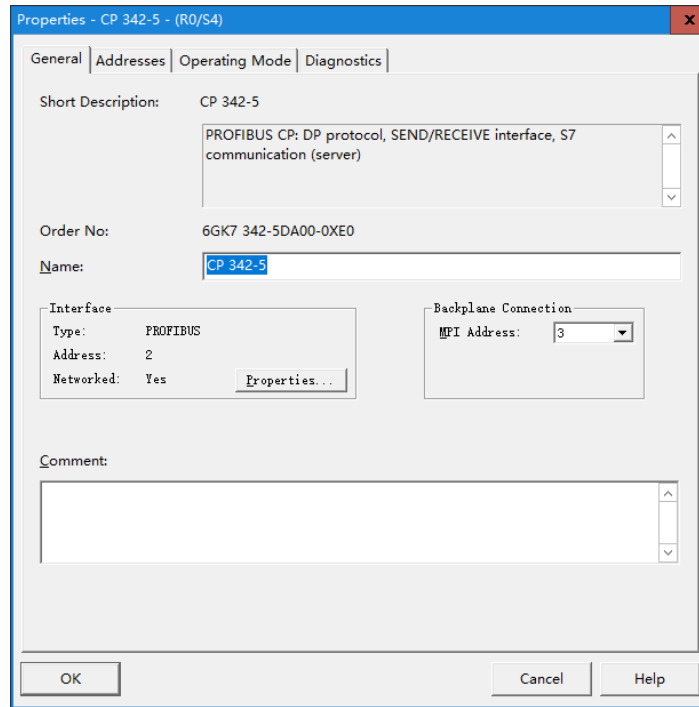


Figure H-10 Set CP 342-5D I/O Address

Set CP 342-5 I/O address

In the **【Address】** of the properties dialog box, the "Input" and "Output" start addresses are set by default (the default is 256 (16#100, the subroutine's CPLADDR parameter use this value)).

Set Operating Mode of CP 342-5

Set CP 342-5 "DP master station" in [Operating Mode] of properties dialog box:

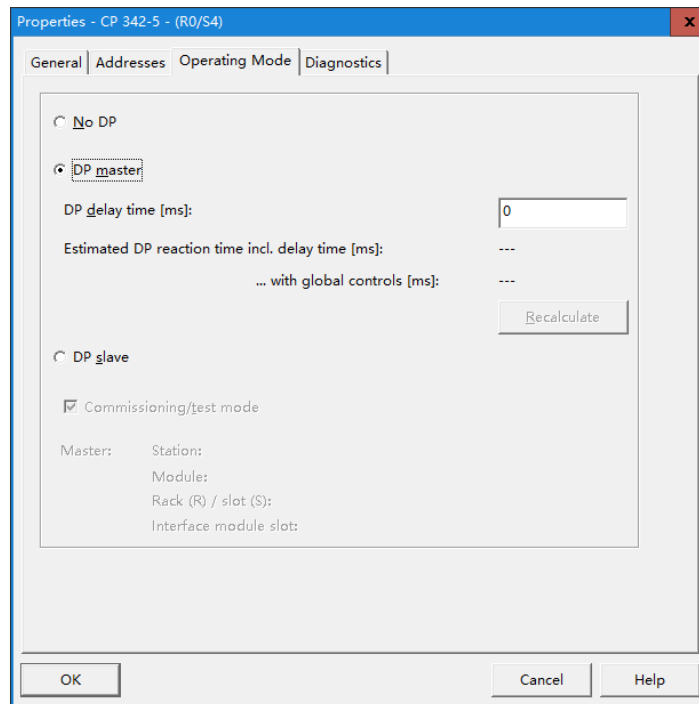


Figure H-11 Set CP 342-5D Work Mode

A dialog box pops up indicating that must call FC1 (DP_SEND) and send FC2 (DP_RECV).

After setting CP342-5 property, a PROFIBUS DP master station system will be automatically added in the blank on the right, as shown in the figure below:

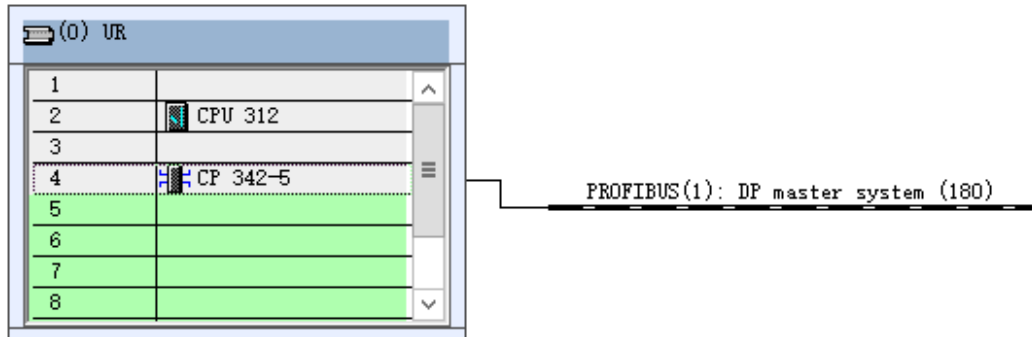


Figure H-12 Complete CP 342-5D Attribute Set

1) Import GSD file of EM277B

First use of EMC277B, you must import GSD file involving device information in STEP 7, otherwise skip this

Install method:

In "HW Config" interface, select menu [Operations] - > [Install GSD Files], click "Browse..." to open the 277 b GSD file directory in the "Import GSD files" dialog, then click "OK", show as second figure, select "ct277B. gsd", then click the "Install", after the installation click the "Close" to complete the EM277B GSD file installation.

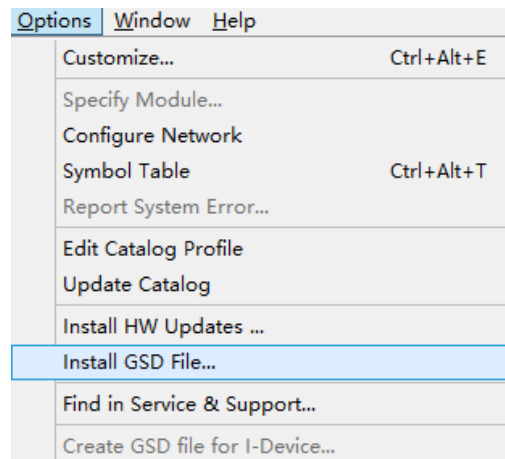


Figure H-13 Import GSD File

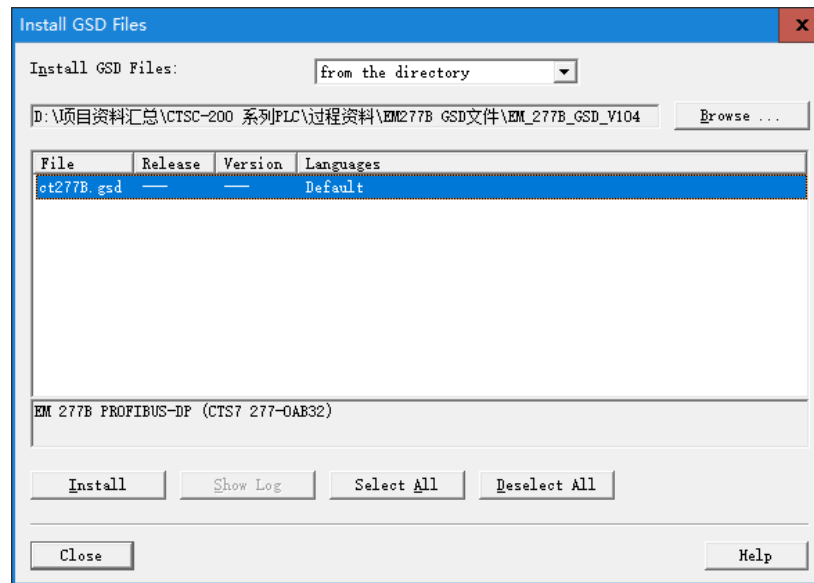


Figure H-14 Install GSD File

Note: You could download the EM 277B GSD files (ct277B.gsd) on www.co-trust.com

After installation, you can see EM277B and expansion module information in HW-Config hardware catalogue:

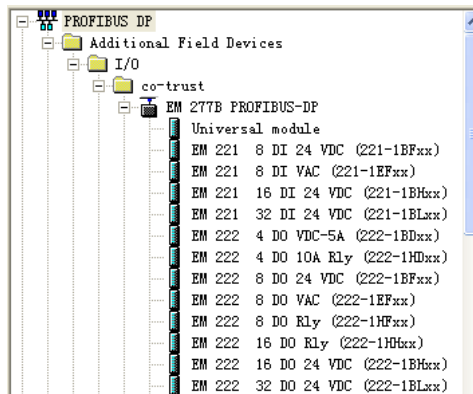


Figure H-15 Complete GSD File Installation

2) Add EM 277B DP slave station

Drag EM 277 PROFIBUS-DP from the hardware directory to the "PROFIBUS (1): DP master station system (180)". As shown in the figure below, set the EM277B slave station and network in the property dialog box that pops up.

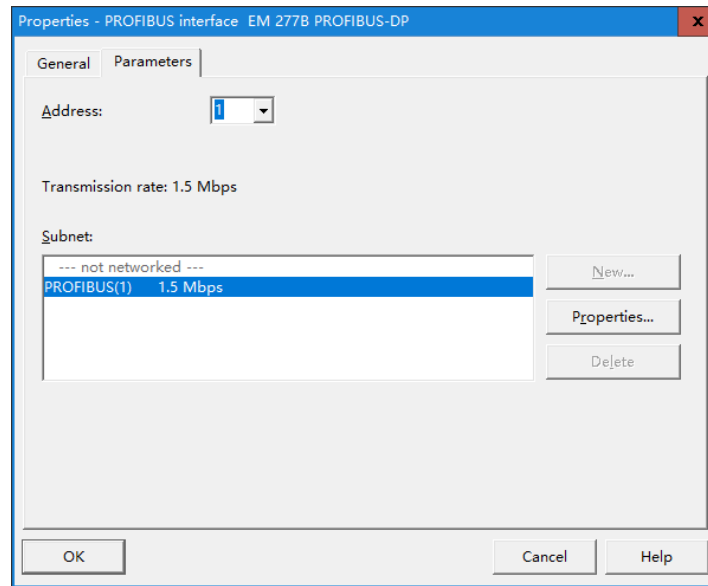


Figure H-16 Add EM 277B DP Slave Station

Set slave station address

Set the EM277B slave address in address under parameters of properties box.



Note

The valid address is 1 to 125, and is unique on PROFIBUS DP, also must be set as the address dialed by the DIP switch of EM277B module.

Set slave station network

Click the "Properties" under [Parameters] of the property box, then pop up a child dialog box as shown in the figure below:

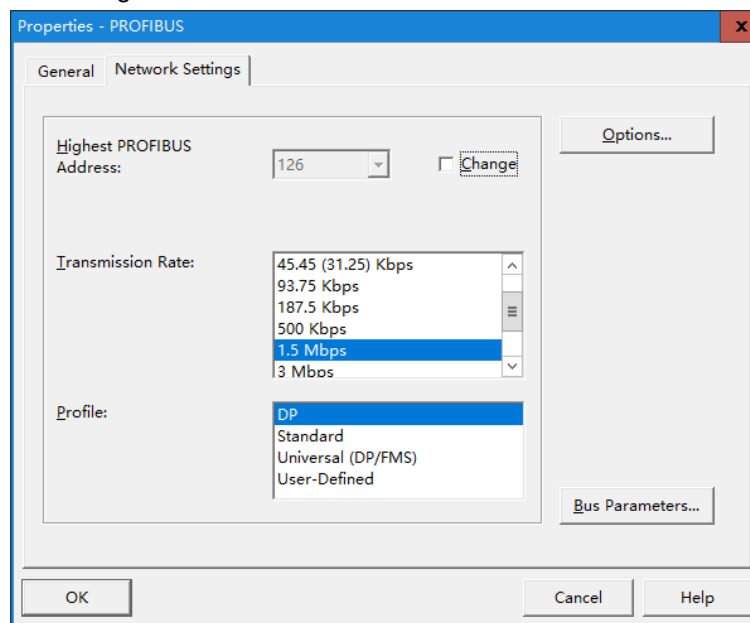


Figure H-17 Set slave station network

Click the [Network Settings] of this dialog box, and set "Transmission Rate" (default as 1.5Mbps) and "Profile" (set as "DP"). Then the addition and configuration of EM277B

slave station is completed, as shown in the following figure:

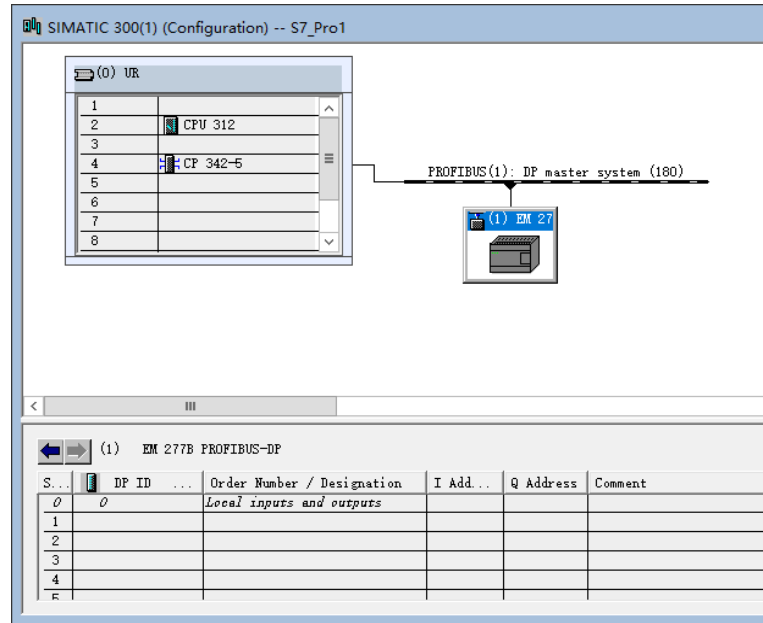


Figure H-18 Complete slave station properties configuration

Note: ensure that the "Hex parameter assignment" of the [Parameter Assignment] of the EM277B slave station property is set by default, as shown in the figure below (00,00,00) (double-click the EM277B icon to see), otherwise there may be a communication failure.

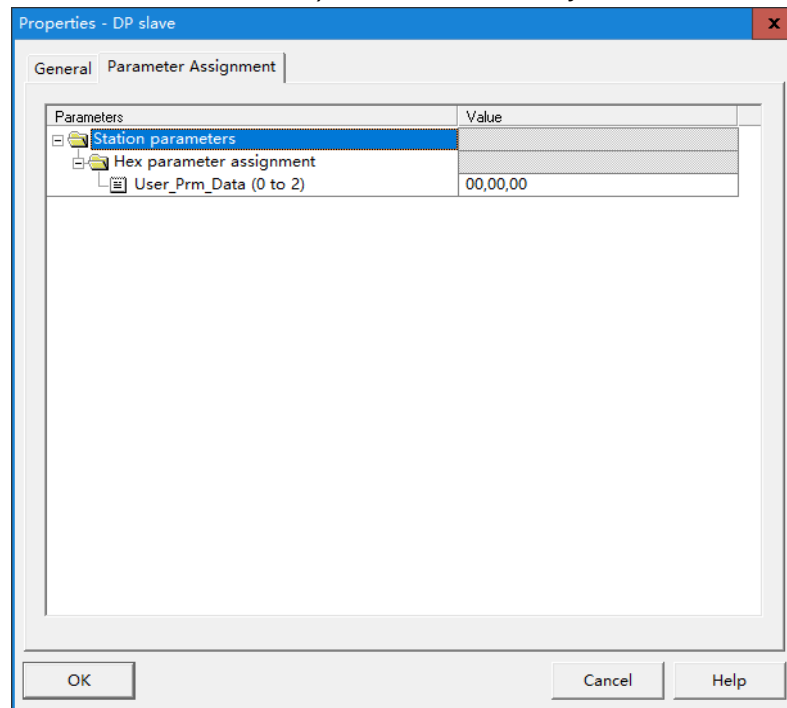


Figure H-19 DP slave station parameter assignment

※ Add expansion modules for EM277B slave station

Drag or double click the required extension module to the EM 277B blank slot number in the order you need from the hardware directory "EM 277B PROFIBUS-DP" to add DP slave station extension module, as shown in the following figure:

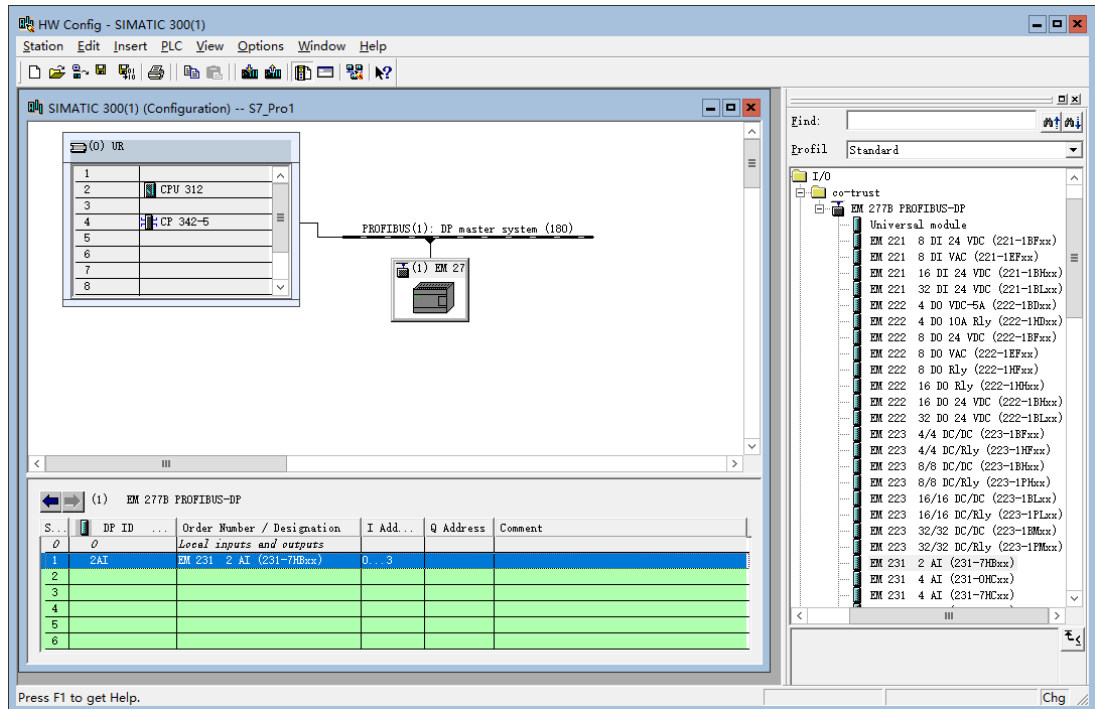


Figure H-20 Add expansion modules for EM 277B

※ Download hardware configuration to CPU of DP master station

Execute "Station" -> "Save and Compile" in of the HW Config interface.

Till now, STEP7 has completed the hardware configuration. Before sending the configuration, confirm that you have established connection between the DP master station and PC/PG (MPI connection), then execute "PLC" -> "Download" in the same interface to download the hardware configuration to the CPU of DP master station.

H.3.2 Programming and Debugging

EM 277B reads the extension module input and makes it available to the DP master, which provides output information to EM277B for writing to the extension module. So create user program in OB 1 using the LAD/STL/FBD editor, and download it to the CPU on the host system.

【Programming】

Steps:

1) Write program to call FC1 (DP_SEND) and FC2 (DP_RECV) in OB1 to realize the DP master station reading and writing 16 bytes of data from slave station, as shown in the following figure:

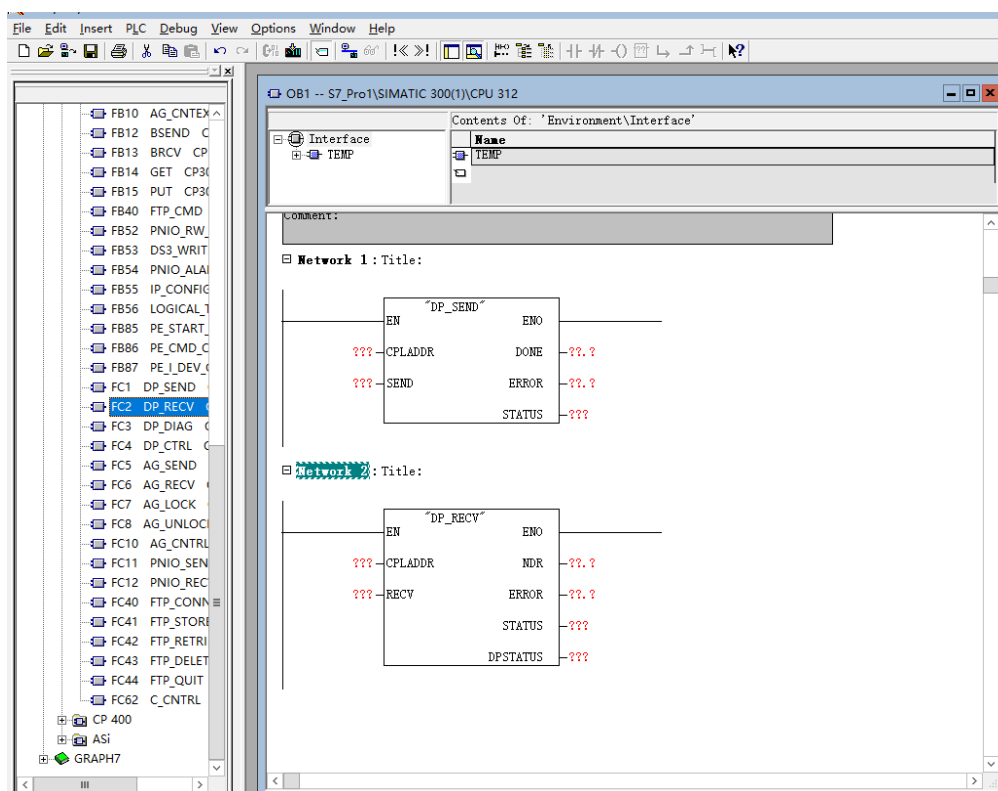


Figure H-21 Program interface

Program function description:

Execute FC1 (DP_SEND) to output CPU312 memory MB0-MB14 data to QB0-QB14 of EM277B extension module; input EM277B extension module IB0-IB14 into the MB15-MB29 memory of CPU312 by performing FC2 (DP_RECV).

CPLADDR: Address 256(16#100) of CP342-5



Note

If you select the CPU with DP communication port (e.g., CPU313C-2DP), no need to call FC1 and FC2, directly access the address assigned by configuration EM277B module (e.g., IB0-IB14, QB0-QB14).

2) Save user program

【debugging】

Steps:

- 1) Connect CP 342-5 and EM277B, set EM277B address and middle resistance, and connect the power supply of all devices in the system.
- 2) STEP7 connect to the main station and download the program to the CPU of the main station system.
- 3) Set the main station as RUN.
- 4) Set and monitor the I/O data to be transmitted and the status results of program execution in the STEP7 variable scale, observe the output status of the EM277B extension module and status indicator light (LED)

Some faults may occur in the debugging process. Please refer to Appendix [H.3.3 Diagnosis](#).

H.3.3 Diagnosis

Diagnose PROFIBUS DP master/slave network through LED status, STEP 7 fault information and specific programming diagnosis.

Diagnose by EM 277B

The LED status for diagnosing EM277B can provide diagnostic information about connection with the main station and self-diagnosis information of EM277B firmware.

After connecting EM 277B power supply, the LED indicator marked "ON" (green) light up.

If the BF and SF LED remain off, EM 277B will function normally.

If the BF or SF LED lights up, indicate an error in the extension module, address setting over range, wiring, or main program configuration.

Function of EM277B LED:

LED	Color	Description
ON	Green	Indicator lights on when EM 277B power on, controlled by EM 277B hardware
SF(system failure)	Red	If EM 277B find expansion module fault or address setting over range, the LED light up
BF(bus failure)	Red	Flicker when no DP exchange with master station

If main system run without fault, bellows are the possible fault reasons:

- STEP 7 hardware configuration didn't match completely with equipment in network. You need to change the hardware configuration and download it to the CPU of DP master.
- STEP 7 configure wrong parameter
- The address STEP7 configured is different with EM277B actual PROFIBUS address, or EM277B actual PROFIBUS address set as 126 127(invalid address). If the EM277B actual address is correct, change the PROFIBUS address of the slave station device in STEP 7 and re-download the configuration to CPU of DP master station. If the actual address of EM277B is wrong, change it, after that EM277B needs to be cut off and re-energized.
- Didn't set the PROFIBUS network terminal resistance correctly. Dial "ON" if you are at the last node in the network, otherwise "OFF".
- PROFIBUS wiring error or PROFIBUS cable damaged. Check wiring and cables to make sure they are correct.

Diagnose by STEP7

【Read diagnosis information in HW-Config】

For all SIMATIC S7/M7 series modules, S7 diagnosis can be called. For the base unit and I/O modules, the structure of S7 diagnostics is the same. To make the diagnosis in STEP 7, read the diagnostic buffer of the main station device or EM 277B.

【Confirm fault】

Confirm faults in HW-Config due to steps below:

- Open project online window through Station >Open ONLINE
- Seek symbols showing diagnostic equipment state and fault condition. Press the F1 to open the help page of the diagnostic symbol to view the comments for the diagnosis.
- Select menu PLC >Faulty Modules to view fault modules list. If there is a failure, the view will not update automatically when open the online view. Select view>update to display the current state.

【Read diagnosis information in master and slave station】

In HW-config, you can open the module information to get detailed diagnostic information. Double-click for displaying the fault modules.

- For DP master, select Diagnostic Buffer symbols to view fault information
- For EM 277B station equipment: select the General symbol to view module status. Select the DP Slave diagnostics TAB, and click the Hex. Format button display the diagnostic bytes of EM277B.

H.3.4 Diagnose EM277B power budget

A power budget is a calculation of whether the power provided is sufficient for the extension module. To make the power budget, you need to calculate the power sources of 5 VDC and 24 VDC

EM277B power	5 VDC power	24 VDC power
Supply	EM 277B offer a 5 VDC logic power for all expansion modules in system	EM 277B offer a 24VDC sensor power for expansion modules or anywhere needed.
Maximum usable DC power	The additional electricity sum of all EM277B expansion modules require less than 660mA	Please confirm power demand less than the budget 400mA

**Warning**

- Do not use EM277B DC sensor and another external power to power for one device.
- Do not use two power at one device
- Connecting an external 24VDC power to EM277B 24VDC sensor power supply will cause conflict. Which may result in shortened service life of the power supply or failure of both power supplies, which may then lead to failure of PLC system operation. Accidental operation may result in serious personal injury or even death, and/or equipment damage.

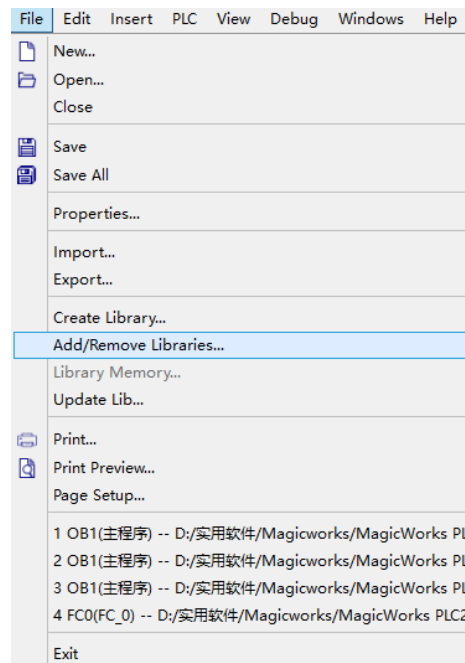
I Using EM253 Motion Control Library

I.1 Function Introduction

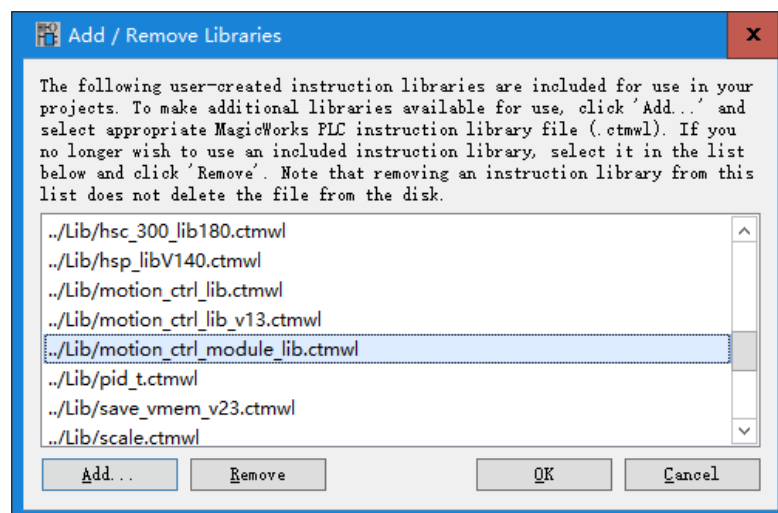
EM253 motion control library is dedicated for EM253 motion control module, able to realize multi-axis independent control.

I.2 Software Installation

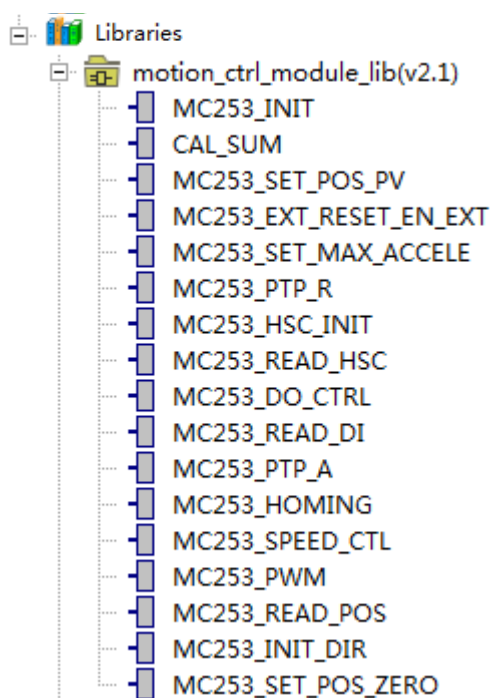
Select menu item “File” -> “Add/Remove Libraries” to find library file “motion_ctrl_module_lib.ctmwl”:



View catalogue where you save library files, select “motion_ctrl_module_lib” and click “Add”:



Click “OK” after addition, check the library node of command tree, you can see the “motion_ctrl_module_lib”:



I.3 “motion_ctrl_module_lib library” Instruction List

TableI-1 Instruction list

Function name	Instruction name	Support or not
MC253_EXT_RESET_EN	External reset coordinate enable instruction	EM253 not support
MC253_INIT_DIR	Configure motor direction instruction	Supported
MC253_READ_POS	Read position instruction	Supported
MC253_PTP_R	Uniaxial relative motion instruction	Supported
MC253_CIRCLE_R	Two-axis arc interpolation motion instruction	EM253 not support
MC253_SPEED_CTL	Speed control instruction	Supported
MC253_SET_POS_ZERO	Software back to zero instruction	Supported
MC253_SET_POS_PV	Set target location instruction	Supported
MC253_LINE_R	Two-axis line interpolation motion instruction	EM253 not support
MC253_EXT_RESET_EN_EXT	External reset coordinate enable instruction II	Supported
MC253_SET_MAX_ACCELE	Set maximum acceleration instruction	Supported
MC253_SET_CI_MODE	Set continuous interpolation instruction	EM253 not support
MC253_PWM	PWM instruction	Supported

MC253_INIT	Initialize motion control module instruction	Supported and must be called first
MC253_DO_CTRL	Control module output instruction	Supported
MC253_READ_DI	Read module input status instruction	Supported
MC253_HSC_INIT	Set module HSC instruction	Supported
MC253_READ_HSC	Read module HSC instruction	Supported
MC253_PTP_A	Uniaxial absolute motion instruction	Supported
MC253_HOMING	Homing instruction	Supported

I.4 “motion_ctrl_module_lib” Library Instruction Illustration

For Micro/Win programming, the instruction format of the operation control module is the same as that of CPU, difference is the instruction library. Prefix of CPU instruction is MC, while the motion control module EM253 is MC253. The axis number of CPU instruction is 0~3, while the motion control module instruction is numbered from 0, which can be continued down. System automatically assigns to each module. For example, the motion control module instruction axis number is 0~1, which corresponds to the 0 and 1 axis of the first module respectively. The motion control module instruction axis number is 2~3, corresponding to the 0 and 1 axis of the second motion control module respectively. By analogy, if the motion control module out of scope, consider the parameter error. For example, there is only one motion control module (with only two axes), motion control module specifies axis 2, but it cannot find the corresponding value, so that it cannot be used.

Similarly, the module HSC is not uniformly numbered with CPU HSC, HSC using modules must call the module instruction. The module HSC numbered from 0 and can be continued down. For example, the HSC of module instruction label 0~1, which corresponds to HSC0 and HSC1 of the first module respectively; HSC of module instruction 2~3 corresponds to HSC0 and HSC1 of the second module respectively.

※ Attention

1 Initialization and library storage area

Using the EM253 motion control module require to initialize the system control variables in the initial call MC253_INIT of the system program (it is needed only once, that is, only be called by SM0.1). Note that the EM253 motion control module use V memory space. Users can specify the library storage area address themselves through "library storage area allocation". Users are not allowed to use the address space occupied by the library in program. (if only one EM253 module hangs after the CPU, the library occupies 218 bytes; 236 bytes for 2. 254 bytes for 3; 272 bytes for 4. 290 bytes for 5.)

2 Analog filter

When using EM253 motion control module, you must clear filtering function at the corresponding channel of EM253 motion control in the upper computer software system

block -> input filter -> analog, and download the system block to the PLC. Otherwise, the EM253 motion control module library instruction cannot be use (check the PLC information, where the 4 in and 4 out analog module channel is corresponding to that of the EM253 motion control module.).

3 Communication status bit

In addition to MC253_INIT, all other instructions of EM253 motion control module have added communication status bit. If this status bit alarms, indicates communication fault that it may not be reliable. At this time check whether the bus connection is normal or the motion control channel of the analog filter is cleared. In addition to MC253_INIT, all other instructions of EM253 operation and control module have added communication status bit. If this status bit alarms, it indicates that the communication is out of order and the communication information may not be reliable. At this time to check whether the bus connection is normal or the operation of the control channel of the analog filter is cleared.

4 Pulse output instruction execution order

Coaxial pulse output instructions (MC253_PTP_R, MC253_SPEED_CTL, MC253_PWM) have up to 3 data buffers, that is, only process 3 instructions at the same time. When there are multiple instructions on the same axis (such as all of them are 0 axis), if the buffer unfell (there are less than 3 enabled output instructions), then the new enabled instructions can be processed in time. Then, the execution order of the instructions is consistent with the time order of the enabled instructions, which is not necessarily related to the instructions position order. If the buffer is full, the enabled output command is more than 3), the new enable instruction cannot be processed in time, and can only be enabled after the end of the instruction and release the buffer before according to place order for processing by the program scan, which is related to order of execution and position, but not necessarily according to the time sequence of instructions. Simply understood as "under 3 by time, full 3 by position" (as shown in example 1).

In order to achieve the trajectory, and if the position order of these motion instructions is consistent with the desired trajectory, all these instructions can be enabled at one time. If not, you must enable the corresponding instruction of the enabled trajectory when the enabled instruction is less than 3 (as shown in example 2).

Case 1) Enable all instructions for one time

The enable instruction (instruction A/B/C) is executed in the enable order when the buffer is less than 3 copies. As following table, enable all the five instructions at one time, and the order is: instruction A--> instruction C--> instruction B--> instruction E-> instruction D, as shown in table 1; Then, the enabled 3 instructions are executed successively in: instruction A--> instruction C--> instruction B, as shown in table 3:

Instruction enable order	Instruction position order		Instruction execute order
1	0—AXIS 1—RUN	Instruction A	1

3	0—AXIS 1—RUN	Instruction B	3
2	0—AXIS 1—RUN	Instruction C	2
5	0—AXIS 1—RUN	Instruction D	--
4	0—AXIS 1—RUN	Instruction E	--

The remaining enabled instructions (instruction D/E) execute in position order when the buffer is filled with 3 portions. After the execution of instruction A, the buffer is released. Because the program scans from top to bottom, instruction D first obtains the buffer. Instruction E obtains the buffer after instruction C finishes executing and releases the buffer.

The instruction enable order: instruction A--> instruction C--> instruction B--> instruction E--> instruction D, as shown in table 1; The order of instruction execution: instruction A--> instruction C--> instruction B--> instruction D--> instruction E, as shown in table 3:

Instruction enable order	Instruction position order		Instruction execute order
1	0—AXIS 1—RUN	Instruction A	1
3	0—AXIS 1—RUN	Instruction B	3
2	0—AXIS 1—RUN	Instruction C	2
5	0—AXIS 1—RUN	Instruction D	4
4	0—AXIS 1—RUN	Instruction E	5

Case 2) Enable 3 instructions first, then enable the rest after instructions complete execute to release buffer

The enable instruction (instruction A/B/C) is executed in the enable order when the buffer is less than 3 copies, as following table: enable 3 instructions first, the order is: instruction A --> instruction C --> instruction B, as shown in table 1, execute in order successively,

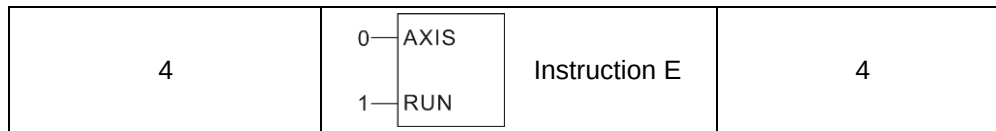
the execution order is: instruction A --> instruction C --> instruction B, as shown in table column 3:

Instruction enable order	Instruction position order		Instruction execute order
1	0— AXIS 1— RUN	Instruction A	1
3	0— AXIS 1— RUN	Instruction B	3
2	0— AXIS 1— RUN	Instruction C	2
--	0— AXIS 0— RUN	Instruction D	--
--	0— AXIS 0— RUN	Instruction E	--

After instruction A completes execution and releases the buffer, enable instruction E first, then it obtains the buffer; Instruction D acquires the buffer after instruction C completes execution and releases the buffer.

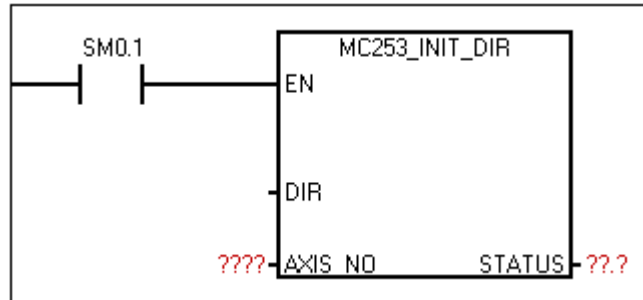
The order of instruction enable is: instruction A--> instruction C--> instruction B--> instruction E-> instruction D, as shown in table 1;The order of instruction execution is: instruction A--> instruction C--> instruction B--> instruction E-> instruction D, as shown in table 3:

Instruction enable order	Instruction position order		Instruction execute order
1	0— AXIS 0— RUN	Instruction A	1
3	0— AXIS 0— RUN	Instruction B	3
2	0— AXIS 1— RUN	Instruction C	2
5	0— AXIS 1— RUN	Instruction D	5



Configure motor direction instruction

- ① Function name: MC253_INIT_DIR



- ② Function: configure motor direction



Note

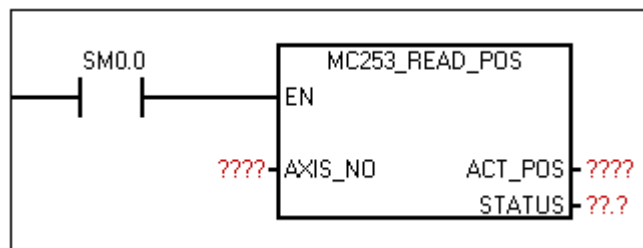
Execute one time on first scan cycle of CPU power on

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
DIR	IN	Configure effective level when the direction signal is positive. DIR=1, set motor forward when corresponding direction axis output "1" DIR=0, set motor reverse when corresponding direction axis output "0"	Bool	0~1	Default: 1, motor forward when direction axis output
AXIS_NO	IN	Set axis number(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255	
STATUS	OUT	Communication status sign bit 1: communication timeout	Bool	0~1	

Read position instruction

- ① Function name: MC253_READ_POS



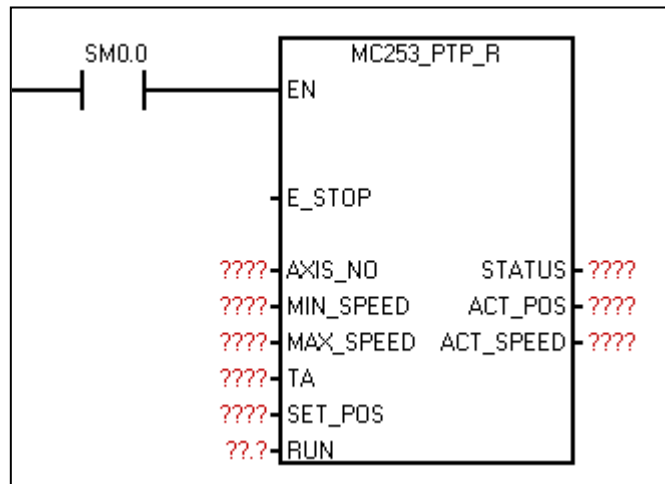
- ② Function: Read the absolute coordinate value of each axis. Once the origin coordinate is set, the value will calculate according to relationship between the output pulse and direction: positive rotation output a pulse: +1, while reverse output a pulse: -1. What you end up with is an absolute coordinate with the set point as the origin.

③ Parameter

Name	I/O	Description	Type	Value range	Note
AXIS_NO	IN	Set axis number(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255	
ACT_POS	OUT	Absolute coordinate of current axis(1 pulse indicate 1 unit coordinate)	Dint	-2147483648~+2147483647	No error status output for this instruct, axis number must set correctly.
STATUS	OUT	Communication status sign bit 1: communication timeout	Bool	0~1	

Uniaxial relative motion instruction

① Function name: MC253_PTP_R



- ② Function: Used for uniaxial point - to - point control (uniaxial fixed - length drive). Call one time output fixed pulse, by setting the maximum, minimum speed and deceleration time, the output pulse will gradually accelerate to the largest speed, when the pulse number is ready to run, cut down pulse frequency automatically to prevent vibration or jammed due to too much inertia when start or stop the machine.

③ Parameter

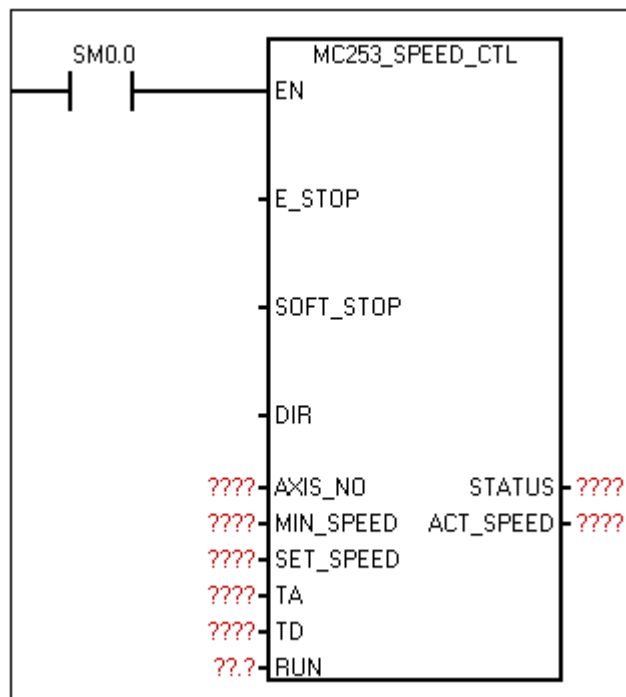
Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	Bool	0/1	1 Run only when Run =1 and E_Stop =0 2 RUN reset internal when E_STOP = 1

AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	0~3	Unchangeable in the process
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	500~200000	1 Set minimum speed less than maximum speed
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	500~200000	2 Changeable in the process
TA	IN	Time of ACC/DEC, Unit: ms	Dword	0~10000	Changeable in the process; if TA=0, no settings for acceleration.
RUN	IN	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 When E_STOP=1, RUN reset internal.
STATUS	OUT	76543210 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1 Judge Only to axis parameter configuration error and homing mode out of range. 2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
SET_POS	IN	Output pulse. The positive output pulse	DInt	-2147483647	Modifiable during operation. When the

		indicates the positive direction along the X-axis, and the negative pulse number indicates the negative direction along the X-axis.		~ +214748364 7	new set value is greater than the number of pulses output, the final output pulse will be subject to the new set value. When the new set value is less than the number of output pulses, the pulse output will be stopped immediately
ACT_POS	OUT	Absolute coordinate of current axis (1 pulse 1 units coordinate)	DInt	-2147483647~ +2147483647	This instruction no error and set the axis number correctly.
ACT_SPEED	OUT	Current run speed	Dword	500~200000	Note: this value may be a little off actual one, at most 5K, related to acceleration time and setting speed

Speed control instruction

- ① Function name: MC253_SPEED_CTL



- ② Function: Control the frequency of uniaxial output pulse, the output pulse frequency

(speed) can be changed any time. When receive the soft stop command, it will automatically slow down to stop. When receive the emergency stop command, the pulse output will stop immediately without any deceleration.

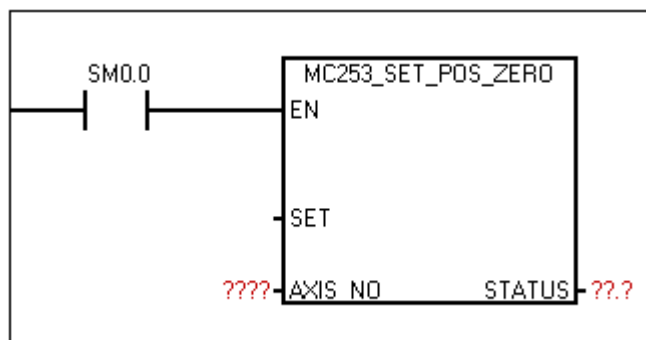
③ Parameter

Name	I/O	Description	Type	Value	Note
RUN	IN	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 when E_STOP=1, RUN reset internal.
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
SOFT_STOP	IN	Soft stop bit 1: valid 0: invalid Output pulse decelerate to stop when receive the instruction	Bool	0/1	
DIR	IN	Pulse direction bit	Bool	0/1	Unmodifiable in the process
AXIS_NO	IN	Set axis number, 0/1/2/3	Byte	0~3	unmodifiable in the process
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	0~20000 0	1 Minimum speed should less than setting speed
SET_SPEED	IN	Setting speed, output pulse will accelerate or decelerate to this speed before stop command	Dword	0~20000 0	2 Modifiable in the process
TA	IN	Time of ACC Unit: ms	Dword	0~10000	Modifiable in the process; Calculate acceleration only when startup and

					TA/TD changes
TD	IN	Time of ACC, from setting time to minimum time Unit: ms	Dword	0~10000	Unmodifiable in the process
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1 Judge Only to axis parameter 2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same
ACT_SPEED	OUT	Current actual speed	DWORD	0~200000	Current actual speed

Software back to zero instruction

- ① Function name: MC253_SET_POS_ZERO



② Function: reset absolute coordinate

**Note**

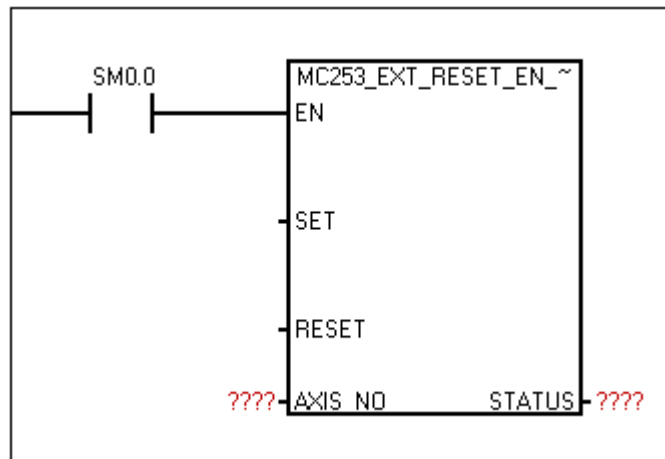
Call the instruction when the machine moves to a certain position, which is equivalent to setting the origin of the axis at that position. Each time you call the "read absolute coordinates" command, you get a coordinate value relative to that point.

③ Parameter

Name	I/O	Description	Type	Value range	Note
SET	IN	Enable reset bit Set absolute coordinate 0 at SET rising edge, SET set 0, then set 1 for each call	Bool	0~1	
AXIS_NO	IN	Set axis number(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255	
STATUS	OUT	Communication status sign bit 1: communication timeout	Bool	0~1	

External reset coordinate enable instruction II

① Function name: MC253_EXT_RESET_EN_EXT



② Function: set if enable external IO reset absolute coordinate

**Note**

Corresponding relation between axis number and external reset signal:

Axis 0 — I0.2 (MC253_HSC0)

Axis 1 — I0.6 (MC253_HSC1)

③ Parameter

Name	I/O	Description	Type	Value range	Note								
SET	IN	Enable reset bit Set absolute coordinate 0 at SET rising edge, set SET bit 0, then set 1 for each call	Bool	0~1									
RESET	IN	RESET rising edge, forbid to enable external reset, RESET then set 0 for each call	Bool	0~1									
AXIS_NO	IN	Set axis number(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255									
STATUS	OUT	Status bit: <table border="1"><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr></table> Bit0: reset status sign bit 1—reset complete 0—reset uncompleted Bit1~Bit6: reserved Bit7: communication status sign bit 1—communication timeout 0—no timeout	7	6	5	4	3	2	1	0	Byte	0~255	
7	6	5	4	3	2	1	0						

④ Illustration

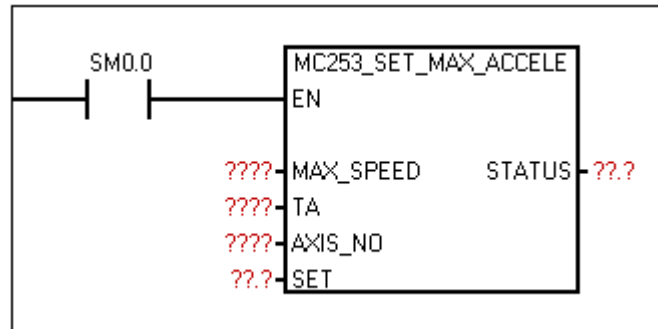
Use 0 axis to call this instruction. After the SET rising edge enables the external reset function and I0.2 detect the "effective reset signal", the system resets the axis 0 absolute coordinate, and also the STATUS setting reset complete. After the RESET rising edge prohibits the external reset function, even if I0.2 detect the "effective reset signal", the system will not reset the 0 axis absolute coordinate, and the STATUS reset instruct is non-reset state.

**Note**

The so-called "effective reset signal", the reset signal of each axis corresponding to an external IO, and the corresponding register set its effective level. For example, 0 axis corresponds to I0.2, and HSCO control register SM37.0 set the effective reset level of 0 axis. When set to 0, the effective reset signal of 0 axis is in I0.2 high level. When set to 1, the effective reset signal of axis 0 is in I0.2 low level; This setting is only effective if and only if the corresponding high-speed counter (0 axis corresponds to HSC0) is enabled, otherwise (no high-speed counter is enabled) the system default high level is the effective reset signal. If 0 axis, I0.2 high level is the effective reset signal. In the same way for other axes, the corresponding relation of the relevant control of each axis is shown in ② of this section.

Set maximum acceleration instruction

- ① Function name: MC253_SET_MAX_ACCELE



- ② Function: set maximum acceleration (= MAX_SPEED/TA) (TA≠0) (if the instruction isn't called, indicate the maximum acceleration has not been set)

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
MAX_SPEED	IN	Maximum speed of long-axis run Unit: HZ	Dword	0~200000	Modifiable in the process
TA	IN	Time of ACC Unit: ms	Dword	0~10000	Modifiable in the process Calculate acceleration only when startup and TA/TD changes
AXIS_NO	IN	Set axis number(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255	
SET	IN	After ensuring parameter above, give SET a rising edge to make it effective.	Bool	0~1	
STATUS	OUT	Communication status sign bit 1: communication timeout	Bool	0~1	

- ④ Illustration

If set the X-axis parameter TA=0, or the X-axis does not call this instruction, it is considered that the X-axis does not set maximum acceleration; Otherwise, $MAX_ACCELE=MAX_SPEED/TA$ is considered to be set as the X-axis maximum acceleration. The significance of this instruction is:

- a. Set an appropriate acceleration to limit the acceleration of each instruction on an axis.

Such as PTP instruction, set $AXIS_NO=0$, $MIN_SPEED=1000$, $MAX_SPEED=11000$, $TA=500$, then in theory the acceleration of PTP motion is 20HZ/ms ($= (MAX_SPEED-MIN_SPEED)/TA$); if 0 axis call MC_SET_MAX_ACCELE and set maximum acceleration 15HZ/ms, the PTP actual acceleration is 15HZ/ms (the same for MC_LINE_R and MC_CIRCLE_R)

- b. An instruction on a certain axis is to obtain the maximum acceleration to run

If the PTP instruction run at the maximum acceleration, call MC_SET_MAX_ACCELE instruction on the coaxes to set the maximum acceleration (the parameter TA of MC_SET_MAX_ACCELE instruction cannot be 0, otherwise cannot obtain the maximum acceleration), and simultaneously set the parameter TA of PTP instruction to 0.

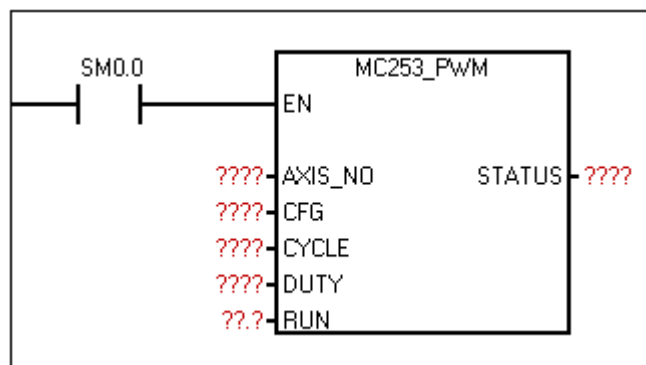
If the maximum acceleration is not set and the PTP instruction TA=0, the alarm parameter fails on the PTP instruction. (Same for MC_LINE_R and MC_CIRCLE_R)

Note: for the biaxial instructions MC_LINE_R and MC_CIRCLE_R, if both set the maximum acceleration, the smaller one will be taken as the maximum acceleration of the biaxial system. If only one axis is set with the maximum acceleration, then take it as the maximum acceleration of the biaxial system. If the maximum acceleration is not set on both axes, the biaxial system has no limitation for acceleration.

Also: after MAX_SPEED, TA, AXIS_NO have been determined, give SET a rising edge, so these parameters are updated and effective.

PWM instruction

- ① Function name: MC253_PWM



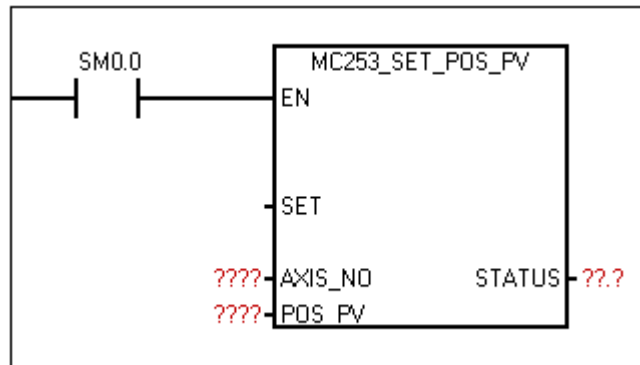
- ② Function: Output different period pulse and duty ratios by setting cycle and duty ratio parameters.

③ Parameter

Name	I/O	Description	Type	Value range	Note
AXIS_NO	IN	Set axis number(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255	Unmodifiable in the process
CFG	IN	Reference time unit 0: 1us, 1: 0.5ms	Byte	0~1	Unmodifiable in the process
CYCLE	IN	Pulse period	Word	2~65535	Unmodifiable in the process
DUTY	IN	Pulse duty factor	Word	0~65535	Unmodifiable in the process
RUN	IN	Enable run	Bool	0~1	
STATUS	OUT	Output status byte: 7 6 5 4 3 2 1 0 Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done. Bit4: modules Estop sign 1—sign is valid, the axis is forbidden run by external condition. 0—sign is invalid Bit5~Bit6: reserved Bit7: instruction communication status sign 1—communication timeout 0—no timeout Byte	0~255	Bit0: ● Judge only to axis parameter configuration error ● CYCLE/DUTY is set to the nearest reasonable value without error reported. ● 253 PLC module is not identified or not connect to the module will prompt parameters configuration errors	

Set target position instruction

- ① Function name: MC253_SET_POS_PV



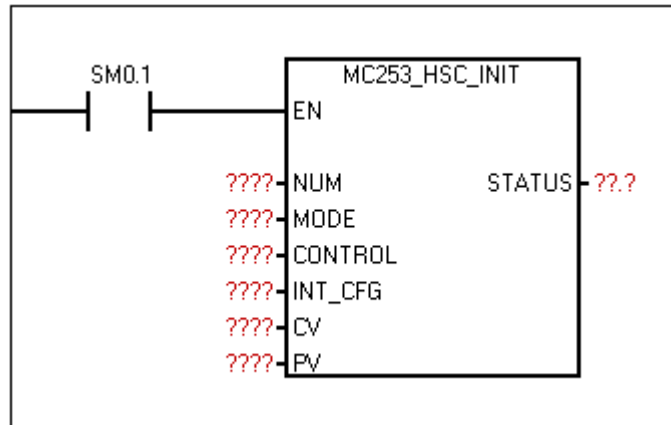
- ② Function: This instruction is used to write the absolute location of the machine to the module. For example, if the power cut off when machine runs to a certain position, save this position. When next time power up, it can be written back to the module, so that the starting point of counting the module absolute position is consistent with the machine actual starting point, no need to return to the origin. If this position happens to be the origin, this instruction has the same effect as MC253_SET_POS_ZERO.

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
AXIS_NO	IN	Set axis number(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255	
SET	IN	After ensuring parameter above, give SET a rising edge to make it effective.	Bool	0~1	
POS_PV	IN	Set target position, in positive and negative. Output positive pulse indicates along X-axis forward direction, negative pulses indicate reverse.	Dint	-2147483648 ~ +2147483647	
STATUS	OUT	Communication status sign 1—communication timeout	Bool	0~1	

Set module HSC instruction

- ① Function name: MC253_HSC_INIT



② Function: Configure HSC



Note

Execute one time on first scan cycle of CPU power on

③ Parameter

Name	I/O	Description	Type	Value range	Note								
NUM	IN	HSC sign(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255									
MODE	IN	Count mode	Byte	0~12	Not support all, refer to attached table below								
CONTROL	IN	Control Word: <table border="1"><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr></table> Bit0: reset effective level control bit 1—low level 0—high level Bit1: reserved Bit2: orthogonal counter speed selection 1—1x velocity 0—4x velocity Bit3: position control bit 1—count addition 0—count decrease Bit4: update direction 1—update direction 0—no update Bit5: update default	7	6	5	4	3	2	1	0	Byte	0~255	
7	6	5	4	3	2	1	0						

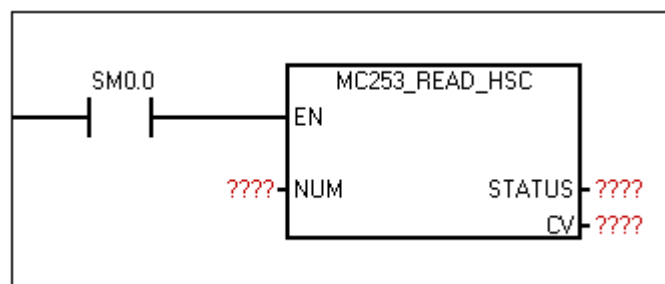
		1—write new default 0—no update Bit6: update current value 1—write new current value 0—no update Bit7: effective bit 1—effective 0—ineffective			
INT_CFG	IN	Interrupt configuration(reserved)	Byte	reserved	reserved
CV	IN	New current value	Dword	-2147483 648 ~ +2147483 647	
PV	IN	New default value	Dword	-2147483 648 ~ +2147483 647	
STATUS	OUT	Communication status sign 1—communication timeout	Bool	0~1	

Attached Table

Mode	Description	Input		
	MC253_HSC0	I0.0	I0.1	I0.2
	MC253_HSC1	I0.4	I0.5	I0.6
0	Single phase counter with internal direction control	Clock		
1		Clock		Reset
3	Single phase counter with external direction control	Clock	Direction	
4		Clock	Direction	Reset
9	A/B phase orthogonal counter	Clock A	Clock B	
10		Clock A	Clock B	Reset

Read modules HSC status instruction

① Function name: MC253_READ_HSC



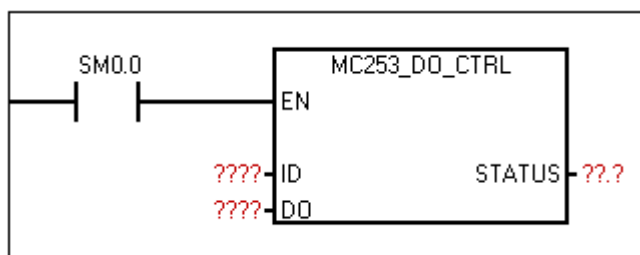
② Function: Read HSC count status and current value

③ Parameter

Name	I/O	Description	Type	Value range								
NUM	IN	HSC sign(2 axes for each EM253 module, axis number range decided by motion control modules number)	Byte	0~255								
STATUS	OUT	Status Word: <table border="1"><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr></table> Bit0: communication status bit 1—communication timeout Bit1~Bit4: reserved Bit5: current count direction bit 1—count addition Bit6: current value equal to default 1—equal Bit7: current value greater than default 1—greater than	7	6	5	4	3	2	1	0	Byte	0~255
7	6	5	4	3	2	1	0					
CV	OUT	New current value	Dword	-2147483648 ~ +2147483647								

Control modules output instruction

① Function name: MC253_DO_CTRL



② Function: Control output of motion control module

③ Parameter

Name	I/O	Description	Type	Value range	Note
ID	IN	Motion control module ID number	Byte	0~255	Apply only to motion control module
DO	IN	Module output port value	Byte	0~255	DO corresponds to 8 Q output, low byte corresponds to Q low bit, such as DO bit0

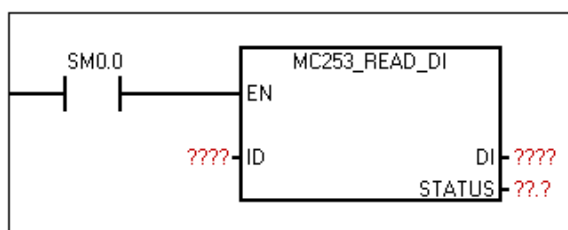
					corresponds to Q0.0
STATUS	OUT	Communication status sign 1—communication timeout	Bool	0~1	

**Note**

- The number of motion control module is only internally unified. If the PLC has 3 modules, among which module 0 and 2 is the motion control module, module 1 is not, then module 0 is ID0, module 1 is uncontrollable for it is not motion control module, and module 2 is ID1.
- If a single motion control module is attached behind the CPU, the delay from enable to execution of the output instruction is about 780us due to transmission delay. If there are up to 5 motion control modules hanging behind the CPU, the delay is about 930us.

Read modules input status instruction

- ① Function name: MC253_READ_DI



- ② Function: Read input of motion control module

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
ID	IN	Motion control module ID number	Byte	0~255	Apply only to motion control module
DI	OUT	Module output port value	Byte	0~255	DI corresponds to 8 I input, low byte corresponds to I low bit, such as DI bit0 corresponds to I0.0
STATUS	OUT	Communication status sign 1—communication timeout	Bool	0~1	

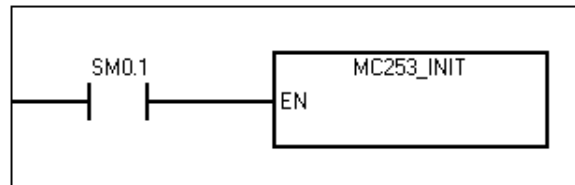
**Note**

- The number of motion control module is only internally unified. If the PLC has 3 modules, among which module 0 and 2 is the motion control module, module 1 is not, then module 0 is ID0, module 1 is uncontrollable for it is not motion control module, and module 2 is ID1.
- When an axis call motion control instruction, if confirm to operating conditions, the

running process of the axis port output pulse or direction according to the instruction (corresponding port is not controlled by the MC253_DO_CTRL), the port can restore normal IO functions after run (accept instruction MC253_DO_CTRL control); use instruction MC253_DO_CTRL carefully so as not to affect the use of motion control.

Initialize motion control module instruction

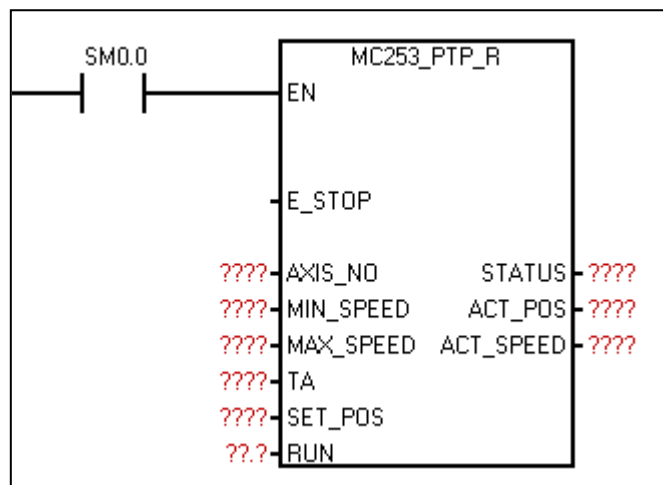
- ① Function name: MC253_INIT



- ② Function: Initialize the system control variable of motion control module
- ③ Parameter: NONE

Uniaxial relative motion instruction

- ① Function name: MC253_PTP_R



- ② Function: Used for uniaxial point - to - point control (uniaxial fixed - length drive). Call one time output fixed pulse, by setting the maximum, minimum speed and deceleration time, the output pulse will gradually accelerate to the largest speed, when the pulse number is ready to run, cut down pulse frequency automatically to prevent vibration or jammed due to too much inertia when start or stop the machine.

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	Bool	0/1	1 Run only when Run =1 and E_Stop =0 2 RUN reset internal when E_STOP = 1
AXIS_NO	IN	Set axis number,	Byte	0~3	Unchangeable in

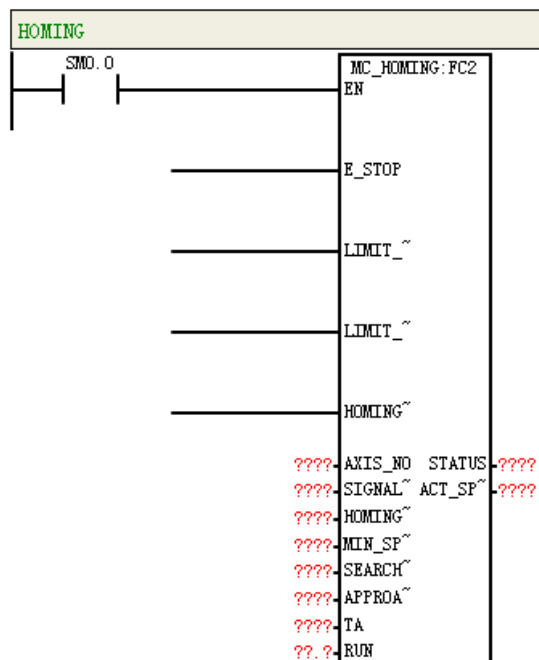
		0/1/2/3			the process
MIN_SPEED	IN	Minimum speed of run or stop. Unit: HZ	Dword	500~200000	1 Set minimum speed less than maximum speed
MAX_SPEED	IN	Maximum speed of run Unit: HZ	Dword	500~200000	2 Changeable in the process
TA	IN	Time of ACC/DEC, Unit: ms	Dword	0~10000	Changeable in the process; if TA=0, no settings for acceleration.
RUN	IN	Run enable bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 RUN reset internal when run complete. 3 when E_STOP=1, RUN reset internal.
STATUS	OUT	7 6 5 4 3 2 1 0 Output status byte: Bit0: parameter configuration error sign 1—parameter configuration error 0—parameter configuration normal Bit1: Run sign 1—Running 0—Do not run Bit2: Completion sign 1—Instruction execution completed 0—Instruction execution do not completed Bit3: Busy sign 1—Valid, the axis occupied by other instruction 0—Invalid, executing instruction or execution done.	BYTE	0~255	Bit0: 1 Judge Only to axis parameter configuration error and homing mode out of range. 2 No error reported for other parameters, it will set to the nearest reasonable value automatically. 3 Report parameter failure if TA=0 and the maximum acceleration is not set; TD the same

SET_POS	IN	Output pulse. The positive output pulse indicates the positive direction along the X-axis, and the negative pulse number indicates the negative direction along the X-axis.	DInt	-2147483647 ~ +2147483647	Modifiable during operation. When the new set value is greater than the number of pulses output, the final output pulse will be subject to the new set value. When new value is less than output pulses number, the pulse output will be stopped immediately
ACT_POS	OUT	Absolute coordinate of current axis (1 pulse 1 units coordinate)	DInt	-2147483647 ~ +2147483647	This instruction no error and set the axis number correctly.
ACT_SPEED	OUT	Current run speed	Dword	500~200000	Note: this value may be a little off actual one, at most 5K, related to acceleration time and setting speed

Homing instruction

- ① Function name: MC253_HOMING

Network 5



- ② Function: Find device origin by setting parameters such as homing mode. The relationship between the axis number and external reset IO signal (such as homing Z pulse):

Axis 0 → I0.2 (MC253_HSC0)

Axis 1 → I0.6 (MC253_HSC1)

If the homing mode refers to the origin switch (mode 3 or 4), connect origin switch signal (the HOMING_SW parameter of the instruction) to the corresponding point above; otherwise, the origin cannot be found.

- ③ Parameter

Name	I/O	Description	Type	Value range	Note
E_STOP	IN	Emergency stop bit 1: valid, 0: invalid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.
LIMIT_SW_CCW	IN	CCW limit switch	BOOL	0~1	
LIMIT_SW-CW	IN	CW limit switch	BOOL	0~1	
HOMING_SW	IN	Origin switch	BOOL	0~1	
AXIS_NO	IN	Axis number	BYTE	0~255	Unchangeable in the process
SIGNAL_TYPE	IN	7 6 5 4 3 2 1 0 Signal type: Bit0: Counterclockwise stroke limit input signal type 0—High level 1—Low level Bit1: Clockwise stroke limit input signal type 0—High level 1—Low level Bit2: Origin switch signal type 0—High level 1—Low level	BYTE	0~255	
HOMING_MODE	IN	Homing mode	BYTE	1~14	Refer to Appendix E.6 Back to Zero Mode Illustration

MIN_SPEED	IN	Minimum speed Unit: Hz	DWORD	0~200000	1 pulse output close when speed < 5Hz. 2 Unchangeable in the process 3 Search speed should not be too large, close speed should be as small as possible
SEARCH_SPEED	IN	Origin search speed Unit: Hz	DWORD	0~20000 0	
APPROACH_SPEED	IN	Origin close speed Unit: Hz	DWORD	0~20000 0	
TA	IN	Time of ACC/DEC. Unit: ms	DWORD	0~10000	Changeable in the process
RUN	IN	Enable run bit 1: valid	BOOL	0~1	1 Run only when RUN =1 and E_STOP =0. 2 When E_STOP=1, RUN reset internal.

④ Illustration

The program detect each input by scanning (the Z phase is not affected by this), so when the switch value changes and process not in time, there may be some delay. If the homing speed (including search and close speed) is too high, the processing delay will be amplified, resulting in inaccurate return.

Note

1 When $TA \neq 0$, $\text{acceleration} = (\text{MAX_SPEED} - \text{MIN_SPEED}) / TA$ (limited by maximum acceleration if it is set before); When $TA=0$, use MC253_SET_MAX_ACCELE to set maximum acceleration, parameter will report error if not set. Same for TD, for two-axis instruction, if both axes have set maximum acceleration, then adopt smaller one as system acceleration.

2 In theory, $\text{acceleration} = \lfloor (\text{MAX_SPEED} - \text{MIN_SPEED}) / TA \rfloor$. If the desired acceleration is too small (less than 1), the instruction default internal acceleration is 1. Users can reasonably set TA/TD according to the expected acceleration.

3 Import "Module emergency stop signal" in module input port: when the module detects this signal, forbid to output pulse, and alarm at the status of pulse output instructions (such as MC253_PTP_R, MC253_SPEED_CTL and MC253_PWM instructions). The corresponding relation between axis number and emergency stop signal:

Axis 0 → I0.3

Axis 1 → I0.7

J Using Permanent V Memory Library “CT_savevmem”

Introduction

CT_SAVEVMEM is provided to the user as a library function. It can save a segment of V memory data the user needs to permanent memory, so that the data will not lost for a long time (about a year).

CTSC -200 CPU supports permanently save V memory library, CPU224 data storage space 8KB can be extended to 108KB; CPU226 data storage space 10KB can be extended to 110KB.



Note

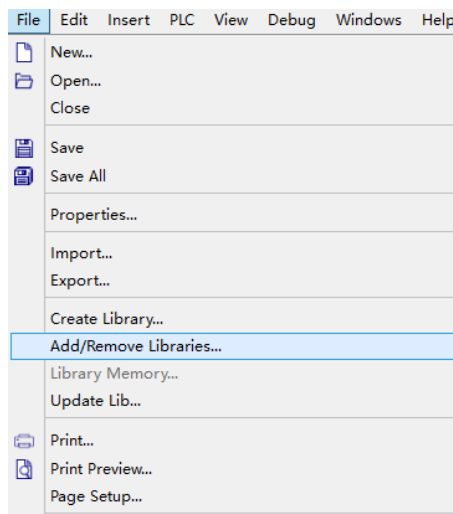
Apply to save parameters, do not save too often.

Do not set power-off hold in the system block for the permanent storage V memory address to be used.

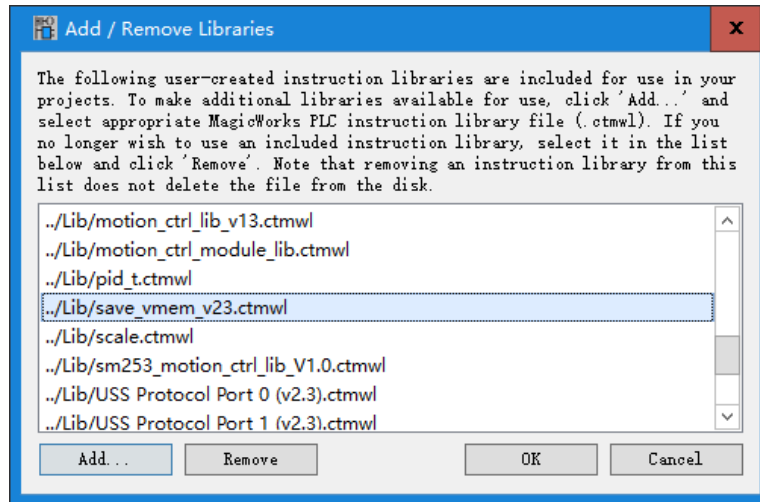
Software Installation

1) Add library file

“File” -> “Add/delete library”, find library file “ct_savevmem” as figure below:



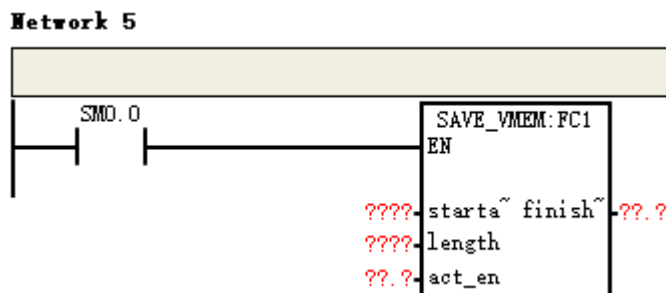
Click “Add” in dialog box popped up, find “ct_savevmem.ctmwl”, click “Save” after you find it, then select “ct_savevmem.ctmwl” in “Add/delete library”, click “OK”



You can see newly added “ct_savevmem” under directory tree after installation:

2) Call CT_SAVEVMEM library

Click “Network” where you add function block, double click “SAVE_VMEM” under “library”, it shows in “Network”



Note

Make sure EN is always ON before complete write, in other Word use sm0.0 or act_en to call.

3) CT_SAVEVMEM library function illustration

Parameter address	Illustration	Type	Note
str_addr	Start address of V memory	WORD	Can be constant or variable. Start address of VW500 is 500
length	Length unit: word	WORD	Permanently store the length of contiguous V memory. Length of VB500-VB4499 is 2000 or 2000 for VW500-VW4498
act_en	Allow write bit	BOOL	When this bit is 1, write permanent memory, automatically reset after write, and the bit is kept at 1 during the write.
finished	End write sign bit	BOOL	Automatically reset at the beginning of writing and set to 1 at the end of writing.
Note: total length of write is integer multiple of the Word.			

K Using CTSC-200 CPU Telemaintenance

Introduction

This chapter is to guide readers to use RS-232/PPI multi-master Cable (RS232/PPI multi-master Cable) to maintain remote PLC (including downloading, uploading and other functions) through modem and telephone network with STEP7-Micro/WIN running on local PC.

To realize telemaintenance, you need:

- Remote: a modem, a CPU to be maintained, a PC with RS232 series port and STEP MICRO/WIN
- Local: modem, a PC with RS232 series port and STEP MICRO/WIN

Configuration to complete CPU telemaintenance:

- Configure remote RS-232 PC/PPI Multi-Master system cable
- Configure “STEP7-Micro/WIN” software run on local PC

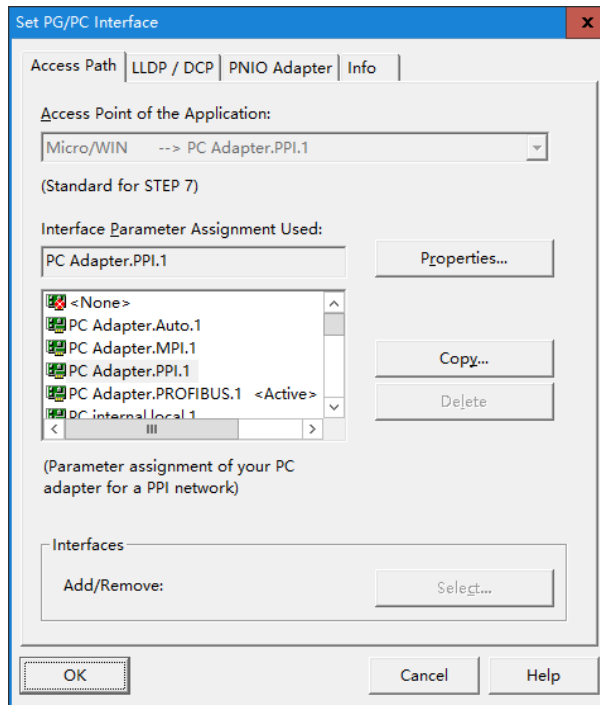
Configuration

Configure remote RS-232 PC/PPI Multi-Master system cable

【Initialize configuration】

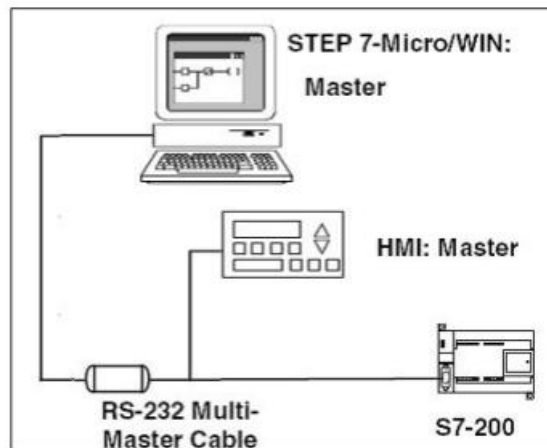
1) Select the COM port for initialize PC/PPI cable

Open “Set PG/PC Interface” of STEP7-Micro/WIN, set port parameter as “PC/PPI cable(PPI)”, click “Properties”, select serial port the cable to be connected in properties box popped up, also select “modem connection”, click “OK” twice to complete initialize configuration as figure shows below:



2) Connect PC/PPI cable

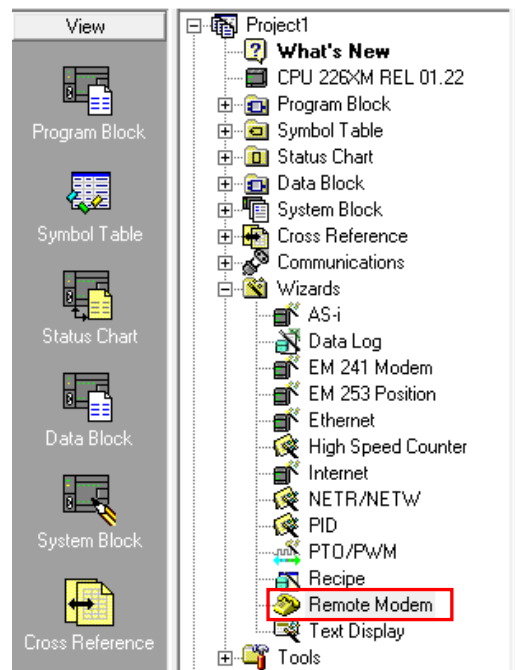
After configuring serial port, set DIP switch of PPI cable as  (white means DIP switch), then connect the cable one end to the serial port of the newly configured PC, and connect the other end to the 485 port of the CPU (the purpose is to power the cable through the CPU) to power the CPU, as shown in the figure:



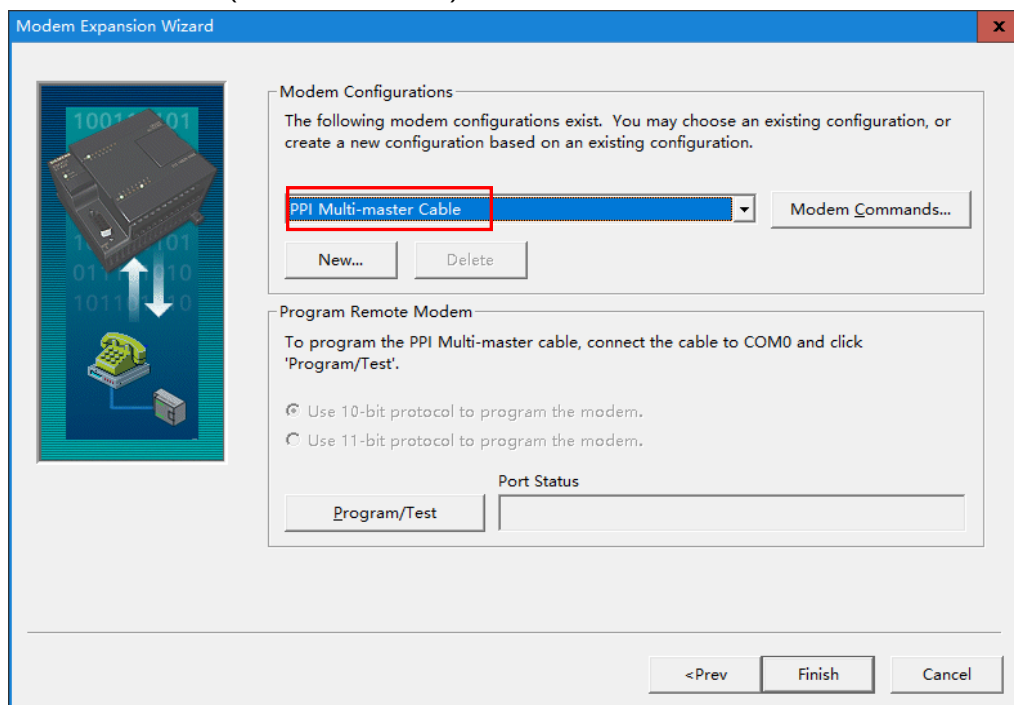
HMI in figure: Master can be both other main station and multi-master station, as well as none

3) Configure "Remote modem" wizard

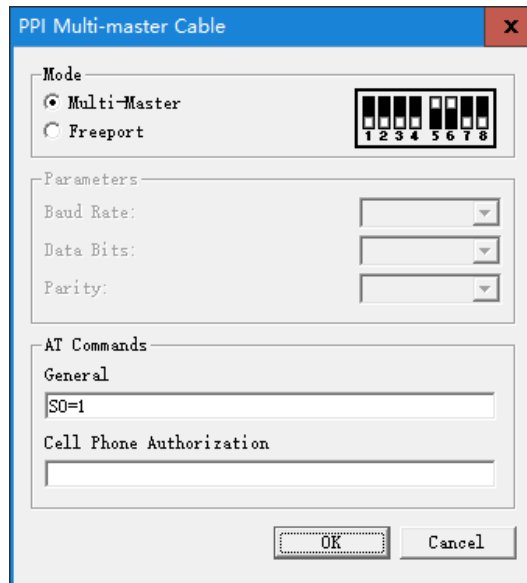
Configure "Remote modem" wizard after connecting PPI cable as shown in the figure:



In “Modem Expansion Wizards”, select “Configure analog modem, mobile phone modem or PPI multi-master station cable” item Click “Next” configure modem, select “PPI Multi-master Cable” (show as red circle)



Select “Modem Command” on interface above to enter dialog box below



Select "Multi-Master" mode in the red circle, AT Commands use default, click "OK"

4) Test PC/PPI cable that configured

After completing the mode configuration of PC/PPI multi-station cables, back to the modem configuration screen and click the "program/test" to test the configured PC/PPI multi-station cables. When the configuration succeeds or fails, a different message is prompted in "Ports Status" in the right.

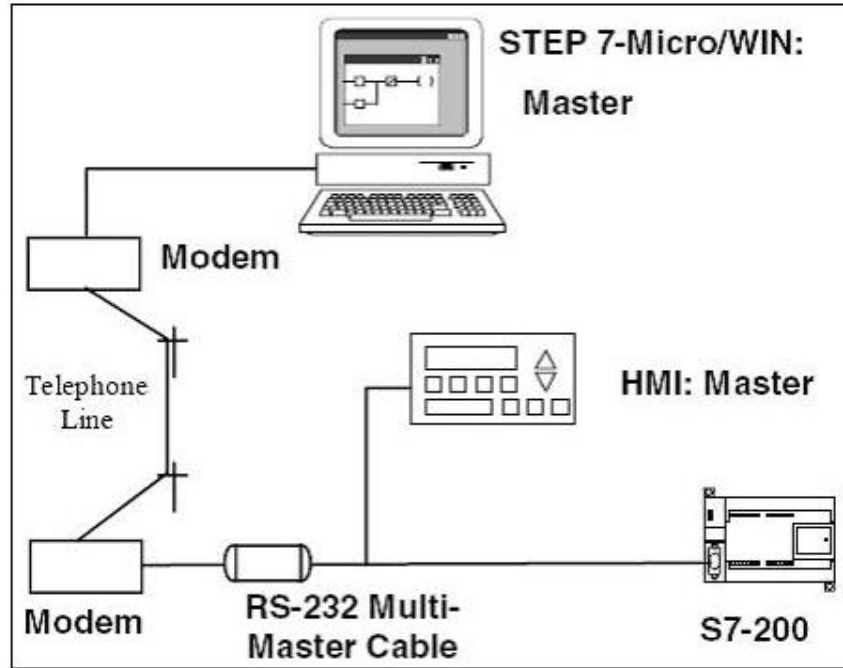
- 1) Configuration succeed (COM3 vary from serial port you use)
- 2) Configuration failed, possible reasons, and 1:DIP switch of cable configuration wrong 2: Connection between cable and PC failure, check connection between DIP switch of cable and PC serial port
- 3) Configuration failed, PG/PC didn't configure as mode use modem, reconfigure "set PG/PC interface"

Complete initialization after configuration succeed

【Use configuration】

After configuration succeed, power off PC/PPI cable, and connect the end remove from PC serial port to modem serial port, then power PC/PPI cable and modem on, configure

PC/PPI cable DIP switch as  (white means DIP switch), show as figure



HMI in the figure: Master can be both other main station and multi- master station, as well as none

**Note**

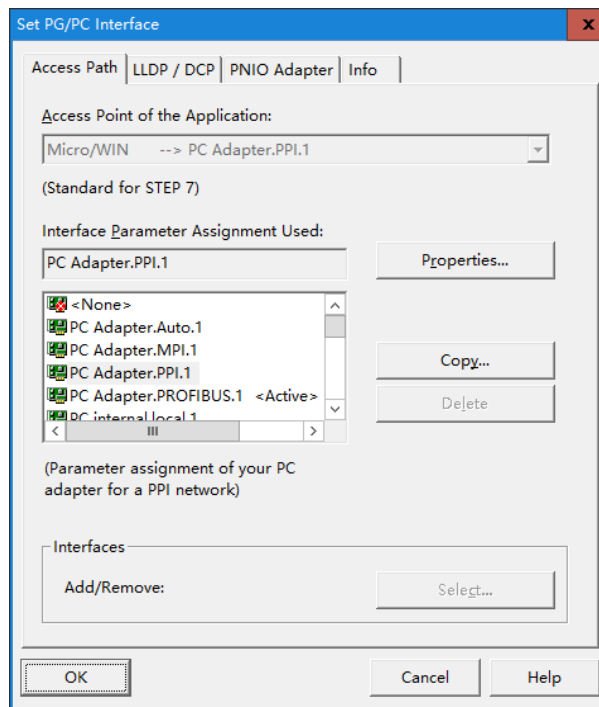
RS-232/PPI multiple primary device cables will be configured with a modem each time power on. Make sure the modem is also powered up before or almost at the same time as the cable is powered up. Also, if the modem power cycles, make sure the cable power cycles as well. This allows cable to properly configure the modem and run at the highest baud rate available.

Configure local PC modem connection

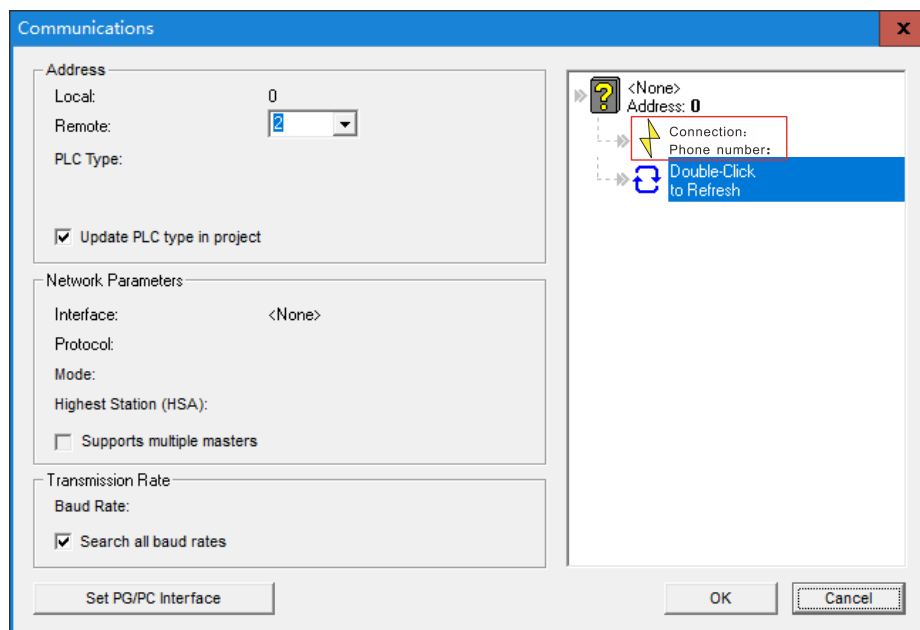
Please install modem drive first

1) Configure modem and PC connected port

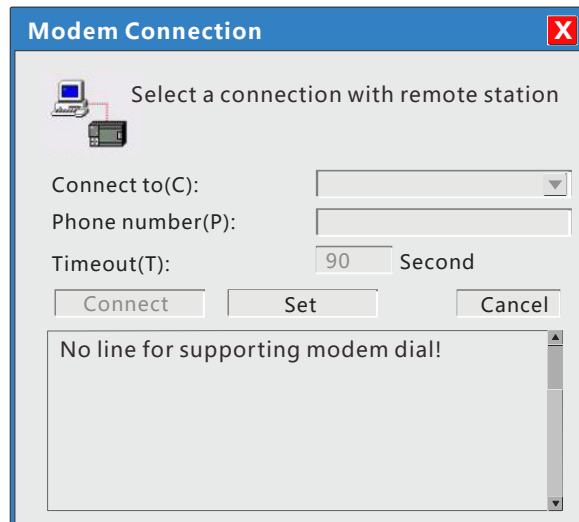
Configure STEP7-Micro/WIN to use Modem mode by selecting Modem connection in the dialog box for configuring STEP7-Micro/WIN to use the serial port, as shown in the figure.



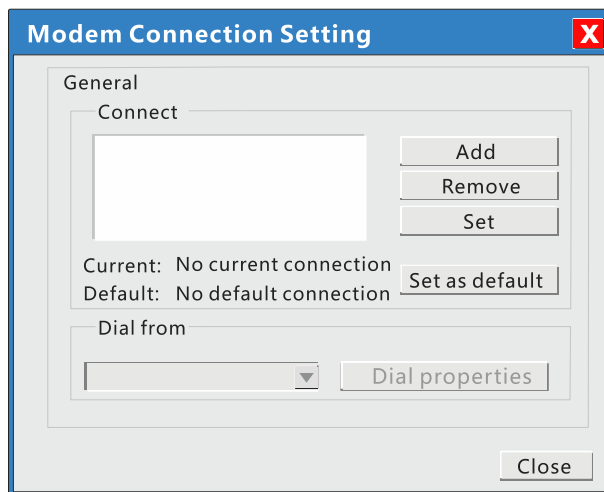
2) Establish connection between modem and PC



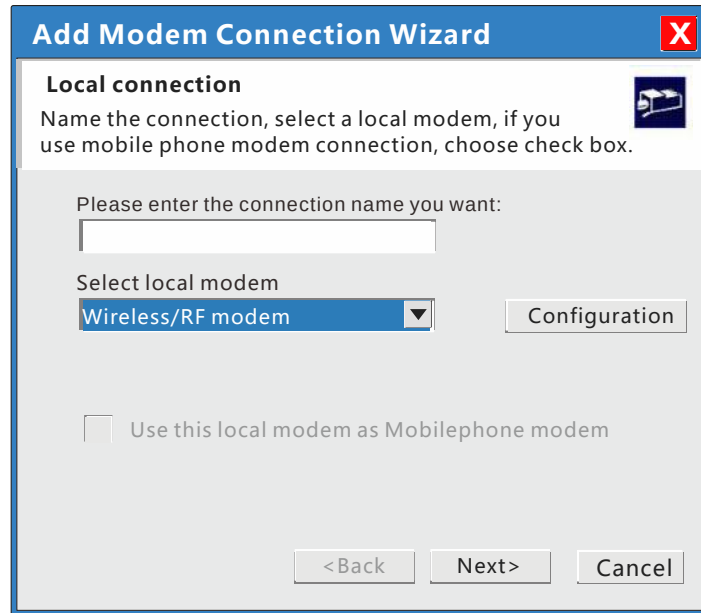
Double click red circle in figure above to enter modem connection configuration as shows below:



Click "Set" to add connection circuit. "Modem Connection Setting" dialog box shows as below:



Click "Add" or "Set" to add modem connection, "Add modem connection wizard" dialog box shows as below:



Add Modem Connection Wizard

Local connection

Name the connection, select a local modem, if you use mobile phone modem connection, choose check box.

Please enter the connection name you want:

Select local modem

Wireless/RF modem

☐ Use this local modem as Mobilephone modem

<Back Next> Cancel

Enter connection name in “Please enter the connection name you want”, then select modem you already installed in “Select local modem” list, shows as below:



Add Modem Connection Wizard

Local connection

Name the connection, select a local modem, if you use mobile phone modem connection, choose check box.

Please enter the connection name you want:

Select local modem

Add a modem

TP-Link ESC 5658V V92 External..

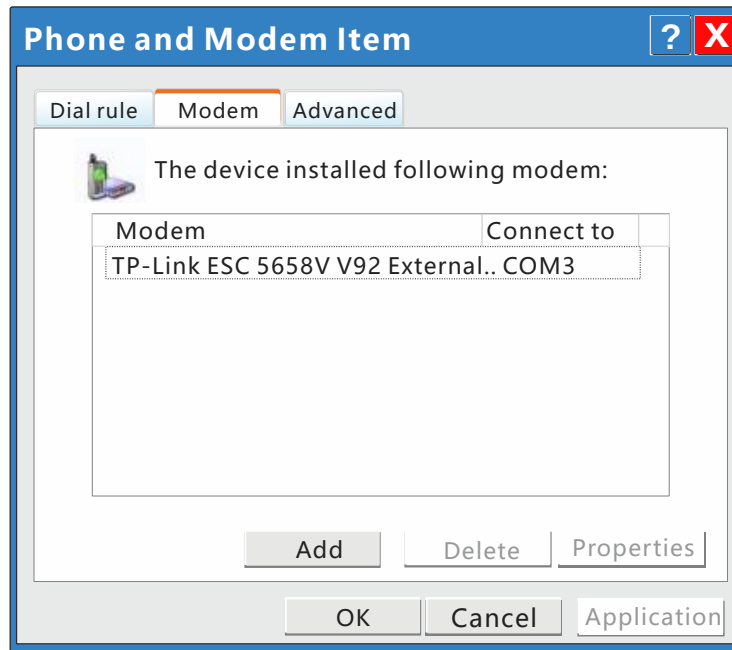
Add a modem

Wireless/RF modem

☐ Use this local modem as Mobilephone modem

<Back Next> Cancel

If you do not find the modem you need, click the "add" in the "configuration" dialog box and follow the wizard to complete the adding process. After that, the information about modem is shown in the red circle below:



After clicking “OK”, the newly added modem is displayed in the list box under select local modem under add modem wizard. Select the desired modem from the list box, as shown below:



Click “Next>”. Set the remote station phone number (remote modem), as shown in the figure, select "Use dial rule", and press the wizard to complete the remote phone number setting.



Add Modem Connection Wizard

Number to dial

You have to appoint the phone number of remote station you want to connect.

Enter phone number of remote station. If you hope PC automatically ensure how to dial from different position, please use dial rule?

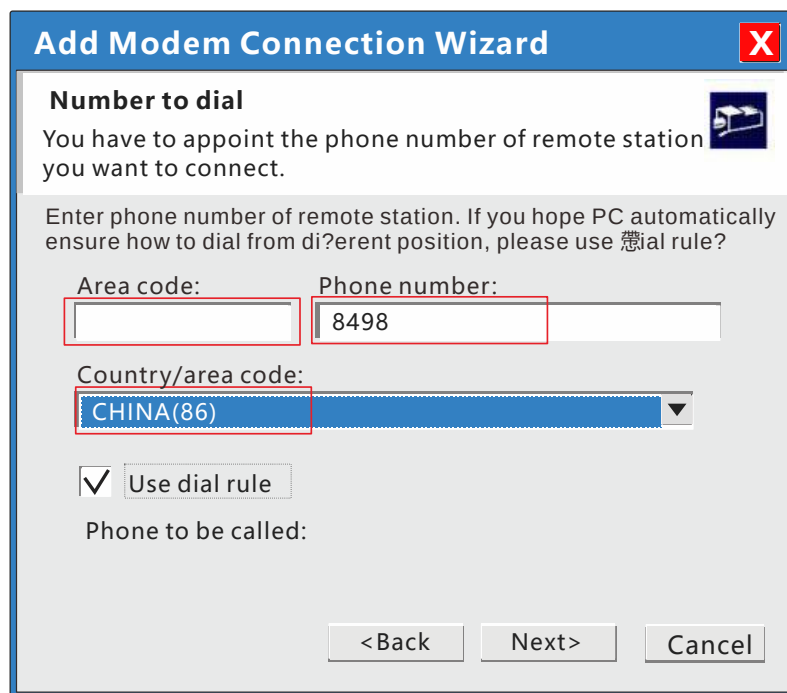
Area code: Phone number:

Country/area code:

☐ Use dial rule

Phone to be called:

<Back Next> Cancel



Add Modem Connection Wizard

Number to dial

You have to appoint the phone number of remote station you want to connect.

Enter phone number of remote station. If you hope PC automatically ensure how to dial from different position, please use dial rule?

Area code: Phone number:

Country/area code:

☒ Use dial rule

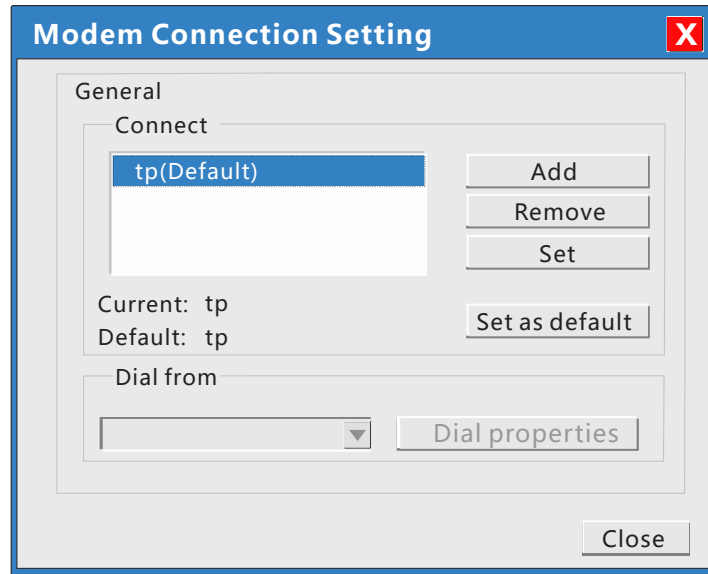
Phone to be called:

<Back Next> Cancel

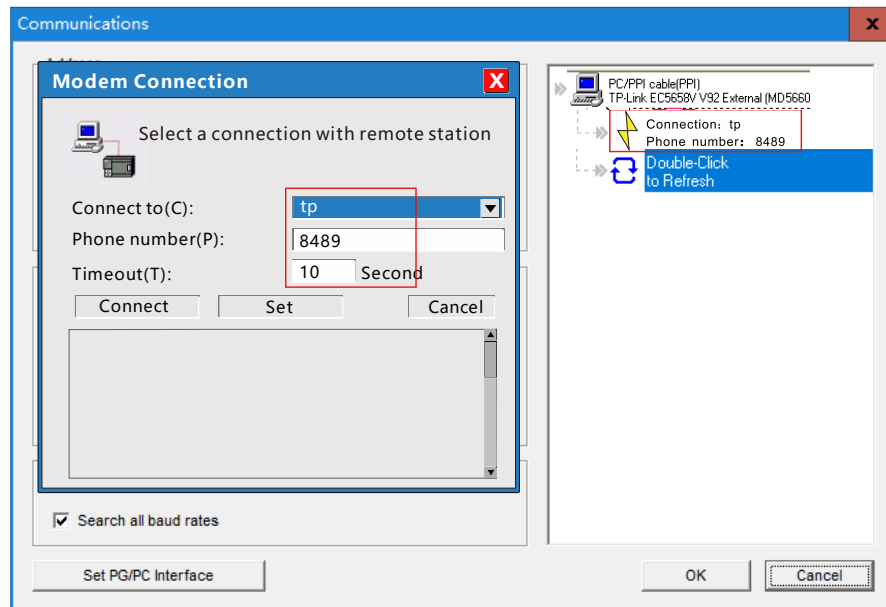
**Note**

If local and remote use different extension Numbers of the same network (i.e. interphone), try to set as the interphone number (if the remote interphone number is 8494, then set the "telephone number" 8489 directly).

After setting the remote phone number, the name will show in "Modem connection setting" as shown in the red circle below:

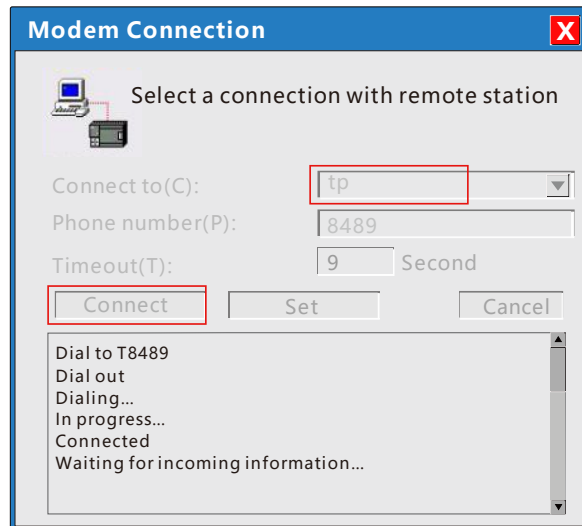


Create multiple modem connections in the same way as above. Select the connection you want to use and click "Close". The connection information in "Communications" and "Modem Connection" are shown as follows:

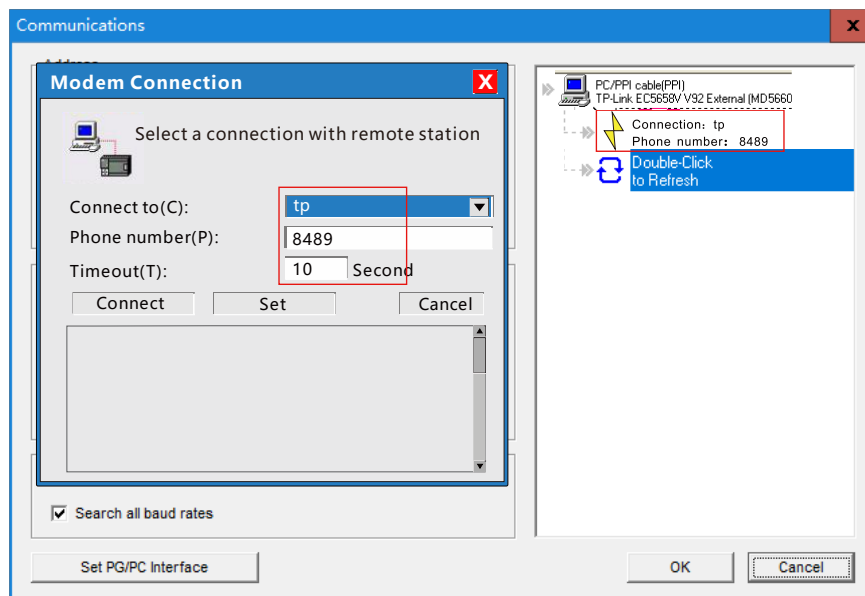


3) Establish connection with remote modem

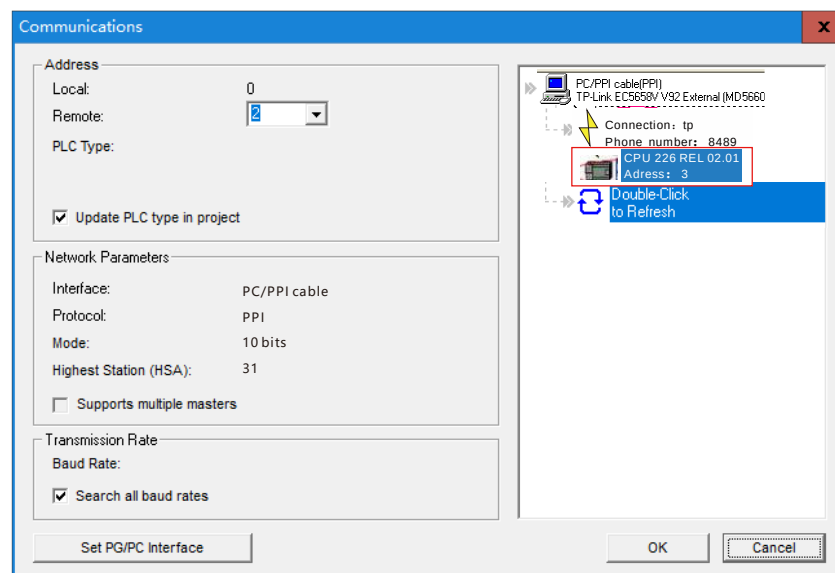
Click the "Connect" in "Modem Connection" to establish connection with the remote station, and the modem will issue a dial-up operation until the dialing is complete. The dialing process is shown in the figure.



There may be some changes in communication dialog box after getting through



Remote connection established as figure below:



L Using CTSC-200 Storage Card and Battery Card

TrustPLC CTSC-200 CPU only support CTSC-200 dedicated storage card and battery card

Type	Description	Order No.
Storage card	Storage card, 64K(user program, recipe and data record)	CTS7 291-MC064
Storage card	Storage card, 256K(user program, recipe and data record)	CTS7 291-MC256
Battery card	Battery card(data-hold time): 200 days, typical	CTS7 291-8BA33



- ✧ The plug-in place of battery, memory card show as left. Lift the box cover to insert the battery card or storage card.
- ✧ COTRUST storage card use the same with Siemens memory card, mainly used for storing user program, recipe and data record, but CTSC - 200 memory card will not automatically copied content to PLC after PLC electrical restart after the memory card.
- ✧ If CTSC-200 battery card backup, the data-hold time is about 200 days

M Instruction Set

Table M-1 CTSC-200 instruction set

Bit logic instruction			
LD	Normally open contact load instruction LD SM0.0	ALD	AND load instruction ALD
A	Normally open contact AND instruction A SM0.0	OLD	OR load instruction OLD
O	Normally open contact OR instruction O SM0.0	LPS	Logic into stack instruction LPS
LDN	Normally close contact load instruction LDN SM0.0	LDS	Load stack instruction LDS 1
AN	Normally close contact AND	LRD	Logic read instruction

	instruction AN SM0.0		LRD
ON	Normally close contact OR instruction ON SM0.0	LPP	Logic out stack instruction LPP
LDI	Normally open contact immediately load instruction LDI IO.0	=	Output instruction = Q0.0
AI	Normally open contact AND immediately instruction AI IO.0	=I	Immediately output instruction =I Q0.0
OI	Normally open contact OR immediately instruction OI IO.0	S	Set instruction S Q0.0 1
LDNI	Normally close contact immediately load instruction LDNI IO.0	SI	Immediately set instruction SI Q0.0 1
ANI	Normally close contact AND immediately instruction ANI IO.0	R	Reset instruction R Q0.0 1
ONI	Normally close contact OR immediately instruction ONI IO.0	RI	Immediately reset instruction RI Q0.0 1
NOT	Logic stack top reverse instruction NOT	AENO	Power flow AND instruction AENO
EU	Rising edge detection instruction EU	NOP	Null operation instruction NOP 1
ED	Falling edge detection instruction ED	--	--
Comparison instruction			
LDBx	Byte comparison (=,<,>,<=,>=,<>) load instruction LDB= 1, VB0	OBx	Byte operand comparison (=, <,>,<=,>=,<>) OR instruction OB= 1, VB0
LDWx	Integer comparison (=,<,>,<=,>=,<>) load instruction LDW= 10000, VW0	OWx	Integer comparison (=,<,>,<=,>=,<>) OR instruction OW= 10000, VW0
LDDx	Long integer comparison (=,<,>,<=,>=,<>) load instruction LDD= 100000, VD0	ODx	Double integer comparison (=,<,>,<=,>=,<>) OR instruction OD= 100000, VD0

LDRx	Floating number comparison (=,<,>,<=,>=,<>) load instruction LDR= 1.0, VD0	ORx	Floating number comparison (=, <, >, <=, >=, <>) OR instruction OR= 1.0, VD0
ABx	Byte operand comparison (=,<,>,<=,>=,<>) AND instruction AB= 1, VB0	LDSx	String comparison (=,<>) load instruction LDS="1234567890",VB0
AWx	Integer comparison (=,<,>,<=,>=,<>) AND instruction AW= 10000, VW0	ASx	String comparison (=,<>) AND instruction AS="1234567890", VB0
ADx	Double integer comparison (=,<,>,<=,>=,<>) AND instruction AD= 100000, VD0	OSx	String comparison (=,<>) OR instruction OS="1234567890",VB0
ARx	Floating number comparison (=,<,>,<=,>=,<>) AND instruction AR= 1.0, VD0	--	--
Transmit instruction			
MOVB	Byte move instruction MOVB 1 VB0	SWAP	High and low byte exchange instruction SWAP VW0
MOVW	Word move instruction MOVW 1000,VD0	BIR	Move byte immediately read instruction BIR IB0, VB0
MOVD	Double Word move instruction MOVD 100000,VD0	BIW	Move byte immediately write instruction BIW VB0, QB0
MOVR	Floating number move instruction MOVR 1.0,VD0	SRCP	Recipe to storage card
BMB	Block move byte instruction BMB VB0,VB100, 1	LRCP	Load recipe from storage card
BMW	Block move Word instruction BMW VW0,VW100, 1	SDL	Record data to storage card
BMD	Block move double Word instruction BMD VD0,VD100, 1	--	--
Integer calculate instruction			
+I	Integer addition instruction +I 10000, VW0	MUL	Integer multiply with double Integer MUL 10000, VD0

-I	Integer subtraction instruction -I 10000, VW0	DIV	Integer divide with double Integer DIV VW0, VD0
*I	Integer multiply instruction *I 10000, VW0	INCB	Byte increasing instruction INCB VB0
/I	Integer division instruction /I 10000, VW0	DECB	Byte decreasing instruction DECB VB0
+D	Double Integer addition instruction +D 100000, VD0	INCW	Integer increasing instruction INCW VW0
-D	Double Integer subtraction instruction -D 100000, VD0	DECW	Integer decreasing instruction DECW VW0
*D	Double Integer multiply instruction *D 100000, VD0	INCD	Long Integer increasing instruction INCD VD0
/D	Double Integer division instruction /D 100000, VD0	DECD	Long Integer decreasing instruction DECD VD0
Float-point calculate instruction			
+R	Real addition instruction +R 1.0, VD0	COS	Cosine calculate instruction COS 1.0, VD0
-R	Real subtraction instruction -R 1.0, VD0	TAN	Tangent calculate instruction TAN 1.0, VD0
*R	Real multiplication instruction *R 1.0, VD0	LN	Nature exponential calculate instruction LN 1.0, VD0
/R	Real division instruction /R 1.0, VD0	EXP	Nature exponential calculate instruction EXP 1.0, VD0
SQRT	Square root instruction	PID	PID circuit calculate instruction PID VB0, 1
SIN	Sine calculate instruction SIN 1.0, VD0	--	--
Convert instruction			
BTI	Convert from byte to Integer BTI 1, VW0	DTR	Convert from Double Integer to real number DTR 100000, VD0
ITB	Convert from Integer to byte ITB 10000, VB0	DTS	Convert from Double Integer to string DTS 100000, VB0, 10
ITD	Convert from Integer to double Integer ITD 10000, VD0	ROUND	Carry integer instruction ROUND 1.0, VD0

ITS	Convert Integer to string ITS 10000, VB0, 10	TRUNC	Cut bit integer instruction TRUNC 1.0, VD0
DTI	Convert from double Integer to Integer DTI 100000, VW0	RTS	Convert from real to string RTS 1.0, VB0, 10
BCDI	Convert from BCD to Integer BCDI VW0	STI	Convert from string to Integer STI"1234567890", 5, VW0
IBCD	Convert from Integer to BCD IBCD VW0	STD	Convert from string to double Integer STD"1234567890", 5, VD0
ITA	Convert from Integer to ASCII ITA 10000,VB0,10	STR	Convert string to real STR"1234567890",5, VD0
DTA	Convert from double Integer to ASCII DTA 100000, VB0, 10	DECO	Decode instruction DECO 1, VW0
RTA	Convert from real to ASCII instruction	ENCO	Coding instruction ENCO 10000, VB0
ATH	Convert from ASCII to 16 system ATH VB0, VB100, 10	SEG	Segment instruction SEG 1, VB0
HTA	Convert from 16 system to ASCII HTA VB0, VB100, 10	--	--
Real-time clock			
TODR	Read real-time clock TODR VB0	TODRX	Read extend real-time clock TODRX VB0
TODW	Set real-time clock TODW VB0	TODWX	Set extend real-time clock TODWX VB0
Shift cycle instruction			
SLB	Left shift byte SLB VB0, 4	RLW	Left rotation Word RLW VW0, 8
SLW	Left shift Word SLW VW0, 8	RLD	Left rotation double Word RLD VD0, 16
SLD	Left shift double Word SLD VD0, 16	RRB	Right rotation byte RRB VB0, 4
SRB	Right shift byte SRB VB0, 4	RRW	Right rotation Word RRW VW0, 8
SRW	Right shift Word	RRD	Right rotation double Word RRD VD0, 16
SRD	Right shift double Word SRD VD0, 16	SHRB	Shifting register bit instruction SHRB I0.0, V0.0, 8
RLB	Left rotation byte RLB VB0, 4	--	--

Logic calculate instruction			
INVB	Reverse byte instruction INVB VB0	ORB	OR byte instruction ORB 1, VB0
INWV	Reverse Word instruction INWV VW0	ORW	OR Word instruction ORW 10000, VW0
INVD	Reverse double Word instruction INVD VD0	ORD	OR double Word instruction OD 100000, VD0
ANDB	AND byte instruction ANDB 1, VB0	XORB	XOR byte instruction XORB 1, VB0
ANDW	AND Word instruction ANDW 10000, VW0	XORW	XOR Word instruction XORW 10000, VW0
ANDD	AND double Word instruction ANDD 100000, VD0	XORD	XOR double Word instruction XORD 100000, VD0
Counter instruction			
CTU	Count-up instruction CTU C1, 10000	HDEF	HSC definition HDEF 0, 0
CTD	Count-down instruction CTD C1, 10000	HSC	HSC instruction
CTUD	Count down and up instruction CTUD C1, 10000	PLS	Pulse output instruction
Timer instruction			
TON	Open delay timer TON T37, 10000	BITIM	Read start timer BITIM VD0
TONR	Reserved open delay timer TONR T31, 10000	R_UTIM	Read current microsecond value instruction
TOF	Turn off delay timer TOF T37, 10000	CITIM	Calculate interval timer CITIM VD0, VD100
String instruction			
SLEN	Get string length instruction SLEN"1234567890", VB0	SCAT	Concatenation string instruction SCAT"1234567890", VB0
SCPY	Copy string instruction SCPY"1234567890", VB0	SFND	Seek string in string instruction SFND"12345678890", "321", VB0
SSCPY	Copy substring from string instruction SSCPY"1234567890", 1, 10, VB0	CFND	Seek a character in string instruction CFND"12345678890", "a", VB0
Table instruction			
FILL	Memory fill instruction FILL 10000, VW0, 10	FIFO	First in-first out instruction FIFO VW200, VW400

ATT	Add to table instruction ATT 10000, VW0	LIFO	Last in-first out instruction LIFO VW200, VW400
FNDx	Seek table (=,<,>,<=,>=,<>) instruction FND=VW0,9999,VW1000	--	--
Interrupt instruction			
CRETI	Return from interrupt routine instruction	ATCH	Add interrupt instruction
ENI	Start interrupt instruction ENI	DTCH	Separate interrupt instruction
DISI	Forbid interrupt instruction DISI	CEVNT	Clear interrupt event instruction CEVNT 10
Program control instruction			
FOR	FOR-NEXT start loop FOR-NEXT FOR VW0, 1, 10 NEXT	LSCR	Load sequence control interrupt
NEXT	FOR-NEXT loop end	SCRE	Sequence control relay end
JMP	Jump to label instruction	SCRT	Sequence control relay transfer
LBL	Label instruction	CSCRE	Conditional Sequence control relay end
WDR	Monitor reset instruction	CRET	Return from subroutine with condition instruction
DLED	Diagnosis LED instruction DLED 1	END	Conditionally end
CALL	Call subroutine instruction	STOP	Stop instruction
Communication instruction			
XMT	FPORT send instruction	SPA	Set port instruction
RCV	FPORT receive instruction	GPA	Get port address instruction
NETR	Network read instruction	EBUSR	Read module information instruction
NETW	Network write instruction	EBUSW	Write module information instruction
UDPNETR	Ethernet read function	EBUSGETDI A	Get module internal diagnosis information instruction
UDPNETW	Ethernet write function	EBUSSNDC MD	Send module operation command instruction
MBSNDMSG	Send Modbus message	--	--

N Special Function Register

SMB0: System status bit

The 8 bits of SMB0 play a special role in the system. At the end of each scan cycle, the PLC will update the 8 bits. Users can use the special functions of these bits in the program to achieve some common operations, the details shows as following table:

SMB0 bit	Function description(read only)
SM0.0	The bit is always on and can be used as enable bit if you want a section of a program to execute every cycle.
SM0.1	This bit is only 1 at first scan and can be used as the enable bit to initialize segment.
SM0.2	If the power-off hold data is lost, the bit opens a scan cycle. This bit can be used as an error memory bit or to activate enable bits in a particular startup order.
SM0.3	When the power is turned on into RUN mode, the bit is switched on for a scan cycle. And it can provide machine warm-up time before starting the operation.
SM0.4	This bit offers a square wave clock pulse period 1 min, 30s on and 30s off.
SM0.5	This bit offers a square wave clock pulse period 1sec, 0.5s on and 0.5s off.
SM0.6	The bit is the scanning clock, set 1 for this scan and 0 for the next scan, and can be used as enable bit of scan counter.
SM0.7	This bit indicates PLC switch (0: TERM, 1: RUN) position. It can be used as the enable bit of the FPORT communication program segment, when the switch is at the RUN position, the FPORT communication mode is effective, and when the switch is at the TERM position, PPI communication is effective.

SMB1: Instruction execution status bit

The 8 bits of SMB1 are used to indicate errors or special states in the execution of certain instructions, as detailed in the following table:

SMB1 bit	Function description(read only)
SM1.0	Set 1 when instruction executed as 0
SM1.1	Set 1 instruction execution results overflow or detect illegal value
SM1.2	Set 1 instruction execution results is negative
SM1.3	Set 1 when the divisor in the instruction is 0
SM1.4	Set 1 if the appended table length exceeds the table range when executing ATT instructions
SM1.5	Set 1 if the table is empty when executing a LIFO or FIFO instruction,
SM1.6	Set 1 when trying to convert a non-BCD number to a binary number
SM1.7	Set 1 When ASCII code cannot be converted to a valid hexadecimal number

SMB2: FPORT receive character

SMB2 is the FPORT receive character buffer, in the FPORT communication mode, every received character is put here, easy for program to read. Detailed description is shown in the following table:

SM address	Function description(read only)
SMB2	This byte including every character of port 0 or 1 of FPORT communication

SMB3: FPORT odd-even parity error

SMB3 is used to display FPORT parity errors. In FPORT communication mode, set SM3.0 1 when detecting checksum error in the received character. Normally, use this bit to detect FPORT information characters for transmission errors before the program receives and reads the information character values stored in SMB2. Detailed description is shown in the following table:

SMB3 bit	Function description(read only)
SM3.0	This bit indicates if there is parity error in port 0 and 1 0: no error, 1: error
SM3.1~SM3.7	Reserved

SMB4: Queue overflow

SMB4 contains interrupt queue overflow bits, whether interrupts allow flag bits, send idle bits, and whether mandatory flag bits exist. Queue overflow indicates either that interrupts occur more frequently than the CPU, or it has been disabled by the global interrupt disable instruction. Detailed description is shown in the following table:

SMB4 bit	Function description(read only)
SM4.0	Turned on when the communication interrupt queue overflows. Reset when the queue is empty and control returns to the main program, use only in interrupt routine
SM4.1	Turned on when the input interrupt queue overflows. Reset when the queue is empty and control returns to the main program, use only in interrupt routine
SM4.2	Turned on when the timing interrupt queue overflows. Reset when the queue is empty and control returns to the main program, use only in interrupt routine
SM4.3	Turn on when detecting run time program error
SM4.4	This bit reflects the global interrupt enabled state, turn on when interrupts are enabled.
SM4.5	Turn on when port 0 transmitter in idle
SM4.6	Turn on when port 1 transmitter in idle
SM4.7	Turn on when any memory position is forced

SMB5: I/O status

SMB5 contains the I/O error status bits found in the system. Which provide information about the I/O errors, as detailed in the following table:

SMB5 bit	Function description(read only)
SM5.0	Set 1 when I/O error

SM5.1	Set 1 when I/O bus connected too many digital I/O
SM5.2	Set 1 when I/O bus connected too many analog I/O
SM5.3	Set 1 when I/O bus connected too many intelligent I/O modules
SM5.4~SM5.7	Reserved

SMB6: CPU identification register

SMB6 is CPU identification (ID) register, use for identify CPU type, as detailed in the following table:

SM address	Function description(read only)
SMB6	CPU ID register
SM6.0~SM6.3	Reserved
SM6.4~SM6.7	use for identify CPU type 0000: CPU212, CPU222; 0010: CPU214, CPU224; 0110: CPU221; 1000: CPU215; 1001: CPU216, CPU226, CPU226XM; 1110: H35-00;

SMB6 format:

MSB				LSB			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
x	x	x	x	r	r	r	r
0000: CPU212, CPU222; 0010: CPU214, CPU224; 0110: CPU221; 1000: CPU215; 1001: CPU216, CPU226, CPU226XM; 1110: H35-00				Reserved			

SMB8 to SMB21: I/O module and error register

SMB8 to SMB21 are prepared in byte pair form (two adjacent bytes) for extension modules 0 to 6. The even-bit byte of each pair of bytes is the module identification register, and the odd-bit byte is the module error register. The former marks the module type, I/O type, input and output points; while the latter is the I/O error detected against corresponding module. Detailed description is shown in the following table:

SM address	Function description(read only)
SMB8	Module 0 identification register
SMB9	Module 0 error register
SMB10	Module 1 identification register
SMB11	Module 1 error register
SMB12	Module 2 identification register
SMB13	Module 2 error register
SMB14	Module 3 identification register
SMB15	Module 3 error register
SMB16	Module 4 identification register
SMB17	Module 4 error register
SMB18	Module 5 identification register
SMB19	Module 5 error register

SMB20	Module 6 identification register
SMB21	Module 6 error register

Even byte (module ID register) format:

MSB				LSB			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
m	t	t	a	i	i	q	q
m: If there is module 0—Module 1—No module	tt: Module type 00—Non-intelligent I/O module 01—Intelligent module 10—Reserved 11-- Reserved		a: I/O type 0—Digital 1—Analog	ii: Inputs 00—No input 01--2 AI or 8 DI 10--4 AI or 16 DI 11--8 AI or 32 DI		qq: Outputs 00—No output 01--2 AQ or 8 DQ 10--4 AQ or 16 DQ 11--8 AQ or 32 DQ	

Odd byte (module error register) format:

MSB				LSB			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
c	0	0	b	r	p	f	t
c: Configuration error 0—No error 1-- Error			b: Bus or parity error 0-- No error 1-- Error	r: Over range error 0-- No error 1-- Error	p: User power error 0-- No error 1-- Error	f: Fuse error 0-- No error 1-- Error	t: Terminal block loose error 0-- No error 1-- Error

SMW22 to SMW26: Scanning time

SMW22, SMW24 and SMW26 offer information of scanning time: minimum scanning time, maximum scanning time and last scanning time calculate by ms

SM address	Function description(read only)
SMW22	Last scanning time
SMW24	Recorded minimum scanning time after entering RUN mode
SMW26	Recorded maximum scanning time after entering RUN mode

SMB28 to SMB29: Analog potentiometer

Special memory bytes 28 and 29 contain values corresponding to analog potentiometer 0 and 1 axes position. The analog potentiometer is located in front of the CPU behind the access gate. Use a screwdriver to adjust the potentiometer (increasing clockwise or decreasing counterclockwise). This type of read-only value can be used by a program to perform various functions, such as updating the current value for a timer or counter, inputting or changing a preset value, or setting limits. It also has a rated range of 0 to 255, and reliable repeats of ± 2 counts. Detailed description is shown in the following table:

SM address	Function description(read only)
------------	---------------------------------

SMB28	Input of memory analog regulator 0. Updating it for every scan in STOP/RUN mode
SMB29	Input of memory analog regulator 1. Updating it for every scan in STOP/RUN mode

SMB30 and SMB130: FPORT control register

SMB30 controls FPORT 0 communication mode, and SMB130 controls FPORT 1 communication mode. You can write and read between SMB30 and SMB130, using these two bytes to set operation mode of FPORT communication and provide the choice between the FPORT and the protocols supported by the system. See the following table for detailed description:

SM address	Function description
SMB30	FPORT 0 control register
SMB130	FPORT 1 control register
SM30.0~SM30.1	Port 0 communication protocol selection
SM130.0~SM130.1	Port 1 communication protocol selection
SM30.2~SM30.4	FPORT 0 baud rate
SM130.2~SM130.4	FPORT 1 baud rate
SM30.5	Each character data bit of FPORT 0
SM130.5	Each character data bit of FPORT 1
SM30.6~SM30.7	FPORT 0 parity selection
SM130.6~SM130.7	FPORT 1 parity selection

FPORT control byte format:

MSB				LSB			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
p	p	d	b	b	b	m	m
Pp: parity selection 00—no parity 01—even parity 10-- no parity 11—odd parity		d: Each character data bit 0--8 bits data bit 1--7 bits data bit		bbb: FPORT baud rate 000--38400 bps 001--19200 bps 010--9600 bps 011--4800 bps 100--2400 bps 101--1200 bps 110--115200 bps 111--57600 bps		Mm: protocol setting 00—point-point port protocol(PPI/slave station mode) 01—FPORT protocol 10--PPI/ master station mode 11—reserved(default PPI/slave station mode)	

<Note> When select mm = 10(PPI master station), PLC will become a network master station, able to execute NETR and NETW instruction. Ignore 2 to 7 bit in PPI mode.

SMB31 and SMW32: Permanent storage (EEPROM) write control

Under the control of the user program, you can store the data in V storage in permanent storage, also known as non-volatile storage. First, store the address of stored data into

SMW32, then store save command into SMB31. Once you issue the storage command, you cannot change the value of V storage until the CPU completes the storage operation and SM31.7 is set to 0.

At the end of each scan cycle, the CPU checks for a command to store data in the permanent storage area. If so, the data is stored in permanent storage. Detailed description is shown in the following table:

SM address	Function description
SMB31	Permanent storage command register
SM31.0~SM31.1	Data size to be saved 00 =byte; 01 =byte; 10 =Word; 11 =double Word
SM31.2~SM31.6	Preserved, always 00000
SM31.7	Store into permanent storage, S7-200 reset the bit after each storage operation. 0 = no request of executing storage 1 = user program applies to permanent storage for data storage
SMW32	The V memory address of stored data, which is offset from V0. When executing a storage command, store data into the corresponding location in permanent storage.

SMB34 and SMB35: Timing interrupt interval register

SMB34 and SMB35 define the interval between timed interrupts 0 and 1. You can specify the time interval (incremented by 1ms) from 1ms to 255ms. If the corresponding timed interrupt event is connected to an interrupt service routine, the CTSC-200CPU gets the interval value. To change the interval, you must either reallocate timed interrupt events to the same or another interrupt program, or terminate timed interrupt events by interrupt detach.

SM address	Function description
SMB34	Time interval value of timing interrupt 0
SMB35	Time interval value of timing interrupt 1

SMB36 to SMB65: HSC0, HSC1 and HSC2 register

SMB36 to SMB65 are used to monitor and control operation of HSC0, HSC1 and HSC2. Detailed description is shown in the following table:

SM address	Function description
HSC0	
SMB36	HSC0 counter status
SM36.0~SM36.4	Reserved
SM36.5	HSC0 current count direction bit: 1=addition count
SM36.6	HSC0 current value equal to preset value bit: 1=equal
SM36.7	HSC0 current value more than preset value bit: 1=more than
SMB37	HSC0 control byte

SM37.0	Reset valid control bit: 0= high level reset valid, 1=low level reset valid
SM37.1	Reserved
SM37.2	Count speed selection of orthography counter: 0=4x count speed; 1=1x count speed
SM37.3	HSC0 direction control bit: 1= addition count
SM37.4	HSC0 update direction: 1= update direction
SM37.5	HSC0 update preset value: 1=write new preset value to HSC0
SM37.6	HSC0 update current value: 1=write new initial value to HSC0
SM37.7	HSC0 valid bit: 1=valid
SMD38	HSC0 new initial value, set SM37.6 as 1 after write new value into SMD38, then the current value will be updated
SMD42	HSC0 new preset value, set SM37.5 as 1 after write new value into SMD42, then the current value will be updated
HSC1	
SMB46	HSC1 counter status
SM46.0~SM46.4	Reserved
SM46.5	HSC1 current count direction bit: 1=addition count
SM46.6	HSC1 current value equal to preset value bit: 1=equal
SM46.7	HSC1 current value more than preset value bit: 1=more than
SMB47	HSC1 control byte
SM47.0	Reset valid control bit: 0= high level reset valid, 1=low level reset valid
SM47.1	HSC1 start valid level control bit: 0= high level, 1=low level
SM47.2	Count speed selection of orthography counter: 0=4x count speed; 1=1x count speed
SM47.3	HSC1 direction control bit: 1= addition count
SM47.4	HSC1 update direction: 1= update direction
SM47.5	HSC1 update preset value: 1=write new preset value to HSC0
SM47.6	HSC1 update current value: 1=write new initial value to HSC0
SM47.7	HSC1 valid bit: 1=valid
SMD48	HSC1 new initial value, set SM47.6 as 1 after write new value into SMD48, then the current value will be updated
SMD52	HSC1 new preset value, set SM47.5 as 1 after write new value into SMD52, then the current value will be updated
HSC2	
SMB56	HSC2 counter status
SM56.0~SM56.4	Reserved
SM56.5	HSC2 current count direction bit: 1=addition count
SM56.6	HSC2 current value equal to preset value bit: 1=equal
SM56.7	HSC2 current value more than preset value bit: 1=more than
SMB57	HSC2 control byte
SM57.0	Reset valid control bit: 0= high level reset valid, 1=low level reset valid
SM57.1	HSC1 start valid level control bit: 0= high level, 1=low level

SM57.2	Count speed selection of orthography counter: 0=4x count speed; 1=1x count speed
SM57.3	HSC2 direction control bit: 1= addition count
SM57.4	HSC2 update direction: 1= update direction
SM57.5	HSC2 update preset value: 1=write new preset value to HSC0
SM57.6	HSC2 update current value: 1=write new initial value to HSC0
SM57.7	HSC2 valid bit: 1=valid
SMD58	HSC2 new initial value, set SM57.6 as 1 after write new value into SMD58, then the current value will be updated
SMD62	HSC2 new preset value, set SM57.5 as 1 after write new value into SMD62, then the current value will be updated

<Note>

- 1) The counter status bit is only valid in interrupt program triggered by HSC event.
- 2) When you use the HSC external reset interrupt event, do not reset new current value in program, and then re-enable the counter in the interrupt program connected to the event, which will cause serious errors.

SMB66 to SMB85: PTO/PWM register

SMB66 to SMB85 are used to monitor and control the pulse chain output and PWM functions of PLC (pulse) commands. Detailed description is shown in the following table:

SM address	Function description
PTO0	
SMB66	PTO0 status byte
SM66.0~SM66.3	Reserved
SM66.4	PTO0 envelope termination: 0=no error, 1=terminate for incremental error
SM66.5	PTO0 envelope termination: 0=terminate not by user command; 1=terminate by user command
SM66.6	PTO0 pipe overflow(clear by system when using external envelope, or by user program): 0=no overflow, 1=overflow
SM66.7	PTO0 free bit: 0=PTO busy, 1=PTO free
SMB67	Control pulse string output and PWM of Q0.0
SM67.0	PTO0/PWM0 update cycle: 1=write new cycle
SM67.1	PWM0 update pulse width: 1=write new pulse width
SM67.2	PTO0 update pulse quantity: 1=write new pulse quantity
SM67.3	PTO0/PWM0 reference time unit: 0=1 μ s/grid, 1=1 ms/grid
SM67.4	Synchronous update PWM0: 0=asynchronous update, 1=synchronous update
SM67.5	PTO0 operation: 0= single segment operation(cycle and pulse store in

	SM storage), 1=multiple segment operation(envelope table involved in V storage area)
SM67.6	PTO0/PWM0 mode selection: 0=PTO, 1=PWM
SM67.7	PTO0/PWM0 valid bit: 1=valid
SMW68	PTO0/PWM0 cycle(2~65, 535 time standard)
SMW70	PWM0 pulse width (0~65, 535 time standard)
SMD72	PTO0 pulse count (1~232 --1)
PTO1	
SMB76	PTO1 status byte
SM76.0~SM76.3	Reserved
SM76.4	PTO1 envelope termination: 0=no error, 1=terminate for incremental error
SM76.5	PTO1 envelope termination: 0=terminate not by user command; 1=terminate by user command
SM76.6	PTO1 pipe overflow(clear by system when using external envelope, or by user program): 0=no overflow, 1=overflow
SM76.7	PTO1 free bit: 0=PTO busy, 1=PTO free
SMB77	Control pulse string output and PWM of Q0.1
SM77.0	PTO1/PWM1 update cycle: 1=write new cycle
SM77.1	PWM1 update pulse width: 1=write new pulse width
SM77.2	PTO1 update pulse quantity: 1=write new pulse quantity
SM77.3	PTO1/PWM1 reference time unit: 0=1 μ s/grid, 1=1 ms/grid
SM77.4	Synchronous update PWM1: 0=asynchronous update, 1=synchronous update
SM77.5	PTO1 operation: 0= single segment operation(cycle and pulse store in SM storage), 1=multiple segment operation(envelope table involved in V storage area)
SM77.6	PTO1/PWM1 mode selection: 0=PTO, 1=PWM
SM77.7	PTO1/PWM1 valid bit: 1=valid
SMW78	PTO1/PWM1 cycle(2~65, 535 time standard)
SMW80	PWM1 pulse width (0~65, 535 time standard)
SMD82	PTO1 pulse count (1~232 --1)

SMB86 to SMB94 and SMB186 to SMB194: receive information control

SMB86 to SMB94 and SMB186 to SMB194 are used to control and read status about receiving information. Detailed description is shown in the following table:

SM address	Function description
P0 received information	
SMB86	P0 receive information status byte
SMB87	P0 receive information control byte
SMB88	P0 start of information character
SMB89	P0 end of information character
SMW90	The P0 free row interval is given in ms. The first character received at

	the end of free line time is the beginning of new information.
SMW92	P0 inter-character/inter-message timer timeout value (represented by ms). If the timeout, stop receiving message.
SMB94	Maximum characters P0 received(1~255 byte) <Note> This area must set as maximum buffer you want, even if character count information is not used to terminate it.
P0 received information	
SMB186	P1 receive information status byte
SMB187	P1 receive information control byte
SMB188	P1 start of information character
SMB189	P1 end of information character
SMW190	The P1 free row interval is given in ms. The first character received at the end of free line time is the beginning of new information.
SMW192	P1 inter-character/inter-message timer timeout value (represented by ms). If the timeout, stop receiving message.
SMB194	Maximum characters P1 received(1~255 byte) <Note> This area must set as maximum buffer you want, even if character count information is not used to terminate it.

Receive information status byte format:

MSB				LSB			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
n	r	E	0	0	t	c	p

n = 1: Receive user order and stop receiving information

r = 1: Input parameter error or no start or end condition result in information receiving end

e = 1: Receive end character

t = 1: Receive information end: timeout

c = 1: Receive information end: over maximum character

p = 1: Receive information end: parity error

Receive information control byte format:

MSB				LSB			
Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
en	sc	ec	il	c/m	tmr	bk	0

en: 0 = Forbid receiving information function

1 = Allow receiving information function

Allow/forbid receive information bit each time execute RCV instruction

sc: 0 = Ignore SMB88 or SMB188

1 = Use value of SMB88 or SMB188 to detect start information

ec: 0 = Ignore SMB89 or SMB189

1 = Use value of SMB89 or SMB189 to detect start information

il: 0 = Ignore SMW90 or SMW190

1 = Use value of SMW90 or SMW190 to detect free status

c/m: 0 = Internal character timer

1 = Inform timer

tmr: 0 = ignore SMW92 or SMW192

1 = Terminate receiving when SMW92 or SMW192 timing timeout.

bk: 0 = Ignore interrupt condition

1 = Use interrupt condition as start of information detection

SMW98: Expansion I/O bus error

SMW98 offer information about expansion I/O bus error. Detailed description is shown in the following table:

SM address	Function description
SMW98	Each time a checksum error occurs on the extended bus, it increases by 1. When the system power on or the user program writes zero, it can be reset.

SMB110 to SMB111: programming card upload control

SMB110 to SMB111 Use for programming card load control:

SM address	Function description
SMB110	Start enforcing load protection function
SMB111	Programming card cover item: 0 – Cover programming block and data block; 1 – Not cover programming block, cover data block 2 - Cover programming block, not cover data block; 3 - Not cover programming block and data block 4 – Same when ADD is 0

SMB114 to SMB115: expansion board information

SMB114: Expansion board module type

SMB114 = 0x20: CAN

0x19: 2AI/1AQ

0x1E: 4AI/2AQ

Detailed description is shown in the following table:

Bti7	Bti6	Bti5	Bti4	Bti3	Bti2	Bti1	Bti0
	E	TT	t	II			OO
E If there is module	1: module do not exist 0: module exist						
TT	00	Non intelligent module					

(module type)	01	Intelligent module	
	10	Special non intelligent module(7PF)	
	11	cotrust intelligent module(PID)	
t (module subtype)	Non intelligent module	0: Digital	
		1: Analog	
	Intelligent module	0: Normal intelligent	
		1: Cotrust intelligent module(PID)	
II(inputs)	Non intelligent module	00	No input
		01	AI/8DI
		10	2AI/16DI
		11	4AI/32DI
	Intelligent module	Reserved	
OO (outputs)	Non intelligent module	00	No output
		01	1AO/8DO
		10	2AO/8DO
		11	4AO/32DO
	Intelligent module	Reserved	

SMB115: Expansion board status

SMB115 = 0x0: Module no error

0x255: Expansion board visit error

Others: Expansion board internal diagnosis

SMW116 to SMW126: Expansion board analog mapping area

Corresponding relation of mapping area:

Expansion board	Analog mapping area
2AI/1AQ	SMW116 corresponds to AIW0; SMW118 corresponds to AIW2; SMW124 corresponds to AQW0
4AI/2AQ	SMW116 corresponds to AIW0; SMW118 corresponds to AIW2; SMW120 corresponds to AIW4 SMW122 corresponds to AIW6; SMW124 corresponds to AQW0; SMW126 corresponds to AQW2

SMB131 CAN expansion board access cycle setting

CAN expansion board visit cycle (ms), default 1ms when set as 0

SMB136 to SMB165: HSC3, HSC4 and HSC5 register

SMB136 to SMB165 use for monitor and control HSC3, HSC4 and HSC5 operation. Detailed description is shown in the following table:

SM address	Function description
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HSC3	
SMB136	HSC3 counter status
SM136.0~4	Reserved
SM136.5	HSC3 current count direction bit: 1=addition count
SM136.6	HSC3 current value equal to preset value bit: 1=equal
SM136.7	HSC3 current value more than preset value bit: 1=more than
SMB137	HSC3 control byte
SM137.0~2	Reserved
SM137.3	HSC3 direction control bit: 1= addition count
SM137.4	HSC3 update direction: 1= update direction
SM137.5	HSC3 update preset value: 1=write new preset value to HSC3
SM137.6	HSC3 update current value: 1=write new initial value to HSC3
SM137.7	HSC3 valid bit: 1=valid
SMD138	HSC3 new initial value, set SM137.6 as 1 after write new value into SMD138, then the current value will be updated
SMD142	HSC3 new preset value, set SM137.5 as 1 after write new value into SMD142, then the current value will be updated
HSC4	
SMB146	HSC4 counter status
SM146.0~4	Reserved
SM146.5	HSC4 current count direction bit: 1=addition count
SM146.6	HSC4 current value equal to preset value bit: 1=equal
SM146.7	HSC4 current value more than preset value bit: 1=more than
SMB147	HSC4 control byte
SM147.0	Reset valid control bit: 0= high level reset valid, 1=low level reset valid
SM147.1	Reserved
SM147.2	Count speed selection of orthography counter: 0=4x count speed; 1=1x count speed
SM147.3	HSC4 direction control bit: 1= addition count
SM147.4	HSC4 update direction: 1= update direction
SM147.5	HSC4 update preset value: 1=write new preset value to HSC4
SM147.6	HSC4 update current value: 1=write new initial value to HSC4
SM147.7	HSC4 valid bit: 1=valid
SMD148	HSC4 new initial value, set SM147.6 as 1 after write new value into SMD148, then the current value will be updated
SMD152	HSC4 new preset value, set SM147.5 as 1 after write new value into SMD152, then the current value will be updated
HSC5	
SMB156	HSC5 counter status
SM156.0~4	Reserved
SM156.5	HSC5 current count direction bit: 1=addition count
SM156.6	HSC5 current value equal to preset value bit: 1=equal
SM156.7	HSC5 current value more than preset value bit: 1=more than

SMB157	HSC5 control byte
SM157.0~2	Reserved
SM157.3	HSC5 direction control bit: 1= addition count
SM157.4	HSC5 update direction: 1= update direction
SM157.5	HSC5 update preset value: 1=write new preset value to HSC5
SM157.6	HSC5 update current value: 1=write new initial value to HSC5
SM157.7	HSC5 valid bit: 1=valid
SMD158	HSC5 new initial value, set SM157.6 as 1 after write new value into SMD168, then the current value will be updated
SMD162	HSC5 new preset value, set SM157.5 as 1 after write new value into SMD162, then the current value will be updated

<Note>

- 1 The counter status bit is only valid in the interrupt program triggered by HSC event.
- 2 When you use the HSC external reset interrupt event, do not reset the new current value in the program, and then re-enable the counter in the interrupt program connected to the event, this operation will cause serious errors.

SM address	Function description
PTO0	
SMB66	PTO0 status byte
SM66.0~SM66.3	Reserved
SM66.4	PTO0 envelope termination: 0=no error, 1=terminate for incremental error
SM66.5	PTO0 envelope termination: 0=terminate not by user command; 1=terminate by user command
SM66.6	PTO0 pipe overflow(clear by system when using external envelope, or by user program): 0=no overflow, 1=overflow
SM66.7	PTO0 free bit: 0=PTO busy, 1=PTO free
SMB67	Control pulse string output and PWM of Q0.0
SM67.0	PTO0/PWM0 update cycle: 1=write new cycle
SM67.1	PWM0 update pulse width: 1=write new pulse width
SM67.2	PTO0 update pulse quantity: 1=write new pulse quantity
SM67.3	PTO0/PWM0 reference time unit: 0=1 μ s/grid, 1=1 ms/grid
SM67.4	Synchronous update PWM0: 0=asynchronous update, 1=synchronous update
SM67.5	PTO0 operation: 0= single segment operation(cycle and pulse store in SM storage), 1=multiple segment operation(envelope table involved in V storage area)
SM67.6	PTO0/PWM0 mode selection: 0=PTO, 1=PWM
SM67.7	PTO0/PWM0 valid bit: 1=valid
SMW68	PTO0/PWM0 cycle(2~65, 535 time standard)
SMW70	PWM0 pulse width (0~65, 535 time standard)
SMD72	PTO0 pulse count (1~232 --1)

SMB166 to SMB185: PTO0, PTO1 envelope definition table

SMB166 to SMB185 use for display quantity of envelope step and envelope table, also the V storage table address (H224/H226L not support; H224X/H226XL support) Detailed description is shown in the following table:

SM address	Function description
PTO0	
SMB166	PTO0 envelope step current count value
SMB167	Reserved
SMW168	PTO0 envelope table V storage address(offset fromV0)
SMB170	Linear PTO0 status byte
SMB171	Linear PTO0 result byte
SMD172	Specifies the frequency of the linear PTO0 generator generates when operating in manual mode. The frequency is a bit integer in Hz.
PTO1	
SMB176	PTO1 envelope step current count value
SMB177	Reserved
SMW178	PTO1 envelope table V storage address(offset fromV0)
SMB180	Linear PTO1 status byte
SMB181	Linear PTO1 result byte
SMD182	Specifies the frequency of the linear PTO0 generator generates when operating in manual mode. The frequency is a bit integer in Hz.

SM195: Communication status

SM195.6	SM195.6=1 after CPU connect to remote server, or set as 0
SM195.7	Control whether heartbeat packets are sent in TCP_Modbus communication over time. SM195.7=1, heartbeat packet sent out after communication timeout; Otherwise do not send; Default is 0.

SMB200 to SMB549: Intelligent module status

SMB200 to SMB549 reserve storage intelligent expansion module information, detailed description is shown in the following table:

0 slot intelligent module	1 slot intelligent module	2 slot intelligent module	3 slot intelligent module	4 slot intelligent module	5 slot intelligent module	6 slot intelligent module	Description
SMB200 ~215	SMB250 ~265	SMB300 ~315	SMB350 ~365	SMB400 ~415	SMB450 ~465	SMB500 ~515	Module name(16 ASCII characters)
SMB216 ~219	SMB266 ~269	SMB316 ~319	SMB366 ~369	SMB416 ~419	SMB466 ~469	SMB516 ~519	S/W revise number (4 ASCII characters)
SMW220	SMW270	SMW320	SMW370	SMW420	SMW470	SMW520	Error code

SMB222 ~249	SMB272 ~299	SMB322 ~349	SMB372 ~399	SMB422 ~449	SMB472 ~499	SMB522 ~549	Information related to specific module
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O Programming Cable

CTSC-200 PLC programming cable with USB-485 port is used for CPU programing, upload, download and monitor, USB end connect to PC/PG, 485 end connect to CPU PPI port.

Table O-1 Physical features of programming cable

	Item	Contents
	Order No.	CTS7 191-USB30
	Supported OS	Windows2000/XP/7/10
	Baud rate	300bps ~ 1Mbps standard baud rate self-adaption
	Work temperature	0 ~ +55 °C
	Cable length	2.5 m
	Cable length each PC supported	1



Note

Please install drive before using programming cable, drive download address: www.co-trust.com

Cable appearance:

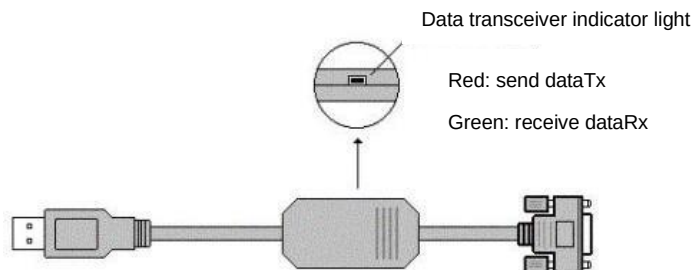


Figure O-1 Programming cable appearance

P How to Use Programming Card

Specification

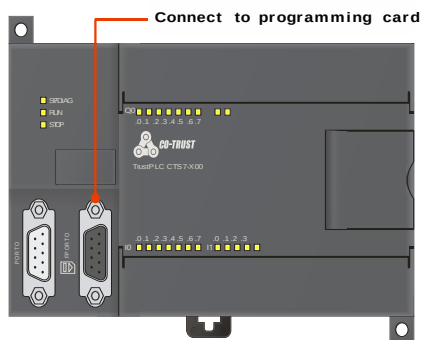
	Specification	
	Type	Portable programming card
	Order No.	CTS7 291-PC001
	Dimension(W×H×D)	63 mm ×33 mm × 17 mm
Note: - Siemens memory card includes two functions: Data storage and portable programming, TrustPLC need use data storage card and portable programming card to complete the corresponding functions. - TrustPLC does not support Siemens storage card.		

【Safety Requirement】

The data of the programming card is encrypted, so as not to be malicious modified.

【Connections】

Connect the inner RS485 port of the PLC.



CPU 224+/224E/224I/224XP/226M/226H/PSC 266

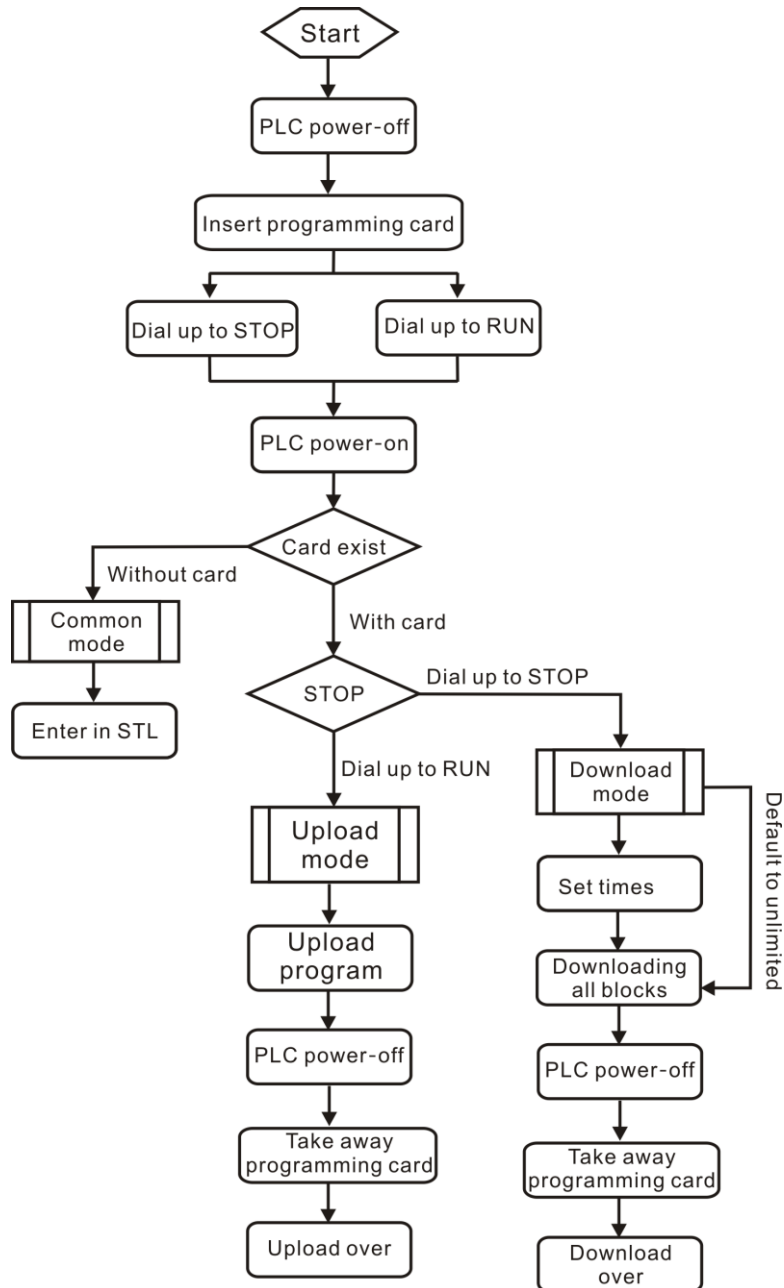
【CPU Requirements】

Applied to all of the CTSC-200 CPU which has two communication ports and above.



Note

Programming card record the CPU type while downloading program from CPU. It is not allowed to upload while the CPU type is inconformity. The same type CPU doesn't differentiate between relay and transistor.

Using programming card**【Flow chart】****【Function of programming card】**

- **Downloading function**

Download the STL program block, STL program library, system block and data block from MicroWIN project to the programming card.

Steps:

Step 1: Turn-off the power of PLC, then insert the programming card to the specified port.

Step 2: Set the dial switch to "Stop" status, and turn on the power for PLC.

Step 3: If programming card is effect, the system would defaulting enter in "Programming downloading mode", the "Stop" light with orange light always on, then click "Download".

Click  button from the tool bar or click "File"->"Download "from the menu bar, that you

can download the program to PLC and programming card.



Note: For the blocks which had not downloaded to programming card, a default block would be generated in programming card, and the default block would be uploaded to CPU.

Step 4: If programming card is invalid, the system would enter in common mode.

Step 5: If users desire to set the limit times of programming card, you must change the value of SMB100 before downloading any blocks, otherwise it would default to unlimited times (The value of SMB100 is 0)

Step 6: If user don't want to cover program block or data block of target PLC, you can change the value of SMB111, otherwise the blocks would defaulted to be covered. The definition of SMB111 see as following table.

Step 7: Download the programming blocks to CPU in turns.

Step 8: CPU send all the downloaded blocks to programming card(When sending is processing, STOP light is flashing orange, after all the sending tasks are finished, orange light would always on).

Step 9: When all of the blocks are downloaded, orange light would changes from flashing to always on and the time continuing for more than 10 seconds, at the time, users can turn-off the power of PLC, and pull out programming card. (If downloading is finished ,and set CPU to RUN mode at the same time, CPU would not run, you can only turn-off the power of CPU, and pull out programming card, then the CPU can work properly).



Note

- If users use 187.5 baud rate to download blocks, the CPU might spend a long time transferring the block to programming card after downloading is finished.
- CPU would clear all the data of programming card and setup the frequency limit for programming card while CPU confirmed the first block has been downloaded.
- If you need coverage function, please set SMB111 first, and then download blocks to CPU.
- For the sake of security, the downloading block must contain system block while users intend to download the blocks to the programming card. You'd better not to turn off the power while the LED is flashing during downloading procedure, otherwise the content of the programming card would be lost.

Light status

	Light-out	Flashing	Light-on
SF light	Normal	Encountering an error	---
STOP light- Orange	---	Transferring data	Normal
SMB100	Set the limitation of displaying times		
SMB111	Coverage function		

Flicker Frequency: About 0.5 second (No precision requirement)

● **Uploading function**

Users could upload all of the blocks in the programming card to CPU, and update the EPROM of the CPU.

Using steps:

Step 1: First turn off the power of PLC, then insert the programming card to the specified port.

Step 2: Set the dial switch to "RUN" status, and turn on the power for PLC.

Step 3: If programming card takes effect, the system would enter in "Programming uploading mode", the "RUN" light would turns on green all the time.

Step 4: If programming card is invalid, the system would enter in common mode.

Step 5: If CPU detected the programming card, it would clear all the original blocks, and read all the blocks into EPPROM.

Note: For the blocks which had not downloaded to programming card, a default block would be generated in programming card, and the default block would be uploaded to CPU.

Step 6: RUN light would flashing green at uploading procedure, after the green light changes from flashing to always on, and the time continues for more than 10 seconds, then users can turn-off the power off Plain pull out the programming card.

Step 7: Turn on the power for PLC again, the new STL program would be performed.

Following cases would result in uploading failure:

1. The using times of Programming card reaches the limited times (It would not affect the original program blocks)
2. The information of the programming card is failed to check (The original program blocks might be deleted)
3. Other blocks in programming card is failed to check (the correct block would be uploaded to CPU, and the original block would be deleted).
4. When uploading the cryptoguard CPU to programming card, if programming card has not been set to coverage block (Bit0 set to 0), the upload would be failed, so as to protect the user program.

When uploading is failed, SF light would flashing red.

The definition of SMB110 to SMB111, see as follows:

SM	Description
SMB110	Start forcing upload protection
SMB111	Coverage selection of programming card:

	0 - Cover program block and data block; 1 - Not cover program block, cover data block. 2 - Cover program block, not cover data block; 3 - Not cover program block and data block. 4 - The same as 0.
--	--

Light status

Lights	Light-out	Flashing	Light-on
RUN light-Green	---	Loading program	Normal
SF light	Normal	Encountering an error.	---
SMB101	The rest of the using times (Including this time), if no limit, would show 255.		

Flicker Frequency: About 0.5 second (No precision requirement)

- **Frequency limitation function**

The frequency limitation of the programming card allows the users to setup the usage times for the programming card, after downloading task is completed, the available times in the programming card would reduce one.

Using steps:

Step1: When PLC is in "Programming card is downloading" status, MicroWIN user set the limit times via SMB100 before download the first block, the data would display in SMB101.

Step2: Once the user upload successful (including the third case which upload failed), the frequency limit would reduce 1.

Meaning of parameters

Value	Meaning
0	Unlimited use
1-255	Available times
Default	0

Q Instructions of ETHERNET_SET

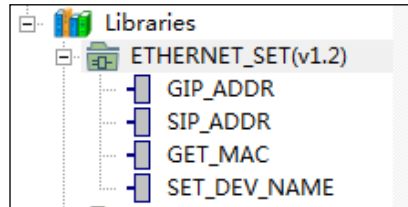
Q.1 Function Introduction

Instructions of ETHERNET_SET(V1.2) is used to set the remote communication parameters of MagicWorks PLC system block, you can set and get the IP address, MAC address and device name of the EtherNET PLC without stop CPU, and allows remote programming and monitoring for remote PLC via MICO.

Q.2 Details of Instructions

It's available for free download the ETHERNET_set library file from CO-TRUST official website: www.co-trust.com.

The library file includes the following parameters:



Get IP address library

① **Library name:** GIP_ADDR

② **Function:** Get IP address.

Parameters	Name	IN/OUT	Data Type	Explanation
	EN	IN	BOOL	Enable, SM0.0 is allowed to invoke.
	STATUS	OUT	BYTE	Status word bit0=1, means successfully acquired the address. bit1=1 means failed to get the address.
	IP_ADDR	OUT	DWORD	IP address, four bytes in total, each byte shows the IP address from low to high.
	MASK	OUT	DWORD	Subnet mask, four bytes in total, each byte shows the subnet mask number from low to high.
	GATE	OUT	DWORD	Gateway, four bytes in total, each byte shows the gateway number from low to high.

Set IP address library

① **Library name:** SIP_ADDR

② **Function:** Set IP address.

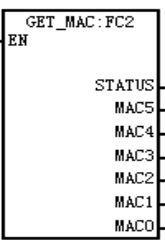
Parameters	Name	IN/OUT	Data Type	Explanation
	EN	IN	BOOL	Enable, edge triggered. The instruction is not allowed to circularly invoke, cause set IP and EEPROM has limits for number of times, and the EtherNET communication would be failed due to frequently write.

	IP_ADDR	IN	DWORD	IP address, four bytes in total, each byte shows the IP address from low to high.
	MASK	IN	DWORD	Subnet mask, four bytes in total, each byte shows the subnet mask number from low to high.
	GATE	IN	DWORD	Gateway, four bytes in total, each byte shows the gateway number from low to high.
	STATUS	OUT	BYTE	Status word bit0=1, means set successful. bit1=1, means illegal IP address. bit2=1, means IP and mask is not match. Bit3=1, means IP and gateway is not match.

Get MAC address library

① **Library name:** GET_MAC

② **Function:** Get MAC address

Parameters	Name	IN/OUT	Data Type	Explanation
	EN	IN	BOOL	Enable, SM0.0 is allowed to invoke.
	STATUS	OUT	BYTE	Status word bit0=1, means successfully acquired the address.
	MAC0~ MAC5	OUT	BYTE	MAC Address xx:xx:xx:xx:xx:xx MAC5-----MAC0

Set device name library

① Library name: SET_DEV_NAME

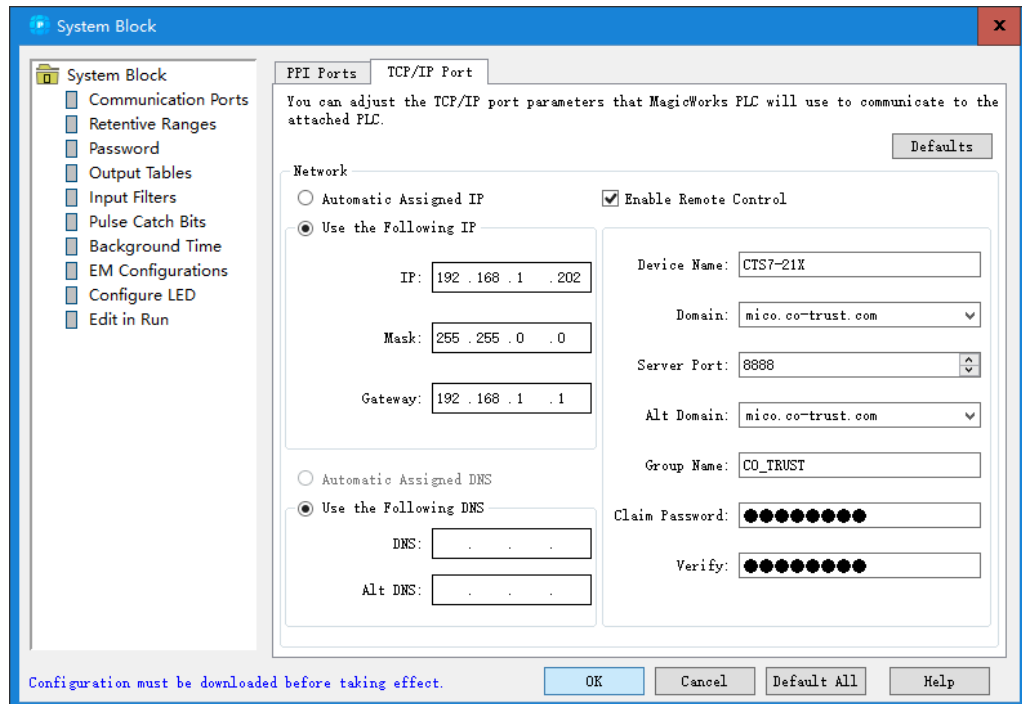
② Function: Set device name.

Parameters	Name	IN/OUT	Data Type	Explanation
	EN	IN	BOOL	Enable, edge triggered. The instruction is not allowed to circularly invoke, cause set IP and EEPROM has limits for number of times, and the EtherNET communication would be failed due to frequently write.
	NAME	IN	DWORD	Pointer of device name. The first byte express the length, the whole string contains 32 length bytes at most.
	STATUS	OUT	BYTE	Status word Bit0=1, means set successful. Bit1=1, means length error. Bit2=1, means save failed. Bit3 means illegal character.

Matters need attention:

It's available for you to set IP via TD4S except the library. If you use TD4S to set IP, please use static IP in the TCP/IP port of system block, which is not allowed to select automatic acquisition IP address mode, otherwise the setting would not be effect.

If you select library, there is no need to set following system block.

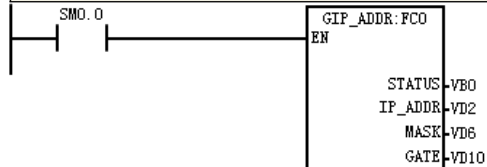


Q.3 Example

The sample application of ETHERNET_SET instruction show as follows:

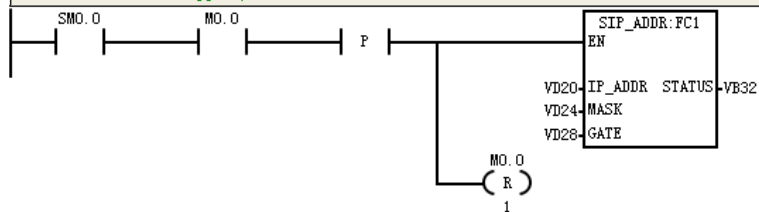
Network 1

PROGRAM COMMENTS:
Get the IP address
STATUS--status Word bit0=1 indicates gain successfully
bit1=1 indicates fail to get
IP_ADDR--IP address
MASK--subnet mask
GATE--gateway



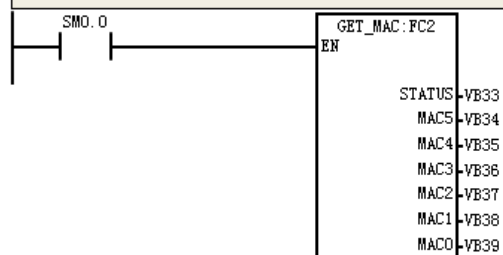
Network 2

PROGRAM COMMENTS:
Set IP address
IP_ADDR--IP address
MASK--subnet mask
GATE--gateway
STATUS--status Word bit0=1 indicates setting completed
bit1=1 indicates illegal IP address
bit2=1 indicates IP is unmatched with mask
bit3=1 indicates IP is unmatched with gateway
other bits are not in use
Note: this instruction cannot be called in cycle, for the IP setting has time limit as EPROM, with trigger.



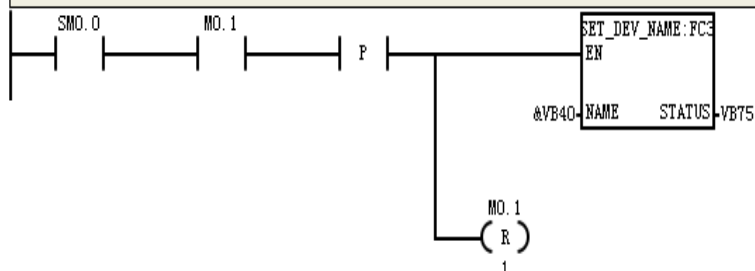
Network 3

PROGRAM COMMENTS:
Get MAC address
STATUS--status Word bit0=1 indicates gain successfully, other bits is not in use
MAC0~5--MAC address
xx:xx:xx:xx:xx:xx
MAC5-----MAC0



Network 4

PROGRAM COMMENTS:
 NAME device name pointer, the pointed first byte of area indicates length. Whole string including maximum length byte is 32 bytes.
 STATUS bit0=1 indicates set successfully
 bit1=1 indicates length error
 bit2=1 indicates fail to save
 bit3 indicates illegal characters



R Product Oder No.

Specification	Order No.
CPU	
TrustPLC CTSC-200 CPU224E, basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB; 24VDC, 14DI / 10DO, Transistor output, 0.75A, one PPI port, one Freeport, 2 axis motion control	CTS7 214-1AE33-0X24
TrustPLC CTSC-200 CPU224E, basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB; 220VAC, 14DI / 10DO, Relay output, 2A, one PPI port, one Freeport (Out of production)	CTS7 214-1BE33-0X24
TrustPLC CTSC-200 CPU224+, basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB; 24VDC, 14DI/10DO, Transistor output, 0.75A, one PPI port, one Freeport	CTS7 214-1AD33-0X24
TrustPLC CTSC-200 CPU224+, basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB; 220VAC, 14 DI/10DO, relay output, 2A, one PPI port, one Freeport	CTS7 214-1BD33-0X24
TrustPLC CTSC-200 CPU224+ V5(Upgrade), basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB (permanent retention) ; 24VDC, 14DI/10DO, Transistor output, 0.75A, one PPI port, one Freeport	CTS7 214-1AD35-0X24
TrustPLC CTSC-200 CPU224+ V5(Upgrade), basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB(permanent retention) ; 220VAC, 14 DI/10DO, relay output, 2A, one PPI port, one Freeport	CTS7 214-1BD35-0X24
TrustPLC CTSC-200 CPU224I, basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB ; 24VDC, 14DI/10DO, Transistor output, 0.75A, one PPI port,	CTS7 214-1AD41-0X24

one Freeport, one EtherCAT port	
TrustPLC CTSC-200 CPU224I, basic program memory 12KB, can be extended to 16KB; Basic data memory 8KB, can be extended to 108KB ; 220VAC, 14 DI/10DO, relay output, 2A, one PPI port, one Freeport, one EtherCAT port	CTS7 214-1BD41-0X24
TrustPLC CTSC-200 CPU224XP, 72KB program memory, 10KB data memory; 24VDC, 14DI/10DO, Transistor output, 2AI/2AQ, two PPI/ freeports, one EtherCAT port, 2 outputs for 50KHz (Pulse/Dir or PTO/PWM)	CTS7 214-2AD41-0324
TrustPLC CTSC-200 CPU224XP, 72KB program memory, 10KB data memory; 220VAC, 14DI/10DO, relay output, 2AI/2AQ, two PPI/ freeports, one EtherCAT port	CTS7 214-2BD41-0324
TrustPLC CTSC-200 CPU226I, basic program memory 24KB, can be extended to 72KB; Basic data memory 10KB, can be extended to 110KB ; 24VDC, 24DI/16DO, Transistor output, 0.75A, two PPI/ freeports, one Freeport	CTS7 216-2AD41-0X40
TrustPLC CTSC-200 CPU226I, basic program memory 24KB, can be extended to 72KB; Basic data memory 10KB, can be extended to 110KB; 220VAC, 24DI/16DO, relay output, 2A, two PPI/ freeports, one Freeport	CTS7 216-2BD41-0X40
TrustPLC CTS7-200 CPU226M, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 220VAC, 14 DI/10 DO, two PPI ports, one Freeport	CTS7 216-1BD33-0X24
TrustPLC CTS7-200 CPU226M, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI/10DO × 24VDC, two PPI ports, one Freeport, support external battery and memory card	CTS7 216-1AD33-0X24
TrustPLC CTS7-200 CPU226M V5(Upgrade), basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 100KB(permanent retention); 24VDC, 14 DI/10 DO, Transistor output, two PPI/Free ports	CTS7 216-1AD35-0X24
TrustPLC CTS7-200 CPU226M V5(Upgrade), basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 100KB(permanent retention); 220VAC, 14 DI/10 DO, relay output, two PPI/Free ports	CTS7 216-1BD35-0X24
TrustPLC CTSC-200 CPU226L, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 24DI/16DO, Transistor output, 0.75A, two PPI ports, one Freeport	CTS7 216-2AD33-0X40
TrustPLC CTSC-200 CPU226L, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 24DI/16DO, Transistor output, 0.75A, one PPI port, 2 freeports (Out of production)	CTS7 216-2AF33-0X40
TrustPLC CTSC-200 CPU226L, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to	CTS7 216-2BD33-0X40

110KB; 220VAC, 24 DI/16 DO, relay outputs, 2A, two PPI ports, one Freeport	
TrustPLC CTSC-200 CPU226L, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 220VAC, 24 DI/16DO, relay outputs, 2A, two PPI ports, two freeports (Out of production)	CTS7 216-2BF33-0X40
TrustPLC CTSC-200 CPU226L V5(Upgrade), basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB(permanent retention); 24VDC, 24 DI/16 DO, Transistor output, two PPI/Free ports	CTS7 216-2AD35-0X40
TrustPLC CTSC-200 CPU226L V5(Upgrade), basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB(permanent retention); 220VAC, 24 DI/16DO, relay outputs, two PPI/Free ports	CTS7 216-2BD35-0X40
Motion control CPU	
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×24VDC, no interpolation, 10DO Transistor output, 2-axis motion control outputs, Pulse + Direction 200kHz for each axis(out of production)	CTS7 216-1AH33-0X24
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×5VDC, no interpolation, 10DO Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis (out of production)	CTS7 216-1AH33-0A24
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×24VDC, no interpolation, 10DO Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis (out of production)	CTS7 216-1AH33-0B24
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×5VDC, Line and arc interpolation, 10DO Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis (out of production)	CTS7 216-1AH33-1A24
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×24VDC, Line and arc interpolation, 10DO Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis (out of production)	CTS7 216-1AH33-1B24
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×24VDC, without interpolation, 10DO Transistor output, 2-axis motion control outputs, Pulse + Direction 200kHz for each axis (out of production)	CTS7 216-1AH34-0X24

TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×24VDC, without interpolation, 10DO Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis	CTS7 216-1AH34-0B24
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×24VDC, Line and arc interpolation, 10DO Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis	CTS7 216-1AH34-1B24
TrustPLC CTSC-200 CPU226H, basic program memory 24KB, can be extended to 72K; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI×24VDC, Line and arc interpolation, 10DO Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis, support CAN communication	CTS7 216-1AH34-2B24
TrustPLC CTSC-200 CPU226H V5(Upgrade), basic program memory 24KB, can be extended to 72KB; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI/10DO, Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis	CTS7 216-1AH35-0B24
TrustPLC CTSC-200 CPU226H V5(Upgrade), basic program memory 24KB, can be extended to 72KB; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI/10DO, Transistor output, 4-axis motion control outputs, Pulse + Direction 200kHz for each axis	CTS7 216-1AH35-2B24
TrustPLC CTSC-200 CPU226M-CAN, basic program memory 24KB, can be extended to 48KB; Basic data memory 10KB, can be extended to 110KB; 24VDC, 14DI/10DO, Transistor output, 0.75A, 2 PPI/Freeports, 2-axis motion control outputs, 2 independent high-speed counters with 30KHz, 2independent high-speed pulse output with 50KHz. Remarks: The firmware of -1 Version CPU (SMW102=16#404 and above) support 6 independent high-speed counter with 30KHz, support high-speed pulse output PTO/PWM2×20KHz or 2×50KHz PTP/Speed_ctrl etc.	CTS7 216-1AC33-0X 24
Dedicated controller	
TrustPLC CTSC-200 PSC266 dedicated controller, 72K program memory/110KB data memory, 24VDC, 20TC, 4AQ, 14DI/10DO, 3 communication ports	CTSC 266-3AT33-2424
Expansion modules	
TrustPLC CTSC-200 EM221 Digital input module, 8DI×24VDC	CTS7 221-1BF32
TrustPLC CTSC-200 EM221 Digital input module, 16DI×24VDC	CTS7 221-1BH32
TrustPLC CTSC-200 EM221 Digital input module, 32DI×24VDC	CTS7 221-1BL32
TrustPLC CTSC-200 EM221 Compact Digital input module, 32DI×24VDC (Out of production)	CTS7 221-2BL32

TrustPLC CTSC-200 EM222 Digital output module, 8DO×24VDC, Transistor output, 0.5A	CTS7 222-1BF32
TrustPLC CTSC-200 EM222 Digital output module, 16DO×24VDC, Transistor output, 0.5A	CTS7 222-1BH32
TrustPLC CTSC-200 EM222 Digital output module, 32DO×24VDC, Transistor output, 0.5A	CTS7 222-1BL32
TrustPLC CTSC-200 EM222, Digital output module, 8DO, Relay output, 2A	CTS7 222-1HF32
TrustPLC CTSC-200 EM222, Digital output module, 16DO, Relay output, 2A	CTS7 222-1HH32
TrustPLC CTSC-200 EM223, Digital input/output module, 4DI/4DO, 24VDC, Transistor output, 0.5A	CTS7 223-1BF32
TrustPLC CTSC-200 EM223 Digital input/output module, 8DI/8DO, 24VDC, Transistor output, 0.5A	CTS7 223-1BH32
TrustPLC CTSC-200 EM223, Digital input/output module, 16DI/16DO, 24VDC, Transistor output, 0.5A	CTS7 223-1BL32
TrustPLC CTSC-200 EM223, Digital input/output module, 4DI/4DO, Relay output, 2A	CTS7 223-1HF32
TrustPLC CTSC-200 EM223 Digital input/output module, 16 inputs/16 Relay outputs, 2A	CTS7 223-1PH32
TrustPLC CTSC-200 EM231 high speed and high precision analog input module 2AI×16BIT, integrated 10VDC power supply output, Sampling rate 1000HZ, Repeatability measurement error is less than 0.02%	CTS7 223-1PL32
TrustPLC CTSC-200 EM231 four-channel analog input module, 4AI×12BIT precision, Current or voltage input, Optocoupler isolation	CTS7 231-7HB32
TrustPLC CTSC-200 EM231 high speed and high precision analog input module 4AI×16BIT, integrated 10VDC power supply output, Sampling rate 500HZ, Repeatability measurement error is less than 0.05%	CTS7 231-0HC32
TrustPLC CTSC-200 EM231 thermal resistance input module, 2AI×RTD, 16BIT Isolation	CTS7 231-7HC32
TrustPLC CTSC-200 EM231 thermal resistance input module, 4AI×RTD, 16BIT Isolation	CTS7 231-7PB32
TrustPLC CTSC-200 EM231 thermocouple input module, 4AI×TC, J/K/R/S/T/E/N, 16BIT Isolation	CTS7 231-7PC32
TrustPLC CTSC-200 EM231 thermocouple input module, 8AI×TC, J/K/R/S/T/E/N, 16BIT Isolation VW addressing	CTS7 231-7PD32
TrustPLC CTSC-200 EM231 NTC input module 2 NTC or PT100, 2 voltage or current inputs, Isolation 16 bit precision	CTS7 231-7PF32
TrustPLC CTSC-200 EM231 high precision analog input module, 8 inputs, 16BIT, Voltage input, Optocoupler isolation	CTS7 231-7PF3A
TrustPLC CTSC-200 EM231 high precision analog current mode input, 8 inputs, 16BIT, Current input, Optocoupler isolation	CTS7 231-7ND32
TrustPLC CTSC-200 EM231 current mode PID module, 8 current	CTS7 231-0HF32

inputs, 0-20mA / 4-20mA, Isolation 16 bit precision, with intelligent PID	
TrustPLC CTSC-200 EM231 analog voltage input module, 8 inputs, $\pm 2.5V, \pm 5, 0\sim 10V, 0\sim 5V$ voltage input, and two channels to be selected, 0-20mA current input, Isolation 12 bit precision(AIW mapping)	CTS7 231-1HF32
TrustPLC CTSC-200 EM231 thermocouple PID module, $4AI \times TC$, 16BIT Isolation	CTS7 231-5HF32
TrustPLC CTSC-200 EM231 thermocouple PID module, $8AI \times TC$, 16BIT Isolation	CTS7 231-7HF32
TrustPLC CTSC-200 EM231 weighting module, Single sensor input , 16bit, 50Hz, 6VDC, Each outputs 150mA excitation power supply	CTS7 231-7TD32
TrustPLC CTSC-200 EM231 weighting module, 2 sensors input, 16bit, 20Hz(Single channel) / 10 (Double channel), 6VDC, Each outputs 150mA excitation power supply	CTS7 231-7TF32
TrustPLC CTSC-200 EM253 motion control module, 8 inputs/ 8 outputs, 24VDC, 2 high-speed counters, Single and double phase 200KHZ and 2 200KHz high-speed pulse output.	CTS7 231-7WA32
TrustPLC CTSC-200 EM232 analog output module, 2 outputs, Voltage/Current mode, 12BIT, Optocoupler isolation	CTS7 231-7WB32
TrustPLC CTSC-200 EM232 analog output module, 4 outputs, Voltage/Current mode, 12BIT, Optocoupler isolation	CTS7 232-0HB32
TrustPLC CTSC-200 EM235 analog input/output module, $4DI/1DO$, 12BIT, Voltage/Current mode, Optocoupler isolation	CTS7 232-0HF32
TrustPLC CTSC-200 EM277C slave station module, CAN slave station module, with $8DI/6DO$	CTS7 235-0KD32
TrustPLC CTSC-200 EM277PN slave station module, PROFINET slave station, 100Mbps, support 8 extension module, do not support EM253, weighing module, communication module etc.(special module).	CTS7 253-1BH32
TrustPLC CTSC-200 EM231 analog input module, $8AI \times Pt100/NTC$, 16BIT Isolation	CTS7 231-7NF32
TrustPLC CTSC-200 EM277A DP Slave module, 12Mbps, opto-isolation	CTS7 277-0AA32
TrustPLC CTSC-200 EM277B DP Slave module, 1.5Mbps, Photoelectric isolation, It can be used as a slave station individually.	CTS7 277-0AB32
TrustPLC CTSC-200 EM223 Digital input/output module, $8DI/8DO$, relay output, 2A	CTS7 277-0AC32
TrustPLC CTSC-200 EM231 thermocouple Input module, $8AI \times TC$, J/K/R/S/T/E/N, 16BIT isolation, AIW addressing	CTS7 277-0PN32
Expansion board	
EBH-CAN:CAN master station communication extension board, 1Mbps, optoelectronic isolation	CTH2 CAN-01S1-EB
EBH-AMS:2*12 bits precision Voltage input, 1*12 bits precision Voltage/current output	CTH2 AMS-03S1-EB

EBH-AMS:4*12 bits precision Voltage input, 2*12 bits precision voltage output	CTH2 AMS-06S1-EB
EBH-AMS:4*12 bits precision Voltage input, 2*12 bits precision Current output	CTH2 AMS-06S2-EB
One channel RS485 communication extension board	CTH2 485-01S1-EB
Fittings	
PLC lithium battery (Apply to CTSC-100 CPU, CTSC-200 CPU)	CTS7 291-8BA33
RS485 programming cable	CTS7 191-USB30
64K memory card (Apply to CTSC-200 CPU)	CTS7 291-MC064
256K memory card (Apply to CTSC-200 CPU)	CTS7 291-MC256
Bus extension wire(Out of production)	CTS7 291-BC001
EC001 encoder conversion module	CTS7 291-EC001
Portable programming card (Apply to CTSC-100 CPU, CTSC-200 CPU)	CTS7 291-PC001
Signal amplify board	CTS7 291-MS001
MagicWorks PLC remote programming module (Apply to CTSC-100 CPU, CTSC-200 CPU)(Out of production)	CTSC 191-GPRS

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